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Definite Integrals

Newton – Leibnitz formula:

Let $\frac{d}{dx}(F(x)) = f(x) \forall x \in (a, b)$. Then $\int_a^b f(x) dx = \lim_{x \rightarrow b^-} F(x) - \lim_{x \rightarrow a^+} F(x)$

Note: 1. If $a > b$, then $\int_a^b f(x) dx = \lim_{x \rightarrow b^+} F(x) - \lim_{x \rightarrow a^-} F(x)$.

2. If $F(x)$ is continuous at a and b , then $\int_a^b f(x) dx = F(b) - F(a)$.

Property (1): $\int_a^b f(x) dx = \int_a^b f(t) dt$ (Definite integral is independent of variable of integration)

Property (2): $\int_a^b f(x) dx = -\int_b^a f(x) dx$

Property (3): $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$ where $a < c < b$

Property (4): $\int_{-a}^a f(x) dx = \int_0^a f(x) + f(-x) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(-x) = f(x) \text{ i.e. } f(x) \text{ is even} \\ 0, & \text{if } f(-x) = -f(x) \text{ i.e. } f(x) \text{ is odd} \end{cases}$

Property (5): $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$ Further $\int_0^a f(x) dx = \int_0^a f(a-x) dx$.

Property (6): $\int_0^{2a} f(x) dx = \int_0^a (f(x) + f(2a-x)) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(2a-x) = f(x) \\ 0, & \text{if } f(2a-x) = -f(x) \end{cases}$

Property (7): If $f(x)$ is a periodic function with period T , then

$$(i) \int_0^{nT} f(x) dx = n \int_0^T f(x) dx, n \in \mathbb{Z} \quad (ii) \int_{mT}^{nT} f(x) dx = (n-m) \int_0^T f(x) dx, m, n \in \mathbb{Z}$$

$$(iii) \int_{nT}^{a+nT} f(x) dx = \int_0^a f(x) dx, n \in \mathbb{Z}, a \in \mathbb{R} \quad (iv) \int_{a+nT}^{b+nT} f(x) dx = \int_a^b f(x) dx, n \in \mathbb{Z}, a, b \in \mathbb{R}$$

Property (8): If $\psi(x) \leq f(x) \leq \phi(x)$ for $a \leq x \leq b$, then $\int_a^b \psi(x) dx \leq \int_a^b f(x) dx \leq \int_a^b \phi(x) dx$

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Property (9): If $m \leq f(x) \leq M$ for $a \leq x \leq b$, then $m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$

If $f(x)$ is monotonically decreasing in (a, b) , then

$$f(b)(b-a) < \int_a^b f(x) dx < f(a)(b-a) \text{ and}$$

If $f(x)$ is monotonically increasing in (a, b) , then

$$f(a)(b-a) < \int_a^b f(x) dx < f(b)(b-a)$$

Property (10): $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$ and

$$\left| \int_a^b f(x)g(x) dx \right| \leq \left(\int_a^b f^2(x) dx \right)^{1/2} \left(\int_a^b g^2(x) dx \right)^{1/2}$$

Property (11): If $f(x) \geq 0$ on $[a, b]$, then $\int_a^b f(x) dx \geq 0$.

Property (12): Leibnitz Theorem:

$$\text{If } F(x) = \int_{g(x)}^{h(x)} f(t) dt, \text{ then } \frac{dF(x)}{dx} = f(h(x))h'(x) - f(g(x))g'(x)$$

Definite Integral as a limit of sum:

$$\int_0^1 f(x) dx = \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \frac{1}{n} f\left(\frac{r}{n}\right). \quad \lim_{n \rightarrow \infty} \sum_{r=1}^{pn} \frac{1}{n} f\left(\frac{r}{n}\right) = \int_0^p f(x) dx$$

Reduction formulae in definite Integrals:

$$1) \quad I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} \cos^n x dx \Rightarrow I_n = \left(\frac{n-1}{n} \right) \left(\frac{n-3}{n-2} \right) \left(\frac{n-5}{n-4} \right) \dots I_0 \text{ or } I_1$$

According as n is even or odd. $I_0 = \frac{\pi}{2}, I_1 = 1$

$$\text{Hence } I_n = \begin{cases} \left(\frac{n-1}{n} \right) \left(\frac{n-3}{n-2} \right) \left(\frac{n-5}{n-4} \right) \dots \left(\frac{1}{2} \right) \cdot \frac{\pi}{2}, & \text{if } n \text{ is even} \\ \left(\frac{n-1}{n} \right) \left(\frac{n-3}{n-2} \right) \left(\frac{n-5}{n-4} \right) \dots \left(\frac{2}{3} \right) \cdot 1, & \text{if } n \text{ is odd} \end{cases}$$

$$2. \quad \text{If } I_{m,n} = \int_0^{\frac{\pi}{2}} \sin^m x \cos^n x dx, \text{ then}$$

$$I_{m,n} = \begin{cases} \frac{(m-1)(m-3)(m-5)\dots\dots 2}{(m+n)(m+n-2)(m+n-4)\dots\dots} \cdot I_n, & \text{if } m \text{ is an odd} \\ \frac{(m-1)(m-3)(m-5)\dots\dots 3}{(m+n)(m+n-2)(m+n-4)\dots\dots} \cdot \left(\frac{1}{n+1}\right), & \text{if } m \text{ is an even} \end{cases}$$

$$3. \quad I_{m,n} = \int_0^1 x^m (1-x)^n dx = \frac{m!n!}{(m+n+1)!}$$

Important Formula:

$$1) \int_0^{\infty} e^{-ax} \sin(bx) dx = \frac{b}{a^2 + b^2}$$

$$2) \int_0^{\infty} e^{-ax} \cos(bx) dx = \frac{a}{a^2 + b^2}$$

$$3) \int_0^{\infty} e^{-ax} x^n dx = \frac{n!}{a^{n+1}}$$

$$4) \int_{\alpha}^{\beta} \frac{dx}{\sqrt{(x-\alpha)(\beta-x)}} = \pi \text{ where } \beta > \alpha$$

$$5) \int_{\alpha}^{\beta} \sqrt{(x-\alpha)(\beta-x)} dx = \frac{\pi}{8} (\beta-\alpha)^2$$

$$6) \int_{\alpha}^{\beta} \sqrt{\frac{x-\alpha}{\beta-x}} dx = \frac{\pi}{2} (\beta-\alpha)$$

$$7) \int_0^{\pi/2} \frac{dx}{\sin x + \cos x} = \sqrt{2} \log(\sqrt{2} + 1)$$

$$8) \int_0^{\pi/2} \frac{dx}{a^2 \sin^2 x + b^2 \cos^2 x} = \frac{\pi}{2ab}$$

$$10) \int_0^{\pi/2} \log \sin x dx = \int_0^{\pi/2} \log \sin 2x dx = \int_0^{\pi/2} \log \cos x dx = -\left(\frac{\pi}{2}\right) \ln 2$$

$$11) \int_0^n [x] dx = \frac{n(n-1)}{2}$$

$$13) \int_{-n}^n [x] dx = -n$$

$$14) \int_0^n \{x\} dx = \frac{n}{2}$$

$$15) \int_0^{n^2} [\sqrt{x}] dx = \frac{n(n-1)(4n+1)}{6}$$

$$16) \int_0^n (x) dx = \frac{n(n+1)}{2}, (.) \text{ is L.I.F}$$

$$17) \int_0^{\infty} \frac{dx}{(x + \sqrt{x^2 + 1})^n} = \int_0^{\frac{\pi}{2}} \frac{\sec^2 x dx}{(\sec x + \tan x)^n} = \frac{n}{n^2 - 1}$$

$$18) \int_0^1 \frac{\log(1+x) dx}{x^2 + 1} = \int_0^{\frac{\pi}{4}} \log(1 + \tan \theta) d\theta = \frac{\pi}{8} \ln 2$$

EXERCISE-I

(Single answer correct type questions)

Topic: Introduction to Definite integration

Sub Topic 1: Definite integral as a limit of an infinite Riemann Sum

1. $\lim_{n \rightarrow \infty} \left(\frac{n}{n^2 + 1^2} + \frac{n}{n^2 + 2^2} + \frac{n}{n^2 + 3^2} + \dots + \frac{1}{5n} \right)$ is equal to:

- 1) $\frac{\pi}{4}$ 2) $\tan^{-1}(3)$ 3) $\frac{\pi}{2}$ 4) $\tan^{-1}(2)$

Key: 4

(Limits using definite integration)2019

Sol: $\lim_{n \rightarrow \infty} \frac{1}{x} \sum_{r=1}^{2x} \frac{1}{1 + \left(\frac{r}{n}\right)^2} = \int_0^2 \frac{1}{1 + x^2} dx = \left[\tan^{-1} x \right]_0^2 = \tan^{-1}(2)$

2. $\lim_{x \rightarrow \infty} \left[\frac{n^2}{(n^2 + 1)(n + 1)} + \frac{n^2}{(n^2 + 4)(n + 2)} + \frac{n^2}{(n^2 + 9)(n + 3)} + \dots + \frac{n^2}{(n^2 + n^2)(n + n)} \right]$ is equal to

- 1) $\frac{\pi}{8} + \frac{1}{4} \log_e 2$ 2) $\frac{\pi}{4} + \frac{1}{8} \log_e 2$ 3) $\frac{\pi}{4} - \frac{1}{8} \log_e 2$ 4) $\frac{\pi}{8} + \log_e \sqrt{2}$

Key: 1

(Limit of a sum in Definite)2021

Sol: $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{n^2}{(n^2 + r^2)(n + r)} = \int_0^1 \frac{dx}{(1 + x^2)(1 + x)}$

$$\frac{1}{(1 + x^2)(1 + x)} = \frac{A}{1 + x} + \frac{Bx + C}{1 + x^2} = \frac{1}{2(1 + x^2)} - \frac{2x}{4(1 + x^2)}$$

$$\therefore \int_0^1 \frac{dx}{(1 + x^2)(1 + x)} = \frac{1}{2} \ln 2 + \frac{\pi}{8} - \frac{1}{4} \ln 2 = \frac{\pi}{8} + \frac{1}{4} \ln 2$$

3. $\lim_{n \rightarrow \infty} \left[\frac{1}{n} + \frac{n}{(n + 1)^2} + \frac{n}{(n + 2)^2} + \dots + \frac{n}{(2n - 1)^2} \right]$ is equal to

- 1) $\frac{1}{4}$ 2) $\frac{1}{2}$ 3) 1 4) $\frac{1}{3}$

Ans:2

(Limit of a sum in Definite)2021

Sol $\lim_{n \rightarrow \infty} \left(\frac{1}{n} + \frac{n}{(n + 1)^2} + \frac{n}{(n + 2)^2} + \dots + \frac{n}{(2n - 1)^2} \right)$

$$\begin{aligned} & \lim_{n \rightarrow \infty} \frac{1}{n^2} \left\{ \sum_{r=0}^{n-1} \frac{1}{\left(1 + \frac{r}{n}\right)^2} \right\} \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{n-1} \frac{1}{\left(1 + \frac{r}{n}\right)^2} = \int_0^1 \frac{1}{(1+x)^2} dx = \left(\frac{-1}{1+x} \right)_0^1 = \frac{-1}{2} + 1 = \frac{1}{2} \end{aligned}$$

4. If $f: R \rightarrow R$ is given by $f(x) = x + 1$, then the value of

$$\lim_{n \rightarrow \infty} \frac{1}{n} \left[f(0) + f\left(\frac{5}{n}\right) + f\left(\frac{10}{n}\right) + \dots + f\left(\frac{5(n-1)}{n}\right) \right], \text{ is :}$$

- 1) $\frac{1}{2}$ 2) $\frac{5}{2}$ 3) $\frac{3}{2}$ 4) $\frac{7}{2}$

(Limit of a sum in Definite) 2021

Key: 4

Sol: $f(x) = x + 1$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \left[f(0) + f\left(\frac{5}{n}\right) + f\left(\frac{10}{n}\right) + \dots + f\left(\frac{5(n-1)}{n}\right) \right]$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \left[1 + \frac{5}{n} + 1 + \frac{10}{n} + 1 + \dots + \frac{5(n-1)}{n} + 1 \right]$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \left[n + 5 \left(\frac{1}{n} + \frac{2}{n} + \frac{3}{n} + \dots + \frac{n-1}{n} \right) \right]$$

$$\lim_{n \rightarrow \infty} 1 + 5 \frac{(n-1)n}{n^2 \times 2}$$

$$1 + \frac{5}{2} = \frac{7}{2}$$

5. The value of $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \frac{(2j-1)+8n}{(2j-1)+4n}$ is equal to

- 1) $3 + 2 \log_e \left(\frac{2}{3} \right)$ 2) $5 + \log_e \left(\frac{3}{2} \right)$ 3) $1 + 2 \log_e \left(\frac{3}{2} \right)$ 4) $2 - \log_e \left(\frac{2}{3} \right)$

Key: 3

(Limit of a sum in Definite) 2021

Sol: $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \left[\frac{(2j-1)+8n}{(2j-1)+4n} \right]$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \left[\frac{\left(\frac{2j}{n} - \frac{1}{n} + 8 \right)}{\left(\frac{2j}{n} - \frac{1}{n} + 4 \right)} \right]$$

$$= \int_0^1 \frac{2x+8}{2x+4} dx$$

$$= \int_0^1 dx + \int_0^1 \frac{4}{2x+4} dx$$

$$= 1 + 4 \cdot \frac{1}{2} [\ln(2x+4)]_0^1$$

$$= 1 + 2 \left[\ln \frac{6}{4} \right] = 1 + 2 \ln \left(\frac{3}{2} \right)$$

6. The value of $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{2n-1} \frac{n^2}{n^2 + 4r^2}$ is:

- 1) $\frac{1}{2} \tan^{-1}(2)$ 2) $\frac{1}{4} \tan^{-1}(4)$ 3) $\frac{1}{2} \tan^{-1}(4)$ 4) $\tan^{-1}(4)$

Key: 3

(Limit of a sum in Definite)2021

Sol: $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{2n-1} \frac{n^2}{n^2 + 4r^2}$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{2n-1} \frac{1}{1 + 4 \left(\frac{r}{n} \right)^2}$$

$$= \int_0^2 \frac{1}{1 + 4x^2} dx = \frac{1}{2} \tan^{-1}(2x) = \frac{1}{2} \tan^{-1}(4) - 0 = \frac{1}{2} \tan^{-1}(4)$$

7. If $U_n = \left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right)^2 \dots \left(1 + \frac{n^2}{n^2}\right)^n$, then $\lim_{n \rightarrow \infty} (U_n)^{\frac{-4}{n^2}}$ is equal to :

- 1) $\frac{16}{e^2}$ 2) $\frac{e^2}{16}$ 3) $\frac{4}{e^2}$ 4) $\frac{4}{e}$

Key: 2

(Limit of a sum in Definite)2021

$$\text{Sol: } -\frac{4}{n^2} \log U_n = \frac{-4}{n^2} \left[\ln \left(1 + \frac{1}{n^2} \right) + 2 \ln \left(1 + \frac{2^2}{n^2} \right) + \dots + n \ln \left(1 + \frac{n}{n^2} \right) \right]$$

$$= -\frac{4}{n} \left[\sum \frac{r}{n} \ln \left(1 + \frac{r^2}{n^2} \right) \right]$$

$$= -\frac{4}{n} \left[\int_0^1 x \ln(1+x^2) dx \right]$$

$$= -4 \left\{ \frac{1}{2} \cdot \int_1^2 \ln t dt - 2 [t \ln t - t]_1^2 \right\}$$

$$= -2 [(2 \ln 2 - 2) - (1 \ln 1 - 1)]$$

$$= -2 [\ln 4 - 2 + 1]$$

$$= -2 [\ln 4 - 1]$$

$$= 2 - \ln 16 = \ln_e x^2 - \ln_e 16 = \ln \left(\frac{e^2}{16} \right)$$

$$\therefore L = \frac{e^2}{16}$$

$$8. \lim_{x \rightarrow \infty} \left[\frac{n^2}{(n^2+1)(n+1)} + \frac{n^2}{(n^2+4)(n+2)} + \frac{n^2}{(n^2+9)(n+3)} + \dots + \frac{n^2}{(n^2+n^2)(n+n)} \right] \text{ is equal to}$$

$$1) \frac{\pi}{8} + \frac{1}{4} \log_e 2$$

$$2) \frac{\pi}{4} + \frac{1}{8} \log_e 2$$

$$3) \frac{\pi}{4} - \frac{1}{8} \log_e 2$$

$$4) \frac{\pi}{8} + \log_e \sqrt{2}$$

Key: 1

(Limit of Sumation) 2022

$$\text{Sol: } \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{n^2}{(n^2+r^2)(n+r)} = \int_0^1 \frac{dx}{(1+x^2)(1+x)}$$

$$\frac{1}{(1+x^2)(1+x)} = \frac{A}{1+x} + \frac{Bx+C}{1+x^2} = \frac{1}{2(1+x^2)} - \frac{2x}{4(1+x^2)}$$

$$\therefore \int_0^1 \frac{dx}{(1+x^2)(1+x)} = \frac{1}{2} \ln 2 + \frac{\pi}{8} - \frac{1}{4} \ln 2 = \frac{\pi}{8} + \frac{1}{4} \ln 2$$

9. $\lim_{n \rightarrow \infty} \left[\frac{1}{1+n} + \frac{2}{2+n} + \frac{1}{3+n} + \dots + \frac{1}{2n} \right]$ is equal to

- 1) $\log_e 2$ 2) $\log_e \left(\frac{2}{3} \right)$ 3) 0 4) $\log_e \left(\frac{3}{2} \right)$

Key: 1

(limit of a sumation)2023

SOL: $\lim_{n \rightarrow \infty} \left[\frac{1}{1+n} + \frac{2}{2+n} + \frac{1}{3+n} + \dots + \frac{1}{2n} \right]$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{1}{r+n} \right) = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} \left(\frac{1}{\frac{r}{n} + 1} \right) = \int_0^1 \frac{dx}{x+1} = \log_e 2$$

10. Among $(S_1): \lim_{n \rightarrow \infty} \frac{1}{n^2} (2 + 4 + 6 + \dots + 2n) = 1$

$(S_2): \lim_{n \rightarrow \infty} \frac{1}{n^{16}} (1^{15} + 2^{15} + 3^{15} + \dots + n^{15}) = \frac{1}{16}$

- 1) Only (S_1) is true 2) Both (S_1) and (S_2) are false
3) Both (S_1) and (S_2) are true 4) Only (S_2) is true

KEY:

3

(limit of a sumation)2023

SOL:

$$S_1: \lim_{n \rightarrow \infty} \frac{1}{n^2} (2 + 4 + 6 + \dots + 2n) = \lim_{n \rightarrow \infty} \frac{1}{n^2} \cdot \frac{n(n+1)}{2} = 1$$

$$S_2: \lim_{n \rightarrow \infty} \frac{1}{n^{16}} (1^{15} + 2^{15} + \dots + n^{15}) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^{15} = \int_0^1 x^{15} dx = \frac{1}{16}$$

11. $\lim_{n \rightarrow \infty} \left[\frac{1}{1+\sqrt{n}} + \frac{1}{2+\sqrt{2n}} + \frac{1}{3+\sqrt{3n}} + \dots + \frac{1}{n+\sqrt{n^2}} \right]$ is,

- 1) 0 2) $\ln 3$ 3) $\ln 4$ 4) $\ln 2$

Key: 3

(Limits using definite integration)

Sol: $L = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{r + \sqrt{rn}} = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} \left(\frac{1}{\frac{r}{n} + \sqrt{\frac{r}{n}}} \right) \Rightarrow \int_0^1 \frac{dx}{x + \sqrt{x}} = \ln 4.$

12. Suppose $S = \frac{1}{n^4} \prod_{r=1}^{2n} (n^2 + r^2)^{1/n}$, then $\lim_{n \rightarrow \infty} \log_e S$ is equal to

- 1) $\int_0^2 \log_e (2 + x^2) dx$ 2) $\int_0^1 \log_e (1 + x^2) dx$
3) $2 \int_0^1 \log_e (1 + x^2) dx$ 4) $\int_0^2 \log_e (x^2 - 4x + 5) dx$

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Key: 4

(Limits using definite integration)

Hint: $S = \frac{1}{n^4} \prod_{r=1}^{2n} (n^2 + r^2)^{1/n}$

$$\log S = \sum_{r=1}^{2n} \frac{1}{n} \log(n^2 + r^2) - 4 \log n = \frac{1}{n} \sum \log \left(n^2 \left(1 + \left(\frac{r}{n} \right)^2 \right) \right) - 4 \log n$$

$$= \frac{1}{n} \left[\sum_{r=1}^{2n} 2 \log n + \sum_{r=1}^{2n} \log \left(1 + \left(\frac{r}{n} \right)^2 \right) \right] - 4 \log(n) = 4 \log(n) + \frac{1}{n} \sum_{r=1}^{2n} \log \left(1 + \left(\frac{r}{n} \right)^2 \right) - 4 \log(n)$$

$$\therefore \lim_{n \rightarrow \infty} \log S = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \log \left(1 + \left(\frac{r}{n} \right)^2 \right)$$

$$= \int_0^2 \log(1+x^2) dx = \int_0^2 \log(x^2 - 4x + 5) dx \quad \left(\because \int_0^a f(x) dx = \int_0^a f(a-x) dx \right)$$

13. If $a = \lim_{n \rightarrow \infty} \left(\frac{(2n)!}{n! n^n} \right)^{\frac{1}{n}}$ and $b = \lim_{n \rightarrow \infty} \left(\left(1 + \frac{1}{n^2} \right)^{\frac{2}{n^2}} \left(1 + \frac{2^2}{n^2} \right)^{\frac{4}{n^2}} \dots - 2^{\frac{2}{n}} \right)$ then $a+b =$

1) $\frac{8}{e}$

2) $\frac{8}{e}$

3) $\frac{2}{e}$

4) $\frac{4}{e}$

Key: 2

(Limits using definite integration)

Sol: $\ln a = \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \frac{1}{n} \ln \left(2 - \frac{r}{n} \right) = \int_0^1 \ln(2-x) dx = \ln 4 - 1 \Rightarrow a = \frac{4}{e}$

$$\ln b = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} \ln \left(1 + \left(\frac{r}{n} \right)^2 \right) = \int_0^1 2x \ln(1+x^2) dx = \ln \frac{4}{e} \Rightarrow b = \frac{4}{e}$$

14. The value of the $\lim_{n \rightarrow \infty} \frac{[(n+1)(n+2) \dots (n+n)]^{1/n}}{\pi}$ is (Limit of a sum in Definite)

A) $\frac{1}{e}$

B) $\frac{2}{e}$

C) $\frac{4}{e}$

D) $\frac{8}{e}$

ANS: C

SOL: let L represent the given limit. We have,

$$\ln L = \frac{1}{n} \ln \left\{ \left(1 + \frac{1}{n} \right) \left(1 + \frac{2}{n} \right) \dots \left(1 + \frac{n}{n} \right) \right\} = \frac{1}{n} \sum_{r=1}^n \ln \left(1 + \frac{r}{n} \right)$$

$$\text{Thus, } \ln L = \int_0^1 \ln(1+x) dx$$

$$= x \ln(1+x) \Big|_0^1 - \int_0^1 \frac{x}{1+x} dx = \ln 2 - \int_0^1 \left(\frac{x+1-1}{1+x} \right) dx$$

$$= \ln 2 - 1 + \ln 2 = 2 \ln 2 - 1 = \ln \left(\frac{4}{e} \right)$$

This implies that

$$L = \frac{4}{e}$$

15. The value of $\lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{6n} \right)$

A) $\log_e 6$

B) $\log_6 e$

C) $\log_e 5$

D) $\log_5 e$

ANS: A

(Limit of a sum in Definite)

SOL: $\lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{6n} \right)$

$$= \lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+5n} \right)$$

$$\sum_{r=0}^{5n} \left(\frac{1}{n+r} \right) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{5n} \left(\frac{1}{1 + \frac{r}{n}} \right)$$

$$\int_0^5 \frac{1}{1+x} dx = \left| \ln(1+x) \right|_0^5$$

16. The value of $\lim_{n \rightarrow \infty} \frac{1^r + 2^r + 3^r + \dots + n^r}{n^{r+1}} (r > -1)$ is

A) $\frac{1}{r}$

B) $\frac{1}{r+1}$

C) $\frac{1}{r-1}$

D) r

ANS: B

(Limit of a sum in Definite)

SOL: $\lim_{n \rightarrow \infty} \frac{1^r + 2^r + 3^r + \dots + n^r}{n^{r+1}} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left(\frac{k}{n} \right)^r \cdot \frac{1}{n}$

$$= \int_0^1 x^r dx = \frac{1}{r+1}, r > -1$$

17. The value of $\lim_{n \rightarrow \infty} \frac{1}{n} \left(\frac{2n!}{n!} \right)^{\frac{1}{n}} =$ (Limit of a sum in Definite)

A) $\ln \frac{4}{e}$

B) $\frac{4}{e}$

C) $\frac{2}{e}$

D) $\frac{e}{4}$

ANS: B

SOL: Let $L = \lim_{n \rightarrow \infty} \left[\frac{n+1}{n} \frac{n+2}{n} \dots \frac{n+n}{n} \right]^{1/n}$

$$= \lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n} \right) \left(1 + \frac{2}{n} \right) \left(1 + \frac{3}{n} \right) \dots \left(1 + \frac{n}{n} \right) \right]^{1/n}$$

$$\begin{aligned}
 \ln L &= \lim_{n \rightarrow \infty} \frac{1}{n} \left[\ln \left(1 + \frac{1}{n} \right) + \ln \left(1 + \frac{2}{n} \right) + \dots + \ln \left(1 + \frac{n}{n} \right) \right] \\
 &= \lim_{x \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \ln \left(1 + \frac{r}{n} \right) = \int_0^1 \ln(1+x) dx \\
 &= \left[x \ln(1+x) \right]_0^1 - \int_0^1 \frac{x}{x+1} dx \\
 &= \ln 2 - \int_0^1 \left(1 - \frac{1}{1+x} \right) dx = \ln 2 - 1 + \ln 2 = \ln \frac{4}{e} \\
 \Rightarrow L &= \frac{4}{e}
 \end{aligned}$$

18. $\lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{4n^2-1}} + \frac{1}{\sqrt{4n^2-4}} + \frac{1}{\sqrt{4n^2-9}} + \dots + \frac{1}{\sqrt{3n^2}} \right] =$ (Limit of a sum in Definite)

A) $\frac{\pi}{3}$ B) $\frac{\pi}{2}$ C) $\frac{\pi}{6}$ D) $\frac{\pi}{4}$

ANS:C

SOL: $L = \lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{4n^2-1}} + \frac{1}{\sqrt{4n^2-4}} + \frac{1}{\sqrt{4n^2-9}} + \dots + \frac{1}{\sqrt{3n^2}} \right]$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{\sqrt{4n^2-r^2}} = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} \frac{1}{\sqrt{4-\frac{r^2}{n^2}}}$$

$$\therefore L = \int_0^1 \frac{dx}{\sqrt{4-x^2}} = \sin^{-1} \frac{x}{2} \Big|_0^1 = \frac{\pi}{6}$$

19. The value of $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{n^2}{(n^2+r^2)^{3/2}}$ is (Limit of a sum in Definite)

A) $\frac{1}{2}$ B) $-\frac{1}{2}$ C) $\frac{1}{\sqrt{2}}$ D) $-\frac{1}{\sqrt{2}}$

ANS:C

SOL: $= \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{n^2}{(n^2+r^2)^{3/2}} = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} \frac{1}{\left(1+\frac{r^2}{n^2}\right)^{3/2}}$

$$= \int_0^1 \frac{1}{(1+x^2)^{3/2}} dx = \int_0^{\frac{\pi}{4}} \left(\frac{\sec^2 t}{\sec^3 t} \right) dt = \frac{1}{\sqrt{2}} \text{ where } \sec t = x$$

20. The value of $\lim_{n \rightarrow \infty} \frac{1}{n} \left[\tan 0 + \tan \frac{\pi}{4n} + \tan \frac{2\pi}{4n} + \dots + \tan \frac{\pi}{4} \right]$ is (Limit of a sum in Definite)

A) $\ln 2$ B) $\frac{1}{\pi} \ln 2$ C) $\frac{3}{\pi} \ln 2$ D) $\frac{2}{\pi} \ln 2$

ANS:D

SOL: $\lim_{n \rightarrow \infty} \frac{1}{n} \left[\tan 0 + \tan \frac{\pi}{4n} + \tan \frac{2\pi}{4n} + \dots + \tan \frac{\pi}{4} \right]$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^n \tan\left(\frac{\pi}{4} \cdot \frac{r}{n}\right) = \int_0^1 \tan \frac{\pi}{4} x dx = \frac{2}{\pi} \ln 2$$

21. $\lim_{n \rightarrow \infty} \frac{(n!)^{1/n}}{n}$ is equal to

1) e

2) $\frac{1}{e}$

3) $e-1$

4) $e-5$

Key: 2 (Limit of a sum in Definite)

Sol: Let $y = \lim_{n \rightarrow \infty} \frac{(n!)^{1/n}}{n} = \lim_{n \rightarrow \infty} \left(\frac{n!}{n^n}\right)^{1/n}$

$$\Rightarrow \log y = \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \log \left[\frac{1}{n} \cdot \frac{2}{n} \cdot \frac{3}{n} \cdots \frac{n}{n} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left[\log \frac{1}{n} + \log \frac{2}{n} + \dots + \log \frac{n}{n} \right]$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} \log \frac{r}{n} = \int_0^1 \log x dx$$

22. $\lim_{n \rightarrow \infty} \left\{ \left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right) \cdots \left(1 + \frac{n^2}{n^2}\right) \right\}^{1/n} =$

1) $e^{(\pi-4)/2}$

2) $2e^{(\pi-4)/2}$

3) $\frac{e^{(\pi-4)/2}}{2}$

4) $e^{(\pi-4)}$

Key: 2 (Limit of a sum in Definite)

Sol: Given series = $e^{\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \log(1 + \frac{r^2}{n^2})}$

23. Let $f(x) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{n^2 + k^2 x^2}$ then $\lim_{x \rightarrow 0} f(x)$ equals (definite integral as a limit of sum)

1) 1

2) e

3) e^2

4) e^{-2}

Key: 1

Sol: $f(x) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{n^2 + k^2 x^2}$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{1}{1 + \left(\frac{kx}{n}\right)^2} = \int_0^1 \frac{dy}{1 + x^2 y^2}$$

$$= \frac{1}{x^2} \cdot x \cdot \tan^{-1} yx \Big|_0^1 = \frac{\tan^{-1} x}{x}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x} = 1.$$

24. The value of $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n-k}{n^2} \cos \frac{4k}{n}$ is: (definite integral as a limit of sum)

1) $\frac{1}{4} \sin 4 + \frac{1}{16} \cos 4 - \frac{1}{16}$

2) $\frac{1}{4} \sin 4 - \frac{1}{16} \cos 4 + \frac{1}{16}$

3) $\frac{1}{16} (1 - \sin 4)$

4) $\frac{1}{16} (1 - \cos 4)$

Key: 4

Sol: $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n-k}{n^2} \cos \frac{4k}{n}$

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{\left(1 - \frac{k}{n}\right)}{n} \cos \left(\frac{4k}{n}\right) = \int_0^1 (1-x) \cos 4x dx$$

$$= -\frac{1}{16} (\cos 4 - 1) = \frac{1}{16} (1 - \cos 4)$$

25. Evaluate $\lim_{n \rightarrow \infty} \left(\frac{1}{n^3 + 1^3} + \frac{2^2}{n^3 + 2^3} + \frac{3^2}{n^3 + 3^3} + \dots + \frac{4}{9n} \right)$ (definite integral as a limit of sum)

1) $\frac{1}{3} \ln 3$

2) $\frac{\ln 9}{3}$

3) $\frac{\ln 4}{3}$

4) $\frac{\ln 6}{3}$

Key: 2

Sol: $\lim_{n \rightarrow \infty} \sum_{r=1}^{2n} \frac{\left(\frac{r}{n}\right)^2}{1 + \left(\frac{r}{n}\right)^3} = \int_0^2 \frac{x^2 dx}{1 + x^3} = \left| \frac{1}{3} \ln |1 + x^3| \right|_0^2 = \frac{1}{3} \ln 9$

26. Consider a parabola $y = \frac{x^2}{4}$ and the point $F(0,1)$.

Let $A_1(x_1, y_1), A_2(x_2, y_2), A_3(x_3, y_3), \dots, A_n(x_n, y_n)$ are 'n' points on the parabola such $x_k > 0$

and $\angle OFA_k = \frac{k\pi}{2n} (k=1, 2, 3, \dots, n)$. Then the value of $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n FA_k$ is equal to

1) $\frac{2}{\pi}$

2) $\frac{4}{\pi}$

(definite integration as limit of sum)

3) $\frac{8}{\pi}$

4) $\frac{23}{\pi}$

Key: 2

Sol: Conceptual

27. $\lim_{n \rightarrow \infty} \frac{1}{n^3} \left(\sqrt{n^2 + 1} + 2\sqrt{n^2 + 2^2} + \dots + n\sqrt{n^2 + n^2} \right) =$ (definite integration as limit of sum)

$$1) \frac{3\sqrt{2}-1}{2}$$

$$\frac{4\sqrt{2}-1}{2}$$

$$2) \frac{2\sqrt{2}-1}{3}$$

$$3) \frac{3\sqrt{2}-1}{3}$$

4)

Key: 2

$$\text{Sol: } \lim_{x \rightarrow n} \sum_{r=1}^n \left(\frac{r\sqrt{n^2+r^2}}{n^2} \right) = \lim_{x \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \frac{r}{n} \sqrt{1 + \left(\frac{r}{n}\right)^2} = \int_0^1 x \sqrt{1+x^2} dx = \left[\frac{(1-x^2)^{3/2}}{3} \right]_0^1$$

28. The value of $\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{k}{n^2 + n + 2k} \right) =$ (definite integration as limit of sum)

$$1) \frac{1}{4}$$

$$2) \frac{1}{3}$$

$$3) \frac{1}{2}$$

4) 1

Key: 3

$$\text{Sol: } \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{k}{n^2 + n + 2k} \right) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left(\frac{\frac{k}{n}}{1 + \frac{1}{n} + \frac{2k}{n^2}} \right) = \int_0^1 x dx$$

29. $\lim_{n \rightarrow \infty} \left(\frac{(n+1)(n+2)\dots 3n}{n^{2n}} \right)^{1/n}$ is equal to (Limit of a summation)

$$1. 3 \log 3 - 2$$

$$2. \frac{18}{e^4}$$

$$3. \frac{27}{e^2}$$

$$4. \frac{9}{e^2}$$

Ans: 3

$$\text{Sol: Given expression } e^{\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \ln \left(1 + \frac{r}{n} \right)} = e^{\int_0^2 \ln(1+x) dx} = \frac{27}{e^2}$$

30. A function f is continuous for all x (and not everywhere zero) such that

$$f^2(x) = \int_0^x f(t) \frac{\cos t}{2 + \sin t} dt, \text{ then } f(x) \text{ is (Leibnitz's rule)}$$

$$A) \frac{1}{2} \ln \left(\frac{x + \cos x}{2} \right); x \neq 0$$

$$B) \frac{1}{2} \ln \left(\frac{3}{2 + \cos x} \right); x \neq 0$$

$$C) \frac{1}{2} \ln \left(\frac{2 + \sin x}{2} \right); x \neq n\pi, n \in \mathbb{Z}$$

$$D) \frac{\cos x + \sin x}{2 + \sin x}; x \neq n\pi + \frac{3\pi}{4}, n \in \mathbb{Z}$$

Key: C

$$\text{Sol: } f^2(x) = \int_0^x f(t) \frac{\cos t}{2 + \sin t} dt$$

Put $x = 0$, then $2f(0) = \log 2 + \log c$, or

31. $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \sin^{2k} \frac{r\pi}{2n}$ is equal to **(Definite Integral limit of a function)**
- A) $\frac{2k!}{2^{2k} (k!)^2}$ B) $\frac{2k!}{2^k (k!)}$ C) $\frac{2k!}{2^k (k!)^2}$ D) none of these

Ans:A

Sol: $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \sin^{2k} \frac{r\pi}{2n}$

$$= \int_0^1 \sin^{2k} \frac{\pi x}{2} dx = \frac{2}{\pi} \int_0^{\pi/2} \sin^{2k} t dt$$

$$\left(\text{putting } \frac{\pi x}{2} = t \Rightarrow dx = \frac{2}{\pi} dt \right)$$

$$= \frac{2}{\pi} \cdot \frac{(2K-1)(2K-3)\dots\dots\dots 1}{2K(2K-2)\dots\dots\dots 2} \cdot \frac{\pi}{2}$$

$$= \frac{(2k)!}{2^{2K} (k!)^2}$$

32. $\lim_{n \rightarrow \infty} \frac{(1^2 + 2^2 + 3^2 + \dots + n^2)(1^3 + 2^3 + 3^3 + \dots + n^3)}{(1^6 + 2^6 + 3^6 + \dots + n^6)}$ **(Substitution)**
- A) $\frac{7}{12}$ B) $\frac{12}{7}$ C) $\frac{5}{12}$ D) none of these

Key:C

Sol: conceptual

Topic: Introduction to Definite integration

Sub Topic 2: Second Fundamental theorem of integral calculus - definition

33. A value of α such that $\int_{\alpha}^{\alpha+1} \frac{dx}{(x+\alpha)(x+\alpha+1)} = \log_e \left(\frac{9}{8} \right)$ is

- 1) -2 2) $-\frac{1}{2}$ 3) 2 4) $\frac{1}{2}$

Ans.1

(Partial Fractions) 2019

Sol. $\left(\log \frac{x+\alpha}{x+\alpha+1} \right)_{\alpha}^{\alpha+1} = \log \left(\frac{9}{8} \right) \log \frac{(2\alpha+1)}{4\alpha(\alpha+1)} = \log \left(\frac{9}{8} \right) \Rightarrow \alpha = -2.$

34. The integral $\int_1^2 e^x \cdot x^x (2 + \log_e x) dx$ equal

- 1) $4e^2 - 1$ 2) $e(4e+1)$ 3) $e(2e-1)$ 4) $e(4e-1)$

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Key: 4 (Substitution Method)2020

$$\text{Sol: } I = \int_1^2 e^x x^x (2 + \log_e x) dx = \int_1^2 (xe)^x (2 + \log_e x) dx$$

$$\text{Put } (xe)^x = t \Rightarrow x(\log x + 1) = \log t$$

$$\Rightarrow \left(x \frac{1}{x} + (\log x + 1)1 \right) dx = \frac{1}{t} dt$$

$$\Rightarrow (xe)^x (2 + \log x) dx = dt$$

$$x = 1 \Rightarrow t = e$$

$$x = 2 \Rightarrow t = (2e)^2 \quad \therefore I = \int_e^{4e^2} dt = 4e^2 - e = e(4e - 1)$$

35. If $I_1 = \int_0^1 (1 - x^{50})^{100} dx$ and $I_2 = \int_0^1 (1 - x^{50})^{101} dx$ such that $I_2 = \alpha I_1$ then α equals to :

- 1) $\frac{5050}{5051}$ 2) $\frac{5051}{5050}$ 3) $\frac{5049}{5050}$ 4) $\frac{5050}{5049}$

Key: 1 (Substitution Method)2020

$$\text{Sol: } I_2 = \int_0^1 (1 - x^{50})^{100} dx$$

$$= \left((1 - x^{50})^{101} x \right)_0^1 - 0 \int_0^1 101(1 - x^{50})^{100} (-50x^{49}) dx$$

$$I_2 = 0 - 5050 \int_0^1 ((-x^{50})^{100} (-x^{50} + 1 - 1)) dx$$

$$I_2 = -5050I_2 + 5050I_1 \Rightarrow I_2 = \frac{5050}{5051} I_1$$

36. The value of $b > 3$ for which $12 \int_3^b \frac{1}{(x^2 - 1)(x^2 - 4)} dx = \log_e \left(\frac{49}{40} \right)$. Then b is,

Key: 6

(Partial fractions)2021

$$\text{Sol: } 4 \int_3^b \frac{3}{(x^2 - 1)(x^2 - 4)} dx = \log_e \left(\frac{49}{40} \right) \Rightarrow 4 \int_3^b \frac{1}{x^2 - 4} - \frac{1}{x^2 - 1} dx = \log_e \left(\frac{49}{40} \right)$$

$$\Rightarrow 4 \left(\frac{1}{2(2)} \log \frac{x-2}{x+2} - \frac{1}{2} \log \frac{x-1}{x+1} \right)_3^b = \log_e \left(\frac{49}{40} \right)$$

$$\Rightarrow 2 \left(\frac{1}{2} \log \frac{x-2}{x+2} - \log \frac{x-1}{x+1} \right)_3^b = \log \left(\frac{49}{40} \right)$$

$$\Rightarrow \log \left(\frac{x-2}{x+2} \right) - 2 \log \left(\frac{x-1}{x+1} \right)_3^b = \log \left(\frac{49}{40} \right)$$

$$\Rightarrow \log \left(\frac{x-2}{x+2} \times \left(\frac{x+1}{x-1} \right)^2 \right)_3^b = \log \left(\frac{49}{40} \right)$$

$$\Rightarrow \log \left(\frac{b-2}{b+2} \times \left(\frac{b+1}{b-1} \right)^2 - \log \left(\frac{1}{5} \times \frac{16}{4} \right) \right) = \log \frac{49}{40}$$

$$\Rightarrow \log \left| \frac{b-2}{b+2} \left(\frac{b+1}{b-1} \right)^2 \right| = \log \left(\frac{49}{40} \times \frac{4}{5} \right)$$

$$\Rightarrow \log \left(\frac{b-2}{b+0} \times \frac{b+1}{b-1} \right)^2 = \log \left(\frac{49}{50} \right)$$

$$\Rightarrow b = 6$$

37. If $\int \frac{(x^2+1)e^x}{(x+1)^2} dx = f(x)e^x + C$, where C is a constant, then $\frac{d^3 f}{dx^3}$ at $x=1$ is equal to:

(A) $-\frac{3}{4}$

(B) $\frac{3}{4}$

(C) $-\frac{3}{2}$

(D) $\frac{3}{2}$

Ans: B

(Standard Result) 2022

Sol: $\int \left(\frac{x^2+1}{(x+1)^2} \right) e^x \cdot dx$

$$= \int \left(\frac{x^2-1+2}{(x+1)^2} \right) e^x dx$$

$$= \int \left(\frac{x-1}{x+1} + \frac{2}{(x+1)^2} \right) e^x dx$$

$$= \int (f(x) + f'(x)) e^x dx$$

$$= f(x)e^x + c$$

Where $f(x) = \frac{x-1}{x+1}$

$$f'(x) = \frac{2}{(x+1)^2}$$

$$f''(x) = \frac{-4}{(x+1)^3}$$

$$= \frac{12}{(x+1)^4}$$

$$f''(1) = \frac{12}{16}$$

$$= \frac{3}{4}$$

38. The value of $\int_1^2 e^x x^{2x} (3 + 2 \log_e x) dx$ is $me^2 + ne$ then $m + n =$ ____

Key: 15

(Derivatives UV rule)

$$\text{Sol: } \int_1^2 [e^x \cdot e^{2x} + e^x 2x^{2x} (1 + \log_e x)] dx$$

$$= e^x x^{2x} \Big|_1^2 = e^2 16 - e \quad \because d(x^{nx}) = x^{nx} [1 + \log_e x] dx$$

$$m + n = 15$$

39. If $2 \int_0^1 \tan^{-1} x dx = \int_0^1 \cot^{-1} (1 - x + x^2) dx$, then $\int_0^1 \tan^{-1} (1 - x + x^2) dx$ is equal to

1) $\log 4$

2) $\frac{\pi}{2} + \log 2$

3) $\log 2$

4) $\frac{\pi}{2} - \log 4$

Key: 3

(By Parts)

$$\text{Sol: Use } \tan^{-1} x = \frac{\pi}{2} - \cot^{-1} x$$

$$40. \int_0^{\frac{\pi}{4}} \frac{x^2 dx}{(x \sin x + \cos x)^2} =$$

1) $\frac{4 + \pi}{4 - \pi}$

2) $\frac{4 + \pi}{2(4 - \pi)}$

3) $\frac{4 - \pi}{4 + \pi}$

4) $2 \left[\frac{4 - \pi}{4 + \pi} \right]$

Key: 3

(By parts)

$$\text{Sol: Given } = \frac{x \cos x}{(x \sin x + \cos x)^2} \times \frac{x}{\cos x} \text{ and using by parts}$$

41. The value of the ratio $\frac{\int_0^{\frac{\pi}{2}} \sin^{\sqrt{2}+1} x dx}{\int_0^{\frac{\pi}{2}} \sin^{\sqrt{2}-1} x dx}$ equals to **(By parts)**

A) $\sqrt{2} - 1$

B) $2 + \sqrt{2}$

C) $2 - \sqrt{2}$

D) $4 - \sqrt{2}$

ANS: C

SOL: Let $I_1 = \int_0^{\frac{\pi}{2}} \sin^{\sqrt{2}+1} x dx$ and $I_2 = \int_0^{\frac{\pi}{2}} \sin^{\sqrt{2}-1} x dx$

Applying integration by parts for I_1 , we get

$$I_1 = \left[-\cos x \sin^{\sqrt{2}} x \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \sqrt{2} \cos^2 x \sin^{\sqrt{2}-1} x dx$$

$$= \sqrt{2} \int_0^{\frac{\pi}{2}} (1 - \sin^2 x) \sin^{\sqrt{2}-1} x dx = \sqrt{2} I_2 - \sqrt{2} I_1$$

$$= (\sqrt{2} + 1) I_1 = \sqrt{2} I_2 \Rightarrow \frac{I_1}{I_2} = \frac{\sqrt{2}}{(\sqrt{2} + 1)} = 2 - \sqrt{2}$$

42. The value of $\int_0^{\pi/2} \frac{x + \sin x}{1 + \cos x} dx$ is

1) π

2) 2π

3) $\frac{\pi}{2}$

4) $\frac{\pi}{4}$

(By parts)

Sol: $I = \int_0^{\pi/2} \frac{x + \sin x}{1 + \cos x} dx$

$$= \int_0^{\pi/2} \frac{1}{2} x \sec^2(x/2) dx + \int_0^{\pi/2} \tan(x/2) dx = \left(x \tan \frac{x}{2} \right)_0^{\pi/2} - \int_0^{\pi/2} \tan \frac{x}{2} dx + \int_0^{\pi/2} \tan \frac{x}{2} dx = \frac{\pi}{2}$$

43. $\int_0^{\pi} x \cdot \log(\sin x) dx$

1) $\pi^2 \log 2$

2) $\frac{\pi^2}{2} \log 2$

3) $\frac{\pi^2}{4}$

4) $\frac{-\pi^2}{2} \log 2$

Key: 4

(By parts)

Sol: Using $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

44. Let $f(x)$ be a function satisfying $f'(x) = f(x)$ with $f(0) = 1$ and $g(x)$ be a function that satisfies $f(x) + g(x) = x^2$. Then, the value of the integral $\int_0^1 f(x)g(x) dx$ is

1) $e + \frac{e^2}{2} - \frac{3}{2}$

2) $e - \frac{e^2}{2} - \frac{3}{2}$

3) $e + \frac{e^2}{2} + \frac{5}{2}$

4) $e - \frac{e^2}{2} - \frac{5}{2}$

Key: 2

(By Parts)

Sol: $f'(x) = f(x) \Rightarrow f(x) = e^x$

$$f(x) + g(x) = x^2 \Rightarrow g(x) = x^2 - e^x$$

$$\int_0^1 f(x)g(x) dx = \int_0^1 (x^2 e^x - e^{2x}) dx$$

45. If $A = \int_0^{\pi} \frac{\cos x}{(x+2)^2} dx$ then $\int_0^{\pi/2} \frac{\sin 2x}{x+1} dx$ is equal to

1) $A - \frac{1}{2} - \frac{1}{\pi+2}$

2) $\frac{1}{2} + \frac{1}{\pi+2} - A$

3) $\frac{1}{\pi+2} - A$

4) $1 + \frac{1}{\pi+2} - A$

Key: 2

(By Parts)

Sol: $\therefore A = \int_0^{\pi} \frac{\cos x}{(x+2)^2} dx$

$$\Rightarrow A = \left\{ \cos x \left(\frac{-1}{x+2} \right) \right\}_0^{\pi} - \int_0^{\pi} (-\sin x) \left(\frac{-1}{x+2} \right) dx$$

$$\Rightarrow A = \frac{1}{\pi+2} + \frac{1}{2} - \int_0^{\pi} \frac{\sin x}{x+2} dx$$

$$A = \frac{1}{\pi+2} + \frac{1}{2} - \int_0^{\pi/2} \frac{\sin 2x}{x+1} dx$$

$$\int_0^{\pi/2} \frac{\sin 2x}{x+1} dx = \frac{1}{\pi+2} + \frac{1}{2} - A$$

46. The value of the definite integral $\int_{-1}^1 e^{-x^4} \left(2 + \ln(x + \sqrt{x^2 + 1}) + 5x^3 - 8x^4 \right) dx$ is equal to

(Integration by parts)

1) $4e$

2) $\frac{4}{e}$

3) $2e$

4) $\frac{2}{e}$

Key: 2

$$\text{Sol: } 2 \left[2 \int_0^1 e^{-x^4} dx - \int_0^1 8x^4 e^{-x^4} dx \right] = 2 \left[2 \left[\left(x e^{-x^4} \right)_0^1 + \int_0^1 4x^4 e^{-x^4} dx \right] - \int_0^1 8x^4 e^{-x^4} dx \right] = \frac{4}{e}$$

47. $\int_0^1 \frac{x^a - 1}{\log x} dx$ where $a > 0$, is equal to **(Partial Differentiation)**
1. $\log(a+1)$ 2. $2\log(a+1)$ 3. $3\log a$ 4. $2\log a$

Ans: 1

Sol: Let $f'(a) = \int_0^1 \frac{x^a \log x}{\log x} dx$

$$\therefore f(a) = \int \frac{1}{a+1} da + C = \log(a+1) + C$$

$$a = 0, f(a) = 0$$

If

$$\text{Hence } f(a) = \log(a+1)$$

48. The value of the integral $\int_0^\pi \frac{x^2 \sin x}{(2x-\pi)(1+\cos^2 x)} dx$ **(Replacing x by a-x - King's property)**
1. $\frac{\pi^2}{4}$ 2. $\frac{\pi^2}{2}$ 3. 0 4. $\frac{\pi^2}{8}$

Ans: 1

Sol: Let $I = \int_0^\pi \frac{x^2 \sin x}{(2x-\pi)(1+\cos^2 x)} dx$

$$I = \int_0^\pi \frac{(\pi^2 - 2\pi x) \sin x}{(\pi - 2x)(1+\cos^2 x)} dx - I$$

$$\therefore I = \frac{\pi^2}{4}$$

49. If $I_m = \int_1^x (\log x)^m dx$ satisfies the relation $I_m = k - lI_{m-1}$, then **(By parts)**
1. $k = e$ 2. $l = m$ 3. $k = \frac{1}{e}$ 4. $l = -m$

Ans: 2

Sol: since $I_m = \int_1^x (\log x)^m dx$

Using ILATE, we have

$$\therefore k = x(\log x)^m \text{ and } l = m$$

50. The value of integral $\int_2^3 \left(\sqrt{2x - \sqrt{5(4x-5)}} + \sqrt{2x + \sqrt{5(4x-5)}} \right) dx$ is equal to

(Definite Integration with surds)

1. $\frac{7\sqrt{7} + 2\sqrt{5}}{3\sqrt{2}}$ 2. $\frac{7\sqrt{7} - 2\sqrt{5}}{3\sqrt{2}}$
3. $\frac{2\sqrt{5} - 7\sqrt{7}}{3\sqrt{2}}$ 4. $\frac{7\sqrt{7} - 2\sqrt{5}}{\sqrt[3]{2}}$

Ans: 2

Sol: Since $\left| \sqrt{2x - \frac{5}{2}} - \sqrt{\frac{5}{2}} \right|^2$

$$= 2x - \frac{5}{2} + \frac{5}{2} - \sqrt{\left(2x - \frac{5}{2}\right) \frac{5}{2}} = 2x - \sqrt{(4x-5)5}$$

$\therefore$ given integral

$$\left[\therefore \sqrt{2x - \frac{5}{2}} > \sqrt{\frac{5}{2}} \Rightarrow 2x - \frac{5}{2} > \frac{5}{2} \Rightarrow 2x > 5 \Rightarrow x > \frac{5}{2} \right]$$

$$= \frac{3\sqrt{5} + 7\sqrt{7} - 5\sqrt{5}}{3\sqrt{2}} = \frac{7\sqrt{7} - 2\sqrt{5}}{3\sqrt{2}}$$

51. If α is a parameter independent of x and $\alpha \neq (2n+1)\pi, n \in \mathbb{Z}$ then the value of the integral

$$\int_0^1 \frac{x^{\cos \alpha} - 1}{\ln x} dx, x > 0, x \neq 1 \text{ is}$$

(Partial Differentiation)

1. $\ln(1 + \cos \alpha)$

2. $\ln(1 - \cos \alpha)$

3. $\ln(\cos \alpha)$

4. $(\cos \alpha)$

Ans: 1

Sol: Let $I(\alpha) = \int_0^1 \frac{x^{\cos \alpha} - 1}{\ln x} dx$

$$\therefore I'(\alpha) = \int_0^1 \frac{x^{\cos \alpha} \ln x (-\sin \alpha)}{\ln x} dx$$

$$(-\sin \alpha) \left\{ \frac{x^{\cos \alpha + 1}}{\cos \alpha + 1} \right\}_0^1$$

$$= \frac{-\sin \alpha}{1 + \cos \alpha}$$

Integrating on both sides w.r.t α , we have $I(\alpha) = \ln(1 + \cos \alpha) + C$

$$\Rightarrow 0 = 0 + C \text{ or } C = 0, \text{ hence } I(\alpha) = \ln(1 + \cos \alpha)$$

52. For any real number x , let $[x]$ denote the largest integer less than or equal to x . let f be a real valued function defined on the interval $[-10, 10]$ by **(Periodic function)**

$$f(x) = \begin{cases} x - [x], & \text{if } [x] \text{ is odd} \\ 1 + [x] - x, & \text{if } [x] \text{ is even} \end{cases}, \text{ then the value of } \frac{\pi^2}{10} \int_{-10}^{10} f(x) \cos \pi x dx \text{ is } \underline{\hspace{2cm}}$$

Key: 4

Sol: $f(x) = \begin{cases} x - 1, & 1 \leq x \leq 2 \\ 1 - x, & 0 \leq x \leq 1 \end{cases}$

$f(x)$ is a periodic with period 2.

$$\therefore I = \int_{-10}^{10} f(x) \cos \pi x dx$$

$$\therefore \frac{\pi^2}{10} I = 4$$

53. The value of the expression $\frac{\int_0^a x^4 \sqrt{a^2 - x^2} dx}{\int_0^a x^2 \sqrt{a^2 - x^2} dx}$ is equal to (By parts)

A) $\frac{a^2}{6}$

B) $\frac{3a^2}{2}$

C) $\frac{3a^2}{4}$

D) $\frac{a^2}{2}$

Key:D

Sol: $\int_0^a x^4 \sqrt{a^2 - x^2} dx$

$$= \left[\frac{-x^3 (a^2 - x^2)^{3/2}}{3} \right] + a^2 \cdot \frac{3}{6} \int_0^a x^2 \sqrt{a^2 - x^2} dx$$

(integrating by parts with x^3 as first function and $x\sqrt{a^2 - x^2}$ as second function)

$$= \frac{a^2}{2} \int_0^a \sqrt{a^2 - x^2} dx$$

$$= \frac{a^2}{2}$$

Topic: Evaluation of definite integrals

Sub Topic 1: Evaluation of definite integrals by substitution

54. If $\int_0^{\pi/3} \frac{\tan \theta}{\sqrt{2k \sec \theta}} d\theta = 1 - \frac{1}{\sqrt{2}}$, where ($k > 0$), then the value of k is:

1) 4

2) 2

3) $\frac{1}{2}$

4) 1

Ans. 4

(Substitution property) 2019

Sol. $\int_0^{\pi/3} \frac{\tan \theta \sec \theta}{\sqrt{2k \sec \theta} \cdot \sec \theta} d\theta$

$$\sec \theta = t^2$$

$$2 \sec \theta \tan \theta = 2t dt$$

$$\int_1^{\sqrt{2}} \frac{t dt}{\sqrt{2k} \cdot t \cdot t^2} = \frac{1}{\sqrt{2k}} \left(\frac{-1}{t} \right)_1^{\sqrt{2}} = \frac{1}{\sqrt{2k}} \left(1 - \frac{1}{\sqrt{2}} \right) = 1 - \frac{1}{\sqrt{2}}$$

55. The Value of integral $\int_{\pi/6}^{\pi/4} \frac{dx}{\sin 2x (\tan^5 x + \cot^5 x)}$ is equal to

1) $\frac{1}{10} \left(\frac{\pi}{4} - \tan^{-1} \left(\frac{1}{9\sqrt{3}} \right) \right)$

2) $\frac{1}{5} \left(\frac{\pi}{4} - \tan^{-1} \left(\frac{1}{3\sqrt{3}} \right) \right)$

3) $\frac{\pi}{10}$

4) $\frac{1}{20} \tan^{-1} \left(\frac{1}{9\sqrt{3}} \right)$

Key: 1

(Substitution property) 2019

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Sol: $I = \int_{\pi/6}^{\pi/4} \frac{dx}{\sin 2x (\tan^5 x + \cot^5 x)}$

$$I = \frac{1}{2} \int_{\pi/6}^{\pi/4} \frac{\tan^4 x \sec^2 x dx}{(1 + \tan^{10} x)} \text{ put } \tan^5 x = t$$

$$I = \frac{1}{10} \int_{\left(\frac{1}{\sqrt{3}}\right)}^1 \frac{dt}{1+t^2} = \frac{1}{10} \left(\frac{\pi}{4} - \tan^{-1} \frac{1}{9\sqrt{3}} \right)$$

56. The value of $\int_0^{\pi/2} \frac{\sin x}{\sin^4 x + \cos^4 x} dx$ is _____

Key:2

(Substitution property) 2019

Sol: $\int_0^{\pi/2} \frac{\sin x}{\sin^4 x + \cos^4 x} dx = \int_0^{\pi/2} \frac{2 \tan x \sec^2 x}{\tan^4 x + 1} dx$

$$= \tan^{-1} \left(\tan^2 x \right) \Big|_0^{\pi/2} = \tan^{-1}(\infty) - \tan^{-1}(0) = \frac{\pi}{2} = \frac{3.14}{2} = 1.57$$

57. The integral $\int_1^e \left\{ \left(\frac{x}{e} \right)^{2x} - \left(\frac{e}{x} \right)^x \right\} \log_e x dx$ is equal to:

1) $\frac{1}{2} - e - \frac{1}{e^2}$ 2) $-\frac{1}{2} + \frac{1}{e} - \frac{1}{2e^2}$ 3) $\frac{3}{2} - \frac{1}{e} - \frac{1}{2e^2}$ 4) $\frac{3}{2} - e - \frac{1}{2e^2}$

Key:4

(Substitution property) 2019

Sol: $\left(\frac{x}{e} \right)^x = t \Rightarrow x(\log x - 1) = \log t \Rightarrow \log x dx = \frac{dt}{t}$

$$\int_1^e \left\{ \left(\frac{x}{e} \right)^{2x} - \left(\frac{e}{x} \right)^x \right\} \log_e x dx = \int_{1/e}^1 \left\{ t - \frac{1}{t^2} \right\} dt = \frac{1}{2} \left(1 - \frac{1}{e^2} \right) + (1 - e) = \frac{3}{2} - \frac{1}{2e^2} - e$$

58. If the value of the integral $\int_0^{1/2} \frac{x^2}{(1-x^2)^{3/2}} dx$ is $\frac{k}{6}$, then k is equal to:

1) $3\sqrt{2} + \pi$ 2) $2\sqrt{3} - \pi$ 3) $3\sqrt{2} - \pi$ 4) $2\sqrt{3} + \pi$

Key: 2

(Substitution Method) 2020

Sol: $I = \int_0^{1/2} \frac{x^2}{(1-x^2)^{3/2}} dx$, put $x = \sin \theta \Rightarrow dx = \cos \theta d\theta$

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$$I = \int_0^{\frac{\pi}{6}} \frac{\sin^2 \theta}{(1 - \sin^2 \theta)^{3/2}} \cos \theta d\theta = \int_0^{\frac{\pi}{6}} \tan^2 \theta d\theta$$

$$= \int_0^{\frac{\pi}{6}} (\sec^2 \theta - 1) d\theta = \tan \theta - \theta \Big|_0^{\frac{\pi}{6}}$$

$$\therefore \frac{k}{6} = \frac{1}{\sqrt{3}} - \frac{\pi}{6} = \frac{6 - \sqrt{3}\pi}{6\sqrt{3}} = 2\sqrt{3} - \pi$$

59. For $x > 0$, if $f(x) = \int_1^x \frac{\log_e t}{(1+t)} dt$, then $f(e) + f\left(\frac{1}{e}\right)$ is equal to

1) -1

2) 0

3) 1

4) $\frac{1}{2}$

Key:4

(Substitution Method)2021

Solution:

$$f(x) = \int_1^x \frac{\log_e t}{1+t} dt \quad f\left(\frac{1}{x}\right) = \int_1^{1/x} \frac{\log t}{1+t} dt$$

$$\text{Put } t = \frac{1}{z}, \int_1^x \frac{-\ln z}{1 + \frac{1}{z}} \left(\frac{-1}{z^2}\right) dz = \int_1^x \frac{\ln z}{z(1+z)} dz$$

$$= \int_1^x \frac{\ln t}{t(1+t)} dt \therefore f(e) + f\left(\frac{1}{e}\right) = \int_1^e \frac{\ln t}{1+t} dt + \int_1^e \frac{\ln t}{t(1+t)} dt$$

$$= \int_1^e \frac{\ln t}{t} dt = \left(\frac{(\ln t)^2}{2} \right)_1^e = \frac{1}{2}$$

60. The value of the integral $\int_0^1 \frac{\sqrt{x} dx}{(1+x)(1+3x)(3+x)}$ is:

1) $\frac{\pi}{8} \left(1 - \frac{\sqrt{3}}{6}\right)$

2) $\frac{\pi}{4} \left(1 - \frac{\sqrt{3}}{2}\right)$

3) $\frac{\pi}{4} \left(1 - \frac{\sqrt{3}}{6}\right)$

4) $\frac{\pi}{8} \left(1 - \frac{\sqrt{3}}{2}\right)$

Key:4

(Partial fractions and Substitution Methods)2021

$$\text{Sol: } \int_0^1 \frac{\sqrt{x} dx}{(1+x)(1+3x)(3+x)} \quad \text{put } x + \sqrt{x} = t \quad x = t^2 \quad dx = 2t dt$$

$$= \int_0^1 \frac{2t^2 dt}{(1+t^2)(1+3t^2)(3+t^2)}$$

$$\begin{aligned}
&= 2 \int_0^1 \frac{(1+t^2-1)dt}{(1+t^2)(1+3t^2)(3+t^2)} \\
&= \int_0^1 \frac{2dt}{(1+3t^2)(3+t^2)} - \int_0^1 \frac{2dt}{(1+t^2)(1+3t^2)(3+t^2)} \\
&= \int_0^1 \frac{2dt}{(1+3t^2)(3+t^2)} - \int_0^1 \frac{dt}{(1+3t^2)(1+t^2)} + \int_0^1 \frac{dt}{(1+3t^2)(3+t^2)} \\
&= \int_0^1 \frac{3dt}{(1+3t^2)(3+t^2)} - \int_0^1 \frac{dt}{(1+3t^2)(1+t^2)} \\
&= \frac{9}{8} \int_0^1 \frac{dt}{1+3t^2} - \frac{3}{8} \int_0^1 \frac{dt}{3+t^2} - \frac{3}{2} \int_0^1 \frac{1}{1+3t^2} + \frac{1}{2} \int_0^1 \frac{dt}{1+t^2} \\
&= \frac{-3}{8} \int_0^1 \frac{dt}{1+3t^2} + \frac{1}{2} \int_0^1 \frac{dt}{1+t^2} - \frac{3}{8} \int_0^1 \frac{dt}{3+t^2} \\
&= -\frac{\sqrt{3}}{8} \tan^{-1}(\sqrt{3}t) \Big|_0^1 + \frac{1}{2} \tan^{-1}(t) \Big|_0^1 - \frac{3}{8\sqrt{3}} \tan^{-1}\left(\frac{t}{\sqrt{3}}\right)
\end{aligned}$$

61. The value of $b > 3$ for which $12 \int_3^b \frac{1}{(x^2-1)(x^2-4)} dx = \log_e \left(\frac{49}{40} \right)$. Then b is,

Key: 6 (DEFINITE INTEGRATION WITH PARTIAL FRACTIONS) 2022

Sol: $4 \int_3^b \frac{3}{(x^2-1)(x^2-4)} dx = \log_e \left(\frac{49}{40} \right) \Rightarrow 4 \int_3^b \frac{1}{x^2-4} - \frac{1}{x^2-1} dx = \log_e \left(\frac{49}{40} \right)$

$$4 \left(\frac{1}{2(2)} \log \frac{x-2}{x+2} - \frac{1}{2} \log \frac{x-1}{x+1} \right) \Big|_3^b = \log_e \left(\frac{49}{40} \right)$$

$$2 \left(\frac{1}{2} \log \frac{x-2}{x+2} - \log \frac{x-1}{x+1} \right) \Big|_3^b = \log \left(\frac{49}{40} \right)$$

$$\log \left(\frac{x-1}{x+2} \right) - 2 \log \left(\frac{x-1}{x+1} \right) \Big|_3^b = \log \left(\frac{49}{40} \right)$$

$$\log \left(\frac{x-2}{x+2} \times \left(\frac{x+1}{x-1} \right)^2 \right) \Big|_3^b = \log \left(\frac{49}{40} \right)$$

$$\log \left(\frac{b-2}{b+2} \times \left(\frac{b+1}{b-1} \right)^2 - \log \left(\frac{1}{5} \times \frac{16}{4} \right) \right) = \log \frac{4}{6}$$

$$\log \left| \frac{b-2}{b+2} \left(\frac{b+1}{b-1} \right)^2 \right| = \log \left(\frac{49}{40} \times \frac{4}{5} \right)$$

$$\log \left(\frac{b-2}{b+2} \times \frac{b+1}{b-1} \right)^2 = \log \left(\frac{40}{82} \right)$$

62. $\int_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}} \frac{48}{\sqrt{9-4x^2}} dx$ is equal to

1) 2π

2) $\frac{\pi}{2}$

3) $\frac{\pi}{3}$

4) $\frac{\pi}{6}$

Key:-

1

(substitution property)2023

Sol:- $\int_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}} \frac{48}{\sqrt{9-(2x)^2}} dx = 48 \cdot \frac{1}{2} \sin^{-1} \left(\frac{2x}{3} \right) \Big|_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}}$

$$= 24 \left(\sin^{-1} \left(\frac{2}{3} \cdot \frac{3\sqrt{3}}{4} \right) - \sin^{-1} \left(\frac{2}{3} \cdot \frac{3\sqrt{2}}{4} \right) \right)$$

$$= 24 \left(\sin^{-1} \left(\frac{\sqrt{3}}{2} \right) - \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) \right)$$

$$= 24 \left(\frac{\pi}{3} - \frac{\pi}{4} \right) = 24 \left(\frac{\pi}{12} \right) = 2\pi$$

63. The integral $16 \int \frac{dx}{x^3(x^2+2)^2}$ is equal to

1) $\frac{11}{6} - \log_e 4$

2) $\frac{11}{6} + \log_e 4$

3) $\frac{11}{126} + \log_e 4$

4) $\frac{11}{126} - \log_e 4$

Key:-

1

(substitution)2023

Sol: $\int \frac{1}{x^7(x^2+2)^2} dx = \int \frac{1}{x^7 \left(1 + \frac{2}{x^2} \right)^2} dx \Rightarrow -\frac{1}{4} \int \left(\frac{t-1}{2} \right)^2 \frac{dt}{t^2}$

$$\frac{1}{4} \int_{3/2}^3 \frac{t^2+1-2t}{4} \frac{1}{t^2} dt \Rightarrow \frac{1}{16} \int_{3/2}^3 \left(1 + \frac{t}{t^2} - \frac{2}{t} \right) dt$$

$$\frac{1}{16} \left(3 - \frac{1}{3} - 2 \log \frac{3}{2} + \frac{2}{3} + 2 \log \frac{3}{2} \right) \Rightarrow \frac{1}{16} \left(\frac{3}{2} + \frac{1}{3} + 2 \log \frac{1}{2} \right)$$

$$\Rightarrow \frac{1}{16} \left(\frac{9+2}{6} + 2 \log t \right) \quad 16I = \frac{11}{6} - \log 4$$

64. Let $f(x) = x + \frac{a}{\pi^2 - 4} \sin x + \frac{b}{\pi^2 - 4} \cos x$, $x \in \mathbb{R}$ be a function which satisfies

$$f(x) = x + \int_0^{\pi/2} \sin(x+y) f(y) dy. \text{ Then } (a+b) \text{ is equal to}$$

1) $-2\pi(\pi-2)$

2) $-\pi(\pi+2)$

3) $-\pi(\pi-2)$

4) $-2\pi(\pi+2)$

Ans:

D

(substitution)2023

$$\begin{aligned} \text{Sol: } f(x) &= x + \int_0^{\pi/2} (\sin x \cos y + \cos x \sin y) f(y) dy \\ &= x + \sin x \cdot \int_0^{\pi/2} \cos y f(y) dy + \cos x \int_0^{\pi/2} \sin y f(y) dy \end{aligned}$$

$$f(x) = x + A \sin x + B \cos x$$

$$A = \int_0^{\pi/2} \cos y f(y) dy$$

$$= \int_0^{\pi/2} \cos y (y + A \sin y + B \cos y) dy$$

$$\int_0^{\pi/2} y \cos y dy + A \int_0^{\pi/2} \sin y \cos y dy + B \int_0^{\pi/2} \cos^2 y dy$$

$$A = \int_0^{\pi/2} y \cdot \cos y dy + \frac{A}{2} \int_0^{\pi/2} \sin 2y + B \int_0^{\pi/2} \cos^2 y dy$$

$$A = \frac{\pi}{2} - 1 + \frac{A}{2} + B \frac{\pi}{4}$$

$$A = \frac{\pi}{2} - 1 + \frac{B}{4} \pi$$

$$A = \pi - 2 + \frac{B}{2} \pi$$

$$B = \int_0^{\pi/2} \sin y f(y) dy$$

$$= \int_0^{\pi/2} \sin y (y + A \sin y + B \cos y) dy$$

$$= \int_0^{\pi/2} \left(y \sin y + A \sin^2 y + \frac{B}{2} \sin 2y \right) dy$$

$$= 1 + A \cdot \frac{1}{2} \cdot \frac{\pi}{2} + \frac{B}{2} (1)$$

$$\frac{B}{2} = 1 + A \cdot \frac{\pi}{4}$$

$$B = 2 + \frac{\pi}{2} A$$

$$A = \pi - 2 + \frac{\pi}{2} \left(2 + \frac{\pi}{2} A \right)$$

$$= \pi - 2 + \pi + \frac{\pi^2}{4} A$$

$$\left(1 - \frac{\pi^2}{4}\right)A = 2\pi - 2 \Rightarrow A = \frac{2(\pi-1)}{(4-\pi^2)/4}$$

$$A = \frac{8(\pi-1)}{4-\pi^2}$$

$$B = 2 + \frac{\pi}{2} \left(\frac{8(\pi-1)}{4-\pi^2} \right)$$

$$= 2(4-\pi^2) + 4\pi^2 - 4\pi$$

$$= \frac{8+2\pi^2-4\pi}{4-\pi^2}$$

$$\frac{4\pi-8-2\pi^2}{\pi^2-4}$$

$$a = -8(\pi-1) = -8\pi+8, b = 4\pi-8-2\pi^2$$

$$a+b = -4\pi-2\pi^2$$

$$= -2\pi(\pi+2)$$

65. Let $\alpha \in (0,1)$ and $\beta = \log_e(1-\alpha)$. let $P_n(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^n}{n}, x \in (0,1)$. Then the

integral $\int_0^\alpha \frac{t^{50}}{1-t} dt$ is equal to

A) $\beta - P_{50}(\alpha)$ B) $P_{50}(\alpha) - \beta$ C) $\beta + P_{50}(\alpha)$ D) $-(\beta + P_{50}(\alpha))$

ANS:D

(definite with partial fractions)2023

$$\text{SOL } \int_0^\alpha \frac{t^{50}}{1-t} dt = \int_0^\alpha \left(\frac{1 - (1-t^{50})}{1-t} \right) dt = \int_0^\alpha \frac{1}{1-t} dt - \int_0^\alpha (1+t+t^2+t^3+\dots+t^{49}) dt$$

$$= -[\ln(1-t)]_0^\alpha - \left[t + \frac{t^2}{2} + \frac{t^3}{3} + \dots + \frac{t^{50}}{50} \right]_0^\alpha = -\ln(1-\alpha) - \left(\alpha + \frac{\alpha^2}{2} + \frac{\alpha^3}{3} + \dots + \frac{\alpha^{50}}{50} \right)$$

$$= -(\beta + P_{50}(\alpha))$$

66. The value of $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{(2+3\sin x)}{\sin x(1+\cos x)} dx$ is equal to

A) $-2 + 3\sqrt{3} + \log_e \sqrt{3}$

B) $\frac{7}{2} - \sqrt{3} - \log_e \sqrt{3}$

C) $\frac{10}{3} - \sqrt{3} - \log_e \sqrt{3}$

D) $\frac{10}{3} - \sqrt{3} + \log_e \sqrt{3}$

ANS:D

(substitution)2023

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$$\text{SOL } I = \int_{\pi/3}^{\pi/2} \frac{2+3\sin x}{\sin x(1+\cos x)} dx = \underbrace{\int_{\pi/3}^{\pi/2} \frac{2}{\sin x(1+\cos x)} dx}_{I_1} + \underbrace{\int_{\pi/3}^{\pi/2} \frac{3}{1+\cos x} dx}_{I_2}$$

$$I_2 = \frac{3}{2} \int_{\pi/3}^{\pi/2} \sec^2 x / 2 dx = \left[\frac{3}{2} (\cancel{2}) \tan x / 2 \right]_{\pi/3}^{\pi/2}$$

$$= 3 - \sqrt{3} \quad I_1 = 2 \int_{\pi/3}^{\pi/2} \frac{\sin x}{\sin^2 x(1+\cos x)} dx \quad \text{put } \cos x = t \text{ then}$$

$$-\sin x dx = dt \therefore I_1 = 2 \int_{\frac{1}{2}}^0 \frac{-dt}{(1+t)(1+t)} = 2 \int_0^{1/2} \frac{dt}{(1+t)(1+t)^2}$$

$$= 2 \int_0^{1/2} \left(\frac{\frac{1}{4}}{1-t} + \frac{\frac{1}{4}}{(1+t)} + \frac{\frac{1}{2}}{(1+t)^2} \right) = \left[-\frac{1}{2} \ln|1-t| + \frac{1}{2} \ln|1+t| - \frac{1}{(1+t)} \right]_0^{1/2}$$

$$= \ln \sqrt{3} + \frac{1}{3} \therefore I_1 + I_2 = \ln \sqrt{3} - \sqrt{3} + \frac{10}{3}$$

67. Let $\alpha > 0$. If $\int_0^{\alpha} \frac{x}{\sqrt{x+\alpha}-\sqrt{x}} dx = \frac{16+20\sqrt{2}}{15}$, then α is equal to:

1) 2

2) $2\sqrt{2}$

3) 4

4) $\sqrt{2}$

Key: 1

(rationalization of function)2023

$$\text{Sol: } \int_0^{\alpha} \frac{x}{\sqrt{x+\alpha}-\sqrt{x}} dx = \frac{16+20\sqrt{2}}{15}$$

$$\frac{1}{\alpha} \int_0^{\alpha} (x\sqrt{x+\alpha} + \sqrt{x}) dx$$

$$\frac{1}{\alpha} \int_0^{\alpha} \left((x+\alpha-\alpha)\sqrt{x+\alpha} + x^{\frac{3}{2}} \right) dx \quad \frac{1}{\alpha} \int_0^{\alpha} \left((x+\alpha)^{\frac{3}{2}} - \alpha(x+\alpha)^{\frac{1}{2}} + x^{\frac{3}{2}} \right) dx$$

$$= \frac{1}{\alpha} \left((x+\alpha)^{\frac{5}{2}} \cdot \frac{2}{5} - \alpha \cdot \frac{2}{3} (x+\alpha)^{\frac{3}{2}} + x^{\frac{5}{2}} \cdot \frac{2}{5} \right)_0^{\alpha} = \alpha \sqrt{\alpha} \frac{(10+4\sqrt{2})}{15} = \frac{16+20\sqrt{2}}{15} \Rightarrow \alpha = 2$$

$$\alpha = 2$$

68. The value of the integral $\int_{-\log_e 2}^{\log_e 2} e^x \left(\log_e \left(e^x + \sqrt{1+e^{2x}} \right) \right) dx$

1) $\log_e \left(\frac{(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}} \right) + \frac{\sqrt{5}}{2}$

2) $\log_e \left(\frac{2(2+\sqrt{5})}{\sqrt{1+\sqrt{5}}} \right) - \frac{\sqrt{5}}{2}$

3) $\log_e \left(\frac{\sqrt{2}(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}} \right) - \frac{\sqrt{5}}{2}$

4) $\log_e \left(\frac{\sqrt{2}(3-\sqrt{5})^2}{\sqrt{1+\sqrt{5}}} \right) + \frac{\sqrt{5}}{2}$

KEY:3

(substitution)2023

SOL: $I = \int_{-\log_e 2}^{\log_e 2} e^x \log_e \left(e^x + \sqrt{1+e^{2x}} \right) dx$

let $e^x = t \Rightarrow e^x \cdot dx = dt$

$x : -\log_e 2 \rightarrow \log_e 2$

$t : \frac{1}{2} \rightarrow 2$

$I = \int_{1/2}^2 \log_e (t + \sqrt{1+t^2}) dt$

$t \cdot \log_e (t + \sqrt{1+t^2}) \Big|_{1/2}^2 - \int_{1/2}^2 \frac{t}{\sqrt{1+t^2}} dt$

$2 \log_e (2+\sqrt{5}) - \frac{1}{2} \log_e \left(\frac{1+\sqrt{5}}{2} \right) - \left(\sqrt{1+t^2} \right) \Big|_{1/2}^2 \log_e (2+\sqrt{5}) + \log \frac{\sqrt{2}}{\sqrt{1+\sqrt{5}}} - \left(\sqrt{5} - \frac{\sqrt{5}}{2} \right)$

$= \log \frac{\sqrt{2} \cdot (2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}} - \frac{\sqrt{5}}{2}$

69. $\int_0^\infty \frac{6}{e^{3x} + 6e^{2x} + 11e^x + 6} dx =$

(Substitution)

1) $\log_e \left(\frac{64}{27} \right)$

2) $\log_e \left(\frac{512}{81} \right)$

3) $\log_e \left(\frac{32}{27} \right)$

4) $\log_e \left(\frac{256}{81} \right)$

KEY:3

SOL: Put $e^x = t \Rightarrow t dx = dt$ $x=0 \Rightarrow t=1$ $x=\infty \Rightarrow t=\infty$ $I = \int_1^\infty \frac{6}{t(t^3 + 6t^2 + 11t + 6)} dt$

$= \int_1^\infty \frac{6}{t(t+1)(t+2)(t+3)} dt = \int_1^\infty \left(\frac{1}{t} - \frac{3}{t+1} + \frac{3}{t+2} - \frac{1}{t+3} \right) dt$

$= \left[\ln t - 3 \ln(t+1) + 3 \ln(t+2) - \ln(t+3) \right]_1^\infty = \left[\ln \frac{t(t+2)^3}{(t+1)^3(t+3)} \right]_1^\infty$

$$= \lim_{t \rightarrow \infty} t \ln \left(\frac{t(t+2)^3}{(t+1)^3(t+3)} \right) - \ln \left(\frac{1 \cdot 3^3}{2^3 \cdot 4} \right) = 0 - \ln \left(\frac{27}{32} \right) = \ln \left(\frac{32}{27} \right)$$

70. The value of $\int_1^e (x^x - x^{-x}) \log \left(\frac{x}{e} \right) dx$

- 1) $e^e - e^{-e} - 2$ 2) $e^e + e^{-e} - 2$ 3) 0 4) 2

Key: 2

(Substitution property)

Sol: $\frac{d}{dx} x^{nx} = nx^{nx} \log \left(\frac{x}{e} \right)$

$$\int_1^e \left[x^x \log \left(\frac{x}{e} \right) + \left(-x^{-x} \log \frac{x}{e} \right) \right] dx = x^x + x^{-x} \Big|_1^e = e^e + e^{-e} - 2$$

71. The value of $\int_1^e \left(\left(\frac{x}{e} \right)^{2x} + \left(\frac{e}{x} \right)^x \right) \log_e^x dx$

- 1) $\frac{1}{2} + e - \frac{1}{e^2}$ 2) $\frac{3}{2} - e + \frac{1}{e^2}$ 3) $\frac{1}{2} - e + \frac{1}{e^2}$ 4) $\frac{3}{2} - e - \frac{1}{e^2}$

Key: 1

(Substitution property)

Sol: $\left(\frac{x}{e} \right)^x \Rightarrow \log t = x(\log x - 1) \Rightarrow \frac{1}{t} dt = \left((\log x - 1) + x \frac{1}{x} \right) dx$

$$\Rightarrow \frac{1}{t} dt = \log_e^x dx \quad I = \int_{1/e}^1 \left(t^2 + \frac{1}{t} \right) \frac{dt}{t} = \int_{1/e}^1 \left(t + \frac{1}{t^2} \right) dt$$

$$= t^2 - \frac{1}{t} \Big|_{1/e}^1 = \frac{1}{2} - \frac{1}{e^2} + e$$

72. $\int_0^{\pi/2} \frac{\cos^2 \theta d\theta}{\cos^2 \theta + 4 \sin^2 \theta} =$

- 1) $\frac{\pi}{6}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{2}$

Key: 1

(Substitution and Partial Fractions Method)

Sol:

$$I = \int_0^{\pi/2} \frac{d\theta}{1 + 4 \tan^2 \theta} = \int_0^{\pi/2} \frac{\sec^2 \theta d\theta}{(1 + \tan^2 \theta)(1 + 4 \tan^2 \theta)}$$

Now put $\tan \theta = x$

$$I = \int_0^\infty \frac{dx}{(1+x^2)(1+4x^2)}$$

$$= \frac{1}{3} \int_0^\infty \left(\frac{-1}{1+x^2} + \frac{4}{1+4x^2} \right) dx = \frac{\pi}{6}$$

73. If $A = \int_0^1 x^{50} (2-x)^{50} dx$, $B = \int_0^1 x^{50} dx$, which of the following is true?

- 1) $A = 2^{50} B$ 2) $A = 2^{-50} B$ 3) $A = 2^{100} B$ 4) $A = 2^{-100} B$

Key: 3 (Substitution Method)

Sol: Given $A = \int_0^1 x^{50} (2-x)^{50} dx$; $B = \int_0^1 x^{50} (1-x)^{50} dx$

In A, Put $x = 2t$ or $dx = 2dt$

$$A = 2 \int_0^{1/2} 2^{50} t^{50} 2^{50} (1-t)^{50} dt$$

$$\text{Now, } B = 2 \int_0^{1/2} x^{50} (1-x)^{50} dx$$

From equations

$$A = 2^{100} B$$

74. Let $F(x)$ be a non-negative continuous function defined on $\mathbb{R}$ such that

$F(x) + F\left(x + \frac{1}{2}\right) = 3$ and the value of $\int_0^{1500} F(x) dx$ is $\frac{9000}{\lambda}$. Then the numerical value of λ is

Ans: 4

(Periodic Property)

Sol: We have $F(x) + F\left(x + \frac{1}{2}\right) = 3$ (i)

Replace x by $x + \frac{1}{2}$ in (i), we get $F\left(x + \frac{1}{2}\right) + F(x+1) = 3$ (2)

$\therefore$ From (1) and (2), we get $F(x) = F(x+1)$ (3) $\Rightarrow F(x)$ is periodic function.

$$\text{Now consider } I = \int_0^{1500} F(x) dx = \int_0^{\frac{1}{2}} F(x) dx + \int_{\frac{1}{2}}^1 F(x) dx + \int_1^{\frac{3}{2}} F(x) dx + \int_{\frac{3}{2}}^2 F(x) dx + \dots + \int_{\frac{1499}{2}}^{\frac{1500}{2}} F(x) dx$$

(Using property of periodic function)

Put $x + y + \frac{1}{2}m2^{\text{nd}}$ integral, we get

$$I = 1500 \left[\int_0^{\frac{1}{2}} F(x) dx + \int_0^{\frac{1}{2}} F\left(y + \frac{1}{2}\right) dy \right] - 1500 \int_0^{\frac{1}{2}} \left(F(x) + F\left(x + \frac{1}{2}\right) \right) dx = 1500 \int_0^{\frac{1}{2}} 3 dx \left| \text{Using (1)} \right|$$

$$\text{Hence } I = 1500(3) \left(\frac{1}{2} \right) = 750 \times 3 = 2250$$

75. The value of the integral $\int_{-\frac{3\pi}{4}}^{\frac{5\pi}{4}} \frac{\cos x + \sin x}{1 + e^{x - \frac{\pi}{4}}} dx$ is **(Substitution Method)**

1. 1

2. 0

3. 2

4. 4

ANS: 2

$$\text{SOL: Let } I = \int_{-\frac{3\pi}{4}}^{\frac{5\pi}{4}} \frac{\sin x + \cos x}{e^{x - \frac{\pi}{4}} + 1} dx = \int_{-\frac{3\pi}{4}}^{\frac{5\pi}{4}} \frac{\sqrt{2} \cos\left(x - \frac{\pi}{4}\right)}{e^{x - \frac{\pi}{4}} + 1} dx$$

$$\text{Put } x - \frac{\pi}{4} = t$$

We get

$$I = \int_{-\pi}^{\pi} \frac{\sqrt{2} \cos t dt}{e^t + 1} = \sqrt{2} \int_0^{\pi} \cos t dt$$

$$\Rightarrow I = 0 \quad (\because \cos(\pi - t) = -\cos t)$$

76. The value of definite integral $\int_0^1 \sin^{-1}(\sqrt{x}) dx$ is **(Substitution Method)**

A) $\frac{\pi}{2}$

B) $-\frac{\pi}{2}$

C) $\frac{\pi}{4}$

D) $-\frac{\pi}{4}$

ANS: C

$$\text{SOL: Put } x = \sin^2 \theta \text{ then } \int_0^1 \sin^{-1} \sqrt{x} dx = \int_0^{\frac{\pi}{2}} \theta \cdot \sin 2\theta d\theta$$

$$= \theta \left(-\frac{\cos 2\theta}{2} \right) + \frac{\sin 2\theta}{4} \Bigg|_0^{\frac{\pi}{2}} = \frac{\pi}{4}.$$

77. If $\int_0^{\pi/3} \frac{\tan \theta}{\sqrt{2k \sec \theta}} d\theta = 1 - \frac{1}{\sqrt{2}}$ ($k > 0$), then the value of k is:

1) 4

2) 2

3) $\frac{1}{2}$

4) 1

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Key: 4

(Substitution Method)

Sol: $\int_0^{\pi/3} \frac{\tan \theta \sec \theta}{\sqrt{2k \sec \theta} \cdot \sec \theta} d\theta \quad \sec \theta = t^2$
 $2 \sec \theta \tan \theta = 2t dt$

$$\int_1^{\sqrt{2}} \frac{t dt}{\sqrt{2k} \cdot t \cdot t^2} = \frac{1}{\sqrt{2k}} \left(\frac{-1}{t} \right)_1^{\sqrt{2}}$$

$$\frac{1}{\sqrt{2k}} \left(1 - \frac{1}{\sqrt{2}} \right) = 1 - \frac{1}{\sqrt{2}}$$

78. The integral $\int_{\pi/6}^{\pi/4} \frac{dx}{\sin 2x (\tan^5 x + \cot^5 x)}$ equals

1) $\frac{1}{10} \left(\frac{\pi}{4} - \tan^{-1} \left(\frac{1}{9\sqrt{3}} \right) \right)$

2) $\frac{1}{5} \left(\frac{\pi}{4} - \tan^{-1} \left(\frac{1}{3\sqrt{3}} \right) \right)$

3) $\frac{\pi}{10}$

4) $\frac{1}{20} \tan^{-1} \left(\frac{1}{9\sqrt{3}} \right)$

Key:1

(Substitution Method)

Sol: $I = \int_{\pi/6}^{\pi/4} \frac{dx}{\sin 2x (\tan^5 x + \cot^5 x)}$

$$I = \frac{1}{2} \int_{\pi/6}^{\pi/4} \frac{\tan^4 x \sec^2 x dx}{(1 + \tan^{10} x)} \text{ put } \tan^5 x = t$$

$$I = \frac{1}{10} \int_{\left(\frac{1}{\sqrt{3}}\right)}^1 \frac{dt}{1+t^2} = \frac{1}{10} \left(\frac{\pi}{4} - \tan^{-1} \frac{1}{9\sqrt{3}} \right)$$

79. The value of the integral $I = \int_1^{\infty} \frac{x^2 - 2}{x^3 \sqrt{x^2 - 1}} dx$ is

1) 0

2) $\frac{2}{3}$

3) $\frac{4}{3}$

4) 1

Key: 1

(Substitution Method)

Sol: $I = \int_1^{\infty} \frac{x^2 - 2}{x^3 \sqrt{x^2 - 1}} dx$ Put $x = \frac{1}{t}$ $dx = -\frac{1}{t^2} dt$

$$I = \int_0^1 \frac{1 - 2t^2}{\sqrt{1-t^2}} dt \text{ Put } t = \sin \theta$$

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$$I = \int_0^{\pi/2} \cos 2\theta d\theta = 0$$

$$80. \int_0^{\pi/2} \frac{\sec^2 x dx}{(\sec x + \tan x)^n} = \quad (n > 2)$$

$$1) \frac{1}{n^2 - 1}$$

$$2) \frac{n}{n^2 - 1}$$

$$3) \frac{n}{n^2 + 1}$$

$$4) \frac{2}{n^2 - 1}$$

Key: 2

(Substitution Method)

Sol: Standard

$$81. \text{ The value of } \int_{-1}^1 \frac{dx}{(2-x)\sqrt{1-x^2}} \text{ is.}$$

(TRIGONOMETRIC SUBSTITUTION)

$$1) 0$$

$$2) \frac{\pi}{\sqrt{3}}$$

$$3) \frac{2\pi}{\sqrt{3}}$$

$$4) \frac{\pi}{3\sqrt{3}}$$

Key: 2

$$\text{Sol: let } I = \int_{-1}^1 \frac{dx}{(2-x)\sqrt{1-x^2}}, \text{ put } x = \sin \theta$$

$$I = \int_{-\pi/2}^{\pi/2} \frac{\cos \theta d\theta}{(2 - \sin \theta) \cos \theta}$$

$$I = \int_0^{\pi/2} \left(\frac{1}{2 - \sin \theta} + \frac{1}{2 + \sin \theta} \right) d\theta$$

$$[\text{ using } \int_{-a}^a f(x) dx = \int_0^a f(x) + f(-x) dx]$$

$$\begin{aligned} \frac{4}{3} \int_0^{\infty} \frac{d\theta}{t^2 + \frac{4}{3}} &= \frac{4}{\sqrt{3}} \cdot \frac{\sqrt{3}}{2} \cdot \tan^{-1} \frac{\sqrt{3}t}{2} \Big|_0^{\infty} \\ &= \frac{2}{\sqrt{3}} \cdot \frac{\pi}{2} = \frac{\pi}{\sqrt{3}} \end{aligned}$$

$$82. \text{ The value of definite integral } \int_0^{\infty} \left(\frac{dx}{(1+x^9)(1+x^2)} \right) \text{ equals to : (Substitution property)}$$

$$1) \frac{\pi}{16}$$

$$2) \frac{\pi}{8}$$

$$3) \frac{\pi}{4}$$

$$4) \frac{\pi}{2}$$

Key: 3

$$\text{Sol: Let } I = \int_0^{\infty} \left(\frac{dx}{(1+x^9)(1+x^2)} \right)$$

$$\text{Put } x = \frac{1}{t} \Rightarrow dx = -\frac{1}{t^2} dt$$

$$2I = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$$

83. $\int_0^{\pi} \frac{x^3 \cos^4 x \sin^2 x}{\pi^2 - 3\pi x + 3x^2} dx = \lambda \int_0^{\pi/2} \sin^2 x dx$, then the value of λ is: **(SUBSTITUTION PROPERTY)**

1. $\frac{\pi}{12}$

2. $\frac{\pi}{8}$

3. $\frac{\pi}{4}$

4. $\frac{\pi}{3}$

Sol: $I = \int_0^{\pi} \frac{x^3 \cos^4 x \sin^2 x}{\pi^2 - 3\pi x + 3x^2} dx = \lambda \int_0^{\pi/2} \sin^2 x dx$

$$\Rightarrow 2I = \int_0^{\pi} \pi \cos^4 x \sin^2 x dx.$$

84. The value of the integral $\int_0^{\pi/2} \frac{1+2\cos x}{(2+\cos x)^2} dx$ is **(Substitution)**

1. $\frac{1}{4}$

2. $\frac{1}{2}$

3. $\frac{-1}{2}$

4. $\frac{-1}{4}$

Ans: 2

Sol: Dividing numerator and denominator with $\sin^2 x$

85. The value of $\int_0^{2\pi} \sqrt{1 + \sin \frac{x}{2}} dx$ is

(Definite Integration)

1. 0

2. 2

3. 8

4. 4

Ans: 3

Sol: $\int_0^{2\pi} \sqrt{1 + \sin \frac{x}{2}} dx$

$$= 4 \left(-\cos \frac{x}{4} + \sin \frac{x}{4} \right) \Big|_0^{2\pi} = 4[(1+1)] = 8$$

86. $\int_0^2 x^3 \sqrt{2x-x^2} dx$ is equal to

(Substitution)

1. $\frac{7\pi}{2}$

2. $\frac{7\pi}{4}$

3. $\frac{7\pi}{8}$

4. $\frac{7\pi}{16}$

Ans: 3

Sol: $\int_0^2 x^3 \sqrt{2x-x^2} dx$

$$\int_0^{\pi/2} 8 \sin^8 \theta \sqrt{4 \sin^2 \theta - 4 \sin^4 \theta} (4 \sin \theta \cos \theta) d\theta$$

$$[\text{Putting } x = \sin^2 \theta \Rightarrow dx = 4 \sin \theta \cdot \cos \theta d\theta]$$

$$= \frac{7\pi}{8}$$

87. If $\int_0^1 \frac{\sin t}{1+t} dt = \alpha$, then the value of the integral $\int_{4\pi-2}^{4\pi} \frac{\sin t/2}{4\pi+2-t} dt$ in terms of α is given by

1. 2α

2. -2α

3. α

4. $-\alpha$

(Substitution)

Ans: 4

$$\begin{aligned}\text{Sol: } \int_{4\pi-2}^{4\pi} \frac{\sin t / 2}{4\pi + 2 - t} dt &= \frac{1}{2} \int_{4\pi-2}^{4\pi} \frac{\sin t / 2}{1 + \left(2\pi - \frac{t}{2}\right)} dt \\ &= \int_0^1 \frac{\sin u}{1+u} du = \int_1^2 \frac{\sin t}{1+t} dt = -\alpha\end{aligned}$$

88. If $\int_0^\infty e^{-ax} dx = \frac{1}{a}$, then $\int_0^\infty x^n e^{-ax} dx$ is

(Substitution property)

1. $\frac{(-1)^n n!}{a^{n+1}}$

2. $\frac{(-1)^n (n-1)!}{a^n}$

3. $\frac{n!}{a^{n+1}}$

4. $\frac{n!}{a^{n-1}}$

Ans: 3

Sol: Let $\int_0^\infty x^n e^{-ax} dx$

$$= -\frac{1}{a} \lim_{x \rightarrow \infty} \frac{x^n}{e^{ax}} + \frac{n}{a} I_{n-1}$$

$$\therefore I_n = \frac{n}{a} I_{n-1}$$

$$= \frac{n!}{a^n} \int_0^\infty e^{-ax} dx = \frac{n!}{a^{n+1}}$$

$$\left[\because \lim_{x \rightarrow \infty} \frac{x^n}{e^{ax}} = 0 \right]$$

89. The value of a positive integer $n \leq 5$ such that $\int_0^1 e^x (x-1)^n dx = 16 - 6e$ is belongs to the interval .

(Substitution)

1. $\left[-\frac{1}{4}, 2\right]$

2. $\left[-\frac{1}{4}, 4\right]$

3. $[0, 5]$

4. $[0, 1]$

Ans: 3

Sol: let $I_n = \int_0^1 e^x (x-1)^n dx$

$$x-1=t \Rightarrow dx=dt$$

$$\therefore I_n = \int_{-1}^0 e^{t+1} \cdot t^n dt$$

$$\text{Hence } n = 3$$

90. $\int_0^1 \frac{x^4 (1-x)^4}{1+x^2} dx$ equals

(Substitution)

1. $\frac{22}{7} - \pi$

2. $\frac{18}{e^4}$

3. $\frac{27}{e^2}$

4. $\frac{9}{e^2}$

Ans: 1

Sol: $\frac{22}{7} - \pi$

91. $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cos 9x}{\sin x} dx =$

(Substitution)

1. $\frac{1}{4} \ln 2 - \frac{1}{12}$

2. $\frac{1}{4} \ln 2 - \frac{1}{6}$

3. $2\left(\frac{1}{4} \ln 2 - \frac{1}{6}\right)$

4. $\frac{1}{2} \ln 2 + \frac{1}{3}$

ANS :C

Sol. We have $\sin 2x + \sin 4x + \dots + \sin 2nx = \frac{\cos x - \cos(2n+1)x}{2 \sin x}$

Put $n=4$ and integrating from $\frac{\pi}{4}$ to $\frac{\pi}{2}$, we get $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\sin 2x + \sin 4x + \sin 6x + \sin 8x) dx$

$$\Rightarrow I = \frac{1}{4} \ln 2 - \frac{1}{6} \Rightarrow \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cos 9x}{\sin x} dx = \frac{1}{2} \ln 2 - \frac{1}{3}$$

92. The value of the integral $\int_{-1}^1 \frac{x^2 + (x-1)\sqrt{x^2+1}}{(\sqrt{x^2+1}+x+1)(\sqrt{x^2+1})} dx$ is equal to (Substitution)

1. 0

2. $\ln(\sqrt{2}+1)$

3. $\ln(\sqrt{2}-1)$

4. $\sqrt{2}+1$

ANSC

Sol Given Integral $I = \int_{-1}^1 \frac{x^2 + 1 + (x-1)\sqrt{x^2+1} - 1}{(\sqrt{x^2+1}+x+1)(\sqrt{x^2+1})} dx$

$$= \int_{-1}^1 \frac{\sqrt{x^2+1} + x - 1}{\sqrt{x^2+1} + x + 1} dx - \int_0^1 \frac{1}{\sqrt{x^2+1} + x + x\sqrt{x^2+1}} dx$$

Clearly first one is odd function for second one

Put $x = \tan \theta, dx = \sec^2 \theta d\theta$

Then $I = - \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\sec^2 \theta d\theta}{(\sec \theta + \tan \theta + 1) \sec \theta}$

$$= -\ln \tan \frac{3\pi}{8} = -\ln(\sqrt{2}+1) = \ln(\sqrt{2}-1)$$

93. If $f(x)$ is function satisfying $f\left(\frac{1}{x}\right) + x^2 f(x) = 0$ for all $x (x \neq 0)$ then the value of the integral $\int_{\tan \theta}^{\cot \theta} f(x) dx$ is **(Substitution)**

- A) $\tan^2 \theta$ B) $2 \tan \theta$ C) 0 D) 1

Ans: C

$$\begin{aligned} \text{Sol: Let } I &= \int_{\tan \theta}^{\cot \theta} f(x) dx = \int_{\tan \theta}^{\cot \theta} \frac{-1}{x^2} f\left(\frac{1}{x}\right) dx \\ \left[\because f(x) &= \frac{-1}{x^2} f\left(\frac{1}{x}\right) \right] \\ &= \int_{\cot \theta}^{\tan \theta} f(x) dx = -I \Rightarrow 2I = 0, \therefore I = 0 \end{aligned}$$

94. The value of the integral $\int_0^{\infty} \frac{dx}{\left(x + \sqrt{x^2 + 1}\right)^n}$, where $n > 1$, is **(Substitution)**

- A) $\frac{n^2}{n^2 - 1}$ B) $\frac{n}{n^2 - 1}$ C) $\frac{n^2}{n^2 + 1}$ D) $\frac{n}{n^2 + 1}$

Key: B

$$\text{Sol: put } \log(x + \sqrt{x^2 + 1}) = t$$

$$\Rightarrow x + \sqrt{x^2 + 1} = e^t$$

$$\Rightarrow \sqrt{x^2 + 1} = e^{2t} - 2xe^t + x^2$$

$$\therefore x = \frac{e^{2t} - 1}{2e^t} = \frac{e^t - e^{-t}}{2}$$

$$dx = \left(\frac{e^t + e^{-t}}{2} \right) dt$$

When $x \rightarrow 0, t \rightarrow 0$ and when $x \rightarrow \infty, t \rightarrow \infty$

$$= \frac{1}{2} \left\{ \frac{2n}{n^2 - 1} \right\} = \frac{n}{n^2 - 1}$$

95. Let f be integrable over $[0, a]$ for any real a . if we define

$$I_1 = \int_0^{\pi/2} \cos \theta f(\sin \theta + \cos^2 \theta) d\theta \text{ and } \textbf{(Trigonometric Substitution)}$$

$$I_2 = \int_0^{\pi/2} \sin 2\theta f(\sin \theta + \cos^2 \theta) d\theta \text{ then}$$

- A) $I_1 = I_2$ B) $I_1 = -I_2$ C) $I_1 = 2I_2$ D) $I_1 = -2I_2$

Ans: A

$$\text{Sol: } I_1 - I_2 = \int_0^{\pi/2} (\cos \theta - \sin 2\theta) f(\sin \theta + \cos^2 \theta) d\theta$$

Put $\sin \theta + \cos^2 \theta = t$

$$\Rightarrow (\cos \theta - \sin 2\theta) d\theta = dt$$

Also if $\theta = 0, t = 0 + 1^2 = 1$

And if $\theta = \frac{\pi}{2}, t = 1 + 0 = 1$

$$\therefore I_1 - I_2 = \int_1^1 f(t) dt = 0 \Rightarrow I_1 = I_2$$

96. $\int_0^{\sqrt{3}} \frac{1}{1+x^2} \cdot \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx$ (Trigonometric Substitution)

A) $\frac{7}{72} \pi^2$ B) $\frac{3}{42} \pi^2$ C) $\frac{17}{72} \pi^2$ D) $7/72$

Key:A

Sol: Let $I = \int_0^{\sqrt{3}} \frac{1}{1+x^2} \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx$

$$\therefore I = \int_0^1 \frac{1}{1+x^2} \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx$$

$$\therefore I = \int_0^1 \frac{1}{1+x^2} \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx + \int_1^{\sqrt{3}} \frac{1}{1+x^2} \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx$$

$$\int_0^1 \frac{2 \tan^{-1} x}{1+x^2} dx + \int_1^{\sqrt{3}} \frac{\pi - 2 \tan^{-1} x}{1+x^2} dx$$

$$= \frac{\pi^2}{16} + \pi \left\{ \frac{\pi}{3} - \frac{\pi}{4} \right\} - \left\{ \frac{\pi^2}{9} - \frac{\pi^2}{16} \right\} = \frac{7}{72} \pi^2$$

97. Let $I = \int_0^{\infty} \frac{2x^2}{e^x - e^{-x}} dx$ and $J = \int_0^{\infty} \frac{x^2}{e^x - 1} dx$ then $\frac{4I}{J}$ equals (Substitution)

A) 4 B) 7 C) 6 D) 8

Key:A

Sol: conceptual

98. A function f is defined by $f(x) = \frac{1}{2^{r-1}}, \frac{1}{2^r} < x \leq \frac{1}{2^{r-1}}, r = 1, 2, 3, \dots$. Then the value of

$$\int_0^1 f(x) dx \text{ is equal to}$$

(Substitution)

A) $1/3$ B) $1/4$ C) $2/3$ D) $1/3$

Key:C

Sol: $\int_0^1 f(x) dx = \sum_{r=1}^{\infty} \int_{\frac{1}{2^r}}^{\frac{1}{2^{r-1}}} \frac{1}{2^{r-1}} dx$

$$\begin{aligned}
 &= \sum_{r=1}^{\infty} 2^{-2(r-1)} - \sum_{r=1}^{\infty} 2^{-2r+1} \\
 &= 2 \cdot \frac{1}{4} \cdot \frac{1}{1 - \frac{1}{4}} = \frac{2}{3}
 \end{aligned}$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 1: Change of variables, interchanging limits, splitting limits Property,

Location of roots in an interval using definite integral,

Improper integral Property, Reflection property, Shifting property

99. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{dx}{[x] + [\sin x] + 4}$, where $[t]$ denotes the greatest integer less than or equal to t , is

- 1) $\frac{1}{12}(7\pi + 5)$ 2) $\frac{1}{12}(7\pi - 5)$ 3) $\frac{3}{20}(4\pi - 3)$ 4) $\frac{3}{10}(4\pi - 3)$

Key: 3

(Step Function and splitting rule) 2019

Sol: $\int_{-\frac{\pi}{2}}^{-3} \frac{dx}{-2-1+4} + \int_{-1}^0 \frac{dx}{-1-1+4} + \int_0^1 \frac{dx}{0+0+4} + \int_1^{\frac{\pi}{2}} \frac{dx}{1+0+4} = \frac{3\pi}{5} - \frac{9}{20}$

100. The value of the integral $\int_{-2}^2 \frac{\sin^2 x}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} dx$ (where $[x]$ denotes the greatest integer less than or equal to x) is:

- 1) 4 2) $4 - \sin 4$ 3) $\sin 4$ 4) 0

Ans. 4

(Step Function and splitting rule) 2019

Sol. $I = \int_{-2}^2 \frac{\sin^2 x}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} dx$ $I = \int_0^2 \left(\frac{\sin^2 x}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} + \frac{\sin^2(-x)}{\left[-\frac{x}{\pi}\right] + \frac{1}{2}} \right) dx$

$$\left(\left[\frac{x}{\pi} \right] + \left[-\frac{x}{\pi} \right] = -1 \text{ as } x \neq n\pi \right) \quad I = \int_0^2 \left(\frac{\sin^2 x}{\left[\frac{x}{\pi} \right] + \frac{1}{2}} + \frac{\sin^2 x}{-1 - \left[\frac{x}{\pi} \right] + \frac{1}{2}} \right) dx = 0$$

101. The value of α for which $4\alpha \int_{-1}^2 e^{-\alpha|x|} dx = 5$, is:

- 1) $\log_e 2$ 2) $\log_e \sqrt{2}$ 3) $\log_e \left(\frac{3}{2} \right)$ 4) $\log_e \left(\frac{4}{3} \right)$

Ans :

1

(Mod Function and splitting rule)2020

$$\begin{aligned} \text{Sol : } 4\alpha \left\{ \int_{-1}^0 e^{\alpha x} dx + \int_0^2 e^{-\alpha x} dx \right\} &= 5 & \Rightarrow 4\alpha \left\{ \left(\frac{e^{\alpha x}}{\alpha} \right)_{-1}^0 + \left(\frac{e^{-\alpha x}}{-\alpha} \right) \right\} &= 5 \\ \Rightarrow 4\alpha \left\{ \left(\frac{1 + e^{-\alpha}}{\alpha} \right) - \left(\frac{e^{-2\alpha} - 1}{\alpha} \right) \right\} &= 5 & \Rightarrow 4(2 - e^{-\alpha} - e^{-2\alpha}) &= 5 \text{ Put } e^{-\alpha} = t \\ \Rightarrow 4t^2 + 4t - 3 &= 0 & \Rightarrow (2t + 3)(2t - 1) &= 0 \Rightarrow e^{-\alpha} = \frac{1}{2} & \Rightarrow \alpha = \ln 2 \end{aligned}$$

102. The integral $\int_0^2 ||x-1|-x| dx$ is equal to

Key: 2

(Mod Function and Splitting rule)2020

$$\begin{aligned} \text{Sol: } \int_0^2 ||x-1|-x| dx &= \int_0^1 |1-2x| dx + \int_1^2 |(-1)| dx \\ &= \int_0^1 (1-2x) dx + \int_{1/2}^1 (2x-1) dx + 1 \\ &= \left(x - x^2 \right)_0^{1/2} + \left(x^2 - x \right)_{1/2}^1 + 1 = 1.5 \end{aligned}$$

103. Let $[t]$ denote the greatest integer less than or equal to t . Then the value of

$$\int_1^2 |2x - [3x]| dx \text{ is } \underline{\hspace{2cm}}$$

Key: 1.00

(Step - 1 Mod Function and splitting rule)2020

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Sol: $\int_1^2 |2x - 3[x]| dx$

$$= \int_1^2 |3x - 3[x] - x| dx$$

$$= \int_1^2 |\{3x\} - x| dx = \int_1^2 x - \{3x\} dx \quad (\text{As } \{3x\} < 1 \quad \forall x \in \mathbb{R})$$

$$\int_1^2 x dx - 3 \int_0^{1/3} 3x dx = \left(2 - \frac{1}{2}\right) - \frac{9}{2} \left(\frac{1}{9} - 0\right) = 1.00$$

104. $\int_{-\pi}^{\pi} |\pi - |x|| dx$ is equal to:

- 1) π^2 2) $\frac{\pi^2}{2}$ 3) $\sqrt{2}\pi^2$ 4) $2\pi^2$

Key: 1 **(Mod Function and splitting rule)2020**

Sol: $2 \int_0^{\pi} |\pi - x| dx = 2 \int_0^{\pi} (\pi - x) dx = 2 \left[\pi x - \frac{x^2}{2} \right]_0^{\pi} = 2 \left(\pi^2 - \frac{\pi^2}{2} \right) = 2 \cdot \frac{\pi^2}{2} = \pi^2$

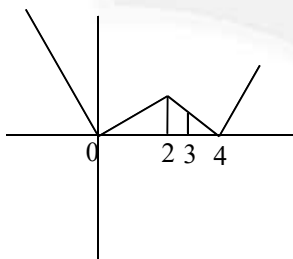
105. Let $f(x) = |x - 2|$ and $g(x) = f(f(x))$, $x \in [0, 4]$. Then $\int_0^3 (g(x) - f(x)) dx$ is equal to:

- 1) 0 2) $\frac{1}{2}$ 3) 1 4) $\frac{3}{2}$

Key: 3 **(Mod Function and Splitting rule)2020**

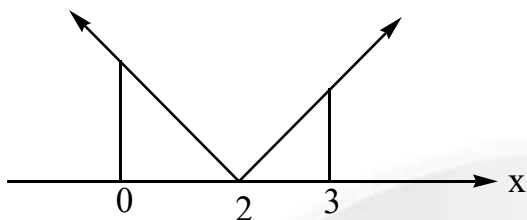
Sol:

$$\int_0^3 f(f(x)) dx - \int_0^3 f(x) dx \quad f(f(x)) = ||x - 2| - 2|$$



$$A_1 = \frac{1}{2} \times 2 \times 2 + \frac{1}{2} [3] = 2 + \frac{3}{2} = \frac{7}{2}$$

$$\text{Reg Area} = \frac{7}{2} - \frac{5}{2} = 1$$



$$A_2 = \frac{1}{2} \times 2 \times 2 + \frac{1}{2} \times 1 \times 1 = 2 + \frac{1}{2} = \frac{5}{2}$$

106. Let $f(x) = \max(|x+1|, |x+2|, |x+3|, \dots, |x+5|)$. Then $\int_{-6}^0 f(x) dx$ is equal to _____.

Key: 21

(Mod Function and Splitting rule)2021

$$\text{Sol. } \int_{-6}^{-3} |x+1| dx + \int_{-3}^0 |x+5| dx$$

$$= -\int_{-6}^{-3} (x+1) dx + \int_{-3}^0 (x+5) dx$$

$$= -\left(\frac{x^2}{2} + x\right)_{-6}^{-3} + \left(\frac{x^2}{2} + 5x\right)_{-3}^0$$

$$= -\left(\frac{9}{2} - 3 - 18 + 6\right) - \left(\frac{9}{2} - 15\right)$$

$$= -\left(\frac{9-30}{2}\right) - \left(\frac{9-30}{2}\right)$$

$$= 21$$

107. The value of the integral $\int_{-2}^2 \frac{|x^3 + x|}{(e^{|x|} + 1)} dx$ is equal to :

(A) $5e^2$

(B) $3e^{-2}$

(C) 4

(D) 6

Ans: D

(Mod Function and Property)2021

$$\text{Sol: } f(x) = \frac{|x^3 + x|}{(e^{|x|} + 1)}$$

$$\int_{-2}^2 f(x) dx = \int_0^2 (f(x) + f(-x)) dx$$

$$= \int_0^2 \left(\frac{|x^3 + x|}{(e^{|x|} + 1)} + \frac{|-x^3 - x|}{(e^{-|x|} + 1)} \right) dx$$

$$= \int_0^2 \left(\frac{|x^3 + x|}{(e^{|x|} + 1)} + \frac{|x^3 + x|}{(e^{-|x|} + 1)} \right) dx$$

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$$\begin{aligned}
 &= \int_0^2 \left(\frac{x^3 + x}{(e^{x^2} + 1)} + \frac{x^3 + x}{(e^{-x^2} + 1)} \right) dx \\
 I &= \int_0^2 \left(\frac{x^3 + x}{1 + e^{x^2}} + \frac{e^{x^2}(x^3 + x)}{1 + e^{x^2}} \right) dx \\
 &= \int_0^2 (x^3 + x) dx \\
 &= \left[\frac{x^4}{4} + \frac{x^2}{2} \right]_0^2 \\
 &= 4 + 2 = 6
 \end{aligned}$$

107. The integral $\int_0^1 \frac{1}{7 \left[\frac{1}{x} \right]} dx$, where $[.]$ denotes the greatest integer function, is equal to

- 1) $1 + 6 \log_e \left(\frac{6}{7} \right)$ 2) $1 - 6 \log_e \left(\frac{6}{7} \right)$ 3) $\log_e \left(\frac{7}{6} \right)$ 4) $1 - 7 \log_e \left(\frac{6}{7} \right)$

Key: 1

(Step Function)2021

Sol: $\int_0^1 \frac{dx}{7 \left[\frac{1}{x} \right]} = - \int_1^0 \frac{dx}{7 \left[\frac{1}{x} \right]}$

$$\begin{aligned}
 &= -1 \left[\int_1^{\frac{1}{2}} \frac{dx}{7^1} + \int_{\frac{1}{2}}^{\frac{1}{3}} \frac{dx}{7^2} + \int_{\frac{1}{3}}^{\frac{1}{4}} \frac{dx}{7^3} + \dots \dots \dots \infty \right] \\
 &= \frac{1}{7} \left(1 - \frac{1}{2} \right) + \frac{1}{7^2} \left(\frac{1}{2} - \frac{1}{3} \right) + \frac{1}{7^3} \left(\frac{1}{3} - \frac{1}{4} \right) + \dots \dots \dots \infty \\
 &= \left(\frac{1}{7} + \frac{1}{2 \cdot 7^2} + \frac{1}{3 \cdot 7^3} + \dots \infty \right) - \left(\frac{1}{7 \cdot 2} + \frac{1}{7^2 \cdot 3} + \frac{1}{7^3 \cdot 4} + \dots \infty \right) \\
 &= -\log \left(1 - \frac{1}{7} \right) - 7 \left(\frac{1}{7^2 \cdot 2} + \frac{1}{7^3 \cdot 3} + \frac{1}{7^4 \cdot 4} + \dots \infty \right) \\
 &= -\log \left(\frac{6}{7} \right) - 7 \left[-\log \left(1 - \frac{1}{7} \right) - \frac{1}{7} \right] = 1 + 6 \log \left(\frac{6}{7} \right)
 \end{aligned}$$

108. Let $[t]$ denote the greatest integer less than or equal to t . Then, the value of the integral

$$\int_0^1 [-8x^2 + 6x - 1] dx \text{ is equal to}$$

A) -1

B) $\frac{-5}{4}$

C) $\frac{\sqrt{17}-13}{8}$

D) $\frac{\sqrt{17}-16}{8}$

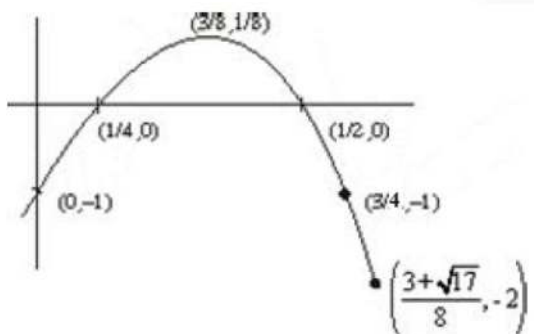
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Key: C

(Step Function and Splitting rule)2021

SOL: $\int_0^1 [-8x^2 + 6x - 1] dx$

$$= \int_0^{1/4} -1 dx + \int_{1/4}^{1/2} 0 dx + \int_{1/2}^{3/4} -1 dx$$



$$+ \int_{3/4}^{\frac{3+\sqrt{17}}{8}} -2 dx + \int_{\frac{3+\sqrt{17}}{8}}^1 -3 dx$$

$$= -\frac{1}{4} - \frac{1}{4} + \frac{-6 - 2\sqrt{17}}{8} + \frac{3}{2} - 3 + \frac{9 + 3\sqrt{17}}{8}$$

$$= \frac{\sqrt{17} - 13}{8}$$

109. $\int_0^5 \cos \left(\pi \left(x - \left[\frac{x}{2} \right] \right) \right) dx$, Where $[t]$ denotes greatest integer less than or equal to t , is equal to:

A) -3

B) -2

C) 2

D) 0

Ans: D

(Step Function and Splitting rule)2021

Sol: $\int_0^5 \cos \left(\pi \left(x - \left[\frac{x}{2} \right] \right) \right) dx$

$$= \int_0^2 \cos \pi x dx + \int_2^4 \cos \pi(x-1) dx + \int_4^5 \cos \pi(x-2) dx$$

$$= \left[\frac{\sin \pi x}{\pi} \right]_0^2 + \left[\frac{\sin \pi(x-1)}{\pi} \right]_2^4 + \left[\frac{\sin \pi(x-2)}{\pi} \right]_4^5$$

$$= 0$$

110. If $\int_0^2 (\sqrt{2x} - \sqrt{2x-x^2}) dx = \int_0^1 \left(1 - \sqrt{1-y^2} - \frac{y^2}{2} \right) dy + \int_0^2 \left(2 - \frac{y^2}{2} \right) dy + I$, then I equals

1) $\int_0^1 (1 + \sqrt{1-y^2}) dy$

2) $\int_0^1 \left(\frac{y^2}{2} - \sqrt{1-y^2} + 1 \right) dy$

$$3) \int_0^1 (1 - \sqrt{1-y^2}) dy$$

$$4) \int_0^1 \left(\frac{y^2}{2} + \sqrt{1-y^2} + 1 \right) dy$$

KEY:3

(Splitting rule)2021

$$\text{SOL: } \int_0^2 (\sqrt{2x} - \sqrt{2x-x^2}) dx = \int_0^1 \left(1 - \sqrt{1-x^2} - \frac{x^2}{2} \right) dx + \int_1^2 \left(2 - (1 + \sqrt{1-x^2}) \right) dx + \int_1^2 \left(2 - \frac{x^2}{2} \right) dx$$

$$\therefore I = \int_0^1 (1 - \sqrt{1-x^2}) dx$$

111. The value of the integral, $\int_1^3 [x^2 - 2x - 2] dx$, where $[x]$ denotes the greatest integer less or equal to x , is

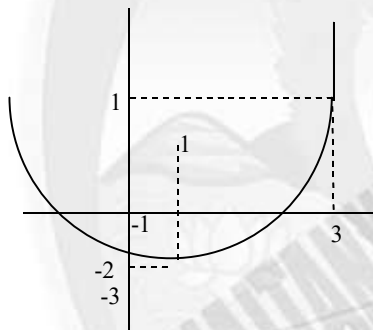
$$1) -\sqrt{2} - \sqrt{3} + 1 \quad 2) -\sqrt{2} - \sqrt{3} - 1 \quad 3) -4$$

$$4) -5$$

Key: 2

(Step Function and Periodic Property)2021

Solution:



$$x^2 - 2x - 2 = -3 \quad x = 1$$

$$= -2 \quad x = 2$$

$$= -1 \quad x = \frac{2 + 2\sqrt{2}}{2}$$

$$= 0 \quad x = \frac{2 + 2\sqrt{3}}{2}$$

$$= 1 \quad x = 3$$

$$\int_1^2 -3 dx + \int_2^{1+\sqrt{2}} -2 dx + \int_{1+\sqrt{2}}^{1+\sqrt{3}} -1 dx + \int_{1+\sqrt{3}}^3 0 dx = -\sqrt{2} - \sqrt{3} - 1$$

112. The value of $\int_{-2}^2 (3x^2 - 3x - 6) dx$ is.....

Ans:19

(Mod Function and Periodic Property)2021

$$\text{Sol } \int_{-2}^2 (3x^2 - 3x - 6) dx = 3 \int_{-2}^2 (x^2 - x - 2) dx$$

$$= 3 \int_{-2}^2 |(x-2)(x+1)| dx = 3 \left\{ \int_{-2}^{-1} (x^2 - x - 2) dx + \int_{-1}^2 (2 + x - x^2) dx \right\} = 19$$

113. If $[x]$ denotes the greatest integer less than or equal to x , then the value of the integral

$$\int_{-\pi/2}^{\pi/2} ([x] - \sin x) dx \text{ is equal to:}$$

1) 1

2) π

3) 0

4) $-\pi$

(Step Function and Splitting rule)2021

Key: 4

Sol: $\int_{-\pi/2}^{\pi/2} ([x] - \sin x) dx =$

$$= \int_0^{\pi/2} ([x] + [-x]) dx + \int_0^{\pi/2} ([-\sin x] + [\sin x]) dx$$

$$= \int_0^{\pi/2} -1 dx + \int_0^{\pi/2} -1 dx$$

$$= (-x)_0^{\pi/2} + (-x)_0^{\pi/2} = -\pi$$

114. Consider the integral $I = \int_0^{10} \frac{[x]e^{[x]}}{e^{x-1}} dx$, where $[x]$ denotes the greatest integer less than or equal to x . Then the value of I is equal to:

1) $45(e-1)$ 2) $9(e+1)$ 3) $45(e+1)$ 4) $9(e-1)$

Key: 1

(Step Function and splitting rule)2021

Sol: $I = \int_0^{10} [x] \cdot e^{[x]+1-x} dx = \int_0^1 0 dx + \int_1^2 e^{2-x} dx + \int_2^3 2e^{3-x} dx + \int_3^4 3e^{4-x} dx + \dots + \int_9^{10} 9e^{10-x} dx$

$$= -\{(1-e) + 2(1-e) + 3(1-e) + \dots + 9(1-e)\} = 45(e-1)$$

115. Let $g(x) = \int_0^x f(t) dt$, where f is continuous function in $[0, 3]$ such that $\frac{1}{3} \leq f(t) \leq 1$

for all $t \in [0, 1]$ and $0 \leq f(t) \leq \frac{1}{2}$ for all $t \in (1, 3]$. The largest possible interval in which $g(3)$ lies is:

1) $\left[-1, -\frac{1}{2}\right]$ 2) $[1, 3]$ 3) $\left[-\frac{3}{2}, -1\right]$ 4) $\left[\frac{1}{3}, 2\right]$

Key:

4

(Splitting rule)2021

Sol: $\int_0^1 \frac{1}{3} dt + \int_1^3 0 \cdot dt \leq g(3) \leq \int_0^1 1 dt + \int_1^3 \frac{1}{2} \cdot dt$ $\frac{1}{3} \leq g(3) \leq 2$

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116. If the value of the integer $\int_0^5 \frac{x + [x]}{e^{x-[x]}} dx = \alpha e^{-1} + \beta$, where $\alpha, \beta \in R, 5\alpha + 6\beta = 0$, & $[x]$ denotes the greatest integer less than or equal to x ; then the value of $(\alpha + \beta)^2$ is equal to

1) 25

2) 36

3) 16

4) 100

Key: 1**(Step Function and splitting rule)2021**

Sol: $I = \int_0^5 \frac{x + [x]}{e^{x-[x]}} dx$

$$\int_0^1 \frac{x}{e^x} dx + \int_1^2 \frac{x+1}{e^{x-1}} dx + \int_2^3 \frac{x+2}{e^{x-2}} dx + \dots + \int_4^5 \frac{x+4}{e^{x-4}} dx$$

$$\int_0^1 \frac{t+2}{e^t} dt + \int_0^1 \frac{z+4}{e^z} dz + \dots + \int_0^1 \frac{y+8}{e^y} dy$$

$$\Rightarrow \int_0^5 \frac{5x+20}{e^x} dx = 5 \int_0^1 \frac{x+4}{e^x} dx$$

$$\Rightarrow 5 \int_0^1 (x+4)e^{-x} dx \Rightarrow 5e^{-x}(-x-5) \Big|_0^1 \Rightarrow \frac{30}{e} + 25$$

$$\alpha = -30, \beta = 25 \Rightarrow 5\alpha + 6\beta = 0 \Rightarrow (\alpha + \beta)^2 = 5^2 = 25$$

117. Let $[t]$ denote the greatest integer $\leq t$. Then the value of $8 \cdot \int_{-\frac{1}{2}}^1 ([2x] + |x|) dx$ IS _____

Key:5**(Step Function and splitting rule)2021**

Sol: $8 \int_{-\frac{1}{2}}^1 ([2x] + |x|) dx$

$$= 8 \int_{-\frac{1}{2}}^1 [2x] dx + 8 \int_{-\frac{1}{2}}^1 |x| dx \quad \text{Put } 2x = t \Rightarrow dx = \frac{1}{2} dt = \lim ts : -1/2, 1 \text{ new limits : } -1, 2$$

$$= 8 \int_{-1}^2 [t] \cdot \frac{1}{2} dt + 8 \left[\int_{-\frac{1}{2}}^0 (-x) dx + \int_0^1 x dx \right]$$

$$= 4 \left(\int_{-1}^0 (-1) dt + \int_0^1 (0) dt + \int_1^2 (1) dt \right)$$

$$+ 8 \left(-\frac{x^2}{2} \right)_{-\frac{1}{2}}^0 + \left(\frac{x^2}{2} \right)_0^1$$

$$= 4(-x)_{-1}^0 + 0 + (x)_1^2 + 8 \left(0 + \frac{1}{8} + \frac{1}{2} \right)$$

$$= 4(0-1+(2-1)) + (1+4) = 0+5 = 5$$

118. If $[x]$ is the greatest integer $\leq x$, then $\pi^2 \int_0^2 \left(\sin \frac{\pi x}{2} \right) (x - [x])^{[x]} dx$ is equal to

- 1) $4(\pi + 1)$ 2) $2(\pi - 1)$ 3) $4(\pi - 1)$ 4) $2(\pi + 1)$

Key : 3

(Step Function and splitting rule)2021

Sol : $\pi^2 \int_0^1 \left(\sin \frac{\pi x}{2} \right) x^0 dx + \pi^2 \int_1^2 \sin \left(\frac{\pi x}{2} \right) (x - 1)^1 dx$

$$= \pi^2 \int_0^1 \sin \left(\frac{\pi}{2} x \right) dx + \pi^2 \int_1^2 (x - 1) \sin \left(\frac{\pi x}{2} \right) dx$$

$$= \pi^2 \left(-\cos \left(\frac{\pi}{2} x \right) \frac{2}{\pi} \right)_0^1 + \pi^2 \left\{ (x - 1) \left[-\cos \left(\frac{\pi}{2} x \right) \right] \left(\frac{2}{\pi} \right) + \sin \left(\frac{\pi x}{2} \right) \left(\frac{2}{\pi} \right)^2 \right\}_1^2$$

$$= \pi^2 \left\{ -\frac{2}{\pi} (0 - 1) + \pi^2 \left(-\frac{2}{\pi} (-1) - \left(\frac{2}{\pi} \right)^2 \right) \right\}$$

$$= 2\pi + 2\pi - 4 = 4\pi - 4 = 4(\pi - 1)$$

119. The integral $\int_0^1 \frac{1}{7^{\left[\frac{1}{x} \right]}} dx$, where $[.]$ denotes the greatest integer function, is equal to

1) $1 + 6 \log_e \left(\frac{6}{7} \right)$ 2) $1 - 6 \log_e \left(\frac{6}{7} \right)$

3) $\log_e \left(\frac{7}{6} \right)$ 4) $1 - 7 \log_e \left(\frac{6}{7} \right)$

Key: 1

(GREATEST INTEGER FUNCTION)2022

Sol: $\int_{\frac{1}{2}}^1 \frac{1}{7^{\left[\frac{1}{x} \right]}} dx + \int_{\frac{1}{3}}^{\frac{1}{2}} \frac{1}{7^{\left[\frac{1}{x} \right]}} dx + \int_{\frac{1}{4}}^{\frac{1}{3}} \frac{1}{7^{\left[\frac{1}{x} \right]}} dx + \dots$

$$= \frac{1}{7} \left(1 - \frac{1}{2} \right) + \frac{1}{7^2} \left(\frac{1}{2} - \frac{1}{3} \right) + \frac{1}{7^3} \left(\frac{1}{3} - \frac{1}{4} \right) + \dots = \left(\frac{1}{7} + \frac{1}{7^2} + \frac{1}{7^3} + \dots \right)$$

$$\begin{aligned}
 -\left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots\right) &= -\log\left(1 - \frac{7}{7}\right) - 7\left(\frac{1}{7} + \frac{\left(\frac{1}{7}\right)^2}{2} + \frac{\left(\frac{1}{7}\right)^3}{3} + \dots\right) \frac{-1}{7} \\
 &= -\log\left(\frac{6}{7}\right) - 7\left[-\log\left(1 - \frac{1}{7}\right) - \frac{1}{7}\right] = 1 + 6 \cdot \log\left(\frac{6}{7}\right)
 \end{aligned}$$

120. Let $[t]$ denote the greatest integer less than or equal to t . Then, the value of the integral

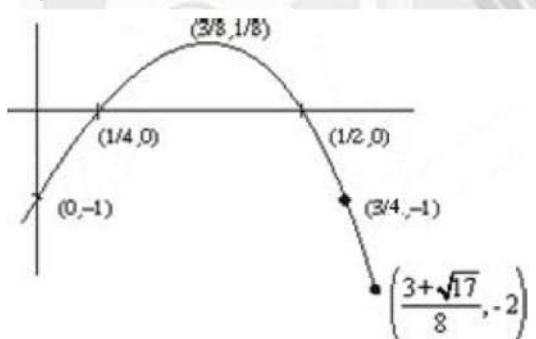
$$\int_0^1 [-8x^2 + 6x - 1] dx \text{ is equal to } \textbf{(GREATEST INTEGER FUNCTION)2022}$$

- A) -1 B) $\frac{-5}{4}$ C) $\frac{\sqrt{17}-13}{8}$ D) $\frac{\sqrt{17}-16}{8}$

Key: C

$$\text{SOL: } \int_0^1 [-8x^2 + 6x - 1] dx$$

$$= \int_0^{1/4} -1 dx + \int_{1/4}^{1/2} 0 dx + \int_{1/2}^{3/4} -1 dx$$



$$\begin{aligned}
 &+ \int_{3/4}^{\frac{3+\sqrt{17}}{8}} -2 dx + \int_{\frac{3+\sqrt{17}}{8}}^1 -3 dx \\
 &= -\frac{1}{4} - \frac{1}{4} + \frac{-6 - 2\sqrt{17}}{8} + \frac{3}{2} - 3 + \frac{9 + 3\sqrt{17}}{8} = \frac{\sqrt{17} - 13}{8}
 \end{aligned}$$

121. $\int_0^5 \cos\left(\pi\left(x - \left[\frac{x}{2}\right]\right)\right) dx$, Where $[t]$ denotes greatest integer less than or equal to t , is equal to:

- A) -3 B) -2 C) 2 D) 0

Ans: D

(GREATEST INTEGER FUNCTION)2022

$$\text{Sol: } \int_0^5 \cos\left(\pi\left(x - \left[\frac{x}{2}\right]\right)\right) dx$$

$$= \int_0^2 \cos \pi x dx + \int_2^4 \cos \pi(x-1) dx + \int_4^5 \cos \pi(x-2) dx$$

$$= \left[\frac{\sin \pi x}{\pi} \right]_0^2 + \left[\frac{\sin \pi(x-1)}{\pi} \right]_2^4 + \left[\frac{\sin \pi(x-2)}{\pi} \right]_4^5$$

$$= 0$$

122. If $\int_0^2 (\sqrt{2x} - \sqrt{2x-x^2}) dx = \int_0^1 \left(1 - \sqrt{1-y^2} - \frac{y^2}{2} \right) dy + \int_0^2 \left(2 - \frac{y^2}{2} \right) dy + I$, then I equals

1) $\int_0^1 (1 + \sqrt{1-y^2}) dy$

2) $\int_0^1 \left(\frac{y^2}{2} - \sqrt{1-y^2} + 1 \right) dy$

3) $\int_0^1 (1 - \sqrt{1-y^2}) dy$

4) $\int_0^1 \left(\frac{y^2}{2} + \sqrt{1-y^2} + 1 \right) dy$

KEY:3

(SPILTING OF LIMITS)2022

SOL: $\int_0^2 (\sqrt{2x} - \sqrt{2x-x^2}) dx = \int_0^1 \left(1 - \sqrt{1-x^2} - \frac{x^2}{2} \right) dx + \int_0^2 \left(2 - (1 + \sqrt{1-x^2}) \right) dx + \int_1^2 \left(2 - \frac{x^2}{2} \right) dx$

$$\therefore I = \int_0^1 (1 - \sqrt{1-x^2}) dx$$

123. If $[t]$ denotes the greatest integer $\leq t$, then the value of $\frac{3(e-1)}{e} \int_1^2 x^2 e^{[x]+[x^3]} dx$ is:

1) $e^8 - e$

2) $e^8 - 1$

3) $e^9 - e$

4) $e^7 - 1$

ANS1

(GREATEST INTEGER FUNCTION)2023

SOL $I = \int_1^2 x^2 e^{[x]+[x^3]} dx = \int_1^2 x^2 e^{1+[x^3]} dx = \frac{e}{3} \int_1^8 e^{[t]} dt$

Where $t=x^3$

$$I = \frac{e}{3} e \left(\frac{e^7 - 1}{e - 1} \right) \therefore \text{Given DI} = \frac{3(e-1)}{e} \cdot \frac{e^2 (e^7 - 1)}{3 (e - 1)} = e^8 - e$$

124. Let the function $f: [0, 2] \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} e^{\min\{x^2, x-[x]\}}, & x \in [0, 1) \\ e^{[x - \log_e x]}, & x \in [1, 2] \end{cases}$$

where $[t]$ denotes the greatest integer less than or equal to t . Then the value of the integral

$$\int_0^2 x f(x) dx \text{ is}$$

1) $(e-1) \left(e^2 + \frac{1}{2} \right)$

2) $2e - \frac{1}{2}$

3) $1 + \frac{3e}{2}$

4) $2e - 1$

KEY:2

(GREATEST INTEGER FUNCTION)2023

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SOL: $\int_0^2 xf(x)dx = \int_0^1 xe^{x^2} dx + \int_1^2 x(e)dx = \frac{1}{2}(e^{x^2})_0^1 + e\left(\frac{x^2}{2}\right)_1^2 = \frac{1}{2}(e-1) + e\left(2-\frac{1}{2}\right) = 2e - \frac{1}{2}$

125. $\int_0^{\pi} ||\sin x| - |\cos x|| dx$

- 1) $4(\sqrt{2}-1)$ 2) $2(\sqrt{2}+1)$ 3) $2(\sqrt{2}-1)$ 4) $4(\sqrt{2}+1)$

Key 1

(Mod Function and splitting rule)

Sol: We have $\Rightarrow |\sin x| - |\cos x| = 0 \Rightarrow |\sin x| = |\cos x| \Rightarrow |\tan x| = 1 \Rightarrow x = \frac{\pi}{4}, \frac{3\pi}{4}$

$$\begin{aligned} \text{Let } I &= \int_0^{\pi} ||\sin x| - |\cos x|| dx = 2 \int_0^{\frac{\pi}{2}} |\sin x - \cos x| dx \\ &= 2 \times \sqrt{2} \int_0^{\frac{\pi}{2}} \left| \frac{1}{\sqrt{2}} \sin x - \frac{1}{\sqrt{2}} \cos x \right| dx = 2\sqrt{2} \int_0^{\frac{\pi}{2}} \left| \sin \left(x - \frac{\pi}{4} \right) \right| dx \\ &= 2\sqrt{2} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} |\sin x| dx = 4\sqrt{2} \int_0^{\frac{\pi}{4}} |\sin x| dx = 4\sqrt{2} \int_0^{\frac{\pi}{4}} \sin x dx \\ &= 4\sqrt{2} (-\cos x)_0^{\pi/4} = 4\sqrt{2} \left(1 - \frac{1}{\sqrt{2}} \right) = 4(\sqrt{2}-1) \end{aligned}$$

126. The value of $\int_0^{2n\pi} [\sin x + \cos x] dx =$

- 1) $n\pi$ 2) $-n\pi$ 3) $-2n\pi$ 4) $\frac{-n\pi}{2}$

Key: 2

(Step Function and splitting rule)

Sol: $\int_0^{2n\pi} [\sin x + \cos x] dx = n \int_0^{2\pi} \left[\sqrt{2} \sin \left(x + \frac{\pi}{4} \right) \right] dx = n \int_{\frac{\pi}{4}}^{\frac{9\pi}{4}} [\sqrt{2} \sin t] dt = n(-\pi) = -n\pi$

127. $[y]$ is the integral part of y , then $\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} [2 \sin x] dx =$

1) $-\pi$

2) 0

3) $\frac{-\pi}{2}$

4) $\frac{\pi}{2}$

Key: 3

(Step Function and Splitting rule)

Sol:

$$= \int_{\frac{\pi}{2}}^{\frac{5\pi}{6}} [2 \sin x] dx + \int_{\frac{5\pi}{6}}^{\pi} [2 \sin x] dx + \int_{\pi}^{\frac{7\pi}{6}} [2 \sin x] dx + \int_{\frac{7\pi}{6}}^{\frac{3\pi}{2}} [2 \sin x] dx$$

128. If $g(x) = \int_0^x \cos^4 t dt$, then $g(x + \pi)$ equals

1) $g(x) + g(\pi)$

2) $g(x) - g(\pi)$

3) $g(x)g(\pi)$

4) $g(x)/g(\pi)$

Key: 1

(Splitting rule)

Sol:

$$g(x) = \int_0^x \cos^4 t dt$$

$$\text{or } g(x + \pi) = \int_0^{x+\pi} \cos^4 t dt$$

$$= \int_0^x \cos^4 t dt + \int_x^{x+\pi} \cos^4 t dt$$

$$= g(x) + \int_0^{\pi} \cos^4 t dt$$

$$= g(x) + g(\pi)$$

129. $\int_4^{10} ([x - 4] + [10 - x]) dx =$ (Where $[x]$ is the largest integer not exceeding x)

1) 26

2) 10

3) 30

4) 20

Key: 3

(Step Function and Splitting rule)

Sol:

$$\begin{aligned}\int_4^{10} \{[x-4] + [10-x]\} dx &= \int_4^{10} \{[x] - 4 + 10 + [-x]\} dx \\ &= \int_4^{10} \{[x] + [-x]\} dx + \int_4^{10} 6 dx = \int_4^{10} -1 dx + \int_4^{10} 6 dx\end{aligned}$$

130. The value of $\int_{-1}^1 \frac{\sin x - x^2}{3-|x|} dx$ is _____

- 1) $17 + 18 \log\left(\frac{2}{3}\right)$ 2) $18 \log\left(\frac{3}{2}\right) - 17$ 3) $17 - 18 \log\frac{2}{3}$ 4) $7 + 18 \log\left(\frac{2}{3}\right)$

Key: 4

(Mod Function and splitting rule)

Sol:

$$\begin{aligned}&\int_{-1}^1 \frac{\sin x}{3-|x|} dx - \int_{-1}^1 \frac{x^2}{3-|x|} dx \\ &= 0(\text{odd}) - 2 \int_0^1 \frac{x^2}{3-|x|} dx \\ &= 2 \int_0^1 \frac{x^2}{|x|-3} dx \\ &= 2 \int_0^1 \frac{(x^2-9)+9}{x-3} dx \\ &= 2 \int_0^1 x+3 + \frac{9}{x-3} dx \\ &= x^2 + 6x + 18 \log|x-3| \Big|_0^1 \\ &= 7 + 18 \log 2 - 18 \log 3 \\ &= 7 + 18 \log\left(\frac{2}{3}\right)\end{aligned}$$

131. $\int_0^2 [x^2 - x + 1] dx, [\cdot]$ is the greatest integer function

- 1) $\left(\frac{5-\sqrt{5}}{2}\right)$ 2) $\left(\frac{5+\sqrt{5}}{2}\right)$ 3) $\left(\frac{3+\sqrt{5}}{2}\right)$ 4) $\left(\frac{3-\sqrt{5}}{2}\right)$

Key: 1

(Step Function and Splitting rule)

Sol:

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$$\text{Let } I = \int_0^2 [x^2 - x + 1] dx$$

$$\text{Let } f(x) = x^2 - x + 1 \Rightarrow f'(x) = 2x - 1$$

$$\begin{aligned} I &= \int_0^{1/2} [x^2 - x + 1] dx + \int_{1/2}^1 [x^2 - x + 1] dx + \int_1^{\frac{1+\sqrt{5}}{2}} [x^2 - x + 1] dx + \int_{\frac{1+\sqrt{5}}{2}}^2 [x^2 - x + 1] dx \\ &= 0 + 0 + 1 \int_1^{\frac{1+\sqrt{5}}{2}} 1 dx + 2 \int_{\frac{1+\sqrt{5}}{2}}^2 1 dx \\ &= \left(\frac{1+\sqrt{5}}{2} - 1 \right) + 2 \left(2 - \frac{1+\sqrt{5}}{2} \right) = \left(\frac{5-\sqrt{5}}{2} \right) \end{aligned}$$

It is clear from the $\int_0^2 [x^2 - x + 1] dx = \text{Area of the bounded region}$

$$\begin{aligned} &= 0 + \left(\frac{1+\sqrt{5}}{2} - 1 \right) \times 1 + \left(2 - \frac{1+\sqrt{5}}{2} \right) \times 2 \\ &= 3 - \left(\frac{1+\sqrt{5}}{2} \right) = \left(\frac{5-\sqrt{5}}{2} \right) \end{aligned}$$

132. $\int_{-3}^3 x^8 \{x^{11}\} dx$ is equal to (where $\{\cdot\}$ is the fractional part of x)

- 1) 3^8 2) 3^7 3) 3^9 4) None of these

Key: 2 (Step Function and Splitting rule)

$$\text{Sol: } I = \int_{-3}^3 x^8 \{x^{11}\} dx$$

$$\text{Replacing } x \text{ by } -x \text{ we have } I = \int_{-3}^3 x^8 \{x^{11}\} dx$$

Adding equation (1) and (2) we get

$$2I = \int_{-3}^3 x^8 (\{x^{11}\} + \{-x^{11}\}) dx = 2 \int_0^3 x^8 dx = 2 \left(\frac{x^9}{9} \right)_0^3 = 2 \cdot 3^7 \text{ or } I = 3^7$$

133. If $[x]$ denotes the greatest integer less than or equal to x , then find the value of the

$$\text{integral } \int_0^2 x^2 [x] dx$$

1) $\frac{7}{3}$

2) $\frac{4}{3}$

3) $\frac{5}{3}$

4) $\frac{11}{3}$

Key: 1

(Step Function and Splitting rule)

Sol:

$$\begin{aligned}\int_0^2 x^2 [x] dx &= \int_0^1 x^2 [x] dx + \int_1^2 x^2 [x] dx \\ &= \int_0^1 x^2 (0) dx + \int_1^2 x^2 (1) dx \\ &= 0 + \int_1^2 x^2 dx = \left[\frac{x^3}{3} \right]_1^2 = \frac{8-1}{3} = \frac{7}{3}\end{aligned}$$

134. The value of $\int_0^{2\pi} [2 \sin x] dx$, where $[.]$ represents the greatest integral function is

1) $-\frac{5\pi}{3}$

2) $-\pi$

3) $\frac{5\pi}{3}$

4) $2 - \pi$

Key: 2

(Step Function and Splitting rule)

Sol:

$$\begin{aligned}I &= \int_{\pi/6}^{2\pi} [2 \sin x] dx \\ &= \int_{\pi/6}^{5\pi/6} 1 dx + \int_{\pi}^{7\pi/6} -1 dx + \int_{7\pi/6}^{11\pi/6} -2 dx + \int_{11\pi/6}^{2\pi} -1 dx \\ &= \left(\frac{5\pi}{6} - \frac{\pi}{6} \right) \left(-\frac{7\pi}{6} + \pi \right) + 2 \left(\frac{11\pi}{6} + \frac{7\pi}{6} \right) + \left(-2\pi + \frac{11\pi}{6} \right) \\ &= \frac{2\pi}{3} - \frac{\pi}{6} - \frac{8\pi}{6} - \frac{\pi}{6} = -\pi\end{aligned}$$

135. If the value of definite integral $\int_1^a x \cdot a^{-[\log_a x]} dx$ where $a > 1$, and $[.]$ denotes the greatest integer, is $\frac{e-1}{2}$ then the value of $5[a]$ is _____ **(Step Function and Splitting rule)**

Ans: 5

Sol. Let $\log_a x = t \Rightarrow a^t = x \Rightarrow dx = a^t \log_e a$

$$\therefore I = \ln a \int_0^1 a^t \cdot a^{-[t]} \cdot a^t dt = \ln a \int_0^1 a^{2t} dt = \frac{a^2 - 1}{2} = \frac{e - 1}{2} \Rightarrow a = \sqrt{e}$$

136. The integral $\int_{\pi/4}^{5\pi/4} (|\cos t| \sin t + |\sin t| \cos t) dt$ has the value equal to

Ans: 0

(Mod Function and Splitting rule)

$$\begin{aligned} \text{Sol. } I &= \int_{\pi/4}^{\pi/2} 2 \sin t \cos t dt + \int_{\pi/2}^{\pi} \{(-\sin t \cos t) + (\sin t \cos t)\} dt + \int_{\pi}^{5\pi/4} -2 \sin t \cos t dt \\ &= \int_{\pi/4}^{\pi/2} \sin 2t dt - \int_{\pi}^{5\pi/4} \sin 2t dt \\ &= 0. \end{aligned}$$

137. If $I = \int_{-\frac{\pi}{2}}^{2\pi} \sin^{-1}(\sin x) dx$ then $-\frac{16}{\pi^2} I =$ (Miscellaneous properties)

Ans: 2

$$\text{Sol; } I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x dx + \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (\pi - x) dx + \int_{\frac{3\pi}{2}}^{2\pi} (x - 2\pi) dx = \frac{-\pi^2}{8}$$

138. The value of $\int_{-n}^n [x] dx$ is – (where $[]$ denotes the GIF) (Step Function)

A)-n

B)n

C)2n

D)-2n

ANS: A

$$\begin{aligned} \text{SOL: } \int_{-n}^n [x] dx &= \int_{-n}^n [-x] dx \\ \int_{-n}^n (-1 - [x]) dx &= \int_{-n}^n (-1) dx - I \\ \therefore 2I &= (-x)_{-n}^n = -2n \\ \therefore I &= \int_{-n}^n [x] dx = -n \end{aligned}$$

139. If $g(x) = \int_0^x (|\sin t| + |\cos t|) dt$, then $g\left(x + \frac{\pi n}{2}\right)$ is equal to – (where, $n \in N$)

A) $g(x) + g(\pi)$

B) $g(x) + g\left(\frac{n\pi}{2}\right)$

C) $g(x) + g\left(\frac{\pi}{2}\right)$

D) $g(x) + g(n\pi)$

ANS: B

(Mod Function)

$$\text{SOL: } g\left(x + \frac{\pi n}{2}\right) = \int_0^{x + \frac{n\pi}{2}} (|\sin t| + |\cos t|) dt$$

$$\begin{aligned}
 &= \int_0^x (|\sin t| + |\cos t|) dt + \int_x^{x+\frac{n\pi}{2}} (|\sin t| + |\cos t|) dt \\
 &= g(x) + \int_0^{\frac{n\pi}{2}} (|\sin t| + |\cos t|) dt = g(x) + g\left(\frac{n\pi}{2}\right) \\
 &\quad (\text{as } |\sin t| + |\cos t| \text{ has a period } \pi/2)
 \end{aligned}$$

140. The value of $\int_0^x [\cos t] dt, x \in \left[(4n+1)\frac{\pi}{2}, (4n+3)\frac{\pi}{2}\right]$ and $n \in \mathbb{N}$, is equal to

- A) $\frac{\pi}{2}(2n-1) - 2x$ B) $\frac{\pi}{2}(2n-1) + x$
 C) $\frac{\pi}{2}(2n+1) - x$ D) $\frac{\pi}{2}(2n+1) + x$

ANS: C

(Step Function)

SOL: $\int_0^{2\pi} [\cos t] dt = \int_0^{\frac{\pi}{2}} 0 dt + \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (-1) dt + \int_{\frac{3\pi}{2}}^{2\pi} 0 dt = -\pi$

Also $[\cos t]$ is periodic with the period 2π .

for $\frac{\pi}{2} < x < \frac{3\pi}{2}$, $[\cos] = -1$

i.e., for $x \in \left(2n\pi + \frac{\pi}{2}, 2n\pi + \frac{3\pi}{2}\right)$, $[\cos x] = -1$

$$\begin{aligned}
 I &= \int_0^x [\cos t] dt = \int_0^{2n\pi} [\cos t] dt + \int_{2n\pi}^x [\cos t] dt \\
 &= n \int_0^{2\pi} [\cos t] dt + \int_{2n\pi}^{2n\pi+\pi/2} [\cos t] dt + \int_{2n\pi+\pi/2}^x [\cos t] dt \\
 &= -n\pi + 0 + \left(x - \left(\frac{2n\pi + \pi}{2}\right)\right)(-1) \\
 &= -n\pi + 2n\pi + \frac{\pi}{2} - x = \frac{(2n+1)\pi - x}{2}
 \end{aligned}$$

141. If $I = \int_{-1}^2 |x \sin \pi x| dx$ then I equals to

- A) $\frac{1}{\pi}$ B) $\frac{2}{\pi}$ C) $\frac{4}{\pi}$ D) $\frac{5}{\pi}$

ANS: D

(Mod Function)

SOL: We can write $I = \int_{-1}^1 |\pi \sin \pi x| dx + \int_1^2 |\pi \sin \pi x| dx$

As $|\pi \sin \pi x|$ is an even function,

$\sin \pi x \geq 0$, for $0 \leq \pi x \leq \pi$ and $\sin \pi x \leq 0$ for $\pi \leq \pi x \leq 2\pi$. $1 \leq x \leq 2$, we get

$$I = 2 \int_0^1 x \sin \pi x dx - \int_1^2 x \sin \pi x dx$$

$$\text{But } \int x \sin \pi x dx = \frac{-x \cos \pi x}{\pi} + \frac{1}{\pi} \int \cos \pi x dx$$

$$= -\frac{x}{\pi} \cos \pi x + \frac{1}{\pi^2} \sin \pi x,$$

$$I = 2 \left(-\frac{1}{\pi} \cos \pi + 0 \right) - \left(-\frac{2}{\pi} \cos 2\pi + \frac{2}{\pi} \cos \pi \right)$$

$$= \frac{5}{\pi}$$

142. The value of $\int_0^{\lfloor x \rfloor} \frac{8^x}{2^{\lfloor 3x \rfloor}} dx$, (where $\lfloor \cdot \rfloor$ denotes the greatest integer function) is equal to

A) $\frac{\lfloor x \rfloor}{\ln 2}$

B) $\frac{\lfloor x \rfloor}{\ln 4}$

C) $\frac{2\lfloor x \rfloor}{\ln 2}$

D) $\frac{\lfloor x \rfloor}{\ln 8}$

ANS:D

(Step Function and Splitting rule)

$$\text{SOL: Let } \lfloor x \rfloor = \int_0^{\lfloor x \rfloor} 2^{\frac{n}{3} \{3x\}} dx = \int_0^{\lfloor x \rfloor} 2^{3x} dx = \frac{n(2^{3x})^{\frac{1}{3}}}{\ln 8} = \frac{\lfloor x \rfloor}{\ln 8}$$

143. $\int_0^x \lfloor \sin t \rfloor dt$, where $x \in (2n\pi, (2n+1)\pi)$, $n \in \mathbb{N}$ and $\lfloor \cdot \rfloor$ denotes the greatest integer function is equal to

(Step Function and Splitting rule)

A) $-n\pi$

B) $-(n+1)\pi$

C) $-2n\pi$

D) $-(2n+1)\pi$

ANS:A

$$\begin{aligned} \text{SOL: } I &= \int_0^x \lfloor \sin t \rfloor dt = \int_0^{2n\pi} \lfloor \sin t \rfloor dt + \int_{2n\pi}^x \lfloor \sin t \rfloor dt \\ &= n \int_0^{2\pi} \lfloor \sin t \rfloor dt + \int_{2n\pi}^x \lfloor \sin t \rfloor dt \quad (\text{as } \lfloor \sin x \rfloor \text{ is periodic with period } 2\pi) \\ &= n \left(\int_0^{\pi/2} \lfloor \sin t \rfloor dt + \int_{\pi/2}^{\pi} \lfloor \sin t \rfloor dt + \int_{\pi}^{2\pi} \lfloor \sin t \rfloor dt \right) + \int_{2n\pi}^x \lfloor \sin t \rfloor dt \\ \Rightarrow I &= n \left(0 + 0 - \int_{\pi}^{2\pi} dt \right) + 0 = -n\pi \end{aligned}$$

144. $\int_0^x |\sin t| dt$, where $x \in (2n\pi, (2n+1)\pi)$, where $n \in \mathbb{N}$ is equal to

A) $4n - \cos x$

B) $4n - \sin x$

C) $4n+1 - \cos x$

D) $4n-1 - \cos x$

ANS:C

(Mod Function and Splitting rule)

$$\text{SOL: } \int_0^x |\sin t| dt = \int_0^{2n\pi} |\sin t| dt + \int_{2n\pi}^x |\sin t| dt$$

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$$= 2n \int_0^{\pi} |\sin t| dt + \int_{2n\pi}^x \sin t dt \quad (\text{as } x \text{ lies in either } 1^{\text{st}} \text{ or } 2^{\text{nd}} \text{ quadrant})$$

$$= 2n(-\cos t)_0^{\pi} + (-\cos t)_{2n\pi}^x = 4n - \cos x + 1$$

$$L = \lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{4n^2 - 1}} + \frac{1}{\sqrt{4n^2 - 4}} + \frac{1}{\sqrt{4n^2 - 9}} + \dots + \frac{1}{\sqrt{3n^2}} \right]$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{\sqrt{4n^2 - r^2}} = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n \sqrt{4 - \frac{r^2}{n^2}}}$$

$$\therefore L = \int_0^1 \frac{dx}{\sqrt{4 - x^2}} = \sin^{-1} \frac{x}{2} \Big|_0^1 = \frac{\pi}{6}$$

145. $\int_0^{20} [x] \{x\} dx = K$ then $\frac{K}{19}$ equals (where $[.]$ and $\{.\}$ respectively denotes G.I.F and F.P.F)

ANS: 5

(Step Function and Splitting rule)

SOL: $K = \int_0^{20} [x] \{x\} dx$

$$\sum_{r=0}^{19} \int_r^{r+1} r \{x\} dx$$

$$= \sum_{r=0}^{19} \int_0^1 r x dx$$

$$= \sum_{r=0}^{19} \frac{r}{2} = \frac{1}{2} \times \frac{19 \times 20}{2} = 95$$

$$\Rightarrow \frac{K}{19} = 5$$

Note: i) $\int_0^P [x] \{x\} dx = \frac{P(P-1)}{4}, P \in \mathbb{N}$

ii) $\int_n^{n+1} [x] \{x\} dx = \frac{n}{2}, n \in \mathbb{Z}$

146. The value of $\int_{-1}^1 [x^2 + 2x] dx$ is

A) $3 - \sqrt{3} - \sqrt{2}$

B) $3 + \sqrt{3} - \sqrt{2}$

C) $-3 + \sqrt{3} + \sqrt{2}$

D) $3 + \sqrt{3} + \sqrt{2}$

ANS: A

(Step Function and Splitting rule)

SOL: Let $y = x^2 + 2x$ for $x \in [-1, 1], y \in [-1, 3]$ to draw the graph of $[x^2 + 2x]$, draw the graph of $y = x^2 + 2x$ and find the values of x such that $y = -1, 0, 1, 2, 3$.

we get $x = -1, 0, \sqrt{2} - 1, \sqrt{3} - 1, 1$

$$\therefore \int_{-1}^1 [x^2 + 2x] dx = \text{Algebraic sum of areas of rectangles}$$

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$$= -1 + 0 + (\sqrt{3} - \sqrt{2})1 + (2 + \sqrt{3}).2$$

$$= 3 - \sqrt{3} - \sqrt{2}$$

147. The value of $\int_0^{2\pi} [2 \sin x] dx$ is **(Step Function and Splitting rule)**

A) $-\frac{5\pi}{3}$ B) $-\pi$ C) $\frac{5\pi}{3}$ D) -2π

ANS: B

SOL: $\int_0^{2\pi} [2 \sin x] dx = \int_0^{\frac{\pi}{6}} 0 dx + \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} 1 dx + \int_{\frac{5\pi}{6}}^{\pi} 0 dx + \int_{\pi}^{\frac{7\pi}{6}} (-1) dx + \int_{\frac{7\pi}{6}}^{\frac{11\pi}{6}} (-2) dx + \int_{\frac{11\pi}{6}}^{2\pi} 0 dx$

$$0 + \frac{2\pi}{3} \times 1 + 0 + \frac{\pi}{6}(-1) + (-2) \times \frac{2\pi}{3} \times (-1) \frac{\pi}{6} = -\pi$$

148. The value of $\int_0^{2\pi} [\sin x + \cos x] dx$ is **(Step Function and Splitting rule)**

A) $-\frac{5\pi}{3}$ B) $-\pi$ C) $\frac{5\pi}{3}$ D) -2π

ANS: B

SOL:

Let $y = \sin x + \cos x$

$$y = \sqrt{2} \sin \left[x + \frac{\pi}{4} \right]$$

$$\Rightarrow y \in [-\sqrt{2}, \sqrt{2}]$$

$$\int_0^{2\pi} [\sin x + \cos x] dx = \text{sum of areas of blocks}$$

$$= 1 \times \frac{\pi}{2} + 0 + \frac{\pi}{4}(-1) + \frac{\pi}{2}(-2) + \frac{\pi}{4}(-1) + 0 = -\pi$$

149. The value of $\int_0^{\infty} [2e^{-x}] dx$ is **(Step Function and Splitting rule)**

A) $\ln \frac{1}{2}$ B) $\ln 2$ C) $\frac{1}{\ln 2}$ D) $-\frac{1}{\ln 2}$

ANS:B

SOL: Let $y = \frac{2}{e^x}$ for $x \in (0, \infty), y \in (0, 2)$

$$y = 1 \Rightarrow e^x = 2 \Rightarrow x = \ln 2$$

for $0 \leq x < \ln 2, y = 1$ and for $\ln 2 \leq x < \infty,$

$$y=0 \text{ then } \int_0^2 [2e^{-x}] dx = 1 \times \ln 2 = \ln 2$$

150. The value of $\int_0^\infty \left[\frac{3}{1+x^2} \right] dx$ is **(Step Function and Splitting rule)**

- A) 3 B) $-\frac{1}{\sqrt{2}}$ C) $\frac{1}{\sqrt{2}}$ D) $\frac{3}{\sqrt{2}}$

ANS:D

SOL: Let $y = \frac{3}{1+x^2}$ when $x=0, y=3$ and as $x \rightarrow \pm\infty,$

$$y \rightarrow 0 \quad \therefore y \in (0, 3]$$

$$\text{Now, } y = 1 \Rightarrow \frac{3}{1+x^2} = 1 \Rightarrow x^2 = 2 \Rightarrow x = \pm\sqrt{2}$$

$$y = 2 \Rightarrow \frac{3}{1+x^2} = 2 \Rightarrow x^2 = \frac{3}{2} - 1 \Rightarrow x = \pm\frac{1}{\sqrt{2}}$$

$$\begin{aligned} \therefore \int_0^\infty \left[\frac{3}{1+x^2} \right] dx &= \int_0^{\frac{1}{\sqrt{2}}} 2 dx + \int_{\frac{1}{\sqrt{2}}}^{\sqrt{2}} 1 dx + \int_{\sqrt{2}}^\infty 0 dx \\ &= 2 \cdot \frac{1}{\sqrt{2}} + \sqrt{2} - \frac{1}{\sqrt{2}} = 2\sqrt{2} - \frac{1}{\sqrt{2}} = \frac{3}{\sqrt{2}} \end{aligned}$$

151. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{dx}{[x] + [\sin x] + 4}$, where $[t]$ denotes the greatest integer less than or equal to t , is

- 1) $\frac{1}{12}(7\pi + 5)$ 2) $\frac{1}{12}(7\pi - 5)$ 3) $\frac{3}{20}(4\pi - 3)$ 4) $\frac{3}{10}(4\pi - 3)$

Key: 3

(Step Function and splitting rule)

Sol: $\int_{-\frac{\pi}{2}}^{-3} \frac{dx}{-2-1+4} + \int_{-1}^0 \frac{dx}{-1-1+4} + \int_0^1 \frac{dx}{0+0+4} + \int_1^{\frac{\pi}{2}} \frac{dx}{1+0+4} = \frac{3\pi}{5} - \frac{9}{20}$

152. The value of the integral $\int_{-2}^2 \frac{\sin^2 x}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} dx$ (where $[x]$ denotes the greatest integer less than

^{20}Cr or equal to (x) is:

1) 4

2) $4 - \sin 4$

3) $\sin 4$

4) 0

Ans. 4

(Step Function and splitting rule)

Sol. $I = \int_{-2}^2 \frac{\sin^2 x}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} dx$ $I = \int_0^2 \left(\frac{\sin^2 x}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} + \frac{\sin^2(-x)}{\left[-\frac{x}{\pi}\right] + \frac{1}{2}} \right) dx$

$\left(\left[\frac{x}{\pi}\right] + \left[-\frac{x}{\pi}\right] = -1 \text{ as } x \neq n\pi \right)$ $I = \int_0^2 \left(\frac{\sin^2 x}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} + \frac{\sin^2 x}{-1 - \left[\frac{x}{\pi}\right] + \frac{1}{2}} \right) dx = 0$

153. $\int_0^{102} [\tan^{-1} x] dx =$ (Where $[x]$ is the largest integer not exceeding x)

1) $102 - \tan 1$

2) 101

3) $102 + \tan 1$

4) $102 - \frac{\pi}{4}$

Key: 1

(Step Function and splitting rule)

Sol: for, $0 \leq x \leq \tan 1$

$$0 \leq \tan^{-1} x < 1 \text{ and } \tan 1 \leq x \leq 102$$

$$1 \leq \tan^{-1} x < 2; \int_0^{102} [\tan^{-1} x] dx = \int_0^{\tan 1} 0 dx + \int_{\tan 1}^{102} 1 dx$$

$$= (x)_{\tan 1}^{102} = 102 - \tan 1$$

154. $\int_1^3 \frac{dx}{x^2 + [x]^2 + 1 - 2x[x]}$ = (where $[x]$ is the largest integer not exceeding x)

- 1) π 2) 3π 3) 2π 4) $\frac{\pi}{2}$

Key: 4

(Step Function and splitting rule)

$$\begin{aligned}\text{Sol: } I &= \int_1^3 \frac{dx}{(x-[x])^2+1} = \int_1^2 \frac{dx}{(x-[x])^2+1} + \int_2^3 \frac{dx}{(x-[x])^2+1} \\ &= \int_1^2 \frac{dx}{(x-1)^2+1} + \int_2^3 \frac{dx}{(x-2)^2+1} \\ &= \left(\tan^{-1}(x-1) \right)_1^2 + \left(\tan^{-1}(x-2) \right)_2^3\end{aligned}$$

155. $\int_0^{\frac{\pi}{4}} \tan^n(x-[x]) + \tan^{n-2}(x-[x]) dx =$ where $[]$ is g.i.f

- 1) $\frac{1}{n}$ 2) $\frac{1}{n-1}$ 3) $\frac{1}{n-2}$ 4) $\frac{1}{n+1}$

Key: 2

(Step Function and splitting rule)

$$\begin{aligned}\text{Sol: } I &= \int_0^{\pi/4} \tan^n x + \tan^{n-2} x \, dx \\ &= \int_0^{\pi/4} \tan^{n-2} x \sec^2 x \, dx\end{aligned}$$

156. $\int_0^{\pi/3} [\sqrt{3} \tan x] dx =$ (where $[]$ is g.i.f.)

- 1) $\frac{5\pi}{6}$ 2) $\frac{5\pi}{6} - \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$
3) $\frac{\pi}{2} - \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$ 4) $\frac{\pi}{2}$

Key: 3

(Step Function and splitting rule)

$$\text{Sol: } \int_0^{\frac{\pi}{6}} 0 dx + \int_{\frac{\pi}{6}}^{\tan^{-1} \frac{2}{\sqrt{3}}} 1 dx + \int_{\tan^{-1} \frac{2}{\sqrt{3}}}^{\frac{\pi}{3}} 2 dx$$

157. The value of $\int_1^a [x] f'(x) dx$, $a > 1$, where $[x]$ denotes the greatest integer not exceeding x is

- 1) $[a] f([a]) - \{f(1) + f(2) + \dots + f(a)\}$ 2) $a f([a]) - \{f(1) + f(2) + \dots + f(a)\}$
3) $a f(a) - \{f(1) + f(2) + \dots + f(a)\}$ 4) $[a] f(a) - \{f(1) + f(2) + \dots + f(a)\}$

Key: 4

(Step Function and splitting rule)

$$\begin{aligned}
 \text{Sol: } I &= \int_1^a [x] f^1(x) dx \\
 &= \int_1^2 [x] f^1(x) dx + \int_2^3 [x] f^1(x) dx + \dots + \int_{[a]}^a [x] f^1(x) dx \\
 &= [a] f(a) - \{f(1) + f(2) + \dots + f[a]\}
 \end{aligned}$$

$$158. \int_{-1/2}^{1/2} \sqrt{\left(\frac{x+1}{x-1}\right)^2 + \left(\frac{x-1}{x+1}\right)^2} - 2 \, dx =$$

1) $8 \log \frac{4}{3}$

2) $8 \log \frac{3}{4}$

3) $4 \log \frac{4}{3}$

4) 0

Key: 3

(Mod Function and Splitting rule)

$$\text{Sol: } \int_{-1/2}^{1/2} \sqrt{\left(\frac{x+1}{x-1} - \frac{x-1}{x+1}\right)^2} \, dx = \int_{-1/2}^{1/2} \left| \frac{x+1}{x-1} - \frac{x-1}{x+1} \right| \, dx$$

$$159. \int_4^{10} ([x-4] + [10-x]) \, dx = (\text{where } \text{ is the largest integer not exceeding } x)$$

1) 26

2) 10

3) 30

4) 20

Key: 3

(Step Function and Splitting rule)

$$\text{Sol: } \int_4^{10} \{[x-4] + [10-x]\} \, dx = \int_4^{10} \{[x] - 4 + 10 + [-x]\} \, dx$$

$$= \int_4^{10} \{[x] + [-x]\} \, dx + \int_4^{10} 6 \, dx = \int_4^{10} -1 \, dx + \int_4^{10} 6 \, dx$$

160. The value of the definite integral $\int_0^1 \frac{x \, dx}{x^3 + 16}$ lies in the interval $[a, b]$. Then smallest such interval is

1) $\left[0, \frac{1}{17}\right]$

2) $[0, 1]$

3) $\left[0, \frac{1}{27}\right]$

4) $\left[\frac{1}{17}, 1\right]$

Key: 1

(Max-min inequality property of definite integral)

Sol: We have

$$(1-0) \left(\frac{0}{0^3+16} \right) \leq \int_0^1 \frac{x}{x^3+16} \, dx \leq (1-0) \left(\frac{1}{1^3+16} \right)$$

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$$0 \leq \int_0^1 \frac{x dx}{x^3 + 16} \leq \frac{1}{17}$$

161. If $I = \int_{-\pi/6}^{\pi/6} \frac{\pi + 4x^5}{1 - \sin\left(|x| + \frac{\pi}{6}\right)} dx$, then I equals to

- 1) 4π 2) $2\pi + \frac{1}{\sqrt{3}}$ 3) $2\pi - \sqrt{3}$ 4) $4\pi + \sqrt{3} - \frac{1}{\sqrt{3}}$

Key: 1

(Mod Function and splitting rule)

Sol: As $\frac{4x^5}{1 - \sin\left(|x| + \frac{\pi}{6}\right)}$ is an odd function, and $\frac{\pi}{1 - \sin\left(|x| + \frac{\pi}{6}\right)}$ is an even function, we get

$$\begin{aligned} I &= 2\pi \int_0^{\pi/6} \frac{dx}{1 - \sin\left(x + \frac{\pi}{6}\right)} \quad \text{Put } x + \frac{\pi}{6} = t \\ &= 2\pi \int_{\pi/6}^{\pi/3} \frac{dt}{1 - \sin t} = 4\pi \end{aligned}$$

162. Let f be a continuous function satisfying $f'(x) = \begin{cases} 1 & \text{for } 0 < x \leq 1 \\ x & \text{for } x > 1 \end{cases}$ and $f(0) = 0$. Then $f(x)$ can be defined as

- 1) $f(x) = \begin{cases} 1 & \text{if } x \leq 0 \\ 1 - e^x & \text{if } x > 0 \end{cases}$
 2) $f(x) = \begin{cases} 1 & \text{if } x \leq 0 \\ e^x - 1 & \text{if } x > 0 \end{cases}$
 3) $f(x) = \begin{cases} x & \text{if } x < 0 \\ e^x & \text{if } x > 0 \end{cases}$
 4) $f(x) = \begin{cases} x & \text{if } x \leq 0 \\ e^x - 1 & \text{if } x > 0 \end{cases}$

(Splitting rule of function)

Sol: Put $\log x = t \Rightarrow \frac{1}{x} dx = dt$

$$\Rightarrow dx = e^t dt$$

For $x > 1$ $f'(t) = e^t, t > 0$ for $0 < x \leq 1$

Integrating $f(t) = e^t + c$

$$f'(x) = 1$$

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$$f(0) = e^0 + c \Rightarrow c = -1$$

$$f^1(t) = 1, t \leq 0$$

$$f(t) = e^t - 1 \text{ for } t > 0$$

$$f^1(t) = t + c$$

$$\therefore f(x) = e^x - 1 \text{ for } x > 0$$

$$163. \quad I = \int_{-1}^1 [x[1 + \sin \pi x]] dx =$$

(Greatest Integer function)

[Where[.] denotes G.I.F]

1) 1

2) $\frac{1}{2}$

3) 0

4) 2

Key: 3

$$\text{Sol: } -1 < x < 0 \Rightarrow -\pi < \pi x < 0 \Rightarrow -1 \leq \sin \pi x < 0$$

$$\Rightarrow 0 \leq 1 + \sin \pi x < 1 \Rightarrow [1 + \sin \pi x] = 0$$

$$0 \leq x \leq 1 \Rightarrow 0 \leq \pi x < \pi \Rightarrow 0 \leq \sin \pi x \leq 1$$

$$\Rightarrow 1 \leq 1 + \sin \pi x \leq 2 \Rightarrow [1 + \sin \pi x] = 1$$

$$I = \int_{-1}^0 [x \cdot 0] dx + \int_0^1 [x \cdot 1] dx = 0$$

$$164. \quad \text{The value of } \int_{-2}^1 \left[x \left[1 + \cos \frac{\pi x}{2} \right] + [x] \right] dx \text{ where [.] denotes the greatest integer function, is}$$

1) -4

2) $\frac{1}{2}$

3) 3

4) -4

(Greatest Integer function)

Key: 1

$$\text{Sol: } -2 < x < -1 \Rightarrow -\pi < \frac{\pi x}{2} < -\frac{\pi}{2} \Rightarrow -1 \leq \cos \frac{\pi x}{2} < 0$$

$$0 \leq 1 + \cos \frac{\pi x}{2} < 1 \Rightarrow \left[1 + \cos \frac{\pi x}{2} \right] = 0$$

$$\left[x \left[1 + \cos \frac{\pi x}{2} \right] \right] = 0$$

$$-1 < x < 0, -\frac{\pi}{2} < \frac{\pi x}{2} < 0 \Rightarrow \cos \frac{\pi x}{2} < 1$$

$$\Rightarrow \left[1 + \cos \frac{\pi x}{2} \right] = 1$$

$$\left[x \left[1 + \cos \frac{\pi x}{2} \right] \right] = -1$$

$$0 < x < 1 \Rightarrow 0 < \frac{\pi x}{2} < \frac{\pi}{2} \Rightarrow 0 < \cos \frac{\pi x}{2} < 1$$

$$\left[x \left[1 + \cos \frac{\pi x}{2} \right] \right] = 0$$

Therefore given interval =

$$\int_{-2}^{-1} -2 dx + \int_{-1}^0 -2 dx + \int_0^1 0 dx = -4$$

$$165. \int_0^2 [x^2 - x + 1] dx$$

(Greatest Integer function)

1) 1

2) $\frac{3-\sqrt{5}}{2}$

3) $\frac{5-\sqrt{5}}{2}$

4) $\frac{7-\sqrt{5}}{2}$

Key: 4

Sol: Draw the graph of $x^2 - x + 1$ and find it's values

$$166. \int_0^1 \frac{1}{9^{\left[\frac{1}{x}\right]}} dx =$$

(greatest integer function)

1. $1 + 8 \log(9/8)$

2. $1 + 8 \log(8/9)$

3. $-1 - 8 \log(6/7)$

4. $-1 + \log(6/7)$

Key: 2

Sol

$$\int_{1/2}^1 \frac{1}{9^1} dx + \int_{1/3}^{1/2} \frac{1}{9^2} dx + \dots$$

$$= \frac{1}{9} \left(1 - \frac{1}{2}\right) + \frac{1}{9^2} \left(\frac{1}{2} - \frac{1}{3}\right) + \dots$$

$$\text{Use } \log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} + \dots$$

$$= \left(\frac{1}{9} + \frac{1}{9^2} + \frac{1}{9^3} + \dots \right) - 9 \left(\frac{1}{9^2} + \frac{1}{9^3} + \dots \right)$$

$$= -\log\left(1 - \frac{1}{9}\right) - 9 \left(-\log\left(1 - \frac{1}{9}\right) - \frac{1}{9} \right)$$

$$= -\log\left(\frac{8}{9}\right) + 9 \log\left(\frac{8}{9}\right) + 1$$

$$= 1 + 8 \log \frac{8}{9}$$

$$167. \int_{-10}^0 \frac{\left| \frac{2[x]}{3x - [x]} \right|}{2[x]} dx \text{ is equal to (where } [*] \text{ denotes greatest integer function.)}$$

(Greatest Integer function)

1) $\frac{28}{3}$

2) $\frac{1}{3}$

3) 0

4) 1

Key: 1

Sol: Split the interval according to value of the function

$$168. \text{ The value of the integral } \int_{e^{-1}}^{e^2} \left| \frac{\ln x}{x} \right| dx \text{ is: (Modulus Function Function)}$$

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1) $\frac{3}{2}$

2) $\frac{5}{2}$

3) 3

4) 5

Key: 2

$$\text{Sol: } -\int_{e^{-1}}^1 \frac{\ln x}{x} dx + \int_1^{e^2} \frac{\ln x}{x} dx = -\left[\frac{(\ln x)^2}{2}\right]_{e^{-1}}^1 + \left[\frac{(\ln x)^2}{2}\right]_1^{e^2} = 2 + \frac{1}{2} = \frac{5}{2}$$

$$169. \text{ If } I_1 = \int_0^{\pi/2} \frac{\cos^2 x}{1 + \cos^2 x} dx, I_2 = \int_0^{\pi/2} \frac{\sin^2 x}{1 + \sin^2 x} dx$$

(Applications Of Inequality)

$$I_3 = \int_0^{\pi/2} \frac{1 + 2 \cos^2 x \sin^2 x}{4 + 2 \cos^2 x \sin^2 x} dx, \text{ then}$$

1) $I_1 = I_2 > I_3$

2) $I_3 > I_1 = I_2$

3) $I_1 = I_2 = I_3$

4) $I_1 = I_2 = 2I_3$

Key: 3

Sol: Use property of definite integrals

$$170. \text{ The value of } \int_0^3 [x^2 - 2x + 2] dx. \text{ Where } [.] \text{ is G.I.F}$$

(Greatest Integer Function)

1) $6 - \sqrt{3} - \sqrt{2}$

2) $6 + \sqrt{3} + \sqrt{2}$

3) $8 - \sqrt{3} - \sqrt{2}$

4) $2 + \sqrt{2}$

Key: 3

Sol: for $0 \leq x \leq 2, x^2 - 2x + 2 \in [2, 5]$

$$y = x^2 - 2x + 2$$

$$2 = x^2 - 2x + 2 \Rightarrow x = 2$$

$$3 = x^2 - 2x + 2 \Rightarrow x = 1 + \sqrt{2}$$

$$4 = x^2 - 2x + 2 \Rightarrow x = 1 + \sqrt{3}$$

Required integral = sum of the areas of the rectangle

$$= 8 - \sqrt{3} - \sqrt{2}$$

$$171. \text{ If } f(x) = \int_0^1 \frac{dt}{1 + |x - t|}, \text{ then } f'\left(\frac{1}{2}\right) \text{ is equal to}$$

(Splitting of limits)

1. 1

2. -1

3. $\frac{1}{2}$

4. 0

Ans: 4

$$\text{Sol: We have } f(x) = \int_0^1 \frac{dt}{1 + |x - t|}$$

$$= \int_0^x \frac{dt}{1 + x - t} + \int_x^1 \frac{dt}{1 - x + t}$$

$$\Rightarrow f'\left(\frac{1}{2}\right) = \frac{1}{1 + 1/2} - \frac{1}{2 - 1/2} = 0$$

$$172. \text{ The value of } \int_{-1}^1 \frac{\sin x}{\left[\frac{x}{\sqrt{2}}\right] + \frac{1}{2}} dx, \text{ where } [x] = \text{greatest integer less than equal to } x, \text{ is}$$

1. 1

2. 0

3. $4 - \sin 4$

4. 2

(Greatest Integer function & Even Odd)

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A) $-\frac{5\pi}{3}$

B) $-\pi$

C) $\frac{5\pi}{3}$

D) -2π

Key:B

Sol: Let $y = \sin x + \cos x$

$$y = \sqrt{2} \sin\left(x + \frac{\pi}{4}\right)$$

$$\Rightarrow y \in [-\sqrt{2}, \sqrt{2}]$$

$$= 1 \times \frac{\pi}{2} + 0 + \frac{\pi}{4}(-1) + \frac{\pi}{2}(-2) + \frac{\pi}{4}(-1) + 0 = -\pi$$

177. $\int_3^{10} [\log[x] dx]$ is equal to (where $[.]$ represents the greatest integer function).

A) 9

B) $16-e$

C) 10

D) $10+e$

Key:

(Greatest integer function)

Sol: conceptual

178. If $f(x) = \int_0^x \frac{t \sin t dt}{\sqrt{1 + \tan^2 x \sin^2 t}}$ for $0 < x < \frac{\pi}{2}$, then

(Kings Rule)

A) $f(0^+) = -\pi$

B) $f\left(\frac{\pi}{4}\right) = \frac{\pi^2}{8}$

C) f is continuous and differentiable in $\left(0, \frac{\pi}{2}\right)$

D) f is continuous but not differentiable in $\left(0, \frac{\pi}{2}\right)$

Key:C

Sol: $f(x) = \int_0^x \frac{t \sin t}{\sqrt{1 + \tan^2 x \sin^2 t}} dt \dots\dots\dots(1)$

Replacing t by $\pi - t$ and then adding $f(x)$ with equation (1) we get

$$\begin{aligned} f(x) &= \frac{\pi}{2} \int_0^{\pi} \frac{\sin t}{\sqrt{1 + \tan^2 x \sin^2 t}} dt \\ &= \pi \int_0^{\pi/2} \frac{\sin t}{\sqrt{\sec^2 x - \tan^2 x \cos^2 t}} dt \end{aligned}$$

Let $y = \cos t$

$\therefore dy = -\sin t dt$

$$\therefore f(x) = \pi \int_0^1 \frac{dy}{\sqrt{\sec^2 x - (\tan^2 x) y^2}}$$

$$= \frac{\pi}{\tan x} \sin^{-1}(\sin x) = \frac{\pi x}{\tan x}$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 2: Replacing x by $a-x$ – King's property, Replacing x by $a+b-x$

179. The value of $\int_0^\pi |\cos x|^3 dx$

1) $2/3$

2) 0

3) $-4/3$

4) $4/3$

Ans. 4

(Replacing x by $a-x$ – King's property) 2019

$$\text{Sol. } \int_0^\pi |\cos x|^3 dx = 2 \int_0^{\pi/2} \cos^3 x dx = 2 \cdot \frac{2}{3} = \frac{4}{3}$$

180. Let f and g be continuous functions on $[0, a]$ such that $f(x) = f(a-x)$ and

$g(x) + g(a-x) = 4$, then $\int_0^a f(x)g(x)dx$ is equal to

1) $4 \int_0^a f(x)dx$

2) $2 \int_0^a f(x)dx$

3) $-3 \int_0^a f(x)dx$

4) $\int_0^a f(x)dx$

Key: 2

(Replacing x by $a-x$ – King's property) 2019

$$\text{Sol: } I = \int_0^a f(x)g(x)dx \quad I = \int_0^a f(a-x)g(a-x)dx$$

$$I = \int_0^a f(x)g(4-g(x))dx$$

$$I = 4 \int_0^a f(x)dx - I$$

$$\Rightarrow I = 2 \int_0^a f(x)dx$$

181. The value of $\int_0^{\pi/2} \frac{\sin^3 x}{\sin x + \cos x} dx$ is:

1. $\frac{\pi-2}{4}$

2. $\frac{\pi-2}{8}$

3. $\frac{\pi-1}{4}$

4. $\frac{\pi-1}{2}$

Ans:

3

(King's property) 2019

$$\text{Sol: use } \int_0^a f(x)dx = \int_0^a f(a-x)dx$$

182. The value of the integral $\int_0^1 x \cot^{-1}(1-x^2+x^4)dx$ is

1. $\frac{\pi}{2} - \frac{1}{2} \log_e 2$

2. $\frac{\pi}{4} - \log_e 2$

3. $\frac{\pi}{4} - \frac{1}{2} \log_e 2$

4. $\frac{\pi}{2} - \log_e 2$

Ans: 3

(King's property) 2019

Sol: $\int_0^1 x \cot^{-1}(1-x^2+x^4) dx$ $x^2 = t$

$$= \frac{1}{2} \int_0^1 \cot^{-1}(1-t+t^2) dt = \frac{1}{2} \int_0^1 \cot^{-1}(1+t(t-1)) dt$$

$$= \frac{1}{2} \int_0^1 \tan^{-1} t + \tan^{-1}(1-t) dt = \int_0^1 \tan^{-1} t dt = \frac{\pi}{4} - \frac{1}{2} \ln 2$$

183. The value of $\int_0^{2\pi} [\sin 2x(1+\cos 3x)] dx$, (where $[t]$ denotes Greatest Integer Function)

1) -2π

2) π

3) 2π

4) $-\pi$

Ans: 4

(f(a + b - x) Property) 2019

Sol. $I = \int_0^{2\pi} [\sin 2x(1+\cos 3x)] dx$ using $f(a+b-x)$ property

$$I = \int_0^{2\pi} [-\sin 2x(1+\cos 3x)] dx \therefore I + I = \int_0^{2\pi} (-1) dx = -x \Big|_0^{2\pi} \Rightarrow I = -\pi$$

184. If $f(a+b+1-x) = f(x)$, for all x , where a and b are fixed positive real numbers, then

$$\frac{1}{a+b} \int_a^b x(f(x)) + f(x+1) dx \text{ is equal to}$$

1) $\int_{a+1}^{b+1} f(x) dx$

2) $\int_{a+1}^{b+1} f(x+1) dx$

3) $\int_{a-1}^{b-1} f(x+1) dx$

4) $\int_{a-1}^{b-1} f(x) dx$

Ans: 1

(f(a + b - x) Property) 2020

Sol: $f(x+1) = f(a+b-x)$ $I = \frac{1}{(a+b)} \int_a^b x(f(x) + f(x+1)) dx \dots (1)$

$$I = \frac{1}{(a+b)} \int_a^b (a+b-x)(f(x+1) + f(x)) dx \dots (2)$$

From 1 and 2

$$2I = \int_a^b (f(x) + f(x+1)) dx \quad 2I = \int_a^b f(a+b-x) dx + \int_a^b f(x+1) dx$$

$$2I = 2 \int_a^b f(a+1) dx \Rightarrow I = \int_a^b f(x+1) dx = \int_{a+1}^{b+1} f(x) dx$$

185. The value of $I = \int_0^{2\pi} \left(\frac{x \sin^8 x}{\sin^8 x + \cos^8 x} \right) dx$ is equal to

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Key: 10

(King's property)2020

$$\text{Sol: } 2I = \int_0^{2\pi} \left(\frac{x \sin^8 x}{\sin^8 x + \cos^8 x} + \frac{(2\pi - x) \sin^8 x}{\sin^8 x + \cos^8 x} \right) dx \quad I = \int_0^{\pi} \frac{2\pi \sin^8 x}{\sin^8 x + \cos^8 x} dx = \int_0^{\pi/2} \frac{4\pi \sin^8 x}{\sin^8 x + \cos^8 x} dx$$

$$2I = 4\pi \int_0^{\pi/2} \left(\frac{\sin^8 x}{\sin^8 x + \cos^8 x} + \frac{\cos^8 x}{\sin^8 x + \cos^8 x} \right) dx$$

$$\Rightarrow I = 2\pi \int_0^{\pi/2} 1 dx = 2\pi \times \frac{\pi}{2} = \pi^2$$

186. The value of $\int_{\pi/6}^{\pi/4} \frac{dx}{1 + e^{\sin x}}$ is:

- 1) π 2) $\frac{3\pi}{2}$ 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{2}$

Key: 4

(f(a + b - x) Property)2020

$$\text{Sol: } I = \int_{-\pi/2}^{\pi/2} \frac{1}{1 + e^{\sin x}} dx$$

$$\Rightarrow I = \int_{-\pi/2}^{\pi/2} \frac{1}{1 + e^{-\sin x}} dx \left(\because \int_{-a}^a f(x) dx = \int_{-a}^a f(-x) dx \right)$$

$$\text{So, } 2I = \int_{-\pi/2}^{\pi/2} dx = \pi \Rightarrow I = \frac{\pi}{2}. \quad \text{So, Option(4) is the correct Keywer}$$

187. The value of the integral

$$\int_{-\pi/2}^{\pi/2} \frac{dx}{(1 + e^x)(\sin^6 x + \cos^6 x)}$$
 is equal to

- 1) 2π 2) 0 3) π 4) $\frac{\pi}{2}$

Key: 3

(f(a + b - x) Property)

$$\text{Sol: } I = \int_{-\pi/2}^{\pi/2} \frac{dx}{(1 + e^x)(\sin^6 x + \cos^6 x)} = \int_{\pi/2}^{\pi} \frac{dx}{(1 + e^{-x})(\sin^6 x + \cos^6 x)}$$

$$\therefore 2I = \int_{-\pi/2}^{\pi/2} \frac{dx}{(\sin^6 x + \cos^6 x)} = 2 \int_0^{\pi/2} \frac{dx}{(\sin^6 x + \cos^6 x)}$$

$$\begin{aligned}
 \therefore 2I &= \int_0^{\frac{\pi}{2}} \frac{dx}{(\sin^6 x + \cos^6 x)} = \int_0^{\frac{\pi}{2}} \frac{dx}{1 - 3\sin^2 x \cos^2 x} \\
 &= \int_0^{\frac{\pi}{2}} \frac{\sec^4 x \, dx}{\sec^4 x - 3\tan^2 x} = \int_0^{\infty} \frac{(1+t^2)dt}{1+t^4-t^2} \\
 &= \int_0^{\infty} \frac{1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}-1} dt = \int_{-\infty}^{\infty} \frac{d\omega}{\omega^2+1} = 2 \cdot \frac{\pi}{2} = \pi
 \end{aligned}$$

188. The value of $\int_0^{\pi} \frac{e^{\cos x} \sin x}{(1+\cos^2 x)(e^{\cos x} + e^{-\cos x})} dx$ is equal to:

A) $\frac{\pi^2}{4}$

B) $\frac{\pi^2}{2}$

C) $\frac{\pi}{4}$

D) $\frac{\pi}{2}$

Ans: C

(King's property)2021

Sol: $I = \int \frac{e^{\cos x} \sin x}{(1+\cos^2 x)[e^{\cos x} + e^{-\cos x}]} dx \dots\dots\dots(1)$

$$= \int \frac{e^{-\cos x} \sin x}{(1+\cos^2 x)[e^{-\cos x} + e^{\cos x}]} dx \left[\int_0^a f(x) = \int_0^a f(a-x) \dots\dots\dots(2) \right]$$

$$(1) + (2)$$

$$\Rightarrow 2I = \int_0^{\pi} \frac{[e^{\cos x} + e^{-\cos x}] \sin x}{[1+\cos^2 x][e^{\cos x} + e^{-\cos x}]} dx, \text{ put } \cos x = t$$

$$= \int_1^{-1} \frac{-dt}{1+t^2} = -\tan^{-1}(t)_1^{-1}$$

$$\Rightarrow 2I = -\left[\frac{-\pi}{4} - \frac{\pi}{4} \right] \Rightarrow 2I = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$$

189. The value of $\int_{-\pi/2}^{\pi/2} \frac{\cos^2 x}{1+3^x} dx$ is :

1) 2π

2) 4π

3) $\frac{\pi}{4}$

4) $\frac{\pi}{2}$

Key : 3

(f(a + b - x) Property)2021

Solution:

$$I = \int_{-\pi/2}^{\pi/2} \frac{\cos^2 x}{1+3^x} dx = \int_{-\pi/2}^{\pi/2} \frac{\cos^2 x}{1+3^{-x}} dx = \int_{-\pi/2}^{\pi/2} \frac{3^x \cos^2 x}{3^x + 1} dx$$

$$\therefore 2I = \int_{-\pi/2}^{\pi/2} \cos^2 x \, dx = 2 \int_0^{\pi/2} \cos^2 x \, dx$$

$$= 2 \times \frac{1}{2} \times \frac{\pi}{2} \quad \therefore I = \frac{\pi}{4}$$

190. The value of the definite Integral $\int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \frac{dx}{1 + \sqrt[3]{\tan 2x}}$ is:

1) $\frac{\pi}{6}$

2) $\frac{\pi}{18}$

3) $\frac{\pi}{3}$

4) $\frac{\pi}{12}$

Key:

4

(f(a + b - x) Property)2021

Sol: $\int_{\pi/24}^{5\pi/24} \frac{dx}{1 + \sqrt[3]{\tan 2x}} = I$

Using $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$

We have $2I = \frac{5\pi}{24} - \frac{\pi}{24} \Rightarrow I = \frac{\pi}{12}$

191. The value of the definite integral $\int_{-\pi/4}^{\pi/4} \frac{dx}{(1 + e^{x \cos x})(\sin^4 x + \cos^4 x)}$ is equal to

1) $-\frac{\pi}{2}$

2) $-\frac{\pi}{4}$

3) $\frac{\pi}{2\sqrt{2}}$

4) $\frac{\pi}{\sqrt{2}}$

Key: 3

(f(a + b - x) Property)2021

Sol: $\int_{-\pi/4}^{\pi/4} \frac{dx}{(1 + e^{x \cos x})(\sin^4 x + \cos^4 x)} = I \text{ (let)}$

$$f(a+b-x) = f(x)$$

$$\Rightarrow 2I = \int_{-\pi/4}^{\pi/4} \left[\frac{1}{1 - \frac{1}{2} \sin^2 2x} \right] dx$$

$$2I = \int_{-\pi/4}^{\pi/4} \left(\frac{2 \sec^2 2x}{\sec^2 2x + 1} \right) dx$$

$$\{ \text{put } \tan 2x = t \Rightarrow 2 \sec^2 2x dx = dt \}$$

$$2I = \int_{-\infty}^{\infty} \frac{dt}{2+t^2} = 2 \int_0^{\infty} \frac{dt}{t^2+2}$$

$$I = \left(\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{t}{\sqrt{2}} \right) \right)_0^{\infty} \Rightarrow I = \frac{\pi}{2\sqrt{2}}$$

192. Let the domain of the function $f(x) = \log_4(\log_5(\log_3(18x - x^2 - 77)))$ be (a, b)

Then the value of the integral $\int_a^b \frac{\sin^3 x}{(\sin^3 x + \sin^3(a+b-x))} dx$ is equal to _____

Key: 1

(f(a + b - x) Property)2021

Sol: Domain of $f(x) = \log_4(\log_5(\log_3(18x - x^2 - 77)))$ be (a, b)

$$\log_5(\log_3(18x - x^2 - 77)) > 0$$

$$\log_3(18x - x^2 - 77) > 1$$

$$18x - x^2 - 77 > 3$$

$$x^2 - 18x + 80 < 0 \Rightarrow 8 < x < 10$$

$$I = \int_8^{10} \frac{\log^3 x}{\sin^3 x + \sin^3(18-x)} du = \int_8^{10} \frac{\sin^3(18-x)}{\sin^3(18-x) + \sin^3 x} dx$$

$$2I = \int_8^{10} du = 2 \Rightarrow I = 1$$

193. Which of the following statements is incorrect for the function $g(\alpha)$ for $\alpha \in R$ such that

$$g(\alpha) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^\alpha x}{\cos^\alpha x + \sin^\alpha x} dx$$

1) $g(\alpha)$ is an even function

2) $g(\alpha)$ is a strictly increasing function

3) $g(\alpha)$ is a strictly decreasing function 4) $g(\alpha)$ has an inflection point $\alpha = -\frac{1}{2}$

Key: 2, 3, 4

(King's property)2021

Sol: $g(x) = \frac{1}{2} \left(\frac{\pi}{3} - \frac{\pi}{6} \right) = \frac{\pi}{12}$ is constant function and has no point of inflection

[Note: Constant function is always even function]

194. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1 + \sin^2 x}{1 + \pi^{\sin x}} \right) dx$ is

1) $\frac{5\pi}{4}$

2) $\frac{\pi}{2}$

3) $\frac{3\pi}{4}$

4) $\frac{3\pi}{2}$

Key: 3

(King's property)2021

Sol: $I = \int_a^b f(x) dx$ $f(a+b-x) = f(x)$

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1 + \sin^2 x}{1 + \pi^{\sin x}} dx \rightarrow (1)$$

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1 + \sin^2 x}{1 + \pi^{\sin x}} dx \rightarrow (2)$$

$$(1) + (2) \quad 2I = \int_{-\pi/2}^{\pi} (1 + \sin^2 x) dx$$

$$2I = \int_{-\pi/2}^{\pi} \left(1 + \frac{1 - \cos 2x}{2} \right) dx$$

$$2I = \int_{-\pi/2}^{\pi} \left(\frac{3 - \cos 2x}{2} \right) dx$$

$$I = \frac{1}{4} \left[3x - \frac{\sin 2x}{2} \right]_{-\pi/2}^{\pi/2} \Rightarrow I = \frac{3\pi}{4}$$

195. $\int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x^2 - 44x + 484)} dx$ is equal to:

1) 6

2) 10

3) 5

4) 8

Key: 3

(King's property) 2021

Sol: Let $I = \int_6^{16} \frac{\log x^2}{\log x^2 + \log_e (x - 22)^2} dx$

$$= \int_6^{16} \frac{\log(22 - x)^2}{\log x^2 + \log_e (22 - x)^2} dx$$

$$2I = \int_6^{16} 1 dx \Rightarrow I = \frac{10}{2} = 5$$

196. The value of the integral $\int_{-\pi/2}^{\pi/2} \frac{dx}{(1 + e^x)(\sin^6 x + \cos^6 x)}$ is equal to

1) 2π

2) 0

3) π 4) $\frac{\pi}{2}$

Key: 3

(Replacing x by a-x - King's property) 2022

Sol: $I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{dx}{(1 + e^x)(\sin^6 x + \cos^6 x)}$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{dx}{(1 + e^{-x})(\sin^6 x + \cos^6 x)}$$

$$\therefore 2I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{dx}{(\sin^6 x + \cos^6 x)} = 2 \int_0^{\frac{\pi}{2}} \frac{dx}{(\sin^6 x + \cos^6 x)}$$

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$$\begin{aligned}
 \therefore 2I &= \int_0^{\frac{\pi}{2}} \frac{dx}{(\sin^6 x + \cos^6 x)} = \int_0^{\frac{\pi}{2}} \frac{dx}{1 - 3\sin^2 x \cos^2 x} \\
 &= \int_0^{\frac{\pi}{2}} \frac{\sec^4 x \, dx}{\sec^4 x - 3\tan^2 x} = \int_0^{\infty} \frac{(1+t^2)dt}{1+t^4-t^2} \\
 &= \int_0^{\infty} \frac{1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}-1} dt = \int_{-\infty}^{\infty} \frac{d\omega}{\omega^2+1} = 2 \cdot \frac{\pi}{2} = \pi
 \end{aligned}$$

197. The value of $\int_0^{\pi} \frac{e^{\cos x} \sin x}{(1+\cos^2 x)(e^{\cos x} + e^{-\cos x})} dx$ is equal to:

- A) $\frac{\pi^2}{4}$ B) $\frac{\pi^2}{2}$ C) $\frac{\pi}{4}$ D) $\frac{\pi}{2}$

Ans: C

(Replacing x by $a-x$ - King's property)2022

$$\begin{aligned}
 \text{Sol: } I &= \int \frac{e^{\cos x} \sin x}{(1+\cos^2 x)[e^{\cos x} + e^{-\cos x}]} dx \dots\dots\dots(1) \\
 &= \int \frac{e^{-\cos x} \sin x}{(1+\cos^2 x)[e^{-\cos x} + e^{\cos x}]} dx \left[\int_0^a f(x) = \int_0^a f(a-x) \dots\dots\dots(2) \right] \\
 (1) + (2) \\
 \Rightarrow 2I &= \int_0^{\pi} \frac{[e^{\cos x} + e^{-\cos x}] \sin x}{[1+\cos^2 x][e^{\cos x} + e^{-\cos x}]} dx, \text{ put } \cos x = t \\
 &= \int_1^{-1} \frac{-dt}{1+t^2} = -\tan^{-1}(t)_1^{-1} \\
 \Rightarrow 2I &= -\left[\frac{-\pi}{4} - \frac{\pi}{4} \right] \Rightarrow 2I = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}
 \end{aligned}$$

198. Let $f(x)$ be a function satisfying $f(x) + f(\pi - x) = \pi^2, \forall x \in R$. Then

$\int_0^{\pi} f(x) \sin x dx$ is equal to

- 1) $\frac{\pi^2}{2}$ 2) $\frac{\pi^2}{4}$ 3) $2\pi^2$ 4) π^2

KEY:4

(Replacing x by $a-x$ - King's property)2023

$$\text{SOL: } I = \int_0^{\pi} f(x) \sin x dx$$

$$= \int_0^{\pi} (\pi^2 - f(\pi - x)) \sin x dx$$

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$$= \pi^2 \int_0^{\pi} \sin x dx - \int_0^{\pi} f(\pi - x) \sin x dx$$

$$= -\pi^2 (\cos x)_0^{\pi} - \int_0^{\pi} f(x) \sin x dx \left(\because \int_0^a f(x) dx = \int_0^a f(a-x) dx \right)$$

$$I = -\pi^2 (-1 - 1) - I$$

$$2I = 2\pi^2$$

$$I = \pi^2$$

199. If $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying $\int_0^{\frac{\pi}{2}} f(\sin 2x) \sin x dx + \alpha \int_0^{\frac{\pi}{4}} f(\cos 2x) \cos x dx = 0$ then the value of α is

1) $-\sqrt{3}$

2) $\sqrt{3}$

3) $-\sqrt{2}$

4) $\sqrt{2}$

KEY: 3

(Replacing x by a-x - King's property) 2023

SOL: Let $I_1 + \alpha I_2 = 0$, where $I_1 = \int_0^{\pi/2} f(\sin 2x) \sin x dx = \int_0^{\pi/2} f(\sin 2x) \cos x dx$

$$I_2 = \int_0^{\pi/4} f(\cos 2x) \cos x dx = \frac{1}{\sqrt{2}} \int_0^{\pi/4} f(\sin 2x) (\cos x + \sin x) dx$$

$$2I_1 = \int_0^{\pi/2} f(\sin 2x) (\sin x + \cos x) dx = 2 \int_0^{\pi/4} f(\sin 2x) (\sin x + \cos x) dx$$

$$= 2\sqrt{2}I_2 \quad \therefore I_1 - \sqrt{2}I_2 = 0 \quad \Rightarrow \alpha = -\sqrt{2}$$

200. If $\int_0^1 \frac{1}{(5+2x-2x^2)(1+e^{(2-4x)})} dx = \frac{1}{\alpha} \log_e \left(\frac{\alpha+1}{\beta} \right)$, $\alpha, \beta > 0$ then $\alpha^4 - \beta^4$ is equal to

1) 0

2) -21

3) 21

4) 19

(Replacing x by a-x - King's property)

KEY: 3

SOL: $I + I = \int_0^1 \frac{dx}{5+2x-2x^2} \quad 2I = \frac{1}{2} \int_0^1 \frac{dx}{\sqrt{\left(\frac{11}{2}\right)^2 - \left(x - \frac{1}{2}\right)^2}}$

$$\Rightarrow I = \frac{1}{2\sqrt{11}} \log \left(\frac{12+2\sqrt{11}}{10} \right) = \frac{1}{\alpha} \log_e \left(\frac{\alpha+1}{\beta} \right) \quad \alpha = \sqrt{11}, \beta = \sqrt{10}$$

$$\alpha^4 - \beta^4 = 121 - 100 = 21$$

201. The value of $\sqrt{\pi \left(\int_0^{2008} x |\sin \pi x| dx \right)}$ is equal to

- 1) $\sqrt{2008}$ 2) $\pi\sqrt{2008}$ 3) 1004 4) $(2008)^2$

Ans. 4

(Replacing x by a-x – King's property)

Sol: Let $t = \pi x$

$$I = \frac{1}{\pi} \int_0^{2008\pi} t |\sin t| dt \quad \text{use } f(a-x) \text{ property}$$

$$I = \frac{2008\pi}{2\pi} \int_0^{2008\pi} |\sin t| dt = \frac{2008}{2} \times 2008 \int_0^{\pi} |\sin t| dt$$

$$= \frac{(2008)^2}{2} [-\cot]_0^{\pi} = (2008)^2$$

202. If $I = \int_0^{\frac{\pi}{2}} \left(\frac{dx}{1 + \sin x} \right)$, then the value of that $\int_0^{\pi} \frac{x^2 \cos x}{(1 + \sin x)^2} dx =$

- 1) $\pi I - \pi^2$ 2) $\pi I + \pi^2$ 3) $-\pi I + \pi^2$ 4) $-\pi I - \pi^2$

Key 1

(By parts and f(a-x) property)

Sol: Given $I = \int_0^{\pi/2} \frac{dx}{1 + \sin x}$ We have $\int_0^{\pi} \left(\frac{x^2 \cos x}{(1 + \sin x)^2} \right) dx =$

$$= \left[x^2 \int \left(\frac{\cos x}{(1 + \sin x)^2} \right) dx \right]_0^{\pi} + \int_0^{\pi} \frac{2x}{1 + \sin x} dx$$

$$= \left(1 \frac{x^2}{(1 + \sin x)} \right)_0^{\pi} + \int_0^{\pi} \frac{x}{1 + \sin x} dx$$

$$= (-\pi^2) + 2 \int_0^{\pi} \frac{x}{1 + \sin x} dx = -\pi^2 + 2 \int_0^{\pi} \frac{(\pi - x)}{1 + \sin x} dx$$

$$2I = -2\pi^2 + 2 \int_0^{\pi} \frac{\pi}{1 + \sin x} dx$$

$$\Rightarrow I = -\pi^2 + \pi \int_0^{\pi} \frac{dx}{1 + \sin x} = -\pi^2 + \pi I = \pi I - \pi^2$$

203. Let $f(x)$ and $g(x)$ be continuous function on $[0, a]$ such that

$$\int_0^a f(x) dx = p, \quad \int_0^a g(x) dx = q, \quad \text{and } f(x) + f(a-x) = 2 \quad \text{and } g(x) + g(a-x) = 4 \quad \text{then}$$

$$\int_0^a f(x) dx \text{ is } \underline{\hspace{2cm}}$$

- 1) $4a + 2p - q$ 2) $2a + 4p - q$ 3) $4a - 2p - q$ 4) $2a - 4p + q$

Key: 3

(King's property)

Sol: $I = \int_0^a f(x) g(x) dx = \int_0^a f(a-x) g(a-x) dx$

$$= \int_0^a (2 - g(x))(4 - g(x)) dx$$

$$= 8 \int_0^a 1 dx - 4p - 2q + I$$

$$I = 4a - 2p - q$$

204. $\int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{\sin x + \cos x} dx =$

- 1) $\sqrt{2} \log(\sqrt{2} + 1)$ 2) $\frac{1}{\sqrt{2}} \log(\sqrt{2} + 1)$
 3) $\log(\sqrt{2} + 1)$ 4) $\frac{1}{\sqrt{2}} \log(\sqrt{2} - 1)$

Key: 2

(King's property)

Sol: Using formula $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

205. Let f and g be continuous function on $[0, a]$ such that $f(x) = f(a - x)$ and

$g(x) + g(a - x) = 4$, then $\int_0^a f(x)g(x)dx$ is equal to

1) $4 \int_0^a f(x)dx$

2) $-3 \int_0^a f(x)dx$

3) $\int_0^a f(x)dx$

4) $2 \int_0^a f(x)dx$

Key: 4

(King's property)

Sol: $I = \int_0^a f(x)g(x)dx \dots (1)$

$$I = \int_0^a f(a-x)g(a-x)dx$$

$$I = \int_0^a f(x)\{4 - g(x)\}dx$$

$$I = \int_0^a 4f(x)dx - \int_0^a f(x)g(x)dx$$

$$I = 4 \int_0^a f(x)dx - I \Rightarrow 2I = 4 \int_0^a f(x)dx \Rightarrow I = 2 \int_0^a f(x)dx$$

206. $\int_0^1 \cot^{-1}(1-x+x^2)dx =$

1) $\frac{\pi}{2}$

2) $\frac{\pi}{2} - \log 2$

3) $\frac{\pi}{2} - \log 3$

4) $\log 2$

Key: 2

(King's property)

Sol: Let $I = \int_0^1 \cot^{-1}(1-x+x^2)dx$

$$I = \int_0^1 \cot^{-1}(1-x(1-x))dx$$

$$\int_0^1 \tan^{-1}\left(\frac{1}{1-x(1-x)}\right)dx$$

$$\begin{aligned}
&= \int_0^1 \tan^{-1} \left(\frac{x + (1-x)}{1-x(1-x)} \right) dx \\
&= \int_0^1 \tan^{-1} x \, dx + \int_0^1 \tan^{-1} (1-x) \, dx \\
&= 2 \int_0^1 \tan^{-1} x \, dx \\
I &= 2 \left[\left[\tan^{-1} x \cdot x \right]_0^1 - \int_0^1 \frac{x}{1+x^2} dx \right] \\
&= 2 \left[\frac{\pi}{4} - \frac{1}{2} \left[\ln |1+x^2| \right]_0^1 \right] = 2 \left[\frac{\pi}{4} - \frac{1}{2} \ln 2 \right]
\end{aligned}$$

207. Let $f(x)$ be a continuous function on $[0, 4]$ satisfying $f(x)f(4-x) = 1$. Find the value of the definite integral $\int_0^4 \frac{1}{1+f(x)} dx =$

1) 3

2) 1

3) 2

4) 7

Key: 3

(King's property)

Sol:

$$\text{Let } I = \int_0^4 \frac{1}{1+f(x)} dx$$

$$I = \int_0^4 \frac{1}{1+f(x)} dx, \text{ put } f(4-x) = \frac{1}{f(x)} \Rightarrow I = \int_0^4 \frac{f(x)}{f(x)+1} dx$$

On adding (1) and (2) we get

$$2I = \int_0^4 dx \Rightarrow I = 2$$

208. If $f(a+1-x) = f(x)$, $\forall x$, when $a > 0$ real number then $\frac{1}{a} \int_0^a x(f(x) + f(x+1)) dx$ is ____

1) $\int_0^a f(x) dx$

2) $\frac{1}{2} \int_0^a f(x) dx$

3) $\int_0^{a+1} f(x) dx$

4) $2 \int_0^{a+1} f(x) dx$

Key: 1

(f(a + b - x) Property)

Sol:

$$f(a+1-x) = f(x)$$

$$x \rightarrow 1+x \Rightarrow f(a-x) = f(x+1)$$

$$I = \frac{1}{a} \int_0^a (a-x)(f(a-x) + f(a-x+1)) dx = \frac{1}{a} \int_0^a (a-x)(f(x+1) + f(x)) dx$$

$$\Rightarrow \int_0^a (f(x) + f(x+1)) dx - \frac{1}{a} \int_0^a x(f(x) + f(x+1)) dx$$

$$I = \frac{1}{4} \int_0^a f(x) + f(x+1) dx = \frac{1}{2} \left(\int_0^a f(x) + f(a-x) dx \right)$$

$$= \frac{1}{2} 2 \int_0^a f(x) = \int_0^a f(x)$$

209. The value of $\int_{\sqrt{\ln 2}}^{\sqrt{\ln 3}} \frac{x \sin x^2}{\sqrt{\ln 2} \sin x^2 + \sin(\ln(6-x^2))} dx$ is _____

1) $\frac{1}{4} \ln\left(\frac{3}{2}\right)$

2) $\frac{1}{2} \ln\left(\frac{3}{2}\right)$

3) $\ln\left(\frac{3}{2}\right)$

4) $\frac{1}{6} \ln\left(\frac{3}{2}\right)$

Key: 1

(f(a + b - x) Property)

Sol:

$$I = \frac{1}{2} \int_{\ln 2}^{\ln 3} \frac{\sin \theta}{\sin \theta + \sin(\ln(6-\theta))} d\theta \quad \text{is } x^2 = \theta$$

$$= \frac{1}{2} \int_{\ln 2}^{\ln 3} \frac{\sin(\ln 6 - \theta)}{\sin(\ln 6 - \theta) + \sin \theta} d\theta \quad f(a+b-x) \text{ property}$$

$$I + I = \frac{1}{2} \int_{\ln 2}^{\ln 3} d\theta$$

$$I = \frac{1}{4} (\ln 3 - \ln 2) = \frac{1}{4} \ln\left(\frac{3}{2}\right)$$

210. If θ_1 and θ_2 are two sol in $[0, 4\pi]$ satisfying $(\sqrt{3} \tan \theta + 1)^2 + (2 \sin \theta - 1)^2 = 0$ then

$$\int_{\theta_1}^{\theta_2} \cos^6\left(\frac{5\pi}{6} + \theta\right) d\theta = \text{_____}$$

1) $\frac{5\pi}{8}$

2) 0

3) $\frac{5\pi}{4}$

4) $\frac{5\pi}{6}$

Key: 1

(Miscellaneous and $f(a + b - x)$ Property)

Sol:

$$\tan \theta = -\frac{1}{3} \text{ and } \sin \theta = \frac{1}{2} \quad \theta \in Q_2$$

$$\theta_1 = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

$$\theta_2 = 2\pi + \frac{5\pi}{6} = \frac{17\pi}{6}$$

$$\int_{\frac{5\pi}{6}}^{\frac{17\pi}{6}} \cos^6 \left(\frac{5\pi}{6} + \theta \right) d\theta$$

$$= \int_0^{2\pi} \cos^6 x dx = 4 \int_0^{\frac{\pi}{2}} \cos^6 x dx = 4I_6$$

$$= 4 \times \frac{5}{6} \times \frac{3}{5} \times \frac{1}{2} \times \frac{\pi}{2}$$

$$= \frac{5\pi}{8}$$

211. If $\cos 2\theta + \lambda \sin \theta = 2\lambda - 6$ possesses a solution and $\lambda \in [a, b]$ then

$$\int_a^b \frac{[x^2 - 16x + 64]}{[x^2] + [x^2 - 16x + 64]} dx = \text{---}$$

1) 2

2) 6

3) 4

4) 3

Key: 1

(Miscellaneous and $f(a + b - x)$ Property)

Sol:

$$\cos 2x + \lambda \sin x = 2a - 7$$

$$\Rightarrow 1 - 2\sin^2 x + \lambda \sin x = 2a - 7$$

$$\Rightarrow \sin x = \frac{\lambda \pm \sqrt{\lambda^2 - 16\lambda + 64}}{4} = \frac{\lambda - 4}{2}$$

$$-1 \leq \frac{\lambda - 4}{2} \leq 1 \Rightarrow 2 \leq \lambda \leq 6 \quad a = 2 \quad b = 0$$

$$I = \int_2^6 \frac{[x^2 - 16x + 64]}{[x^2] + [x^2 - 16x + 64]} dx$$

$$I = \frac{6-2}{2} = 2$$

212. If $\int_{-2\pi}^{2\pi} \frac{\sin^6 x}{\sin^6 x + \cos^6 x} \times \frac{dx}{1 + e^{-x}} = \text{---}$

- 1) 2π 2) π 3) $\frac{\pi}{2}$ 4) 4π

Key: 2

(f(a + b - x) Property)

Sol: $I = \int_{-2\pi}^{2\pi} \frac{\sin^6 x}{\sin^6 x + \cos^6 x} \times \frac{dx}{1 + e^{-x}}$

$$I = \int_{-2\pi}^{2\pi} \frac{\sin^6 x}{\sin^6 x + \cos^6 x} \times \frac{e^x dx}{1 + e^x} \because f(a + b - x) \text{ property}$$

$$I + I = \int_{-2\pi}^{2\pi} \frac{\sin^6 x}{\sin^6 x + \cos^6 x} \times \frac{e^x + 1}{e^x + 1} dx$$

$$2I = 2 \int_0^{2\pi} \frac{\sin^6 x}{\sin^6 x + \cos^6 x} dx \Rightarrow I = 4 \int_0^{\frac{\pi}{4}} \frac{\sin^6 x dx}{\sin^6 x + \cos^6 x} = 4 \left[\frac{\pi}{4} \right] = \pi$$

213. The value of $I = \int_{-\pi}^{\pi} x \sin^5 x dx$ is _____

- 1) 0 2) $\frac{16\pi}{45}$ 3) $\frac{32\pi}{45}$ 4) $\frac{3\pi}{15}$

Key: 3

(King's property and Even Odd Property)

Sol: $f(x) = x \sin^5 x$ is even fraction

$$I = 2 \int_0^{\pi} x \sin^5 x dx, f(a - x) = 2 \int_0^{\pi} (\pi - x) \sin^5 x dx$$

$$= 2\pi \int_0^{\pi} \sin^5 x dx - 2I$$

$$I = \frac{2\pi}{3} \times 2 \int_0^{\frac{\pi}{2}} \sin^5 x dx = \frac{4\pi}{3} I_5 = \frac{4\pi}{3} \times \frac{4}{5} \times \frac{2}{3} \times 1 = \frac{32\pi}{45}$$

214. The value of $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1 + a^x} dx$, where $a > 0$, is

- 1) π 2) $a\pi$ 3) $\pi / 2$ 4) 2π

Key: 3

(f(a + b - x) Property)

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Sol:

$$I = \int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x} dx = \int_{-\pi}^{\pi} \frac{\cos^2(0-x)}{1+a^{(0-x)}} dx \Rightarrow I = \int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x} dx$$

$$2I = \int_{-\pi}^{\pi} \cos^2 x dx = 2 \int_0^{\pi} \cos^2 x dx = 4 \int_0^{\pi/2} \cos^2 x dx = 4 \int_0^{\pi/2} \sin^2 x dx$$

$$\Rightarrow 4I = 4 \int_0^{\pi/2} 1 dx \text{ or } I = \pi / 2$$

215. If $I_1 = \int_{-100}^{101} \frac{dx}{(5+2x-2x^2)(1+e^{2-4x})}$ and $I_2 = \int_{-100}^{101} \frac{dx}{5+2x-2x^2}$, then $\frac{I_1}{I_2}$ is

1) 2

2) $\frac{1}{2}$

3) 1

4) $-\frac{1}{2}$

Key: 2

(f(a + b - x) Property)

Sol: $I_1 = \int_{-100}^{101} \frac{dx}{(5+2x-2x^2)(1+e^{2-4x})}$

$$I_1 = \int_{-100}^{101} \frac{dx}{(5+2(1-x)-2(1-x)^2)(1+e^{2-4(1-x)})}$$

$$\text{or } 2I_1 = \int_{-100}^{101} \frac{dx}{5+2x-2x^2} = I_2$$

$$\text{or } \frac{I_1}{I_2} = \frac{1}{2}$$

216. If $I_1 = \int_0^{\pi/2} \frac{\cos^2 x}{1+\cos^2 x} dx$, $I_2 = \int_0^{\pi/2} \frac{\sin^2 x}{1+\sin^2 x} dx$, $I_3 = \int_0^{\pi/2} \frac{1+2\cos^2 x \sin^2 x}{4+2\cos^2 x \sin^2 x} dx$, then

1) $I_1 = I_2 > I_3$

2) $I_3 < I_1 = I_2$

3) $I_1 = I_2 = I_3$

4) $I_1 = I_3$

Key: 3

(f(a - x) Property)

Sol: $I_1 = \int_0^{\pi/2} \frac{\cos^2 x}{1+\cos^2 x} dx$

$$= \int_0^{\pi/2} \frac{\cos^2(\pi/2 - x)}{1+\cos^2(\pi/2 - x)} dx = \int_0^{\pi/2} \frac{\sin^2 x}{1+\sin^2 x} dx = I_2$$

$$\text{Also } I_1 + I_2 = \int_0^{\pi/2} \left(\frac{\sin^2 x}{1+\sin^2 x} + \frac{\cos^2 x}{1+\cos^2 x} \right) dx$$

$$= \int_0^{\pi/2} \frac{\sin^2 x + \sin^2 \cos^2 x + \cos^2 x + \sin^2 x \cos^2 x}{1 + \sin^2 x + \cos^2 x + \sin^2 x \cos^2 x} dx$$

$$= \int \frac{1 + 2\sin^2 x \cos^2 x}{2 + \sin^2 x \cos^2 x} dx = 2I_3$$

$$2I_1 = 2I_3 \Rightarrow I_1 = I_3 \Rightarrow I_1 = I_2 = I_3$$

217. If $I = \int_0^{\pi} x(\sin^2(\sin x) + \cos^2(\cos x)) dx$, then $[I] = \underline{\hspace{1cm}}$, where $[.]$ denotes the greatest integer function (King's property)

Ans: 4

Sol. $I = \int_0^{\pi} (\pi - x)((\sin^2(\sin x)) + \cos^2(\cos x)) dx \Rightarrow 2I = 2\pi \int_0^{\frac{\pi}{2}} (\sin^2(\sin x) + \cos^2(\cos x)) dx$

$$\Rightarrow I = \pi \int_0^{\frac{\pi}{2}} (\sin^2(\sin x) + \cos^2(\cos x)) dx = \pi \int_0^{\frac{\pi}{2}} (\sin^2(\cos x) + \cos^2(\sin x)) dx$$

$$\Rightarrow 2I = \pi \int_0^{\frac{\pi}{2}} 2 dx \Rightarrow I = \frac{\pi^2}{2}$$

218. If $m, n \in \mathbb{N}$, the value of $I_{m,n} = \int_0^1 x^m (1-x)^n dx$ is

A) $\frac{m!n!}{(m+n)!}$ B) $\frac{(m+1)!(n+1)!}{(m+n)!}$ C) $\frac{m!n!}{(m+n+1)!}$ D) $\frac{(m-1)!(n-1)!}{(m+n+1)!}$

ANS: C

(King's property)

SOL: It should be obvious that we require a recursive relation involving $I_{m,n}$ and a lower order integral (with respect to one of the two variables m or n)

$$I_{m,n} = \int_0^1 \underbrace{(1-x)^n}_{\text{1st function}} \underbrace{x^m}_{\text{2nd Function}} dx = \frac{(1-x)^n x^{m+1}}{m+1} \Big|_0^1 + \frac{n}{m+1} \int_0^1 (1-x)^{n-1} x^{m+1} dx = 0 + \frac{n}{m+1} I_{m+1,n-1}$$

Thus, the required recursive relation is

$$I_{m,n} = \frac{n}{m+1} I_{m+1,n-1}$$

We use this relation repeatedly now till n reduces to 0;

$$I_{m,n} = \frac{n}{m+1} I_{m+1,n-1} = \frac{n(n-1)}{(m+1)(m+2)} I_{m+2,n-2}$$

$$= \frac{n(n-1)\dots 1}{(m+1)(m+2)\dots(m+n)} I_{m+n,0}$$

$$= \frac{m!n!}{(m+n)!} I_{m+n,0}$$

$$I_{m,n} = \frac{m!n!}{(m+n+1)!}$$

219. The value of $\int_{-a}^a \sqrt{\frac{a-x}{a+x}} dx$ is

(f(a + b - x) Property)

A) πa

B) a

C) π

D) $-\pi a$

ANS: A

SOL:

$$I = \int_{-a}^a \sqrt{\frac{a-x}{a+x}} dx = \int_{-a}^a \frac{a-x}{\sqrt{a^2-x^2}} dx$$

$$= a \int_{-a}^a \frac{dx}{\sqrt{a^2-x^2}} - \int_{-a}^a \frac{x dx}{\sqrt{a^2-x^2}}$$

($\because \frac{x}{\sqrt{a^2-x^2}}$ is an odd function)

$$I = a \cdot 2 \int_0^a \frac{dx}{\sqrt{a^2-x^2}} - 0 = 2a \left[\sin^{-1} \frac{x}{a} \right]_0^a$$

$$\Rightarrow 2a \left[\sin^{-1}(1) - \sin^{-1}(0) \right] = 2a \left[\frac{\pi}{2} - 0 \right] = \pi a$$

220. Let $f, g: [-a, a] \rightarrow R$ be two continuous functions, and f is odd g is even then

$$= \int_{-a}^a \frac{g(x)}{1+f(x)+\sqrt{1+f^2(x)}} dx =$$

(f(a + b - x) Property)

A) $2 \int_0^a g(x) dx$

B) $\frac{1}{2} \int_0^a g(x) dx$

C) $\int_0^a g(x) dx$

D) $\int_{-a}^0 g(x) dx$

ANS: C

SOL: $I = \int_{-a}^a \frac{g(x) dx}{1+f(x)+\sqrt{1+f^2(x)}}$

$$I = \int_{-a}^a \frac{g(-x)}{1+f(-x)+\sqrt{1+f^2(-x)}} dx$$

$$I = \int_{-a}^a \frac{g(x)}{1-f(-x)+\sqrt{1+f^2(x)}} dx$$

$$2I = \int_{-a}^a \frac{2g(x) \left[1 + \sqrt{1 + f^2(x)} \right]}{\left[1 + \sqrt{1 + f^2(x)} \right]^2 - f^2(x)} dx$$

$$I = \int_{-a}^a \frac{g(x) \left[1 + \sqrt{1 + f^2(x)} \right]}{2 \left(1 + \sqrt{f^2(x) + 1} \right)} dx$$

$$I = \int_{-a}^a g(x) dx \quad \text{OR Let } h(x) = \frac{g(x)}{1 + f(x) + \sqrt{1 + f^2(x)}} \text{ then}$$

$$h(x) + h(-x) = g(x)$$

221. The value of the integral $\int_0^1 \frac{\sin^{-1} \sqrt{x}}{x^2 - x + 1} dx$ is (King's property)

A) $\frac{\pi^2}{6\sqrt{3}}$

B) $\frac{\pi}{6\sqrt{3}}$

C) $\frac{\pi^2}{3\sqrt{3}}$

D) $\frac{\pi^2}{12\sqrt{3}}$

ANS: A

SOL:

$$I = \int_0^1 \frac{\sin^{-1} \sqrt{x}}{x^2 - x + 1} dx$$

$$I = \int_0^1 \frac{\sin^{-1} \sqrt{1-x}}{(1-x)^2 - (1-x) + 1} dx$$

$$I = \int_0^1 \frac{\sin^{-1} \sqrt{1-(\sqrt{x})^2}}{x^2 - x + 1} dx = \int_0^1 \frac{\cos^{-1} \sqrt{x}}{x^2 - x + 1} dx$$

On adding both the integrals, we get

$$2I = \int_0^1 \frac{\frac{\pi}{2}}{\left(x - \frac{1}{2} \right)^2 + \left(\frac{\sqrt{3}}{2} \right)^2} dx$$

$$\Rightarrow I = \frac{\pi}{4} \times \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2x-1}{\sqrt{3}} \right) \Big|_0^1 \Rightarrow I = \frac{\pi^2}{6\sqrt{3}}$$

222. The value of $I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} [x \sec x] dx$ is (f(a + b - x) Property)

A) $-\frac{\pi}{4}$

B) 0

C) $\frac{\pi}{4}$

D) $\frac{\pi}{2}$

ANS: A

SOL: $I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} [x \sec x] dx$

$$I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} [-x \sec x] dx$$

$$I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} (-1 - [x \sec x]) dx$$

$$I = -\frac{\pi}{2} - I \Rightarrow 2I = -\frac{\pi}{2} \Rightarrow I = -\frac{\pi}{4}$$

223. If $I_1 = \int_0^2 \frac{[x^2]}{[x^2 - 4x - 4] + [x^2]} dx$, and $I_2 = \int_0^2 \frac{4}{[x^2 - 4x] + [x^2 - 4]} dx$ (where $[.]$ denotes G.I.F.)
then $I_1 - I_2 =$ (f(a + b - x) Property)

ANS: 1

SOL: $I_1 = \int_0^2 \frac{[x^2]}{[x^2 - 4x - 4] + [x^2]} dx$

$$I_1 = \int_0^2 \frac{[(2-x)^2]}{[(2-x)^2 - 4(2-x) - 4] + [(2-x)^2]} dx$$

$$= \int_0^2 \frac{[x^2 - 4x] + 4}{[x^2 - 8] + [x^2 - 4x + 4]} dx$$

$$= \int_0^2 \frac{[x^2 - 4x] + 4}{[x^2 - 4] + [x^2 - 4x]} dx = \int_0^2 \frac{[x^2 - 4x] + 4}{[x^2 - 4] + [x^2 - 4x]} dx$$

$$= \int_0^2 \frac{[x^2 - 4x]}{[x^2 - 4] + [x^2 - 4x]} dx = I_2 \Rightarrow I_1 - I_2 = 1$$

224. If $I_1 = \int_0^{\pi/2} \frac{\cos^2 x}{1 + \cos^2 x} dx$, $I_2 = \int_0^{\pi/2} \frac{\sin^2 x}{1 + \sin^2 x} dx$, $I_3 = \int_0^{\pi/2} \frac{1 + 2 \cos^2 x \sin^2 x}{4 + 2 \cos^2 x \sin^2 x} dx$, then

A) $I_1 = I_2 > I_3$ B) $I_3 > I_1 = I_2$ C) $I_1 = I_2 = I_3$ D) $I_1 = I_2 = 2I_3$

ANS: C

(King's property)

$$\begin{aligned} \text{SOL: } I_1 &= \int_0^{\pi/2} \frac{\cos^2 x}{1 + \cos^2 x} dx \\ &= \int_0^{\pi/2} \frac{\cos^2(\pi/2 - x)}{1 + \cos^2(\pi/2 - x)} dx = \int_0^{\pi/2} \frac{\sin^2 x}{1 + \sin^2 x} dx = I_2 \end{aligned}$$

$$\begin{aligned} \text{Also } I_1 + I_2 &= \int_0^{\pi/2} \left(\frac{\sin^2 x}{1 + \sin^2 x} + \frac{\cos^2 x}{1 + \cos^2 x} \right) dx \\ &= \int_0^{\pi/2} \frac{\sin^2 x + \sin^2 \cos^2 x + \cos^2 x + \sin^2 x \cos^2 x}{1 + \sin^2 x + \cos^2 x + \sin^2 x \cos^2 x} dx \\ &= \int_0^{\pi/2} \frac{1 + 2\sin^2 x \cos^2 x}{2 + \sin^2 x \cos^2 x} dx = 2I_3 \end{aligned}$$

$$2I_1 = 2I_3 \Rightarrow I_1 = I_3 \Rightarrow I_1 = I_2 = I_3$$

225. $f(x) > 0 \forall x \in R$ and is bounded. If

(King's property)

$$\lim_{x \rightarrow \infty} \left[\int_0^a \frac{f(x) dx}{f(x) + f(a-x)} + a \int_0^{2a} \frac{f(x) dx}{f(x) + f(3a-x)} + a^2 \int_0^{3a} \frac{f(x) dx}{f(x) + f(5a-x)} + \dots + a^{n-1} \int_{(n-1)a}^{na} \frac{f(x) dx}{f(x) + f((2n-1)a-x)} \right] = \frac{7}{5} \text{ (where } a < 1), \text{ then } a \text{ is equal to}$$

A) $\frac{7}{2}$

B) $\frac{1}{7}$

C) $\frac{14}{19}$

D) $\frac{9}{14}$

ANS: C

$$\text{SOL: We have } \int_0^a \frac{f(x)}{f(x) + f(a-x)} = \frac{a}{2} \text{ and}$$

$$\int_a^b \frac{f(x) dx}{f(x) + f(a+b-x)} = \frac{b-a}{2}$$

There fore given sum reduces to

$$\lim_{x \rightarrow \infty} \left[\frac{a}{2} + \frac{a^2}{2} + \frac{a^3}{2} + \dots + \frac{a^n}{2} \right] = \frac{7}{5}$$

$$\Rightarrow \frac{a}{1-a} = \frac{14}{5} \Rightarrow a = \frac{14}{19}$$

226. $A = \int_0^1 x^{50} (2-x)^{50} dx; B = \int_0^1 x^{50} (1-x)^{50} dx$, which of the following is true?

A) $A = 2^{50} B$

B) $A = 2^{-50} B$

C) $A = 2^{100} B$

D) $A = 2^{-100} B$

ANS: C

(King's property)

$$\text{SOL: Given } A = \int_0^1 x^{50} (2-x)^{50} dx; B = \int_0^1 x^{50} (1-x)^{50} dx$$

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In A, put $x=2t \Rightarrow dx=2dt$

$$\Rightarrow A = 2 \int_0^{1/2} 2^{50} t^{50} 2^{50} (1-t)^{50} dt \quad \dots\dots(1)$$

$$B = \int_0^{1/2} x^{50} (1-x)^{50} dx \quad \dots\dots\dots(2)$$

$$[\text{using } \int_0^{2\pi} f(x) dx = 2 \int_0^a f(x) dx \text{ if } f(2a-x) = f(x)]$$

From equations (1) and (2), we get

$$A = 2^{100} B$$

$$227. \int_{\pi}^{2\pi} \frac{\sin^4 t + \sin^4(2-t)}{\sin^4(2-t) + 2\sin^4 t + \sin^4(2+t)} dt =$$

- A) π B) $\frac{\pi}{2}$ C) 2π D) $\frac{\pi^2}{2}$

ANS:B

(King's property)

$$\text{SOL: If } I = \int_a^b \frac{f(x) + f(a-x)}{2f(a-x) + f(x) + f(a+b-x)} dx$$

$$\text{Where } f(a-x) = f(x-b) \text{ then } I = \frac{b-a}{2}$$

$$\text{or } I = \int_{+\pi}^{2\pi} \frac{\sin^4 t + \sin^4(2-t)}{\sin^4(2-t) + 2\sin^4 t + \sin^4(2+t)} dt$$

$$I = \int_{\pi}^{2\pi} \frac{\sin^4(3\pi-t) + \sin^4(2-(3\pi-t))}{\sin^4(2-(3\pi-t)) + 2\sin^4(3\pi-t) + \sin^4(2+3\pi-t)} dt$$

$$= \int_{\pi}^{2\pi} \frac{\sin^4 t + \sin^4(2-t)}{\sin^4(2+t) + 2\sin^4 t + \sin^4(2-t)} dt$$

$$2I = \int_{\pi}^{2\pi} dt = \pi \Rightarrow I = \frac{\pi}{2}$$

$$228 \int_0^1 \tan^{-1}(1-x+x^2) dx = \underline{\hspace{2cm}}$$

- (A) $\ln 2$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{2} - \ln 2$

Key: A

(f(a + b - x) Property)

$$\text{Sol: } \int_0^1 \tan^{-1}(1-x+x^2) dx = \int_0^1 \left(\frac{\pi}{2} - \tan^{-1} \left(\frac{x+(1-x)}{1-x(1-x)} \right) \right) dx$$

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$$= \frac{\pi}{2} - \int_0^1 (\tan^{-1}(1-x) + \tan^{-1} x) dx$$

$$\Rightarrow \frac{\pi}{2} - 2 \int_0^1 \tan^{-1}(x) dx = \ln 2$$

229. $\int_0^{\frac{\pi}{4}} (\pi x - 4x^2) \ln(1 + \tan x) dx = \underline{\hspace{2cm}}$

(A) $\frac{\pi^3}{192}$

(B) $\frac{\pi^3}{192} \ln 2$

(C) $\frac{\pi^2}{96}$

(D) $\frac{\pi^2}{96} \ln 2$

Key: B

(f(a + b - x) Property)

Sol: Let $I = \int_0^{\frac{\pi}{4}} 4x \left(\frac{\pi}{4} - x \right) \ln(1 + \tan x) dx = \int_0^{\frac{\pi}{4}} 4 \left(\frac{\pi}{4} - x \right) x \ln \left(1 + \tan \left(\frac{\pi}{4} - x \right) \right) dx$

$$= 4 \ln 2 \int_0^{\frac{\pi}{4}} x \left(\frac{\pi}{4} - x \right) dx - I$$

$$\Rightarrow I = 2 \ln 2 \int_0^{\frac{\pi}{4}} x \left(\frac{\pi}{4} - x \right) dx$$

$$= \frac{\pi^3}{192} \ln 2$$

230. $\int_{-\pi/4}^{\pi/4} \log(\cos x + \sin x) dx =$

1) $\pi \log 2$

2) $-\pi \log 2$

3) $-\frac{\pi}{2} \log 2$

4) $\frac{\pi}{2} \log 2$

Key:

(f(a + b - x) Property)

Sol: Using formula $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$

231. $\int_{2-\sqrt{3}}^{2+\sqrt{3}} \frac{x dx}{(1+x)(1+x^2)} =$

1) $\frac{\pi}{4}$

2) $\frac{\pi}{6}$

3) $\frac{\pi}{12}$

4) $\frac{\pi}{24}$

Key: 2

(Substitution and $f(a + b - x)$ Property)Sol: Put $x = \tan \theta$

232.
$$\int_2^5 \frac{[x^2 + 49 - 14x]}{[x^2 - 14x + 49] + [x^2]} dx =$$

(where $[]$ is g.i.f.)

1) $\frac{1}{2}$

2) $\frac{3}{2}$

3) $\frac{5}{2}$

4) $\frac{7}{2}$

Key: 2

(f(a + b - x) Property)

Sol: Apply $f(a + b - x)$ formula

$$I = \int_2^5 \frac{[(7-x)^2]}{[(7-x)^2] + [x^2]} dx$$

233.
$$\int_{-\pi}^{\pi} \frac{2x(1 + \sin x)}{1 + \cos^2 x} dx =$$

1) $\frac{\pi^2}{4}$

2) π^2

3) 0

4) $\frac{\pi}{2}$

Key: 2

(Removal of x and $f(a + b - x)$ Property)

Sol: Given =
$$\int_{-\pi}^{\pi} \frac{2x}{1 + \cos^2 x} dx + \int_{-\pi}^{\pi} \frac{2x \sin x}{1 + \cos^2 x} dx$$

$$= 4 \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$$

$\therefore \frac{2x}{1 + \cos^2 x}$ is odd and $\frac{x \sin x}{1 + \cos^2 x}$ is even $\int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx$ apply $f(2a - x) = f(x)$ if
and put $\cos x = t$

234.
$$\int_0^{\pi} [\cot x] dx, \text{ where } [.] \text{ denotes the greatest integer function, is equal to (AIEEE 2009)}$$

1) 1

2) -1

3) $-\frac{\pi}{2}$

4) $\frac{\pi}{2}$

Key: 3

(f(a + b - x) Property)

Sol: Apply $f(a + b - x)$ formula.

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$$I = \int_0^{\pi} [\cot x] dx, I = \int_0^{\pi} [-\cot x] dx$$

$$I + I = \int_0^{\pi} [\cot x] dx + \int_0^{\pi} [-\cot x] dx = \int_0^{\pi} -1 dx$$

235. If $I_1 = \int_0^{\pi/2} f(\sin 2x) \sin x dx$ and $I_2 = \int_0^{\pi/4} f(\cos 2x) \cos x dx$ then $\frac{I_1}{I_2} =$

1) $\sqrt{2}$

2) $2\sqrt{2}$

3) $3\sqrt{2}$

4) $4\sqrt{2}$

Key: 1

(King's property)

Sol: $I_1 + I_1 = \int_0^{\pi/2} f(\sin 2x)(\sin x + \cos x) dx$

$$= \sqrt{2} \int_0^{\pi/4} f(\sin 2x) \cos\left(x - \frac{\pi}{4}\right) dx$$

$$= \sqrt{2} \int_{-\pi/4}^{\pi/4} f(\cos 2t) \cos t dt \text{ where } t = x - \frac{\pi}{4}$$

236. $\int_0^{\pi} \frac{x \sin 2x \sin\left(\frac{\pi \cos x}{2}\right)}{2x - \pi} dx =$

1) $\frac{2}{\pi^2}$

2) $\frac{8}{\pi^2}$

3) $\frac{3}{\pi^2}$

4) $\frac{4}{\pi^2}$

Key: 2

(King's property)

Sol: $I = \int_0^{\pi} \frac{(\pi - x)(\sin 2x) \left(-\sin\left(\frac{\pi}{2} \cos x\right)\right)}{\pi - 2x} dx$

then $2I = \int_0^{\pi} \left(\sin 2x \cdot \sin\left(\frac{\pi}{2} \cos x\right) \right) dx$

$$= 4 \int_0^{\pi/2} \cos x \sin\left(\frac{\pi}{2} \cos x\right) \sin x dx$$

$$\Rightarrow I = 2 \int \left(\frac{2}{\pi} t\right) \sin t \left(\frac{2}{\pi} dt\right)$$

(where $t = \frac{\pi}{2} \cos x$)

$$\Rightarrow I = \frac{8}{\pi^2} \int_0^{\pi/2} t \sin t \, dt$$

$$= \frac{8}{\pi^2} (-t \cos t + \sin t) \Big|_0^{\pi/2} = \frac{8}{\pi^2}$$

237. The integral $\int_{\pi/4}^{5\pi/4} (|\cos t| \sin t + |\sin t| \cos t) \, dt$ has the value equal to

Key: 0

(f(a + b - x) Property)

$$\begin{aligned} \text{Sol: } I &= \int_{\pi/4}^{\pi/2} 2 \sin t \cos t \, dt + \int_{\pi/2}^{\pi} \{(-\sin t \cos t) + (\sin t \cos t)\} \, dt + \int_{\pi}^{5\pi/4} -2 \sin t \cos t \, dt \\ &= \int_{\pi/4}^{\pi/2} \sin 2t \, dt - \int_{\pi}^{5\pi/4} \sin 2t \, dt \\ &= 0 \end{aligned}$$

238. Value of the integral $I = \int_0^1 x(1-x)^n \, dx =$

$$\frac{1}{n+1} + \frac{1}{n+2} \quad \begin{matrix} 1) \frac{1}{n+2} \\ 2) \frac{1}{n+1} - \frac{1}{n+2} \\ 3) \frac{1}{n+1} \\ 4) \frac{1}{n+2} \end{matrix}$$

Key: 2

(King's property)

Sol: Using formula $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$

239. The integral $\int_0^a \frac{g(x)}{f(x) + f(a-x)} \, dx$ vanishes, if

$$1) g(x) \text{ is odd} \quad 2) f(x) = f(a-x) \quad 3) g(x) = -g(a-x) \quad 4) f(a-x) = -g(x)$$

Key: 3

(King's property)

Sol: Conceptual

Clearly I vanishes if $g(a-x) = -g(x)$

240. If $\frac{d}{dx} f(x) = g(x)$ then $\int_a^b f(x) g(x) \, dx =$

$$1) \frac{f(b) - f(a)}{2} \quad 2) \frac{f(a) - f(b)}{2} \quad 3) \frac{f^2(b) - f^2(a)}{2} \quad 4) \frac{f^2(a) - f^2(b)}{2}$$

Key: 3

(King's property)

Sol: $\int_a^b f(x) f'(x) dx = \left[f^2(x) \right]_a^b$

241. $I = \int_0^{\frac{\pi}{2}} \frac{x \sin x \cos x dx}{\sin^4 x + \cos^4 x}$ equals **(Replacing x by a-x - King's property)**

1) $\frac{\pi^2}{8}$

2) $-\frac{\pi^2}{8}$

3) $\frac{\pi^2}{16}$

4) $-\frac{\pi^2}{16}$

Key: 3

Sol: $I = \frac{\pi}{4} \int_0^{\pi/2} \frac{\sin x \cos x}{1 - \frac{1}{2} \sin^2 2x} dx = \frac{\pi}{4} \int_0^{\pi/2} \frac{\sin 2x}{1 + \cos^2 2x} dx$
 $= -\frac{\pi}{8} \tan^{-1}(\cos 2x) \Big|_0^{\pi/2} = -\frac{\pi}{8} \left(-\frac{\pi}{4} - \frac{\pi}{4} \right) = \frac{\pi^2}{16}$

242. The value of $\int_{-\pi/2}^{\pi/2} \frac{x^2 \cos x}{1+e^x} dx$ is equal to **(Replacing x by a + b - x)**

1) $\frac{\pi^2}{4} - 2$

2) $\frac{\pi^2}{4} + 2$

3) $\pi^2 - e^{\frac{\pi}{2}}$

4) $\pi^2 + e^{\frac{\pi}{2}}$

Key: 1

Sol: Let $I = \int_{-\pi/2}^{\pi/2} \frac{x^2 \cos x}{1+e^x} dx$

Using $\int_a^b f(x) dx = \frac{1}{2} \int_a^b [f(x) + f(a+b-x)] dx$

$\therefore I = \frac{\pi^2}{4} - 2$

243. The value of $\int_0^{\pi} \frac{5^{\cos x} (1 + \sin x \cdot \sin 3x + \sin^2 x + \sin^3 x \cdot \sin 3x)}{1 + 5^{\cos x}} dx$ **(Kings Property)**

1. $\frac{13\pi}{16}$

2. $\frac{11\pi}{16}$

3. $\frac{9\pi}{16}$

4. $\frac{7\pi}{16}$

ANS : 2

SOL

$$I = \int_0^{\pi} \frac{5^{\cos x} (1 + \sin x \cdot \sin 3x + \sin^2 x + \sin 3x \cdot \sin^2 x)}{1 + 5^{\cos x}} dx$$

$$I = \int_0^{\pi} \frac{5^{-\cos x} (1 + \sin x \cdot \sin 3x + \sin^2 x + \sin 3x \cdot \sin^2 x)}{1 + 5^{-\cos x}} dx$$

Adding above two equations

$$2I = 2 \int_0^{\pi/2} \left(1 + \frac{1}{2} (\cos 2x - \cos 4x) + \sin^2 x + 3 \sin^4 x - 4 \sin^6 x \right) dx$$

$$I = \frac{\pi}{2} + \frac{\pi}{4} + \frac{9\pi}{16} - \frac{5\pi}{8} = \frac{8\pi + 4\pi + 9\pi - 10\pi}{16}$$

$$I = \frac{11\pi}{16}$$

244. If $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1 + \pi^x) \sin x} dx$, $n = 0, 1, 2, 3, \dots$ then (Replacing x by $a + b - x$)

- 1) $I_n = I_{n+1}$ 2) $\sum_{m=1}^{10} I_{2m+1} = 10\pi$ 3) $\sum_{m=1}^{10} I_{2m} = 10\pi$ 4) none

Key: 2

Sol: Using $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$, we have

$$I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1 - \pi^{-x}) \sin x} dx = \int_{-\pi}^{\pi} \frac{\pi^x \sin nx}{(1 + \pi^x) \sin x} dx$$

$$2I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{\sin x} dx = 2 \int_0^{\pi} \frac{\sin nx}{\sin x} dx$$

$$\left(\because f(x) = \frac{\sin nx}{\sin x} \text{ is an even function} \right)$$

$$\Rightarrow I_n = \int_0^{\pi} \frac{\sin nx}{\sin x} dx = \begin{cases} 0 & \text{if } n \text{ is even} \\ \pi & \text{if } n \text{ is odd} \end{cases}$$

245. If $A = \int_0^1 X^{50} (2-x)^{50} dx$; $B = \int_0^1 X^{50} (1-x)^{50} dx$, which of the following is true?

(Replacing X By $A-X$ - King S Property)

- 1) $A = 2^{50} B$ 2) $A = 2^{-50} B$ 3) $A = 2^{100} B$ 4) $A = 2^{-100} B$

Key: 3

Sol: Given $A = \int_0^1 x^{50} (2-x)^{50} dx$; $B = \int_0^1 x^{50} (1-x)^{50} dx$

In A, put $x = 2t \Rightarrow dx = 2dt$

$$\Rightarrow A = 2 \int_0^{1/2} 2^{50} t^{50} \cdot 2^{50} (1-t)^{50} dt \dots \dots \dots (1)$$

$$\text{Now, } B = 2 \int_0^{1/2} x^{50} (1-x)^{50} dx \dots \dots \dots (2)$$

$$\left[u \sin g \int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx \text{ if } f(2a-x) = f(x) \right]$$

From equations (1) and (2) we get

$$A = 2^{100} B$$

246. The value of the integral $\int_0^{\pi} \frac{x^2 \sin x}{(2x-\pi)(1+\cos^2 x)} dx$ (Replacing x by a-x - King's property)

1. $\frac{\pi^2}{4}$

2. $\frac{\pi^2}{2}$

3. 0

4. $\frac{\pi^2}{8}$

Ans: 1

Sol: Let $I = \int_0^{\pi} \frac{x^2 \sin x}{(2x-\pi)(1+\cos^2 x)} dx$

$$I = \int_0^{\pi} \frac{(\pi^2 - 2\pi x) \sin x}{(\pi - 2x)(1+\cos^2 x)} dx - I$$

$$\therefore I = \frac{\pi^2}{4}$$

247. The value of the integral $I = \int_0^1 x(1-x)^n dx$ is (Replacing x by a-x - King's property)

1. $\frac{1}{n+1}$

2. $\frac{1}{n+2}$

3. $\frac{1}{n+1} - \frac{1}{n+2}$

4. $\frac{1}{n+1} + \frac{1}{n+2}$

Ans: 3

Sol: $\int_0^1 x(1-x)^n dx = \int_0^1 x^n(1-x) dx$

$$\int_0^1 (x^n - x^{n+1}) = \frac{1}{n+1} - \frac{1}{n+2}$$

248. $\int_a^b \frac{f(x)}{f(x)+f(a+b-x)} dx =$

1. $\frac{b-a}{2}$

2. $\frac{b-a}{4}$

(Replacing x by a + b - x)

3. $\frac{b-a}{8}$

4. $\frac{b-a}{16}$

Ans: 1

Sol: Let $I = \int_a^b \frac{f(x)}{f(x)+f(a+b-x)} dx$ then

249. If $I_1 = \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx, I_2 = \int_0^{\pi} x \sin^4 x dx$ then $I_1 : I_2 =$ (Replacing x by a-x - King's property)

1. 3:4

2. 1:2

3. 4:3

4. 2:3

ANS: C

Sol: Using $\int_0^{\pi} f(x) dx = \frac{1}{2} \int_0^{\pi} (f(x) + f(a-x)) dx \dots (1)$

$$I_1 = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1+\cos^2 x} dx$$

$$\Rightarrow I_1 = \pi \int_0^{\pi/2} \frac{\sin x}{1 + \cos^2 x} dx (\because f(2a-x) = f(x))$$

$$\therefore I_1 : I_2 = \frac{1}{4} : \frac{3}{16} = 4 : 3$$

250. The value of the integral $\int_{-3\pi/4}^{5\pi/4} \frac{(\sin x + \cos x)}{e^{x-\pi/4} + 1} dx$

A) 0

B) 1

C) 2

D) 3

Key:A

Sol: let $I = \int_{-3\pi/4}^{5\pi/4} \frac{(\sin x + \cos x)}{e^{x-\pi/4} + 1} dx$

$$= \int_{-3\pi/4}^{5\pi/4} \frac{\sqrt{2} \cos\left(x - \frac{\pi}{4}\right)}{e^{x-\pi/4} + 1} dx$$

Putting $x - \frac{\pi}{4} = t$ or $dx = dt$

$$\therefore I = \int_{-\pi}^{\pi} \frac{\sqrt{2} \cos t}{e^t + 1} dt$$

$$2I = \sqrt{2} \int_{-\pi}^{\pi} \cos t dt \text{ or } I = 0$$

251. If $f(x) = \int_0^{\pi} \frac{t \sin t dt}{\sqrt{1 + \tan^2 x \sin^2 t}}$ for $0 < x < \frac{\pi}{2}$, then

(Kings Rule)

A) $f(0^+) = -\pi$

B) $f\left(\frac{\pi}{4}\right) = \frac{\pi^2}{8}$

C) f is continuous and differentiable in $\left(0, \frac{\pi}{2}\right)$

D) f is continuous but not differentiable in $\left(0, \frac{\pi}{2}\right)$

Key:C

Sol: $f(x) = \int_0^{\pi} \frac{t \sin t}{\sqrt{1 + \tan^2 x \sin^2 t}} dt \dots\dots\dots(1)$

Replacing t by $\pi - t$ and then adding $f(x)$ with equation (1) we get

$$f(x) = \frac{\pi}{2} \int_0^{\pi} \frac{\sin t}{\sqrt{1 + \tan^2 x \sin^2 t}} dt$$

$$= \pi \int_0^{\pi/2} \frac{\sin t}{\sqrt{\sec^2 x - \tan^2 x \cos^2 t}} dt$$

Let $y = \cos t$

$\therefore dy = -\sin t dt$

$$\begin{aligned}\therefore f(x) &= \pi \int_0^1 \frac{dy}{\sqrt{\sec^2 x - (\tan^2 x)y^2}} \\ &= \frac{\pi}{\tan x} \sin^{-1}(\sin x) = \frac{\pi x}{\tan x}\end{aligned}$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 3: Even - odd property

252. Let $f(x) = \int_0^x g(t) dt$, where g is a non-zero even function. If $f(x+5) = g(x)$, then

$$\int_0^x f(t) dt \text{ equals:}$$

$$1) \int_{x+5}^5 g(t) dt \quad 2) 2 \int_5^{x+5} g(t) dt \quad 3) \int_5^{x+5} g(t) dt \quad 4) 5 \int_{x+5}^5 g(t) dt$$

Ans: 1

(King's property and even odd Property) 2019

Sol: $g(x) = f'(x)$ & $g(-x) = g(x) = f(x+5) = f(5-x)$ and also $f(x)$ is odd.

$$\text{Put } t+5 = y \Rightarrow \int_0^x f(x) dt = \int_5^{5+x} f(y-5) dy = - \int_5^{x+5} g(y) dy = \int_{x+5}^5 g(t) dt$$

253. If $f(x) = \frac{2-x \cos x}{2+x \cos x}$ and $g(x) = \log_e x$, ($x > 0$) then the value of the integral

$$\int_{-\pi/4}^{\pi/4} g(f(x)) dx \text{ is:}$$

$$1. \log_e 3 \quad 2. \log_e 1 \quad 3. \log_e 2 \quad 4. \log_e e$$

Ans: 2

(Even-Odd Property) 2019

$$\text{Sol; } I = \int_{-\pi/4}^{\pi/4} g(f(x)) dx = \int_{-\pi/4}^{\pi/4} \log \left(\frac{2-x \log x}{2+x \cos x} \right) dx = \int_{-\pi/4}^{\pi/4} \log \left(\frac{2+x \cos x}{2-x \cos x} \right) dx = -I \Rightarrow I = 0$$

254. The value of the integral $\int_{-1}^1 \log_e(\sqrt{1-x} + \sqrt{1+x}) dx$ is equal to :

1) $2 \log_e 2 + \frac{\pi}{2} - \frac{1}{2}$ 2) $2 \log_e 2 + \frac{\pi}{4} - 1$ 3) $\log_e 2 + \frac{\pi}{2} - 1$ 4) $\frac{1}{2} \log_e 2 + \frac{\pi}{4} - \frac{3}{2}$

Key:

3

(Even and Odd Property)2021

Sol: Let $I = \int_{-1}^1 \log(\sqrt{1-x} + \sqrt{1+x}) dx$

Let $f(x) = \log(\sqrt{1-x} + \sqrt{1+x})$

$f(-x) = \log(\sqrt{1+x} + \sqrt{1-x}) = f(x)$ = even function

$\Rightarrow I = 2 \int_0^1 \log(\sqrt{1-x} + \sqrt{1+x}) dx$ put $x = \cos 2\theta$

$= 2 \int_{\frac{\pi}{4}}^0 \log(\sqrt{1-\cos 2\theta} + \sqrt{1+\cos 2\theta}) (-2 \sin 2\theta) d\theta = 4 \int_0^{\pi/4} \log[(\sin \theta + \cos \theta)\sqrt{2}] \sin 2\theta d\theta$

$= 4 \int_0^{\pi/4} \log(\sin \theta + \cos \theta) \sin 2\theta d\theta + 4 \int_0^{\pi/4} \log \sqrt{2} \cdot \sin 2\theta d\theta$

$= 4 \left[-\ln(\sin \theta + \cos \theta) \frac{\cos 2\theta}{2} \right]_0^{\pi/4} - \int_0^{\pi/4} \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta} \cdot \frac{\cos 2\theta}{2} d\theta + \left[4 \ln \sqrt{2} \left(-\frac{\cos 2\theta}{2} \right) \right]_0^{\pi/4}$

$= 4 \left[0 + \frac{1}{2} \int_0^{\pi/4} (\cos \theta - \sin \theta)^2 d\theta \right] + 4 \ln \sqrt{2} (0 + 1/2) (\because \cos 2\theta = \cos^2 \theta - \sin^2 \theta)$

$= 4 \left[0 + \frac{1}{2} \int_0^{\pi/4} (1 - \sin 2\theta) d\theta \right] + 2 \ln \sqrt{2}$

$= +2 \left(\theta + \frac{\cos 2\theta}{2} \right)_0^{\pi/4} + \ln 2 = +2 \left(\frac{\pi}{4} - \frac{1}{2} \right) + \ln 2 = \ln 2 + \frac{\pi}{2} - 1$

255. Let $g(t) = \int_{-\pi/2}^{\pi/2} \cos\left(\frac{\pi}{4}t + f(x)\right) dx$, where $f(x) = \log_e(x + \sqrt{x^2 + 1})$, $x \in R$.

Then which one of the following is correct?

1) $g(1) + g(0) = 0$

2) $g(1) = g(0)$

3) $g(1) = \sqrt{2}g(0)$

4) $\sqrt{2}g(1) = g(0)$

(Even and Odd Property)2021

Key: 4

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Sol: $g(t) = \int_{-\pi/2}^{\pi/2} \cos\left(\frac{\pi}{4}t + f(x)\right) dx$

where $f(x) = \log_e(x + \sqrt{x^2 + 1})$

$f(x) = \log_e(-x + \sqrt{x^2 + 1}) = -\log_e(x + \sqrt{x^2 + 1})$

$g(t) = \int_{-\pi/2}^{\pi/2} \cos\left(\left(\frac{\pi}{4}t + f(x)\right) + \cos\left(\frac{\pi}{4}t - f(x)\right)\right) dx$

$= 2 \int_0^{\pi/2} \cos \frac{\pi}{4}t \cos f(x) dx$

$g(0) = 2 \int_0^{\pi/2} \cos f(x) dx$

$g(1) = \sqrt{2} \int_0^{\pi/2} \cos f(x) dx \quad g(0) = \sqrt{2}g(1)$

256. The value of the integral $\int_{-1}^1 \log(x + \sqrt{x^2 + 1}) dx$ is:

1) 0

2) -1

3) 1

4) 2

Key: 1

(Even and Odd Property)2021

SOL:

$I = \int_{-1}^1 \log(x + \sqrt{x^2 + 1}) dx$

$f(x) = \log(\sqrt{x^2 + 1} + x)$

$f(-x) = \log(\sqrt{x^2 + 1} - x)$

$= -f(x), \quad \text{So } f(x) \text{ is an odd function} \Rightarrow I = 0$

257. The value of $\int_{-1/\sqrt{2}}^{1/\sqrt{2}} \left(\left(\frac{x+1}{x-1} \right)^2 + \left(\frac{x-1}{x+1} \right)^2 - 2 \right)^{1/2} dx$ is:

1) $2\log_e 16$

2) $\log_e 4$

3) $\log_e 16$

4) $4\log_e(3 + 2\sqrt{2})$

Key:3

(Splitting rule and Even function)2021

Sol: $\int_{-1/\sqrt{2}}^{1/\sqrt{2}} \left(\left(\frac{x+1}{x-1} \right)^2 + \left(\frac{x-1}{x+1} \right)^2 - 2 \right)^{1/2} dx$

$$= \left(\left(\frac{x+1}{x-1} - \frac{x-1}{x+1} \right)^2 \right)^{\frac{1}{2}} = \left(\frac{x+1}{x-1} - \frac{x-1}{x+1} \right)$$

$$f(x) = \text{even}$$

$$f(x) = \int_{-1/\sqrt{2}}^{1/\sqrt{2}} \left| \frac{4x}{x^2-1} \right| dx = 2 \int_0^{1/\sqrt{2}} \frac{4x}{1-x^2} dx$$

$$= -4 \left[\log(1-x^2) \right]_0^{1/\sqrt{2}}$$

$$= -4 \log\left(\frac{1}{2}\right)$$

$$= \log_e 16$$

258. The value of the integral $\int_{-2}^2 \frac{|x^3 + x|}{(e^{x|x|} + 1)} dx$ is equal to :

(A) $5e^2$

(B) $3e^{-2}$

(C) 4

(D) 6

Ans: D

(Even And Odd Function) 2022

Sol: $f(x) = \frac{|x^3 + x|}{(e^{x|x|} + 1)}$

$$\int_{-2}^2 f(x) dx = \int_0^2 (f(x) + f(-x)) dx$$

$$= \int_0^2 \left(\frac{|x^3 + x|}{(e^{x|x|} + 1)} + \frac{|-x^3 - x|}{(e^{-x|-x|} + 1)} \right) dx$$

$$= \int_0^2 \left(\frac{|x^3 + x|}{(e^{x|x|} + 1)} + \frac{|x^3 + x|}{(e^{x|x|} + 1)} \right) dx$$

$$= \int_0^2 \left(\frac{x^3 + x}{(e^{x^2} + 1)} + \frac{x^3 + x}{(e^{x^2} + 1)} \right) dx$$

$$I = \int_0^2 \left(\frac{x^3 + x}{1 + e^{x^2}} + \frac{e^{x^2}(x^3 + x)}{1 + e^{x^2}} \right) dx$$

$$= \int_0^2 (x^3 + x) dx$$

$$= \left[\frac{x^4}{4} + \frac{x^2}{2} \right]_0^2$$

$$= 4 + 2 = 6$$

259. The value of the integral $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{x + \frac{\pi}{4}}{2 - \cos 2x} dx$ is :

1) $\frac{\pi^2}{6\sqrt{3}}$

2) $\frac{\pi^2}{12\sqrt{3}}$

3) $\frac{\pi^2}{6}$

4) $\frac{\pi^2}{3\sqrt{3}}$

Key:

1

(Even And Odd)2023

Sol:

$$\text{Sol : } I = \int_{-\pi/4}^{\pi/4} \frac{x + \frac{\pi}{4}}{2 - \cos 2x} dx$$

$$I = \int_{-\pi/4}^{\pi/4} \frac{-x + \frac{\pi}{4}}{2 - \cos 2x} dx \quad (\because \int_{-a}^a f(x) dx = \int_{-a}^a f(-x) dx)$$

$$\Rightarrow 2I = \int_{-\pi/4}^{\pi/4} \frac{2 \times \frac{\pi}{4}}{2 - \cos 2x} dx$$

$$\Rightarrow I = \int_{-\pi/4}^{\pi/4} \frac{\frac{\pi}{4} dx}{2 - \cos 2x} = 2 \times \frac{\pi}{4} \int_0^{\pi/4} \frac{1}{2 - \cos 2x} dx \quad (\because \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ if } f(x) \text{ even})$$

$$\Rightarrow I = \frac{\pi}{2} \int_0^{\pi/4} \frac{1}{2 - \cos 2x} dx$$

$$\Rightarrow I = \frac{\pi}{2} \int_0^{\pi/4} \frac{1}{2 - \frac{(1 - \tan^2 x)}{(1 + \tan^2 x)}} dx$$

put $\tan x = t$

$$\Rightarrow I = \frac{\pi}{2} \int_0^1 \frac{dt}{3t^2 + 1} = \frac{\pi}{2\sqrt{3}} \tan^{-1}(\sqrt{3}t) \Big|_0^1 = \frac{\pi}{2\sqrt{3}} \times \frac{\pi}{3}$$

$$\therefore I = \frac{\pi^2}{6\sqrt{3}}$$

260. If $\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{\pi + 4x^3}{2 - \cos\left(|x| + \frac{\pi}{3}\right)} dx = \frac{4\pi}{\sqrt{k_1}} \tan^{-1} \frac{1}{k_2}$ then $(k_1 + k_2) =$

1) 3

2) 10

3) 5

4) 7

Key: 3

(Splitting rule and Even Odd function)

Sol: Let I

$$= \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{\pi dx}{2 - \cos\left(|x| + \frac{\pi}{3}\right)} + 4 \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{x^3 dx}{2 - \cos\left(|x| + \frac{\pi}{3}\right)}$$

Put $x + \frac{\pi}{3} = t \Rightarrow dx = dt$

$$I = 2\pi \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{dt}{2 - \cos t} = 2\pi \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{\sec^2 \frac{t}{2} dt}{1 + 3 \tan^2 \frac{t}{2}}$$

Put $\tan \frac{t}{2} = u \Rightarrow \sec^2 \frac{t}{2} dt = 2du$

$$\Rightarrow I = 2\pi \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{2du}{1 + 3u^2} = \frac{4\pi}{3} \left[\sqrt{3} \tan^{-1} \sqrt{3}u \right]_{\frac{1}{\sqrt{3}}}^{\sqrt{3}}$$

$$= \frac{4\pi}{\sqrt{3}} \left(\tan^{-1} 3 - \tan^{-1} 1 \right) = \frac{4\pi}{\sqrt{3}} \tan^{-1} \left(\frac{1}{2} \right)$$

$$\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{\pi + 4x^3}{2 - \cos\left(|x| + \frac{\pi}{3}\right)} dx = \frac{4\pi}{\sqrt{3}} \tan^{-1} \left(\frac{1}{2} \right)$$

261. If $f(x) = \frac{x^7 - 3x^5 + 7x^3 - x + 1}{\cos^2 x}$ then, evaluate $\int_{-\pi/4}^{\pi/4} f(x) dx$

1) 3

2) 7

3) 4

4) 2

Key: 4

(Even and Odd Properties)

Sol:

$$\begin{aligned}
 f(x) &= \frac{x^7 - 3x^5 + 7x^3 - x + 1}{\cos^2 x} \\
 &= \left[\frac{x^7 - 3x^5 + 7x^3 - x}{\cos^2 x} \right] + \sec^2 x \\
 &= (\text{odd function}) + (\text{even function}) \\
 \int_{-\pi/4}^{\pi/4} f(x) dx &= \int_{-\pi/4}^{\pi/4} \sec^2 x \, dx + \int_{-\pi/4}^{\pi/4} \left[\frac{x^7 - 3x^5 + 7x^3 - x}{\cos^2 x} \right] dx \\
 &= 2 + 0 = 2
 \end{aligned}$$

262. Suppose f is continuous and satisfies $f(x) + f(-x) = x^2$ then find the value of the integral $\int_{-1}^1 f(x) dx =$

- 1) $\frac{1}{3}$ 2) $\frac{2}{3}$ 3) $\frac{4}{3}$ 4) 3

Key: 1

(Even and Odd Properties)

Sol:

$$\begin{aligned}
 \int_{-1}^1 f(x) dx &= \int_{-1}^1 f(-x) dx \\
 2I &= \int_{-1}^1 f(x) + f(-x) = \int_{-1}^1 x^2 dx \\
 2I &= 2 \int_0^1 x^2 dx \\
 \Rightarrow I &= \int_0^1 x^2 dx = \frac{1}{3}
 \end{aligned}$$

263. $\int_{-\pi}^{\pi} \frac{2x(1 + \sin x)}{1 + \cos^2 x} dx$ is equal to

(even and odd function)

- 1) π 2) π^2 3) 0 4) $\frac{\pi}{2}$

Key: 2

$$\text{Sol: } I = 0 + 2 \int_0^{\pi} \frac{2x \sin x}{1 + \cos^2 x} = 4 \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = 4 \frac{\pi^2}{4} = \pi^2$$

264. The value of $I = \int_{-\frac{1}{2}}^{\frac{1}{2}} \left([x] \ln \left(\frac{1+x}{1-x} \right) \right) dx$ is

(Even and Odd Property)

- A) $-\frac{1}{2}$ B) 0 C) 1 D) $\ln \frac{1}{2}$

ANS: A

SOL:
$$I = \int_{-\frac{1}{2}}^{\frac{1}{2}} \left([x] \ln \left(\frac{1+x}{1-x} \right) \right) dx$$

$$I = \int_{-\frac{1}{2}}^{\frac{1}{2}} [x] dx + \int_{-\frac{1}{2}}^{\frac{1}{2}} \ln \left(\frac{1+x}{1-x} \right) dx$$

Clearly secondary integral is zero since $\ln \left(\frac{1+x}{1-x} \right)$ is an odd function

$$I = \int_{-\frac{1}{2}}^{\frac{1}{2}} [x] dx \quad I = \int_{-\frac{1}{2}}^{\frac{1}{2}} [-x] dx = \int_{-\frac{1}{2}}^{\frac{1}{2}} (-1 - [x]) dx$$

$$I = -1 - I$$

$$I = -\frac{1}{2}$$

265. The value of the integral $\frac{8}{\pi} \int_{-1}^1 \frac{dx}{(x^2+x+1)\sqrt{x^4+3x^2+1}}$ is equal to

A)1

B)1/2

C)2

D)3

ANS: B

(Even and Odd Property)

SOL:
$$I = \int_{-1}^1 \frac{(x^2+x+1) - \sqrt{x^4+3x^2+1}}{(x^2+x+1)^2 - (x^4+3x^2+1)} dx$$

$$= \int_{-1}^1 \frac{x^2+x+1}{2x(x^2+1)} dx - \int_{-1}^1 \frac{\sqrt{x^4+3x^2+1}}{2x(x^2+1)} dx$$

Clearly second one is odd function

$$\therefore \text{the required integrated } I = \int_{-1}^1 \frac{x^2+x+1}{2x(x^2+1)} dx$$

$$= \frac{1}{2} \left[\int_{-1}^1 \frac{1}{x} + \int_{-1}^1 \frac{dx}{1+x^2} \right] = \frac{1}{2} \times 2 \times \frac{\pi}{4} = \frac{\pi}{4} \Rightarrow \frac{8}{\pi} I = 2$$

266. $\left(\sum_{n=1}^{10} \int_{-2n-1}^{-2n} \sin^{27} x dx \right) + \left(\sum_{n=1}^{10} \int_{2n}^{2n+1} \sin^{27} x dx \right) =$

(Even and Odd Property)

A)27²

B)54

C)-54

D)0

ANS: D

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$$\begin{aligned} \text{SOL: } &= \sum_{n=1}^{10} \int_{2n-1}^{2n} \sin^{27}(-x) d(-x) + \sum_{n=1}^{10} \int_{2n}^{2n+1} \sin^{27} x dx \\ &= \sum_{n=1}^{10} \left[\int_{2n+1}^{2n+1} \sin^{27} x dx \right] = 0 \end{aligned}$$

267. $\int_{-1/2}^{1/2} \sqrt{\left\{ \left(\frac{x+1}{x-1} \right)^2 + \left(\frac{x-1}{x+1} \right)^2 - 2 \right\}} dx$ equals, (If f(x) is even and definite integral of f(t) from 0 to x is odd)

- 1) $4 \ln\left(\frac{4}{3}\right)$ 2) $4 \ln\left(\frac{3}{4}\right)$ 3) $-\ln\left(\frac{81}{256}\right)$ 4) $\ln\left(\frac{256}{81}\right)$

Key: 3

$$\begin{aligned} \text{Sol: } I &= \int_{-1/2}^{1/2} \sqrt{\left(\frac{x+1}{x-1} - \frac{x-1}{x+1} \right)^2} dx \\ &= \int_{-1/2}^{1/2} \left| \frac{4x}{x^2-1} \right| dx = -2 \int_0^{1/2} \frac{4x}{(x^2-1)} dx = -\ln\left(\frac{81}{256}\right) \end{aligned}$$

268. $\int_{-3\pi/2}^{-\pi/2} [(x+\pi)^3 + \cos^2(x+3\pi)] dx$ is equal to (Even or Odd function)

1. $\frac{\pi^4}{32}$ 2. $\frac{\pi^4}{32} + \frac{\pi}{2}$ 3. $\frac{\pi}{2}$ 4. $\frac{\pi}{4} - 1$

Ans: 3

Sol: Given integral $I = \int_{-3\pi/2}^{-\pi/2} [(x+\pi)^3 + \cos^2(x+3\pi)] dx$

Put $x+\pi = t$

$$\begin{aligned} I &= \int_{-\pi/2}^{\pi/2} [t^3 + \cos^2(x+3\pi)] dx \\ &= \int_0^{\pi/2} (1 + \cos 2t) dt = \frac{\pi}{2} + 0 \end{aligned}$$

269. The value of the integral $\int_{-1}^1 \frac{x^2 e^{\tan^{-1} x}}{\sqrt{1+x^2}} dx$ is equal to

(-a to a property - odd and even functions - half the integral limit property)

1. $\frac{e^{-\pi/4}}{\sqrt{2}}$ 2. $\sqrt{2}e^{-\pi/4}$ 3. $\sqrt{2}e^{3\pi/4}$ 4. $\sqrt{2}e^{\frac{\pi}{4}}$

ANS: B

Sol Denote $I_1 = \int \frac{e^{\tan^{-1} x}}{\sqrt{1+x^2}} dx$ and $I = \int \frac{x e^{\tan^{-1} x}}{\sqrt{1+x^2}} dx$

We have $I_1 = \int \sqrt{1+x^2} \frac{e^{\tan^{-1}x}}{1+x^2} dx = \sqrt{1+x^2} e^{\tan^{-1}x}$

$$I_1 + I_2 = \sqrt{1+x^2} e^{\tan^{-1}x}$$

On the other hand $I_2 = \int \frac{x e^{\tan^{-1}x}}{\sqrt{1+x^2}} = \int x \sqrt{1+x^2} d(e^{\tan^{-1}x})$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 4: Half the integral limit property (Splitting of limits 0 to 2a as 0 to a and a to 2a in f(x)), f(2a-x) property

270. Find the value of $\int_0^{\pi} |\cos^3 x| dx$ is _____

1) $\frac{1}{3}$

2) $\frac{2}{3}$

3) $\frac{4}{3}$

4) $\frac{3}{4}$

Key 3

(f(2a-x) - property)

Sol:

$$I = \int_0^{\pi} |\cos^3 x| dx \quad f(2a-x) \text{ property}$$

$$I = 2 \int_0^{\frac{\pi}{2}} \cos^3 x = 2I_3 = 2 \times \frac{2}{3} \times 1 = \frac{4}{3}$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 5: Periodic function property

271. Let a be a positive real number such that

$$\int_0^a e^{x-[x]} dx = 10e - 9 \quad \text{where } [x] \text{ is the greatest integer less than or equal to } x \text{ then a is equal to :}$$

1) $10 + \log_e(1+e)$ 2) $10 - \log_e(1+e)$ 3) $10 + \log_e 2$ 4) $10 + \log_e 3$

Key: 3

(Periodic Property)

Sol: 'a' be a +ve real number

$$\int_0^a e^{x-[x]} dx = 10e - 9$$

Since RHS $10e - 9$

Take $a = 10 + K$ $0 \leq k \leq 1$

$$\text{Take } \int_0^{10+k} e^{x-[x]} dx = \int_0^{10} e^{x-[x]} dx + \int_{10}^{10+k} e^{x-[x]} dx$$

Period of $x - [x]$ is 1

$$= 10 \int_0^1 e^{x-[x]} dx + \int_0^k e^{x-[x]} dx = 10 \int_0^1 e^x dx + \int_0^k e^x dx$$

$$= 10 \left(e^x \right)_0^1 + \left(e^x \right)_0^k$$

$$= 10(e - 1) + (e^k - 1)$$

$$= 10e - 10 + e^k - 1$$

$$= 10e + e^k - 11$$

$$= 10e - 9 \text{ (given)}$$

$$\Rightarrow e^k = 2 \Rightarrow k = \log_e 2$$

$$a = 10 + \log_e 2$$

272. Let $f: R \rightarrow R$ be a continuous function satisfying $f(x) + f(x+k) = n$ for all $x \in R$ where

$k > 0$ and n is a positive integer. If $I_1 = \int_0^{4nk} f(x) dx$ and $I_2 = \int_{-k}^{3k} f(x) dx$, then:

A. $I_1 + 2I_2 = 4nk$ B. $I_1 + 2I_2 = 2nk$ C. $I_1 + nI_2 = 4n^2k$ D. $I_1 + nI_2 = 6n^2k$

Key: C

(Periodic Property) 2021

Solution:

Given condition is $f(x) + f(x+k) = n$

$$I_1 = \int_0^{4nk} f(x) dx \quad I_2 = \int_{-k}^{3k} f(x) dx$$

f is a periodic function with period $2k$

$$I_1 = \int_0^{4nk} f(x) dx = 2n \int_0^{2k} f(x) dx$$

$$I_2 = \int_{-k}^{3nk} f(x) dx = 2 \int_0^{2k} f(x) dx$$

$$f(x) + f(x+k) = n$$

take definite integration on both sides

$$\int_0^k f(x) dx + \int_0^k f(x+k) dx = \int_0^k n dx$$

$$\int_0^k f(x) dx + \int_0^{2k} f(x) dx = nk$$

$$\int_0^{2k} f(x) dx = nk$$

$$\frac{I_1}{2n} = nk \quad I_1 = 2n^2k$$

$$\frac{I_2}{2} = nk \quad I_2 = 2nk$$

$$I_1 + nI_2 = 2n^2k + 2n^2k = 4n^2k$$

273. The value of $\sum_{n=1}^{100} \int_{n-1}^n e^{x-[x]} dx$, where $[x]$ is the greatest integer $\leq x$, is :

- 1) $100(1-e)$ 2) $100(1+e)$ 3) $100e$ 4) $100(e-1)$

Key : 4

(Periodic Property)2021

Solution:

$$\sum_{n=1}^{100} \int_{n-1}^n e^{x-(x)} dx = \sum_{n=1}^{100} \int_0^1 e^x dx \quad (\because x-(x) \text{ is periodic with period 1}) = 100(e-1)$$

274. If the integral $\int_0^{10} \frac{[\sin 2\pi x]}{e^{x-[x]}} dx = \alpha e^{-1} + \beta e^{-\frac{1}{2}} + \gamma$, where α, β, γ are integers and $[x]$ denotes the greatest integer less than or equal to x , then the value of $\alpha + \beta + \gamma$ is equal to:

- 1) 25 2) 10 3) 20 4) 0

Key: 1

(Step Function and Periodic Property)2021

Sol: Integrand is periodic with period 1

$$I = 10 \int_0^1 \frac{[\sin 2\pi x]}{e^{[x]}} dx$$

$$= 10 \left[\int_0^{1/2} 0 dx + \int_{1/2}^1 \frac{-1}{e^x} dx \right]$$

$$= 10e^{-1} - 10e^{-1/2}$$

$$\alpha = 10, \beta = -10, \gamma = 0 \Rightarrow \alpha + \beta + \gamma = 0$$

275. Let $f: R \rightarrow R$ be a continuous function satisfying $f(x) + f(x+k) = n$ for all $x \in R$ where

$k > 0$ and n is a positive integer. If $I_1 = \int_0^{4nk} f(x) dx$ and $I_2 = \int_{-k}^{3k} f(x) dx$, then:

A. $I_1 + 2I_2 = 4nk$ B. $I_1 + 2I_2 = 2nk$ C. $I_1 + nI_2 = 4n^2k$ D. $I_1 + nI_2 = 6n^2k$

Key: C (Periodic function property - Application 1 of periodic functions) 2022

Solution:

Given condition is $f(x) + f(x+k) = n$

$$I_1 = \int_0^{4nk} f(x) dx \quad I_2 = \int_{-k}^{3k} f(x) dx$$

f is a periodic function with period $2k$

$$I_1 = \int_0^{4nk} f(x) dx = 2n \int_0^{2k} f(x) dx$$

$$I_2 = \int_{-k}^{3nk} f(x) dx = 2 \int_0^{2k} f(x) dx$$

$$f(x) + f(x+k) = n$$

take definite integration on both sides

$$\int_0^k f(x) dx + \int_0^k f(x+k) dx = \int_0^k n dx$$

$$\int_0^k f(x) dx + \int_0^{2k} f(x) dx = nk$$

$$\int_0^{2k} f(x) dx = nk$$

$$\frac{I_1}{2n} = nk \quad I_1 = 2n^2k$$

$$\frac{I_2}{2} = nk \quad I_2 = 2nk$$

$$I_1 + nI_2 = 2n^2k + 2n^2k = 4n^2k$$

276. $\int_{-1}^2 |x^3 - x| dx$ is equal to

1) $\frac{7}{4}$

2) $\frac{9}{4}$

3) $\frac{11}{4}$

4) $\frac{13}{4}$

Key: 3

(Mod Function and Periodic Property)

Sol:

$$f(x) = x^3 - x = x(x^2 - 1) = x(x-1)(x+1)$$

$$\int_{-1}^2 |x^3 - x| dx = \int_{-1}^0 (x^3 - x) dx + \int_0^1 -(x^3 - x) dx + \int_1^2 (x^3 - x) dx$$

$$= \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_{-1}^0 + \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 + \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_1^2$$

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$$\begin{aligned}
 &= -\left(\frac{1}{4} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + (4-2) - \left(\frac{1}{4} - \frac{1}{2}\right) \\
 &= -\frac{1}{4} + \frac{1}{2} + \frac{1}{2} - \frac{1}{4} + 2 - \frac{1}{4} + \frac{1}{2} = \frac{3}{2} - \frac{3}{4} + 2 = \frac{11}{4}
 \end{aligned}$$

277. $\int_0^2 |2x-1| dx$ is equal to

1) $\frac{5}{2}$

2) $\frac{3}{2}$

3) $\frac{7}{2}$

4) $\frac{1}{2}$

Key: 1

(Mod Function and Periodic Property)

Sol: Using the definition of absolute value we rewrite integrand as follows

$$|2x-1| = \begin{cases} -(2x-1), & x < \frac{1}{2} \\ 2x-1, & x \geq \frac{1}{2} \end{cases}$$

Using this we can rewrite the integral in two parts

$$\begin{aligned}
 \int_0^2 |2x-1| dx &= \int_0^{1/2} -(2x-1) dx + \int_{1/2}^2 (2x-1) dx \\
 &= \left[-x^2 + x\right]_0^{1/2} + \left[x^2 + x\right]_{1/2}^2 \\
 &= \left(-\frac{1}{4} + \frac{1}{2}\right) - 1(0+0) + (4-2) - \left(\frac{1}{4} - \frac{1}{2}\right) = \frac{5}{2}
 \end{aligned}$$

278. $\int_0^{\frac{16\pi}{3}} |\sin x| dx$ is equal to

1) $\frac{7}{2}$

2) $\frac{13}{2}$

3) $\frac{19}{2}$

4) $\frac{21}{2}$

Key: 4

(Periodic Property)

Sol:

$$\begin{aligned}
 \int_0^{16\pi/3} |\sin x| dx &= \int_0^{5\pi} |\sin x| dx + \int_{5\pi}^{5\pi+\pi/3} |\sin x| dx \\
 &= 5 \int_0^{\pi} |\sin x| dx + \int_0^{\pi/3} |\sin x| dx \\
 &= 5 \int_0^{\pi} \sin x dx + \int_0^{\pi/3} \sin x dx \\
 &= 5 \times 2 + \left(-\frac{1}{2} + 1 \right) = \frac{21}{2}
 \end{aligned}$$

279. Let the function f be defined by $f(x) = |x-1| - \frac{1}{2}$, $0 \leq x \leq 2$, $f(x+2) = f(x) \forall x \in \mathbb{R}$.

A) $\int_0^{2016} f(x) dx = 0$

B) $\int_0^1 |f(2x)| dx = \frac{1}{8}$

C) $\int_0^{2016} f(x) dx = 2016$

D) $\int_0^1 |f(2x)| dx = \frac{1}{2}$

ANS: A

(Mod Function and Periodic Property)

SOL: It is clear that $\int_0^{\frac{1}{2}} f(x) dx + \int_{\frac{1}{2}}^0 f(x) dx = 0 \therefore \int_0^2 f(x) dx = 0$ and

f is periodic with the period 2.

A) $\int_0^{2016} f(x) dx = 1008 \int_0^2 f(x) dx = 0$

B) $\int_0^1 |f(2x)| dx = \frac{1}{2} \int_0^2 |f(x)| dx$
 $= \frac{1}{2} \times 4 \times \text{area of } \triangle OAB$
 $= 2 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

280. Let $f(x)$ be a non-negative continuous function defined on $\mathbb{R}$ such that

$$f(x) + f\left(x + \frac{1}{2}\right) = 3, \frac{1}{1008} \int_0^{2016} f(x) dx = \text{(Periodic function and Substitution Methods)}$$

ANS: 3

SOL: Given that $f(x) + f\left(x + \frac{1}{2}\right) = 3$ (1)

$$\Rightarrow f\left(x + \frac{1}{2}\right) + f(x+1) = 3 \left(\text{replace } x \text{ by } x + \frac{1}{2} \right) \text{(2)}$$

$$(1) - (2) \Rightarrow f(x) - f(x+1) = 0 \Rightarrow f(x+1) = f(x)$$

$\therefore f$ is periodic function with the period 1.

$$\begin{aligned}
 I &= \int_0^{2016} f(x) dx = 2016 \int_0^1 f(x) dx \\
 I &= 2016 \left[\int_0^{\frac{1}{2}} f(x) dx + \int_{\frac{1}{2}}^1 f(x) dx \right] \\
 &= 2016 \left[\int_0^{\frac{1}{2}} f(x) dx + \int_0^{\frac{1}{2}} f\left(x + \frac{1}{2}\right) dx \right] \\
 &= 2016 \int_0^{\frac{1}{2}} \left[f(x) + f\left(x + \frac{1}{2}\right) \right] dx = 2016 \int_0^{\frac{1}{2}} 3 dx \\
 &= 2016 \times \frac{3}{2} = 1008 \times 3.
 \end{aligned}$$

281. $f(x)$ is a continuous function such that $f(x+4) = f(x+2) - f(x)$. The $\int_{\lambda}^{\lambda+12} f(x) dx$ is equal to

(Periodic Property)

A) $\int_0^{12} f(x) dx$

B) $\int_2^6 [f(x+6) + f(x)] dx$

C) 1

D) $2 \int_0^6 f(x) dx$

ANS: A

SOL: $f(x+4) = f(x+2) - f(x)$

$$\Rightarrow f(x+6) = f(x+4) - f(x+2)$$

$$\Rightarrow f(x+2) - f(x) - f(x+2) = -f(x)$$

$$f(x+12) = f(x) \Rightarrow f \text{ is periodic with the period 12}$$

$$\int_{\lambda}^{\lambda+12} f(x) dx = \int_0^{12} f(x) dx = \int_0^4 f(x) dx + \int_4^8 f(x) dx + \int_8^{12} f(x) dx$$

$$= \int_0^4 f(x) dx + \int_0^4 f(x+4) dx + \int_0^4 f(x+8) dx$$

$$= \int_0^4 [f(x+2) + f(x+8)] dx$$

$$= \int_2^6 [f(x) + f(x+6)] dx = 0$$

282. $\int_{2\pi}^{2\pi+\frac{\pi}{4}} \sqrt{\cos x + \{[\tan x] + \{\sin x\}\}} dx$ equals (where $[\cdot]$ and $\{\cdot\}$ denotes G.I.F and F.P.F)

A) 0

B) $\frac{\sqrt{2}\pi}{4}$

C) $\frac{\pi}{2}$

D) $\frac{\pi}{4}$

ANS: D

(Periodic Property)

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SOL: $f(x) = [\cos x + \{\tan x\} + \{\sin x\}]$
 $= [\cos x + \{\{\sin x\}\}] \quad (\because \{x+n\} = \{x\}, n \in \mathbb{Z})$
 $= [\cos x + \{\sin x\}]$

The period of $f(x)$ is 2π

$\therefore$ Given integral $= \int_0^{\frac{\pi}{4}} \sqrt{f(x)} dx = \int_0^{\frac{\pi}{4}} \sqrt{[\cos x + \sin x]} dx$

$\int_0^{\frac{\pi}{4}} \sqrt{2} \sin\left(x + \frac{\pi}{4}\right) dx = \frac{\pi}{4}$

$\left(\because \text{For } x \in \left[0, \frac{\pi}{4}\right], \sqrt{2} \sin\left(x + \frac{\pi}{4}\right) \in [1, \sqrt{2}]\right)$

283. If $f(x+1) + f(x+7) = 0, \forall x \in \mathbb{R}$, then possible value of t for which $\int_a^{a+t} f(x) dx$ is independent of a , is

(a) 6

(b) 8

(c) 12

(d) None of these

Key: D

(Periodic Property)

Sol: $f(x+1) + f(x+7) = 0 \Rightarrow f(x) = -f(x+6) = f(x+12)$

Now let $F(a) = \int_a^{a+t} f(x) dx$. Then $F'(a) = f(a+t) - f(a) = 0$ for $t = 12$.

284. The period of $f(x)$ is 1 and $f\left(\frac{1}{2}\right) = \frac{1}{2}$. If $\phi(x) = \int_0^x f(t+n) dt, n \in \mathbb{N}$ then $\phi^1(3/2) =$

1) $\frac{1}{2}$

2) 1

3) $3/2$

4) 2

Key: 3

(Periodic Property)

Sol: period of $f(x) = 1 \Rightarrow f(x) = f(1+x)$

For $n \in \mathbb{N}, f(x+n) = f(x)$

$\therefore \phi^1(x) = f(x+n) = f(x)$

$\Rightarrow \phi^1(3/2) = f\left(\frac{3}{2}\right) = f\left(1 + \frac{1}{2}\right) = f\left(\frac{1}{2}\right)$

285. The value of the definite integral $\int_0^{\pi/2} \sin|2x - \alpha| dx$ where $\alpha \in [0, \pi]$

1) 1

2) $\cos \alpha$ 3) $\frac{1 + \cos \alpha}{2}$ 4) $\frac{1 - \cos \alpha}{2}$

Key: 1

(Periodic Property)

Sol: $\int_0^{\pi/2} \sin|2x - \alpha| dx$

$$= \frac{1}{2} \int_{-\alpha}^{\pi/\alpha} \sin|t| dt = \frac{1}{2} \int_{-\alpha}^0 (-\sin t) dt + \frac{1}{2} \int_0^{\pi-\alpha} \sin t dt = \frac{1}{2} [\cos t]_{-\alpha}^0 - \frac{1}{2} [\cot t]_0^{\pi-\alpha}$$

286. If $f(x)$ is a differentiable function on $\mathbb{R}$ such that

$$f(10+x) = f(10-x), f(8+x) = f(8-x) \text{ and } f(10) = 1 \text{ and}$$

$$\int_k^{k+4} f(x) dx = 10 (k > 0) \text{ then } \int_{10}^{14} x f'(x) dx =$$

(Periodic functions)

1) -10

2) 10

3) 6

4) -6

Key: 4

Sol: $f(10-x) = f(10+x), f(8+x) = f(8-x)$

$$f(x+4) = f(x), \forall x \in \mathbb{R}$$

$\therefore f(x)$ is periodic function with period = 4

$$\int_{10}^{14} x f'(x) dx = 14 f(14) - 10 f(10) - \int_{10}^{14} f(x) dx$$

$$= 14(1) - 10(1) - 10 = -6$$

287. The value of $\int_0^{\pi/4} \ln\left(\frac{1-\tan x}{1+\tan^2 x}\right) dx$ is (Some Important Expansions and Results to**Remember)**

1) $\frac{-\pi}{4} \ln 2$

2) $\frac{3\pi}{8} \ln 2$

3) $\frac{\pi}{8} \ln 2$

4) $-\frac{3\pi}{8} \ln 2$

Key: 4

Sol: $I = \int_0^{\pi/4} \ln\left(\frac{1-\tan^2 x}{1+\tan^2 x} \times \frac{1}{1+\tan x}\right) dx$

$$= \int_0^{\pi/4} \ln \cos 2x - \int_0^{\pi/4} \ln(1+\tan x) dx$$

$$I = I_1 - I_2 = -\frac{3\pi}{8} \ln 2$$

Topic: Evaluation of definite integrals**Sub Topic 2:** Properties of evaluation of definite integrals**Mini Topic 6:** Applications of inequality, Sandwich property in comparison of functions in definite, Boundedness and continuity property of definite integral,

Max-min inequality property of definite integral, Root mean square value - RMSV of a definite integral

288. Let $I = \int_a^b (x^4 - 2x^2) dx$. If I is minimum then the ordered pair (a, b) is

1) $(0, \sqrt{2})$

2) $(-\sqrt{2}, 0)$

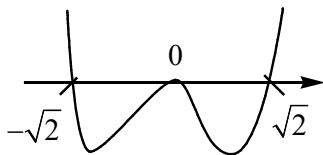
3) $(\sqrt{2}, -\sqrt{2})$

4) $(-\sqrt{2}, \sqrt{2})$

Key: 4

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Sol: Let $f(x) = x^4 - 2x^2$



$\therefore$ If $\int_a^b f(x)dx$ is minimum, then $(a, b) = (-\sqrt{2}, \sqrt{2})$

$(\because f(x) < 0 \text{ for } (-\sqrt{2}, \sqrt{2}))$

289. If $I = \int_1^2 \frac{dx}{\sqrt{2x^3 - 9x^2 + 12x + 4}}$, then:

- 1) $\frac{1}{8} < I^2 < \frac{1}{4}$ 2) $\frac{1}{16} < I^2 < \frac{1}{9}$ 3) $\frac{1}{6} < I^2 < \frac{1}{2}$ 4) $\frac{1}{9} < I^2 < \frac{1}{8}$

Ans: 4

(Max-Min Inequality Property of Definite Integral) 2020

Sol: Let $f(x) = \frac{1}{\sqrt{2x^3 - 9x^2 + 12x + 4}}$

$$f'(x) = \frac{-1}{2} \frac{(6x^2 - 18x + 12)}{(2x^3 - 9x^2 + 12x + 4)^{\frac{3}{2}}}$$

$$= \frac{-6(x-1)(x-2)}{2(2x^3 - 9x^2 + 12x + 4)^{\frac{3}{2}}}$$

$$f(1) = \frac{1}{3}, \quad f(2) = \frac{1}{\sqrt{8}}$$

$$\frac{1}{3} < I < \frac{1}{\sqrt{8}}$$

290. Let $f: (a, b) \rightarrow \mathbb{R}$ be twice differentiable function such that $f(x) = \int_a^x g(t)dt$ for a

differentiable function $g(x)$. If $f(x) = 0$ has exactly five distinct roots in (a, b) , then $g(x)g'(x) = 0$ has at least:

- 1) three roots in (a, b) 2) five roots in (a, b)
3) twelve roots in (a, b) 4) seven roots in (a, b)

Key: 4

(Location of roots in an interval using definite integral) 2021

Sol: $f(x) = \int_0^x g(t)dt$

$g(x) = f'(x)$

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$$g'(x) = f''(x)$$

$$g(x) \cdot g'(x) = f(x) f''(x)$$

Graph of $f(x)$ can be drawn as



If $f(x) = 0$ has at least 5 real roots

$f''(x) = 0$ has at least 4 real roots

$f''(x) = 0$ has at least 3 real roots

$f'(x)f''(x) = 0$ has at least 7 real roots

$g(x)g'(x) = 0$ has at least 7 real root.

291. Let $f: R \rightarrow R$ be defined as $f(x) = e^{-x} \sin x$. If $F: [0, 1] \rightarrow R$ is a differentiable function such that $F(x) = \int_0^x f(t) dt$, then the value of $\int_0^x (F'(x) + f(x)) e^x dx$ lies in the interval

1) $\left[\frac{330}{360}, \frac{331}{360}\right]$

2) $\left[\frac{331}{360}, \frac{334}{360}\right]$

3) $\left[\frac{335}{360}, \frac{336}{360}\right]$

4) $\left[\frac{337}{360}, \frac{339}{360}\right]$

Key: 16

(By parts and inequality)2021

Sol: $F(x) = \int_0^x f(t) dt \Rightarrow F'(x) = f(x)$

$$I = \int_0^1 2f(x) e^x dx = 2 \int_0^1 \sin x dx = 2(-1 \cos 1) = 0.9194 \in \left(\frac{360}{360}, \frac{331}{360}\right)$$

292. The minimum value of the function $f(x) = \int_0^2 e^{|x-t|} dt$ is

1) 2

2) $2(e-1)$

3) $2e-1$

4) $e(e-1)$

KEY:2

(Spilting Of Limits)2023

SOL: For $x > 2$

$$f(x) = \int_0^2 e^{x-t} dt \Rightarrow e^x (-e^{-t}) \Big|_0^2 \Rightarrow e^x (1 - e^{-2})$$

For $x < 0$ $f(x) = \int_0^2 e^{x-t} dt \Rightarrow e^{-x} e^t \Big|_0^2 \Rightarrow e^{-x} (e^2 - 1)$

For $0 \leq x \leq 2$ $f(x) = \int_0^x e^{x-t} dt + \int_x^2 e^{t-x} dt$

$$= -e^x e^{-t} \Big|_0^x + e^{-x} e^t \Big|_x^2 = -e^x (e^{-x} - 1) + e^{-x} (e^2 - e^x)$$

$$= -1 + e^x + e^{2-x} - 1 = e^{2-x} + e^x - 2$$

$$f(x) = \begin{cases} e^x(1-e^{-2})x > 2 \\ e^{2-x} + e^x - 2 & 0 \leq x \leq 2 \\ e^{-x}(e^x - 1)x < 0 \end{cases} \quad \text{For } x > 2, \quad f(x)_{\min} = e^2 - 1$$

$$\begin{aligned} \text{For } 0 \leq x \leq 2 \quad f'(x) &= -e^{2-x} + e^x = 0 \Rightarrow e^x = e^{2-x} \Rightarrow e^{2x} = e^2 \\ \Rightarrow x &= 1. \quad f(x)_{\min} = 2e - 2 = 2(e - 1) \end{aligned}$$

293. Let $[x]$ denote the greatest integer $\leq x$. Consider the function $f(x) = \max\{x^2, 1 + [x]\}$. Then the value of the integral $\int_0^2 f(x) dx$ is

1) $\frac{4+5\sqrt{2}}{3}$

2) $\frac{1+5\sqrt{2}}{3}$

3) $\frac{8+4\sqrt{2}}{3}$

4) $\frac{5+4\sqrt{2}}{3}$

Ans: D

(Maxima And Minima)2023

Sol: $\int_0^2 f(x) dx$

$$= \int_0^1 (1 + [x]) dx + \int_1^{\sqrt{2}} (1 + [x]) dx + \int_{\sqrt{2}}^2 x^2 dx$$

$$= 1 + 2(\sqrt{2} - 1) + \left(\frac{x^3}{3}\right)_{\sqrt{2}}^2$$

$$= 1 + 2\sqrt{2} - 2 + \frac{1}{3}(8 - 2\sqrt{2})$$

$$= 1 + 2\sqrt{2} - 2 + \frac{8 - 2\sqrt{2}}{3}$$

$$= 2\sqrt{2} - 1 + \frac{8 - 2\sqrt{2}}{3}$$

$$= \frac{6\sqrt{2} - 3 + 8 - 2\sqrt{2}}{3}$$

$$= \frac{4\sqrt{2} + 5}{3}$$

294. If θ_1 and θ_2 are min and max arguments of the complex number z satisfying

$$|z - a - ai| \leq a\sqrt{2}, a > 0 \text{ then } \int_{\theta_1}^{\theta_2} \frac{\sin^{\frac{3}{2}} x dx}{\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x} \text{ is } \underline{\hspace{2cm}}$$

1) $\frac{\pi}{2}$

2) $\frac{\pi}{3}$

3) $\frac{\pi}{4}$

4) $\frac{\pi}{6}$

Key: 3

(Max-Min Inequality Property of Definite Integral)Sol: Locus of z is a circle transfer axis is Q_1

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$$\text{Min arg} = \theta_1 = 0$$

$$\text{Max arg} = \theta_2 = \frac{\pi}{2}$$

$$\int_0^{\pi/2} \frac{\sin^{\frac{3}{2}} x dx}{\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x} = \frac{\pi}{4}$$

295. Let $I = \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}}$

P) $I < 1$

Q) $I < \frac{\pi}{4}$

R) $I > \frac{1}{2}$

1) P, Q are False

2) Q, R are False

3) P, R are False

4) P, Q, R are True

Key: 4

(Max-Min Inequality Property of Definite Integral)

Sol:

$$0 < x < 1 \Rightarrow 4 - x^2 > 4 - x^2 - x^3 > 4 - 2x^2$$

$$\Rightarrow \frac{1}{\sqrt{4-x^2}} < \frac{1}{\sqrt{4-x^2-x^3}} < \frac{1}{\sqrt{4-2x^2}}$$

$$\sin^{-1}\left(\frac{x}{2}\right) \Big|_0^1 < I < \frac{1}{\sqrt{2}} \sin^{-1} \frac{x}{2} \Big|_0^1$$

$$\frac{1}{2} < \frac{\pi}{6} < I < \frac{1}{\sqrt{2}} \frac{\pi}{4} < \frac{\pi}{4}$$

296. The range of the function $f(x) = \int_0^x |t-1| dt$, where $0 \leq x \leq 2$.

1) $[-2, -1]$

2) $[-1, 0]$

3) $[0, 1]$

4) $(-7, 0)$

Key: 3

(Max-Min Inequality Property of Definite Integral)

Sol:

$$\begin{aligned}
 f(x) &= \int_0^x |t-1| dt \\
 &= \begin{cases} \int_0^x (1-t) dt, 0 \leq x \leq 1 \\ \int_0^1 (1-t) dt + \int_1^x (t-1) dt, 1 < x \leq 2 \end{cases} \\
 &= \begin{cases} x - \frac{x^2}{2}, 0 \leq x \leq 1 \\ \frac{x^2}{2} - x + 1, 1 < x \leq 2 \end{cases}
 \end{aligned}$$

$\Rightarrow$ The range of the function $f(x)$ is $[0, 1]$

297. The range of the function $f(x) = \int_{-1}^1 \frac{\sin x dt}{(1 - 2t \cos x + t^2)}$

- 1) $\left\{0, \frac{\pi}{4}\right\}$ 2) $\left\{-\frac{\pi}{3}, \frac{\pi}{4}\right\}$ 3) $\left\{-\frac{\pi}{2}, \frac{\pi}{2}\right\}$ 4) $\left\{-\frac{\pi}{4}, \frac{\pi}{4}\right\}$

Key: 3

(Max-Min Inequality Property of Definite Integral)

Sol: We have $f(x) = \int_{-1}^1 \frac{\sin x dt}{(1 - 2t \cos x + t^2)}$

$$\begin{aligned}
 &= \frac{\sin x}{\sin x} \tan^{-1} \left(\frac{t - \cos x}{\sin x} \right) \Big|_{-1}^1 \\
 &= \tan^{-1} \left(\frac{1 - \cos x}{\sin x} \right) - \tan^{-1} \left(\frac{-1 - \cos x}{\sin x} \right) \\
 &= \tan^{-1} (\tan x / 2) + \tan^{-1} (\cot x / 2)
 \end{aligned}$$

Case-I: When $0 < x < \pi$

$$0 < \frac{x}{2} < \frac{\pi}{2} \text{ and } 0 < \frac{\pi}{2} - \frac{x}{2} < \frac{\pi}{2}$$

$$f(x) = \tan^{-1} (\tan x / 2) + \tan^{-1} \left(\tan \left(\frac{\pi}{2} - \frac{x}{2} \right) \right)$$

$$= x / 2 + \pi / 2 - x / 2$$

$$= \pi / 2$$

Case-II: When $\pi < x < 2\pi$

$$\frac{\pi}{2} < \frac{x}{2} < \pi \quad \text{and} \quad -\frac{\pi}{2} < \frac{\pi}{2} - \frac{x}{2} < 0$$

$$f(x) = x/2 - \pi + \pi/2 - x/2 = -\pi/2$$

Hence, the range of $f(x)$ is $\left\{-\frac{\pi}{2}, \frac{\pi}{2}\right\}$

298. If $f(x) = \int_{-1}^x |t| dt$, then for any $x \geq 0$, $f(x)$ equals

- 1) $\frac{1}{2}(1-x^2)$ 2) $\frac{1}{2}x^2$ 3) $\frac{1}{2}(1+x^2)$ 4) None of these

Key: 3 (Max-Min Inequality Property of Definite Integral)

Sol:

$$f(x) = \begin{cases} \int_{-1}^x -tdt, & -1 \leq x \leq 0 \\ \int_{-1}^0 -tdt + \int_0^x tdt, & x \geq 0 \end{cases}$$

$$= \begin{cases} \frac{1}{2}(1-x^2), & -1 \leq x \leq 0 \\ \frac{1}{2}(1+x^2), & x \geq 0 \end{cases}$$

299. The value of $\int_{-1}^1 [x[1+\sin \pi x]+1] dx$ is ([.] denote the greatest integer function)

Ans: 2

(Step Function and Splitting rule)

Sol: Let $I = \int_{-1}^1 [x[1+\sin \pi x]+1] dx = \int_{-1}^0 [x[1+\sin \pi x]+1] dx + \int_0^1 [x[1+\sin \pi x]+1] dx$

Now, $-1 < x < 0 \Rightarrow [1+\sin \pi x] = 0$

and $0 < x < 1 \Rightarrow [1+\sin \pi x] = 1$

$$\therefore I = \int_{-1}^0 1 \cdot dx + \int_0^1 [x+1] dx$$

$$= \int_{-1}^0 1 \cdot dx + \int_0^1 [x] dx + \int_0^1 1 \cdot dx$$

$$= [0 - (-1)] + 0 + (1 - 0) = 2$$

300. The value of the definite integral $\int_0^{2n\pi} \max.(\sin x, \sin^{-1}(\sin x)) dx$ equals

(Where $n \in \mathbb{N}$)

- A) $\frac{n(\pi^2 - 4)}{2}$ B) $\frac{n(\pi^2 - 4)}{4}$ C) $\frac{n(\pi^2 - 8)}{4}$ D) $\frac{n(\pi^2 - 2)}{4}$

ANS: C

(Max-Min Inequality Property of Definite Integral)

SOL: $\int_0^{2n\pi} \max.(\sin x, \sin^{-1}(\sin x)) dx$

$$= n \left[\int_0^{\pi/2} x dx + \int_{\pi/2}^{\pi} (\pi - x) dx + \int_{\pi}^{2\pi} (\sin x) dx \right]$$

$$= n \left[\frac{\pi^2}{8} + \frac{\pi^2}{2} - \frac{1}{2} \left(\pi^2 - \frac{\pi^2}{4} \right) - 2 \right]$$

301. If $f(x) + f(x+2) + f(x+4) + f(x+6) = 10$ and $I = \int_0^T f(x) dx$ and $I_1 = \int_{10}^{12+6T} f(4x) dx$ where

T is period of f then which of the following is/are correct

- A) The period of f(x) is 8 B) The period of f(x) is 4
C) $\sqrt{\frac{I_1}{I}} = 10$ D) $\sqrt{\frac{I_1}{I}} = 25$

ANS: A

(Max-Min Inequality Property of Definite Integral)

SOL: $f(x+2) + f(x+4) + f(x+6) + f(x+8) = 10$

$$\Rightarrow f(x) = f(x+8) \Rightarrow T = 8$$

$$I_1 = \int_{10}^{60} f(4x) dx = \int_{10}^{10+50} f(4x) dx = \int_0^{50} f(4x) dx$$

$$25 \int_0^2 f(4x) dx = 25I \Rightarrow \frac{I_1}{I} = 25 \Rightarrow \sqrt{\frac{I_1}{I}} = 5$$

302. Let f(x) be a positive differentiable function on $[0, a]$ such that

$f(0) = 1$ and $f(a) = 3^{1/4}$. If $f'(x) \geq (f(x))^3 + (f(x))^{-1}$, then, maximum value of a is

- a) $\frac{\pi}{12}$ b) $\frac{\pi}{24}$ c) $\frac{\pi}{36}$ d) $\frac{\pi}{48}$

Key: B (Max-min inequality property of definite integral)

Sol: $f'(x) f(x) \geq (f(x))^4 + 1$

$$\Rightarrow \frac{2f^1(x)f(x)}{\{(f(x))^2\}^2 + 1} \geq 2$$

$$\Rightarrow \int_0^a \frac{2f^1(x)f(x)}{\{(f(x))^2\}^2 + 1} \geq 2 \int_0^a 1 dx$$

$$\Rightarrow \left| \tan^{-1}(f(x)) \right|_0^a \geq 2a \Rightarrow \frac{\pi}{3} - \frac{\pi}{4} \geq 2a$$

303. $\int_0^1 \sqrt{(1+x)(1+x^3)} dx$ can not exceed

1) $\sqrt{\frac{15}{8}}$

2) $\sqrt{\frac{13}{8}}$

3) $\sqrt{\frac{11}{8}}$

4) $\sqrt{\frac{9}{8}}$

Key: 1

(RMSV of a definite integral)

Sol: $I \leq \sqrt{\int_0^1 (1+x) dx \int_0^1 (1+x^3) dx}$

304. Let $f(x) = x - x^2 + 1$ and $g(x) = \max\{f(t) : 0 \leq t \leq x\}$.

(Increasing and Decreasing functions)

$$S_1: \int_0^1 g(x) = \frac{29}{24}$$

$S_2: f(x)$ is increasing in $\left(0, \frac{1}{2}\right)$ and decreasing in $\left(\frac{1}{2}, 1\right)$

- 1) Both S_1 and S_2 are true
- 2) Both S_1 and S_2 are false
- 3) S_1 is true and S_2 is false
- 4) Only at $a = 0$

Key: 1

Sol: $\because f(x) = x - x^2 + 1 \quad \therefore f'(x) = 1 - 2x$

$$f'(x) > 0 \Rightarrow 1 - 2x > 0$$

$$\Rightarrow x < \frac{1}{2} \text{ and } f'(x) < 0$$

$$\Rightarrow 1 - 2x < 0 \Rightarrow x > \frac{1}{2}$$

$\Rightarrow f(x)$ is increasing in $\left(0, \frac{1}{2}\right)$ and decreasing in $\left(\frac{1}{2}, 1\right)$

Now, $g(x) = \max\{f(t); 0 \leq t \leq x\}$

$$= \begin{cases} x - x^2 + 1, 0 \leq x \leq \frac{1}{2} \\ \frac{1}{2} - \frac{1}{4} + 1, \frac{1}{2} \leq x \leq 1 \end{cases}$$

$$\begin{aligned}
 &= \begin{cases} x - x^2 + 1, & 0 \leq x \leq \frac{1}{2} \\ \frac{5}{4}, & \frac{1}{2} \leq x \leq 1 \end{cases} \\
 &\int_0^1 g(x) dx = \int_0^{1/2} (x - x^2 + 1) dx + \int_{1/2}^1 \frac{5}{4} dx \\
 &= \left[\frac{x^2}{2} - \frac{x^3}{3} + x \right]_0^{1/2} + \frac{5}{4} [x]_{1/2}^1 \\
 &= \left(\frac{1}{8} - \frac{1}{24} + \frac{1}{2} \right) + \frac{5}{4} \left(1 - \frac{1}{2} \right) \\
 &= \frac{1}{8} - \frac{1}{24} + \frac{1}{2} + \frac{5}{8} = \frac{3-1+12+15}{24} = \frac{29}{24}
 \end{aligned}$$

305. Let n be a positive integer then $\int_0^{n+1} \min\{|x-1|, |x-2|, |x-3|, \dots, |x-n|\} dx$ equals

(MAXIMA AND MINIMA)

1) $\frac{(n+1)}{4}$

2) $\frac{(n+2)}{4}$

3) $\frac{(n+3)}{4}$

4) $\frac{(n+4)}{4}$

Key: 3

Sol: $\int_0^{n+1} \min\{|x-1|, |x-2|, |x-3|, \dots, |x-n|\} dx = \frac{1}{2}(1) + \frac{1}{2} \times \frac{1}{2} \times (n-1) + \frac{1}{2} \times (1) = \frac{n+3}{4}$

306. $\int_{-2}^2 \min\{|x|, [x]\} dx$. (where $[.]$ denotes the greatest integer function) is equal to

(MAXIMA AND MINIMA)

1) -2

2) -1

3) 0

4) 1

Key: 1

Sol: $\int_{-2}^2 \min\{|x|, [x]\} dx$

$$\begin{aligned}
 &= \int_{-2}^{-1} [x] dx + \int_{-1}^0 [x] dx + \int_0^1 [x] dx + \int_1^2 [x] dx \\
 &= -(2 \times 1) - (1 \times 1) + 0 \times 1 + 1 \times 1 \\
 &= -2
 \end{aligned}$$

307. If $I = \int_a^b (x^4 - 5x^2 + 4) dx$ is minimum then the no. of ordered pairs (a, b) is _____

1) 1

2) 2

3) 3

4) 4

Key: 2

Sol: $I_{\min} = \int_1^2 f(x) dx \text{ or } \int_{-2}^{-1} f(x) dx$

308. Let $I = \int_a^b (\sin x - \cos x) dx$ and $a, b \in [-\pi, \pi]$. If I is minimum then the value of $|a + b|$ is ____

- 1) π 2) $\frac{\pi}{2}$ 3) 2π 4) $\frac{3\pi}{2}$

Key: 2

Sol: $(a, b) = \left(-\frac{3\pi}{4}, \frac{\pi}{4}\right) \Rightarrow |a + b| = \frac{\pi}{2}$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 7: Leibnitz's rule for differentiation under the integral sign

309. If $\int_0^x f(t) dt = x^2 + \int_x^1 t^2 f(t) dt$, then $f'\left(\frac{1}{2}\right)$ is

- 1) $\frac{24}{25}$ 2) $\frac{18}{25}$ 3) $\frac{4}{5}$ 4) $\frac{6}{25}$

Key: 1

(Leibnitz's rule) 2019

Sol: $f(x) = 2x - x^2 f'(x)$ $f'(x) = 2 - 2xf'(x) - x^2 f''(x)$ $\Rightarrow f'\left(\frac{1}{2}\right) = \frac{24}{25}$

310. If $f: R \rightarrow R$ is a differentiable function and $f(2) = 6$, then $\lim_{x \rightarrow 2} \int_6^{f(x)} \frac{2t dt}{(x-2)}$ is

1. 0 2. $24f'(2)$ 3. $12f'(2)$ 4. $2f'(2)$

Ans: 3

(Leibnitz's rule) 2019

Sol: $\lim_{x \rightarrow 2} \frac{\int_6^{f(x)} 2 dt}{x-2} = \lim_{x \rightarrow 2} \frac{2 \cdot f(x) - 12}{x-2} = \lim_{x \rightarrow 2} \frac{2 \cdot f(x) - 12}{x-2} = 2 \times 6 \times f'(2) = 12f'(2)$

311. Let $f: R \rightarrow R$ be a continuously differentiable function such that $f(2) = 6$ and $f'(2) = \frac{1}{48}$.

If $\int_6^{f(x)} 4t^3 dt = (x-2)g(x)$, then $\lim_{x \rightarrow 2} g(x)$ is equal to:

1. 24 2. 18 3. 36 4. 12

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Ans: 2

(Leibnitz's rule) 2019

$$\text{Sol: } \int_6^{f(x)} 4t^2 d(x-2)g(x) \Rightarrow g(x) = \frac{\int_6^{f(x)} 4t^2 dt}{x-2}$$

$$\lim_{x \rightarrow 2} g(x) = \lim_{x \rightarrow 2} \frac{4(f(x))^3 \cdot f'(x)}{1} = 4 \cdot 6^3 \cdot \frac{1}{48} = 18$$

$$312. \lim_{x \rightarrow 1} \left(\frac{\int_0^{(x-1)^2} t \cos(t^2) dt}{(x-1) \sin(x-1)} \right)$$

1) is equal to $\frac{1}{2}$ 2) is equal to $-\frac{1}{2}$

3) does not exist

4) is equal to 1

Key: BONUS

(Leibnitz's rule) 2020

$$\text{Sol: } \lim_{x \rightarrow 1} \frac{\int_0^{(x-1)^2} t \cdot \cos(t)^2 dt}{(x-1) \sin(x-1)} \quad \lim_{x \rightarrow 1} \frac{\int_0^{(x-1)^2} t \cdot \cos(t)^2 dt}{(x-1)^2}$$

$$\lim_{x \rightarrow 1} \frac{(x-1)^2 \cos(x-1)^4 \cdot 2(x-1)}{2(x-1)} = 0$$

313. Let f be a real valued continuous function on $[0,1]$ and $f(x) = x + \int_0^1 (x-t)f(t) dt$.

Then, which of the following points (x, y) lies on the curve $y = f(x)$?

1) (2,4)

2) (1,2)

3) (4,17)

4) (6,8)

KEY:4

(Leibnitz's rule.) 2021

$$\text{SOL: } f(x) = x + \int_0^1 (x-t)f(t) dt, x \in [0,1]$$

$$f(x) = x + x \int_0^1 f(t) dt - \int_0^1 t f(t) dt$$

$$f(x) = x + ax - b \text{ where } a = \int_0^1 f(t) dt \text{ and } b = \int_0^1 t f(t) dt$$

$$a = \int_0^1 (t + at - b) dt \Rightarrow a + 2b = 1 \quad (1)$$

$$\text{and } b = \int_0^1 t(t + at - b) dt \Rightarrow 2a - 9b = -2 \quad (2)$$

$$\text{Solving 1 and 2, we get } a = 5/13 \text{ and } b = 4/13 \therefore f(x) = \frac{18x}{13} - \frac{4}{13}$$

314. Let $f(x)$ be a differentiable function defined on $[0, 2]$ such that $f'(x) = f'(2-x)$ for all

$x \in (0, 2), f(0) = 1$ and $f(2) = e^2$. Then the value of $\int_0^2 f(x) dx$ is

1) $1 - e^2$

2) $1 + e^2$

3) $2(1 - e^2)$

4) $2(1 + e^2)$

Key: 2

(Leibnitz's rule) 2020

Solution:

$$f'(x) = f'(2-x)$$

$$\forall x \in (0, 2)$$

$$f(0) = 1, f(2) = e^2$$

$$\int_0^2 f(x) dx = I$$

$$I = x f(x) \Big|_0^2 - \int_0^2 f'(x) - x$$

$$I = 2e^2 - \int_0^2 (2-x) f'(2-x)$$

$$= 2e^2 - \int_0^2 (2-x) f'(x) = 2e^2 - 2(e^2 - 1) + \int_0^2 x f'(x)$$

$$2I = 2e^2 + 2 \Rightarrow I = 1 + e^2$$

315. Let $f(x) = \int_0^x e^t f(t) dt + e^x$ be a differentiable function for all $x \in \mathbb{R}$. Then $f(x)$ equals:

1) $e^{(e^x-1)}$

2) $2e^{(e^x-1)} - 1$

3) $e^{e^x} - 1$

4) $2e^{e^x} - 1$

Key:2

(Leibnitz's rule) 2021

Solution:

$$f(x) = \int_0^x e^t f(t) dx + e^x$$

$$f'(x) = e^x f(x) + e^x$$

$$= e^x (f(x) + 1) \Rightarrow \frac{f'(x)}{f(x) + 1} = e^x \Rightarrow \int \frac{f'(x)}{f(x) + 1} dx = \int e^x dx$$

$$\Rightarrow \ln(1 + f(x)) = e^x + c \text{ but } f(0) = 1 \therefore \ln 2 = 1 + c$$

$$\ln(1 + f(x)) = e^x + \ln 2 - 1 = \ln(2 \cdot e^{e^x - 1})$$

$$1 + f(x) = 2 \cdot e^{(e^x - 1)} \quad f(x) = 2e^{(e^x - 1)} - 1$$

316. If $x\phi(x) = \int_5^x (3t^2 - 2\phi'(t)) dt, x > -2$, and $\phi(0) = 4$, then $\phi(2)$ is _____

Key:4

(Leibnitz's rule)2021

Sol: $x\phi(x) = \int_5^x (3t^2 - 2\phi'(t)) dt, x > -2$ and $\theta(0) = 4$

Diff w.r.t x

$$\Rightarrow x\phi'(x) + \phi(x) = 3x^2 - 2\phi'(x)$$

$$\Rightarrow (x+2)\phi'(x) + \phi(x) = 3x^2$$

$$\Rightarrow (x+2) \frac{dy}{dx} + y = 3x^2$$

$$\Rightarrow \frac{dy}{dx} + \frac{1}{x+2} y = \frac{3x^2}{x+2}$$

$$\text{I.F } e^{\int \frac{1}{x+2} dx} = e^{\log(x+2)} = x+2$$

Solutions,

$$y(x+2) = \int (x+2) \cdot \frac{3x^2}{(x+2)} dx + c$$

$$\Rightarrow y(x+2) = 3 \cdot \frac{x^3}{3} + C$$

$$\phi(0) = 4 \Rightarrow 4(0+2) = O + C \Rightarrow C = 8$$

$$y(x+2) = x^3 + 8 \Rightarrow y = \frac{x^3 + 8}{x+2}$$

$$\phi(x) = \frac{x^3 + 8}{x+2}$$

$$\phi(2) = \frac{8+8}{2+2} = \frac{16}{4} = 4$$

317. Let $f: R \rightarrow R$ be a continuous function. Then $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\frac{\pi}{4} \int_{\frac{\pi}{4}}^{\sec^2 x} f(x) dx}{x^2 - \frac{\pi^2}{16}}$ is equal to

1) $4f(2)$ 2) $2f(\sqrt{2})$ 3) $2f(2)$ 4) $f(2)$

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Key: 3**(Leibnitz's rule)2021**

$$\text{Sol: } \lim_{x \rightarrow \frac{\pi}{4}} \frac{\frac{\pi}{4} f(\sec^2 x) 2 \sec^2 x \tan x}{2x} = \frac{\frac{\pi}{4} f(2) \cdot 2(2)(1)}{2 \times \frac{\pi}{4}} = 2f(2)$$

318. Let f be a real valued continuous function on $[0,1]$ and $f(x) = x + \int_0^1 (x-t)f(t)dt$.

Then, which of the following points (x,y) lies on the curve $y = f(x)$?

- 1) (2,4) 2) (1,2) 3) (4,17) 4) (6,8)

KEY:4**(LEIBNITZ'S RULE)2022**

$$\text{SOL: } f(x) = x + \int_0^1 (x-t)f(t)dt, x \in [0,1]$$

$$f(x) = x + x \int_0^1 f(t)dt - \int_0^1 t f(t)dt$$

$$f(x) = x + ax - b \text{ where } a = \int_0^1 f(t)dt \text{ and } b = \int_0^1 t f(t)dt$$

$$a = \int_0^1 (t + at - b)dt \Rightarrow a + 2b = 1 \text{ ---- (1)}$$

$$\text{and } b = \int_0^1 t(t + at - b)dt \Rightarrow 2a - 9b = -2 \text{ ---- (2)}$$

$$\text{Solving 1 and 2, we get } a = 5/13 \text{ and } b = 4/13 \therefore f(x) = \frac{18x}{13} - \frac{4}{13}$$

319. Let f be a differentiable function such that $x^2 f(x) - x = 4 \int_0^x t f(t)dt$, $f(1) = \frac{2}{3}$. Then

$18f(3)$ is equal to

- 1) 210 2) 150 3) 160 4) 180

KEY:3**(LEIBNITZ'S RULE)2023****SOL:** Differentiable on both sides

$$x^2 f'(x) - 2x f(x) = 1$$

$$\frac{x^2 f'(x) - 2x f(x)}{x^4} = \frac{1}{x^4}$$

$$\Rightarrow \frac{f(x)}{x^2} = -\frac{1}{3x^3} + c$$

$$\text{Put } x=1 \Rightarrow c=1$$

$$\therefore 18f(3) = 160$$

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320. If $\int_0^{t^2} x f(x) dx = \frac{2}{5} t^5$, then $f\left(\frac{4}{25}\right)$ is

1) $\frac{2}{5}$

2) $-\frac{5}{2}$

3) 1

4) $\frac{5}{2}$

Key: 1

(Leibnitz's rule)

Sol:

$$\int_0^{t^2} x f(x) dx = \frac{2}{5} t^5$$

Similarity w.r.to x

$$t^2 f(t^2) 2t = 2t^4 \Rightarrow f(t^2) = t \Rightarrow f(x) = \sqrt{x}$$

$$f\left(\frac{4}{25}\right) = \frac{2}{5}$$

321. If $\int_0^x f(t) dt = x + \int_x^1 t f(t) dt$, then the value of $\int_0^{e-1} f(x) dx$ _____

1) 2

2) $\frac{1}{2}$

3) 1

4) 0

Key: 1

(Leibnitz's rule)

Sol:

$$\int_0^{e-1} f(x) dx = x + \int_x^1 t f(t) dt$$

similarity w.r.to x

$$f(x) = 1 + 0 - x f(x)$$

$$f(x) = \frac{1}{1+x}$$

$$\int_0^{e^2-1} \frac{dx}{1+x} = |dy| |1+x|$$

$$= 2 - 0 = 2$$

322. Let $f(x)$ be a differentiable function on $\mathbb{R}$ and satisfying the integral equation

$$\int_0^x f(t) dt + \int_0^x t f(x-t) dt = -1 + e^{-x}, \text{ for all } x \in \mathbb{R}, \text{ then}$$

1) $f(2) = e^{-2}$

2) $f(0) + f'(0) = 1$

3) $f'(0) = 2$

4) $f'(0) = 1$

Key: 2

(Leibnitz's rule)

Sol:

$$\int_0^x f(x) dx + \int_x^x t f(x-t) dt = -1 + e^x$$

 $f(a-x)$ property

$$\int_0^x f(t) dt + \int_0^x (x-t) f(t) dt = -1 + e^x$$

$$\int_0^x f(t) dt + x \int_0^x f(t) dt - \int_0^x t f(t) dt = e^x - 1$$

Similarly w.r to x

$$f(x) + 1 \int_0^x f(t) dt + x f(x) - x f(x) = e^x \Rightarrow \int_0^x f(t) dt = e^x - f(x)$$

$$f(x) = e^x - f'(x) \Rightarrow f(x) + f'(x) = e^x \Rightarrow f(0) + f'(0) = e^0 = 1$$

$$323. \lim_{x \rightarrow \infty} \frac{\int_0^x (\tan^{-1} t)^2 dt}{\sqrt{x^2 + 1}}$$

1) $\frac{\pi}{4}$

2) $\frac{\pi^2}{2}$

3) $\frac{\pi^2}{4}$

4) π

Key: 3

(Leibnitz's rule)

Sol: Using Leibnitz rule

$$324. \text{ If } \int_0^x f(t) dt = x^2 + \int_x^t t^2 f(t) dt \text{ then } f^t\left(\frac{1}{2}\right) \text{ is}$$

1) $\frac{18}{25}$

2) $\frac{6}{25}$

3) $\frac{24}{25}$

4) $\frac{4}{5}$

Key: 3

(Leibnitz's rule)

Sol:

$$f'(x) = 2x - x^2 f'(x)$$

$$f'(x) = \frac{2x}{1+x^2}$$

$$f'(x) = \frac{(1+x^2)^2 - 2x(2x)}{(1+x^2)^2} = 2 \left[\frac{1-x^2}{(1+x^2)^2} \right]$$

$$f'\left(\frac{1}{2}\right) = \frac{2\left(1-\frac{1}{4}\right)}{\left(1+\frac{1}{4}\right)} = \frac{2\left(\frac{3}{4}\right)}{\frac{25}{16}} = \frac{24}{25}$$

$$325. \lim_{x \rightarrow \infty} \frac{\left(\int_0^x e^{t^2} dt \right)^2}{\int_0^x e^{2t^2} dt} =$$

1) 2

2) 3

3) 0

4) 1

Key: 3

(Leibnitz's rule)

$$\text{Sol: } \lim_{x \rightarrow \infty} \frac{\left(\int_0^x e^{t^2} dt \right)^2}{\int_0^x e^{2t^2} dt} \left(\frac{\infty}{\infty} \text{ form} \right)$$

$$= \lim_{x \rightarrow \infty} \frac{2 \int_0^x e^{t^2} dt \cdot e^{x^2}}{1 \cdot e^{2x^2}} \text{ Applying L' Hospital Rule}$$

$$= \lim_{x \rightarrow \infty} \frac{2 \int_0^x e^{t^2} dt}{e^{x^2}} = \lim_{x \rightarrow \infty} \frac{2 \cdot e^{x^2}}{2x \cdot e^{x^2}} = 0$$

$$326. \lim_{x \rightarrow 3} \left(\frac{x}{x-3} \int_3^x \frac{\sin t}{t} dt \right) =$$

1) sin 2

2) sin 3

3) sin 4

4) sin 1

Key: 2

(Leibnitz's rule)

Sol:

$$\lim_{x \rightarrow 3} \left(\frac{\int_3^x \frac{\sin t}{t} dt}{1 - \frac{3}{x}} \right) = \lim_{x \rightarrow 3} \frac{\frac{\sin x}{x}}{\frac{3}{x^2}} = \lim_{x \rightarrow 3} \frac{x \sin x}{3} = \sin 3$$

327. let $f(x) = \int_0^x |2t - 3| dt$. Then $f(x)$ at $x = \frac{3}{2}$ is _____

- 1) Continuous but not differentiable 2) Not continuous not differentiable
3) Continuous and differentiable 4) Not continuous and differentiable

Key: 3

(Leibnitz's rule)

Sol:

$$f(x) = \int_0^x |2t - 3| dt$$

$$\begin{cases} \int_0^x (3 - 2t) dt, & x < \frac{3}{2} \end{cases}$$

$$\begin{cases} \int_0^{3/2} (3 - 2t) dt + \int_{3/2}^x (2t - 3) dt, & x \geq \frac{3}{2} \end{cases}$$

$$\begin{cases} 3x - x^2, & x < \frac{3}{2} \\ x^2 - 3x + \frac{9}{2}, & x \geq \frac{3}{2} \end{cases}$$

Clearly, $f(x)$ is continuous at $x = \frac{3}{2}$ as $f\left(\frac{3^-}{2}\right) = f\left(\frac{3^+}{2}\right) = \frac{9}{4}$

$$f'(x) = \begin{cases} 3 - 2x, & x < \frac{3}{2} \\ 2x - 3, & x > \frac{3}{2} \end{cases}$$

$$f'\left(\frac{3^-}{2}\right) = f'\left(\frac{3^+}{2}\right) = 0$$

Hence, $f(x)$ is differentiable at $x = \frac{3}{2}$

328. If $f'(x) = f(x) + \int_0^1 f(x) dx$, given $f(0) = 1$, then the value of $f(\log_e^2)$ is (Leibnitz's rule)

- 1) $\frac{1}{3+e}$ 2) $\frac{5-e}{3-e}$ 3) $\frac{2+e}{e-2}$ 4) $\frac{e-5}{3-e}$

Key: 2

Sol: Differentiating the given equations w.r.t x, we get $f''(x) = f'(x)$

$$\int \frac{df'(x)}{f'(x)} = \int dx \Rightarrow \ln f'(x) = x + c \Rightarrow f'(x) = Ae^x \text{ -----(1)}$$

$$\Rightarrow \int f'(x) dx = \int Ae^x dx \Rightarrow f(x) = Ae^x + B \text{ -----(2)}$$

$$\text{Now, } f(0) = 1 \Rightarrow A + B = 1$$

$$\Rightarrow f(\log_e 2) = \frac{4}{3-e} + \frac{1-e}{3-e} = \frac{5-e}{3-e}$$

329. Let $f: R^+ \rightarrow R$ be differentiable function which $f(1) = 3$ and satisfying: **(Leibnitz's rule)**

$$\int_1^{xy} f(t) dt = y \int_1^x f(t) dt + x \int_1^y f(t) dt \forall x, y \in R^+, \text{ then } f(e) =$$

1) 3

2) 4

3) 1

4) none of these

Key: 4

Sol: Use Leibnitz rule

330. Given function 'g' continues everywhere such that $\int_0^1 g(t) dt = 2$ and $g(1) = 5$

If $f(x) = \frac{1}{2} \int_0^x (x-t)^2 g(t) dt$, then the value of $f'''(1) - f''(1)$ is: **(Leibnitz's rule)**

1) 0

2) 1

3) 2

4) 3

Key: 2

$$\text{Sol: } 2f(x) = \int_0^x (x^2 - 2xt + t^2) g(t) dt$$

$$f''(x) = \int_0^x g(t) dt$$

$$f''(x) = g(x)$$

331. The value of function $f(x) = 1 + x + \int_1^x (In^2 t + 2Int) dt$ where $f'(x)$ vanishes is:

(LEIBNITZ'S RULE)

1) $\frac{1}{e}$

2) 0

3) $\frac{2}{e}$ 4) $1 + \frac{2}{e}$

Key: 4

$$\text{Sol: } f'(x) = 1 + In^2 x + 2In x = 0 \Rightarrow x = \frac{1}{e}$$

$$f\left(\frac{1}{e}\right) = 1 + \frac{1}{e} + \int_1^{1/e} (In^2 t + 2Int) dt$$

$$\text{Let } I = \int_1^{1/e} (In^2 t + 2Int) dt$$

$$Int = x \Rightarrow t = e^x; dt = e^x dx = \int_0^{-1} (x^2 + 2x) e^x dx = [e^x \cdot x^2]_0^{-1} = \frac{1}{e}$$

332. Let f be a differentiable function on R and satisfies

$$f(x) = x^2 + \int_0^x e^{-t} f(x-t) dt : \text{ then } \int_0^1 f(x) dx \text{ is equal to}$$

(LEIBNITZ'S RULE)

1) $\frac{1}{3}$

2) $\frac{1}{4}$

3) $\frac{7}{12}$

4) $\frac{5}{12}$

Key: 4

Sol: $f(x) = x^2 + \int_0^x e^{-t} f(x-t) dt$

$$\Rightarrow f(x) = 2x + x^2 \Rightarrow f(x) \frac{x^3}{3} + x^2$$

$$\Rightarrow y = \frac{1}{4}(-2x^2 + 6x - 1)$$

333. The function $f(x) = \int_0^{x^2} \frac{t^2 - 5t + 4}{2 + e^t} dt$ has**(LEIBNITZ'S RULE)**1) Local maxima at $x = -1, 1$ 2) Local maxima at $x = -2, 3, 2$ 3) Local maxima at $x = -2, 0, 2$ 4) Local maxima at $x = -2, 0, 2$

Key: 1

Sol: Use Leibnitz rule

334. For $x > 0$, let $f(x) = \int_1^x \frac{\ln t}{1+t} dt$ then the value of $f(e) + f\left(\frac{1}{e}\right)$ is **(Leibnitz rule)**

1. 1

2. 2

3. $\frac{1}{2}$

4. 0

Ans: 3

Sol: We have $f(x) = \int_1^x \frac{\ln t}{1+t} dt, x > 0 \dots \dots \dots (1)$

$$\Rightarrow f\left(\frac{1}{x}\right) = \int_1^{1/x} \frac{\ln t}{1+t} dt$$

$$\text{put } y = \frac{1}{t} \Rightarrow dt = \frac{-1}{y^2} dy$$

$$\therefore f\left(\frac{1}{x}\right) = \int_1^x \frac{\ln\left(\frac{1}{y}\right)}{1+\frac{1}{y}} \left(\frac{-1}{y^2}\right) dy$$

$$= \int_1^x \frac{\ln t}{(1+t)} dt \dots \dots \dots (2)$$

From (1) and (2)

$$\Rightarrow f(e) + f\left(\frac{1}{e}\right) = \frac{(\ln e)^2}{2} = \frac{1}{2}$$

335. $f(x)$ is a continuous function for all real values of x and satisfies

$$\int_0^x f(t) dt = \int_x^1 t^2 f(t) dt + \frac{x^{16}}{8} + \frac{x^6}{3} + \alpha. \text{ Then}$$

the value of α is equal to**(Leibnitz's rule)**

A) $-\frac{1}{24}$

B) $\frac{17}{168}$

C) $\frac{1}{7}$

D) $-\frac{167}{840}$

Key; D

$$\text{Sol: } \int_0^x f(t) dt = \int_x^1 t^2 f(t) dt = 0 + \frac{1}{8} + \frac{1}{3} + \alpha = \frac{11}{24} + \alpha \dots\dots\dots(1)$$

$$\text{for } x=1, \int_0^1 f(t) dt = 0 + \frac{1}{8} + \frac{1}{3} + \alpha = \frac{11}{24} + \alpha$$

Differentiating both sides of equation (1) w.r.t, x, we get

$$f(x) = 0 - x^2 f(x) + 2x^{15} + 2x^5$$

$$= \frac{2(x^{15} + x^5)}{1 + x^2}$$

$$\text{or } \alpha = -\frac{167}{840}$$

Topic: Evaluation of definite integrals**Sub Topic 2:** Properties of evaluation of definite integrals**Mini Topic 8:** Miscellaneous properties336. Let f be a differentiable function from $\mathbb{R}$ to $\mathbb{R}$ such that $|f(x) - f(y)| \leq 2|x - y|^{3/2}$, for all
 $x, y \in \mathbb{R}$, If $f(0) = 1$ then $\int_0^1 f^2(x) dx$ is equal to:

1) 1

2) 0

3) 2

4) $\frac{1}{2}$

Ans. 1

(Miscellaneous and Mixed properties) 2019

$$\text{Sol. } \lim_{x \rightarrow y} \left| \frac{f(x) - f(y)}{x - y} \right| \leq \lim_{x \rightarrow y} 2|x - y|^{1/2}$$

$$|f'(y)| \leq 0 \Rightarrow f'(y) = 0 \Rightarrow f(y) = k$$

$$\text{Given } f(0) = 1 \Rightarrow f(x) = 1$$

$$\int_0^1 f^2(x) dx = \int_0^1 1 dx = (x)_0^1 = 1$$

337. If θ_1 and θ_2 be respectively the smallest and the largest values of $\theta \ln(0, 2\pi) - \{\pi\}$ which

satisfy the equation $2 \cot^2 \theta - \frac{5}{\sin \theta} + 4 = 0$ then $\int_{\theta_1}^{\theta_2} \cos^2 3\theta d\theta$ is equal to:

1) $\frac{\pi}{3}$ 2) $\frac{2\pi}{3}$ 3) $\frac{\pi}{9}$ 4) $\frac{\pi}{3} + \frac{1}{6}$

Ans: 1

(Miscellaneous) 2020

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Sol : $2 \cot^2 \theta - \frac{5}{\sin \theta} + 4 = 0 \quad \frac{2 \cos^2 \theta}{\sin^2 \theta} - \frac{5}{\sin \theta} + 4 = 0$

$$2 \cos^2 \theta - 5 \sin \theta + 4 \sin^2 \theta = 0, \sin \theta \neq 0 \quad 2 \sin^2 \theta - 5 \sin \theta + 2 = 0 \quad (2 \sin \theta - 1)(\sin \theta - 2) = 0$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\therefore \int_{\pi/6}^{5\pi/6} \cos^2 3\theta d\theta = \int_{\pi/6}^{5\pi/6} \frac{1 + \cos 6\theta}{2} d\theta$$

$$= \frac{1}{2} \left[\theta + \frac{\sin 6\theta}{6} \right]_{\pi/6}^{5\pi/6} = \frac{1}{2} \left[\frac{5\pi}{6} - \frac{\pi}{6} + \frac{1}{6}(0 - 0) \right] = \frac{1}{2} \cdot \frac{4\pi}{6} = \frac{\pi}{3}$$

338. If for all real triplets (a, b, c), $f(x) = a + bx + cx^2$; then $\int_0^1 f(x) dx$ is equal to:

1) $2 \left\{ 3f(1) + 2f\left(\frac{1}{2}\right) \right\}$ 2) $\frac{1}{2} \left\{ f(1) + 3f\left(\frac{1}{2}\right) \right\}$

3) $\frac{1}{3} \left\{ f(0) + f\left(\frac{1}{2}\right) \right\}$

4) $\frac{1}{6} \left\{ f(0) + f(1) + 4f\left(\frac{1}{2}\right) \right\}$

Key: 4

(Miscellaneous)2020

Sol: $\int_0^1 (a + bx + cx^2) dx = \left[ax + \frac{bx^2}{2} + \frac{cx^3}{3} \right]_0^1 = a + \frac{b}{2} + \frac{c}{3}$

We have $f(1) = a + b + c$, $f(0) = a$ and $f\left(\frac{1}{2}\right) = a + \frac{b}{2} + \frac{c}{4}$

Now $\frac{1}{6} \left(f(1) + f(0) + 4f\left(\frac{1}{2}\right) \right) = \frac{1}{6} \left(a + b + c + a + 4 \left(a + \frac{b}{2} + \frac{c}{4} \right) \right)$
 $= \frac{1}{6} (6a + 3b + 2c) = a + \frac{b}{2} + \frac{c}{3}$

339. The integral

$\int_{\pi/6}^{\pi/3} \tan^3 x \cdot \sin^2 3x (2 \sec^2 x \cdot \sin^2 3x + 3 \tan x \cdot \sin 6x) dx$ is equal to :

1. $-\frac{1}{18}$

2. $-\frac{1}{9}$

3. $\frac{9}{2}$

4. $\frac{7}{18}$

Key: 1

(Miscellaneous)2020

Sol:

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \tan^3 x \sin^2 3x (2 \sec^2 x \sin^2 3x + 3 \tan x \sin 6x) dx$$

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \left(\frac{4 \tan^3 x \sec^2 x}{2} \sin^4 3x dx + 6 \tan^4 x \sin^3 3x \cos x \right) dx$$

$$= \frac{1}{2} \left(\tan^4 x \sin^4 3x \right)_{\frac{\pi}{6}}^{\frac{\pi}{3}} = -\frac{1}{18}$$

340. Let $J_{n,m} = \int_0^{\frac{1}{2}} \frac{x^n}{x^m - 1} dx, \forall n > m \text{ and } n, m \in \mathbb{N}$. Consider a matrix $A = [a_{ij}]_{3 \times 3}$ where

$$a_{ij} = \begin{cases} J_{6+i} - J_{i+3,3}, & i \leq j \\ 0, & i > j \end{cases}. \text{ Then } |adj A^{-1}| \text{ is :}$$

1) $(105)^2 \times 2^{36}$

2) $(105)^2 \times 2^{38}$

3) $(15)^2 \times 2^{34}$

4) $(15)^2 \times 2^{42}$

Key: 2

(Miscellaneous properties)2021

Sol: $J_{6+i,3} - J_{i+3,3}; i \leq j$

$$\Rightarrow \int_0^{\frac{1}{2}} \frac{x^{6+i}}{x^3 - 1} - \int_0^{\frac{1}{2}} \frac{x^{i+3}}{x^3 - 1}$$

$$\Rightarrow \int_0^{\frac{1}{2}} \frac{x^{i+3}(x^3 - 1)}{x^3 - 1}$$

$$\frac{x^{3+i+1}}{3+i+1} = \left(\frac{x^{4+i}}{4+i} \right)^{\frac{1}{2}} \quad a_{ij} = j_{6+i,3} - j_{i+3,3} = \frac{\left(\frac{1}{2} \right)^{4+i}}{4+i}$$

$$a_{11} = \frac{1}{5.25}$$

$$a_{22} = \frac{1}{6.26}$$

$$a_{33} = \frac{1}{7.27}$$

$$|A| = a_{11}a_{22}a_{33} \quad (\because \text{Below diagonal elements are 0's}) = \frac{1}{5.6.7.2^{18}}$$

$$|Adj A^{-1}| = \frac{1}{|A|^2} = (105)^2 \times 2^{38}$$

341. The function $f(x)$, that satisfies the condition $f(x) = x + \int_0^{\frac{\pi}{2}} \sin x \cdot \cos y f(y) dy$, is:

1) $x + (\pi + 2) \sin x$

2) $x + \frac{\pi}{2} \sin x$

3) $x + (\pi - 2) \sin x$

4) $x + \frac{2}{3}(\pi - 2) \sin x$

Key: 1

(Miscellaneous properties)2021

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Sol: $f(x) = x + k \sin x$ where $K = \int_0^{\frac{\pi}{2}} \cos y \cdot f(y) dy$

$$\therefore K = \int_0^{\frac{\pi}{2}} \cos y \cdot (y + K \sin y) dy$$

$$= \int_0^{\frac{\pi}{2}} y \cos y dy + K \int_0^{\frac{\pi}{2}} \sin y \cdot \cos y dy$$

$$K = \frac{\pi}{2} - 1 + \frac{K}{2} \Rightarrow \frac{K}{2} = \frac{\pi - 2}{2}$$

$$\Rightarrow K = \pi - 2$$

$$\therefore f(x) = x + (\pi - 1) \sin x$$

342. Let $5f(x) + 4f\left(\frac{1}{x}\right) = \frac{1}{x} + 3, x > 0$. Then $18 \int_1^2 f(x) dx$ is equal to

- 1) $10 \log_e 2 + 6$ 2) $10 \log_e 2 - 6$
 3) $5 \log_e 2 - 3$ 4) $5 \log_e 2 + 3$

KEY:2

(FUNCTIONAL EQUATIONS)2023

SOL: $5f(x) + 4f\left(\frac{1}{x}\right) = \frac{1}{x} + 3 \rightarrow (1)$

Let $x \rightarrow \frac{1}{x}$ $5f\left(\frac{1}{x}\right) + 4f(x) = x + 3 \rightarrow (2)$

Solve (1) & (2)

$$20f(x) + 16f\left(\frac{1}{x}\right) = 12 + \frac{4}{x}$$

$$20f(x) + 25f\left(\frac{1}{x}\right) = 5x + 15$$

$$-9f\left(\frac{1}{x}\right) = \frac{4}{x} - 5x - 3$$

$$f\left(\frac{1}{x}\right) = \frac{5x}{9} + \frac{1}{3} - \frac{4}{9x}$$

Again $x \rightarrow \frac{1}{x} \Rightarrow f(x) = \frac{5}{9x} + \frac{1}{3} - \frac{4x}{9}$

$$18 \int_1^2 f(x) dx = 18 \int_1^2 \left(\frac{5}{9x} + \frac{1}{3} - \frac{4x}{9} \right) dx = 18 \left(\frac{5}{9} \ln x + \frac{x}{3} - \frac{2x^2}{9} \right)_1^2$$

$$= 18 \left[\frac{5}{9} \ln + \frac{2}{3} - \frac{1}{3} - \frac{2}{9} (2^2 - 1) \right]$$

$$18 \int_1^2 f(x) dx = 10 \ln 2 - 6$$

343. If $\frac{1}{n+1} {}^nC_n + \frac{1}{n} {}^nC_{n-1} + \dots + \frac{1}{2} {}^nC_1 + {}^nC_0 = \frac{1023}{10}$ then n is equal to

1) 6

2) 7

3) 9

4) 8

KEY: 3

(IMPORTANT EXPANSIONS) 2023

SOL:

$$\int_0^1 (1+x)^n dx = \int_0^1 ({}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n) dx$$

$$\Rightarrow \left[\frac{(1+x)^{n+1}}{n+1} \right]_0^1 = \left[{}^nC_0 x + {}^nC_1 \frac{x^2}{2} + {}^nC_2 \frac{x^3}{3} + \dots + {}^nC_n \frac{x^{n+1}}{n+1} \right]_0^1$$

$$\Rightarrow \frac{2^{n+1} - 1}{n+1} = {}^nC_0 + \frac{{}^nC_1}{2} + \frac{{}^nC_2}{3} + \dots + \frac{{}^nC_n}{n+1}$$

$$= \frac{1023}{10} = \frac{2^{10} - 1}{10}$$

$$\therefore n = 9$$

344. If $\log_3^2, \log_3(2^a - 5), \log_3\left(2^a - \frac{7}{2}\right)$ are in A.P then $\int_{-a}^a e^{-|x|} dx = \lambda - \frac{\lambda}{e^M}$

Key: 5

(Miscellaneous)

Sol:

$$\log_3^2 + \log_3\left(2^a - \frac{7}{2}\right) = \log_3\left(2^a - 5\right)^2$$

$$\Rightarrow 2\left(2^a - \frac{7}{2}\right) = \left(2^a - 5\right)^2, 2^a = t$$

$$\Rightarrow 2t - 7 = (t - 5)^2$$

$$\Rightarrow t > 5 \quad t = 8 \Rightarrow a = 3$$

$$\int_{-a}^a e^{-|x|} dx = 2 \int_0^3 e^{-x} dx = \left[\frac{-2}{e^x} \right]_0^3 = \frac{-2}{e^3} + 2$$

$$\lambda + M = 5$$

345. Let $f(x) = x - x^2; 0 \leq x \leq 1$ and $f(x+1) = f(x) \forall x \in R$. Then the function $g(x)$ defined as

$$g(x) = \int_0^x f(t) dt \text{ is equal to}$$

(Miscellaneous properties)

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- A) $\frac{n}{6} + \frac{(x-n)^2}{2} - \frac{(x-n)^3}{3}$ where $n \leq x < n+1$
 B) $\frac{n}{6} + \frac{(x+n)^2}{2} - \frac{(x-n)^3}{3}$ where $n \leq x < n+1$
 C) $\frac{n}{6} + \frac{(x-n)^2}{2} + \frac{(x+n)^3}{3}$ where $n \leq x < n+1$
 D) $\frac{n}{6} + \frac{(x-n)^2}{2} + \frac{(x-n)^3}{3} \quad \forall x \in R$

ANS: A

SOL: Let $x \in R$, then there exists an integer n such that

$$n \leq x < n+1. \text{ Now } g(x) = \int_0^x f(t) dt$$

$$= \int_0^n f(t) dt + \int_n^x f(t) dt = n \int_0^1 f(t) dt + \int_0^{x-n} f(t) dt$$

$$= \frac{n}{6} + \frac{(x-n)^2}{2} - \frac{(x-n)^3}{3} \text{ when } n \leq x < n+1$$

346. If $I = \int_0^1 x^{70} (1-x)^{30} dx$. then the value of $({}^{100}C_{30} I) \times 202$ is

- A)1 B)2 C)3 D)4

ANS: B

(Miscellaneous properties)

SOL: Using $\int_0^1 x^m (1-x)^n dx = \frac{m!n!}{(m+n+1)!}$

$$I = \frac{70!30!}{101 \times 100!} = \frac{1}{101} \cdot \frac{1}{{}^{100}C_{30}}$$

$$\therefore (I \cdot {}^{100}C_{30}) \times 202 = 2$$

347. If $\int_0^1 \frac{(x-x^2)^{10} \cdot x}{(1+x)^{23}} dx = \frac{1}{11({}^mC_n) \cdot 2^k}$ then $K - 3\left[\frac{m}{n}\right]$ equals $(m, n, K \in N)$ where $[.]$

denotes G.I.F

- A)2 B)3 C)4 D)5

ANS: D

(Miscellaneous properties)

SOL: $I = \frac{(x-x^2)^{10} x}{(1+x)^{23}} = \left(\frac{1-x}{1+x}\right)^{10} \left(\frac{x}{1+x}\right)^{11} \frac{1}{(1+x)^2}$

Let $t = \frac{1-x}{1+x}$ then $1-t = \frac{2x}{1+x} \Rightarrow \frac{x}{1+x} = \frac{1-t}{2}$ and

$$dt = \frac{-2}{(1+x)^2} dx \Rightarrow \frac{dx}{1+x^2} = \frac{dt}{2}$$

$$\therefore I = \int_2^0 t^{10} \cdot \left(\frac{1-t}{2}\right)^{11} \left(-\frac{dt}{2}\right) = \frac{1}{2^{12}} \int_0^1 t^{10} (1-t)^{11} dt$$

$$= \frac{1}{2^{12}} \times \frac{10!11!}{(22)!} = \frac{1}{2^{12} \cdot 22 \times {}^{21}C_{10}} = \frac{1}{11 \cdot 2^{13} \cdot {}^{21}C_{10}}$$

$$\therefore K = 13, m = 21, n = 10 \text{ or } 11$$

$$\therefore K - 3 \left\lfloor \frac{m}{n} \right\rfloor = 5$$

348. $\lim_{x \rightarrow 0} \frac{1}{x} \left(\int_y^4 e^{\cos^2 t} dt - \int_{x+y}^4 e^{\cos^2 t} dt \right)$ is equal to

(A) $e^{\cos^2 y} (-\sin^2 y)$ (B) $e^{\cos^2 y}$

(C) 0 (D) $e^4 - e^{\cos^2 y}$

Key: B

(Miscellaneous properties)

Sol: $\lim_{x \rightarrow 0} \frac{\int_y^4 e^{\cos^2 t} dt - \int_{x+y}^4 e^{\cos^2 t} dt}{x} \left(\frac{0}{0} \text{ form} \right)$

$$= \lim_{x \rightarrow 0} 0 - \left[0 - e^{\cos^2(x+y)} \right] = e^{\cos^2 y}$$

349. Let $f: R \rightarrow R$ be a continuous function given by $f(x+y) = f(x) + f(y)$ for all $x, y \in R$.

If $\int_0^2 f(x) dx = \alpha$, then $\int_{-2}^2 f(x) dx$ is equal to

1) 2α 2) α 3) 0 4) $\alpha - 2$

Key: 3

(Functional Equations)

Sol: we have $f(x+y) = f(x) + f(y)$ for all $x, y \in R$

$$\Rightarrow f(x) = \lambda x \text{ for all } x \in R \text{ where } \lambda = f(1)$$

$$\Rightarrow \int_{-2}^2 f(x) dx = 0$$

350. If $f(y) = e^y$, $g(y) = y$; $y > 0$ and $F(t) = \int_0^t f(t-y) g(y) dy$, then

1) $F(t) = e^t - (1+t)$ 2) $F(t) = te^t$ 3) $F(t) = te^{-t}$ 4) $F(t) = 1 - e^{-t}(1+t)$

Key: 1

(Miscellaneous properties)

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Sol: $F(t) = \int_0^t f(t-y)g(y)dy$

$$\int_0^t e^{t-y} \cdot y dy$$

$$= e^t - t - 1$$

351. If $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$ then $\int_0^\infty \frac{\sin^3 x}{x} dx$ is equal to

1) $\frac{\pi}{2}$

2) $\frac{\pi}{4}$

3) $\frac{\pi}{6}$

4) $3\frac{\pi}{2}$

Key: 2

(Miscellaneous)

Sol: $I = \int_0^\infty \frac{3 \sin x - \sin 3x}{4x} dx = \frac{3}{4} \frac{\pi}{2} - \frac{1}{4} \int_0^\infty \frac{\sin 3x}{x} dx$

$$= \frac{3\pi}{8} - \frac{1}{4} \int_0^\infty \frac{\sin t}{\frac{t}{3}} dt = \frac{3\pi}{8} - \frac{1}{4} \frac{\pi}{2}$$

352. If $\int_{-1}^1 \frac{d}{dx} \left(\frac{1}{1+e^{1/x}} \right) dx = \lambda + \left(\frac{1-e}{1+e} \right)$ then λ equals

Key: 1

(Miscellaneous)

Sol: $\int_{-1}^1 \frac{d}{dx} \left(\frac{1}{1+e^{1/x}} \right) dx$

$$= \left[\frac{1}{1+e^{1/x}} \right]_{-1}^0 + \left[\frac{1}{1+e^{1/x}} \right]_0^1$$

$$= \left[1 - \frac{e}{e+1} \right] + \left[\frac{1}{1+e} - 0 \right]$$

$$= 1 + \left(\frac{1-e}{1+e} \right)$$

353. The value of the definite integral $\int_0^{\pi/2} \frac{dx}{\tan x + \cot x + \cos x + \sec x}$ is: (Trigonometric

Simplification)

1) $1 - \frac{\pi}{4}$

2) $\frac{\pi}{4} + 1$

3) $\pi + \frac{1}{4}$

4) $\frac{\pi}{4} + 2$

Key: 1

Sol: $I = \int_0^{\pi/2} \frac{\sin x \cos x dx}{1 + \sin x + \cos x} = \int_0^{\pi/2} \sin \frac{x}{2} \left(\cos \frac{x}{2} - \sin \frac{x}{2} \right) dx$

$$= \frac{1}{2} \int_0^{\pi/2} (\sin x + \cos x - 1) dx = 1 - \frac{\pi}{4}$$

354. The value of the definite integral $\int_0^{\pi/3} \ln(1 + \sqrt{3} \tan x) dx$ equals

(Trigonometric Simplification)

1) $\frac{\pi}{3} \ln 2$

2) $\frac{\pi}{3}$

3) $\frac{\pi^2}{6} \ln 2$

4) $\frac{\pi}{2} \ln 2$

Key: 1

Sol: $\int_0^{\pi/3} (\ln(\cos x + \sqrt{3} \sin x) - \ln \cos x) dx$
 $= \int_0^{\pi/2} \left\{ \ln \left(2 \cos \left(x - \frac{\pi}{3} \right) \right) - \ln \cos x \right\} dx = \frac{\pi}{3} \ln 2$

355. The value of the definite integral $\int_{-\infty}^0 \frac{ze^{-z}}{\sqrt{1-e^{-2z}}} dz$ **(Important expansions)**

A) $-\frac{\pi}{2} \ln 2$

B) $\frac{\pi}{2} \ln 2$

C) $-\pi \ln 2$

D) $\pi \ln 2$

Key: A

Sol: put $e^{-z} = \sin \theta$

Then $\int_{-\infty}^0 \frac{ze^{-z}}{\sqrt{1-e^{-2z}}} dz = \int_0^{\pi/2} \frac{\ln \sin \theta}{\sqrt{1-\sin^2 \theta}} \cos \theta d\theta$
 $= \int_0^{\pi/2} \ln \sin \theta d\theta = -\frac{\pi}{2} \ln 2$

356. The value of the integral $\int_0^{2\pi} e^{\cos \theta} \cos(\sin \theta) d\theta$ **(Miscellaneous properties)**

A) 0

B) π

C) 2π

D) 4π

Key: C

Sol: $\int_0^{2\pi} e^{\cos \theta} \cos(\sin \theta) d\theta$
 $= \text{Real part of } \int_0^{2\pi} e^{\cos \theta} \{ \cos(\sin \theta) + i \sin(\sin \theta) \} d\theta$
 $= \text{Real part of } \int_0^{2\pi} e^{\cos \theta} e^{i \sin \theta} d\theta$
 $= \text{Real part of } \int_0^{2\pi} e^{\cos \theta} e^{i \sin \theta} d\theta$

$$= \text{Real part of } \int_0^{2\pi} e^{e^{i\theta}} d\theta$$

$$= \text{Real part of } \int_0^{2\pi} \left[1 + (\cos \theta + i \sin \theta) + \frac{1}{2!} (\cos 2\theta + i \sin 2\theta) + \dots \right] d\theta$$

$$= \left[\theta + \sin \theta + \frac{\sin 2\theta}{2 \cdot 2!} + \dots \right]_0^{2\pi} = 2\pi$$

357. suppose the function $g_n(x) = x^{2n+1} + a_n x + b_n$ ($n \in \mathbb{N}$) satisfies the equation

$$\int_{-1}^1 (px + q) g_n(x) dx = 0 \text{ for all linear functions } (px + q) \text{ then (Miscellaneous properties)}$$

A) $a_n = b_n = 0$

B) $b_n = 0; a_n = -\frac{3}{2n+3}$

C) $a_n = 0, b_n = -\frac{3}{2n+3}$

D) $a_n = \frac{3}{2n+3}; b_n = -\frac{3}{2n+3}$

Key:B

Sol: we have $\int_{-1}^1 (px + q)(x^{2n+1} + a_n x + b_n) dx =$

$$\Rightarrow \int_{-1}^1 (px^{2n+2} + pa_n x^2 + Pb_n + qx^{2n+1} + qa_n x + qb_n) dx = 0$$

$$\Rightarrow P \frac{x^{2n+3}}{2n+3} + pa_n \frac{x^3}{3} + (pb_n + qb_n) \Big|_{-1}^1 = 0$$

$$\Rightarrow b_n = 0 \text{ and } a_n = -\frac{3}{2n+3}$$

358. If $\int_0^x \ln \sin \theta d\theta + \int_{-x}^x \ln \left(\cos \frac{\theta}{2} + \sin \frac{\theta}{2} \right) d\theta + \int_{-x}^x \ln (\cos \theta + \sin \theta) d\theta + \int_{-x}^x \ln (\cos 2\theta + \sin 2\theta) d\theta + 3x \ln 2 = 0$

Then $\int_0^x \ln (\sin 8\theta) d\theta$ is equal to (Standard result)

Key:0

Sol: $\int_{-x}^x \ln (\cos \theta / 2 + \sin \theta / 2) d\theta =$

$$= \int_0^x \ln \cos \theta d\theta$$

So the given definite integral

$$\Rightarrow \int_0^x \ln \sin 8\theta d\theta = 0$$

359. If $2f(x) + 3f\left(\frac{1}{x}\right) = \frac{1}{x} - 2, x \neq 0$, then $\int_1^2 f(x) dx$ is equal to

Yashpatil TG~ @bohring_bot (Functional Equations) **152**

- A) $\frac{-2}{5} \log 2 + \frac{1}{2}$ B) $\frac{-2}{5} \log 2 - \frac{1}{2}$ C) $\frac{2}{5} \log 2 + \frac{1}{2}$ D) none of these

Key: A

Sol: We have, $2f(x) + 3f\left(\frac{1}{x}\right) = \frac{1}{x} - 2$

$$\Rightarrow 2f\left(\frac{1}{x}\right) + 3f(x) = x - 2$$

Solving the above two equations, we get

$$f(x) = \frac{-2}{5x} + \frac{3x}{5} - \frac{2}{5}$$

$$\therefore \int_1^2 f(x) dx = \int_1^2 \left(\frac{-2}{5x} + \frac{3x}{5} - \frac{2}{5} \right) dx$$

$$= \left(\frac{-2}{5} \log x + \frac{3x^2}{10} - \frac{2}{5}x \right) \Big|_1^2$$

$$\left(\frac{-2}{5} \log 2 + \frac{1}{2} \right)$$

360. A differentiable function $f(x)$ satisfies the condition

$$f(x+y) = f(x) + f(y) + xy, \forall x, y \in R \text{ and } \lim_{x \rightarrow 0} \frac{1}{h} f(x) = 3 \text{ then } \int_1^2 f(x) dx = \underline{\hspace{2cm}}$$

Key: 15.67

(Functional Equations)

Sol:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(h)}{h} + x = 3 + x$$

$$f(x) = 3x + \frac{x^2}{2} + c$$

$$x = y = 0 \Rightarrow f(0) = 0 \Rightarrow f(x) = 3x + \frac{x^2}{2}$$

$$\int_1^2 f(x) dx = \int_1^2 \left(3x + \frac{x^2}{2} \right) dx$$

$$= \left[\frac{3x^2}{2} + \frac{x^3}{6} \right]_1^2 = 6 + \frac{4}{3} - \frac{3}{2} - \frac{1}{6} = \frac{17}{3} = 15.67$$

361. Let $f\left(\frac{x+y}{2}\right) = \frac{f(x) + f(y)}{2} \forall x, y \in R$. If $f'(0) = -1$ and $f(0) = 1$, then

$$\int_0^{e-1} \frac{f(x)}{1-x^2} dx = \underline{\hspace{2cm}}$$

1) 1

2) $e - 1$

3) e

4) $e^2 - 1$

Key: 1 (Functional Equations)

Sol: $f\left(\frac{x+y}{2}\right) = \frac{f(x) + f(y)}{2}$

If w. r, to x

$$f'\left(\frac{x+y}{2}\right)\left(\frac{1}{2}\right) = \frac{f'(x) + 0}{2}$$

$$x=0, y=2x \quad f'(x) = f'(0)$$

$$= -1$$

$$f(x) = -x + c$$

$$f(0) = 1 \Rightarrow f(x) = 1 - x$$

$$\int_0^{e-1} \frac{1-x}{1-x^2} dx = \int_0^{e-1} \frac{dx}{1+x} = \ln(1+x) \Big|_0^{e-1}$$

$$= 1 - \ln(1) = 1 - 0 = 1$$

362. If $f(x) + f(y) = f\left(\frac{x+y}{1-xy}\right) \forall x, y \in R$ and $xy < 1, \lim_{x \rightarrow 0} \frac{f(x)}{x} = 2$ then $\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} f'(x) dx$

1) $\frac{\pi}{6}$

2) $\frac{2\pi}{3}$

3) $\frac{5\pi}{6}$

4) $\frac{\pi}{3}$

Key: 4 (Functional Equations)

Sol: $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$ if $xy < 1$

$$\lim_{x \rightarrow 0} \frac{f(x)}{x} = 2 \Rightarrow f'(0) = 2$$

$$f(x) = 2 \tan^{-1}(x) \Rightarrow f'(x) = \frac{2}{1+x^2}$$

$$\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{2dx}{1+x^2} = 2 \left[\pi - \frac{1}{x^2} \right]_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} = 2 \left[\frac{\pi}{3} - \frac{\pi}{6} \right] = \frac{\pi}{3}$$

Topic: Evaluation of definite integrals**Sub Topic 3:** Reduction formulae in definite integrals (Wallis' Formula)

363. If $b_n = \int_0^{\pi/2} \frac{\cos^2 nx}{\sin x} dx, n \in N$ then

1. $b_3 - b_2, b_4 - b_3, b_5 - b_4$ are in an A.P with common difference -2

2. $\frac{1}{b_3 - b_2}, \frac{1}{b_4 - b_3}, \frac{1}{b_5 - b_4}$ are in an A.P with common difference 2

3. $b_3 - b_2, b_4 - b_3, b_5 - b_4$ are in a G.P

4. $\frac{1}{b_3 - b_2}, \frac{1}{b_4 - b_3}, \frac{1}{b_5 - b_4}$ are in an A.P with common difference -2

Key:

4

(Reduction formula)2021

Sol: If $b_n = \int_0^{\pi/2} \frac{\cos^2 nx}{\sin x} dx, n \in \mathbb{N}$

Then

$$b_n - b_{n-1} = \int_0^{\pi/2} \frac{\cos^2 nx - \cos^2 (n-1)x}{\sin x} dx$$

$$\int_0^{\pi/2} \frac{\sin^2 (n-1)x - \sin^2 nx}{\sin x} dx$$

By $\sin^2 A - \sin^2 B = \sin(A+B)\sin(A-B)$

$$\int_0^{\pi/2} \frac{\sin(nx-x+x)\sin(nx-x-nx)}{\sin x} dx = - \int_0^{\pi/2} \sin(2n-1)x dx$$

$$= - \frac{[-\cos(2n-1)x]_0^{\pi/2}}{2n-1} = - \frac{1}{2n-1}$$

i.e $\frac{1}{b_n - b_{n-1}} = -(2n-1)$

$n = 3, 4, 5 \rightarrow \frac{1}{b_3 - b_2} = -5$

$\frac{1}{b_4 - b_3} = -(7), \frac{1}{b_5 - b_4} = -9 =$

A.P with C.D = -2

364. If $I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x dx$, then:

1) $I_2 + I_4, I_3 + I_5, I_4 + I_6$ are in AP

2) $I_2 + I_4, (I_3 + I_5)^2, I_4 + I_6$ are in GP

3) $\frac{1}{I_2 + I_4}, \frac{1}{I_3 + I_5}, \frac{1}{I_4 + I_6}$ are in GP

4) $\frac{1}{I_2 + I_4}, \frac{1}{I_3 + I_5}, \frac{1}{I_4 + I_6}$ are in AP

Ans: 4

(Miscellaneous properties)2021

Sol: $I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x dx$

Consider $I_n + I_{n+2} = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\cot^n x + \cot^{n+2} x) dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x \cdot \operatorname{cosec}^2 x dx$

$$= -\frac{\cot^{n+1} x}{n+1} \Bigg|_{\frac{\pi}{4}}^{\frac{\pi}{2}} = \frac{1}{n+1}$$

$$\frac{1}{I_2 + I_4} = 3; \frac{1}{I_3 + I_5} = 4; \frac{1}{I_4 + I_6} = 5 \text{ are in A.P.}$$

365. If $b_n = \int_0^{\pi/2} \frac{\cos^2 nx}{\sin x} dx, n \in N$ then

1. $b_3 - b_2, b_4 - b_3, b_5 - b_4$ are in an A.P with common difference -2
2. $\frac{1}{b_3 - b_2} \cdot \frac{1}{b_4 - b_3} \cdot \frac{1}{b_5 - b_4}$ are in an A.P with common difference 2
3. $b_3 - b_2, b_4 - b_3, b_5 - b_4$ are in a G.P
4. $\frac{1}{b_3 - b_2} \cdot \frac{1}{b_4 - b_3} \cdot \frac{1}{b_5 - b_4}$ are in an A.P with common difference -2

Key:

4

(Reduction)2022

Sol: If $b_n = \int_0^{\pi/2} \frac{\cos^2 nx}{\sin x} dx, n \in N$

Then

$$b_n - b_{n-1} = \int_0^{\pi/2} \frac{\cos^2 nx - \cos^2 (n-1)x}{\sin x} dx$$

$$= \int_0^{\pi/2} \frac{\sin^2 (n-1)x - \sin^2 nx}{\sin x} dx$$

By $\sin^2 A - \sin^2 B = \sin(A+B)\sin(A-B)$

$$\int_0^{\pi/2} \frac{\sin(nx-x)\sin(nx+x)}{\sin x} dx = - \int_0^{\pi/2} \sin(2n-1)x dx$$

$$= - \frac{[-\cos(2n-1)x]_0^{\pi/2}}{2n-1} = - \frac{1}{2n-1}$$

$$\text{i.e } \frac{1}{b_n - b_{n-1}} = -(2n-1)$$

$$n = 3, 4, 5 \rightarrow \frac{1}{b_3 - b_2} = -5$$

$$\frac{1}{b_4 - b_3} = -(7), \frac{1}{b_5 - b_4} = -9 =$$

A.P with C.D = -2

366. Let $I_n = \int_0^{\pi/2} (\sin x + \cos x)^n dx$ ($n \geq 2$). Then the value of $nI_n - 2(n-1)I_{n-2}$ is

Ans: 2

(Reduction property)

$$\text{Sol: } I_n = \int_0^{\pi/2} (\sin x + \cos x)^{n-1} (\sin x - \cos x)' dx$$

$$= 2 + (n-1) \int_0^{\pi/2} (\sin x + \cos x)^{n-2} (\cos x - \sin x)^2 dx$$

$$= 2 + (n-1) \int_0^{\pi/2} (\sin x + \cos x)^{n-2} [2 - (\sin x + \cos x)^2] dx$$

$$= 2 + 2(n-1)I_{n-2} - (n-1)I_n$$

$$\Rightarrow nI_n - 2(n-1)I_{n-2} = 2$$

367. Let $I_n = \int_0^{\pi/2} (\sin x + \cos x)^n dx$ ($n \geq 2$). Then the value of $nI_n - 2(n-1)I_{n-2}$ is

Ans: 2

(Reduction property)

$$\text{Sol: } I_n = \int_0^{\pi/2} (\sin x + \cos x)^{n-1} (\sin x - \cos x)' dx$$

$$= 2 + (n-1) \int_0^{\pi/2} (\sin x + \cos x)^{n-2} (\cos x - \sin x)^2 dx$$

$$= 2 + (n-1) \int_0^{\pi/2} (\sin x + \cos x)^{n-2} [2 - (\sin x + \cos x)^2] dx$$

$$= 2 + 2(n-1)I_{n-2} - (n-1)I_n$$

$$\Rightarrow nI_n - 2(n-1)I_{n-2} = 2$$

368. If $I_{(m,n)} = \int_0^1 t^m (1+t)^n dt$, then the expression for $I_{(m,n)}$ in terms of $I_{(m+1,n-1)}$ is

1) $\frac{2^n}{m+1} - \frac{n}{m+1} I(m+1, n-1)$

2) $\frac{n}{m+1} I(m+1, n-1)$

3) $\frac{2^n}{m+1} + \frac{n}{m+1} I(m+1, n-1)$

4) $\frac{m}{m+1} I(m+1, n-1)$

Key: 1

(By Parts and Successive Reduction)

Sol: Use by parts

369. If $I_k = \int_0^{\pi} \frac{\sin^2 kx}{\sin^2 x} dx$ then

(Reduction formula)

1) $I_1, I_2, I_3, \dots$ form an A.P2) common difference of $I_1, I_2, I_3, \dots$ is π 3) $I_k = k\pi$ 4) $I_{k+2} + I_k = 2I_{k+1}$

Key: (1,2,3,4)

Sol: $I_k = \int_0^{\pi} \frac{\sin^2 kx}{\sin^2 x} dx$ or $\int_0^{\pi} \frac{1 - \cos 2kx}{1 - \cos 2x} dx$

$$I_{k+1} - I_k = \int_0^{\pi} \frac{\sin^2 (k+1)x - \sin^2 kx}{\sin^2 x} dx$$

$$I_1 = \int_0^{\pi} \frac{\sin^2 x}{\sin^2 x} dx = \pi$$

$$I_2 = 2\pi, I_3 = k\pi$$

Therefore $I_k = k\pi$

370. If $I_n = \int_0^{\pi} \frac{\sin(2nx)}{\sin 2x} dx$ then the value of $I_{n+\frac{1}{2}}$ is equal to ($n \in \mathbb{I}$):

(Reduction Formula)

1) $\frac{n\pi}{2}$

2) π

3) $\frac{\pi}{2}$

4) 0

Key: 4

Sol: $I_{n+\frac{1}{2}} = \int_0^{\pi} \frac{\sin(2nx+x)}{\sin 2x} dx = \int_0^{\pi} \frac{\sin 2nx \cos x}{\sin 2x} dx + \int_0^{\pi} \frac{\cos 2nx \sin x}{\sin 2x} dx$

$$= \frac{1}{2} \int_0^{\pi} \frac{\sin 2nx}{\sin x} dx + \frac{1}{2} \int_0^{\pi} \frac{\cos 2nx}{\cos x} dx$$

371. If $u_{10} = \int_0^{\pi/2} x^{10} \sin x dx$, then the value of $u_{10} + 90u_8$ is (Reduction formula)

1. $9\left(\frac{\pi}{2}\right)^9$

2. $10\left(\frac{\pi}{2}\right)^9$

3. $\left(\frac{\pi}{2}\right)^9$

4. $9\left(\frac{\pi}{2}\right)^8$

Ans: 2

Sol: $u_{10} = \int_0^{\pi/2} x^{10} \sin x dx$

$$= 10\left(\frac{\pi}{2}\right)^9 - 90u_8$$

$$\therefore u_{10} + 90u_8 = 10\left(\frac{\pi}{2}\right)^9$$

372. If $I_n = \int_0^{\pi/2} \cos^n \cos nx dx$, then I_1, I_2, I_3 are in

(Reduction formula)

1. AP

2. GP

3. HP

4. AGP

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Ans: 2

Sol: $I_n = \int_0^{\pi/2} \cos^n x \cos nx dx$

$$I_n = \left[\cos^n x \cdot \frac{\sin nx}{n} \right]_0^{\pi/2} + \int_0^{\pi/2} n \cos^{n-1} x (-\sin x) \left(\frac{\sin nx}{n} \right) dx$$

$$\therefore \frac{I_n}{I_{n-1}} = \frac{1}{2} \Rightarrow I_1, I_2, I_3 \text{ are in GP}$$

373. If $I_n = \int_0^1 (1-x^\alpha)^n dx$, then $\frac{I_n}{I_{n+1}} = 1 + \frac{1}{k}$ where $k =$ **(Reduction formula)**

1. $(n+1)\alpha$

2. $n\alpha$

3. $(n-1)\alpha$

4. $(n-1)$

Ans: 1

Sol: We have $I_{n+1} = \int_0^1 (1-x^\alpha)^{n+1} dx$

$$= (n+1)\alpha \int_0^1 (1-x^\alpha) dx - (n+1)\alpha \int_0^1 (1-x^\alpha)^{n+1} dx$$

$$\therefore k = (n+1)\alpha$$

374. If $I(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx, (m, n \in \mathbb{I}, m, n \geq 0)$, then **(Reduction formula)**

A) $I(m, n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m-n}} dx$

B) $I(m, n) = \int_0^\infty \frac{x^m}{(1+x)^{m+n}} dx$

C) $I(m, n) = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}} dx$

D) $I(m, n) = \int_0^\infty \frac{x^n}{(1+x)^{m+n}} dx$

Key: C

Sol: putting $x = \frac{1}{1+y}, dx = \frac{1}{(1+y)^2} dy$, we get

$$\begin{aligned} I_{(m,n)} &= \int_0^1 x^{m-1} (1-x)^{n-1} dx \\ &= \int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n}} dy = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}} dx \end{aligned}$$

375. If $L(m, n) = \int_0^1 t^m (1+t)^n dt$, then the expression for $L(m, n)$ in terms of $(m+1, n-1)$ is,

$(m, n \in \mathbb{N})$

(Reduction formula)

A) $\frac{2^n}{m+1} - \frac{n}{m+1} L(m+1, n-1)$

- B) $\frac{n}{m+1}L(m+1, n-1)$
 C) $\frac{2^n}{m+1} + \frac{n}{m+1}L(m+1, n-1)$
 D) $\frac{m}{n+1}L(m+1, n-1)$

Key:A

Sol: let $I = \int_0^{\pi/4} \ln(1 + \tan x) dx$
 $= \int_0^{\pi/4} \ln(1 + \tan(\pi/4 - x)) dx$
 $\left[\because \int_0^a f(x) dx = \int_0^a f(a-x) dx \right]$
 $I = \frac{\pi}{8} \ln 2$

EXERCISE-II (Integer or Numeric value questions)

Topic: Introduction to Definite integration

Sub Topic 1: Definite integral as a limit of an infinite Reimann Sum

1. Let $f : (0, 2) \rightarrow R$ be defined as $f(x) = \log_2 \left(1 + \tan \left(\frac{\pi x}{4} \right) \right)$. Then,

$\lim_{n \rightarrow \infty} \frac{2}{n} \left(f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \dots + f(1) \right)$ is equal to _____

Key: 1

(Limit of a sum in Definite)2021

Sol: $l = \lim_{n \rightarrow \infty} \frac{2}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right) = 2 \int_0^1 f(x) dx = \int_0^1 \log_2 \left(1 + \tan \frac{\pi x}{4} \right) dx$

$l = 2 \int_0^1 \log_2 \left(1 + \tan \left(\frac{\pi}{4} - \frac{\pi}{4} x \right) \right) dx$

$2l = 2 \int_0^1 \log_2 \left(\left(1 + \tan \frac{\pi}{4} x \right) \left(\frac{2}{1 + \tan \frac{\pi}{4} x} \right) \right) dx$

$2l = 2 \cdot 1 \Rightarrow l = 1$

2. If $\lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n \sqrt{r} \sum_{r=1}^n \frac{1}{\sqrt{r}}}{\sum_{r=1}^n r} = \frac{k}{3}$ then the value of k is (Limit of a sum in Definite)

Ans: 8

Sol: $\lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n \sqrt{r} \sum_{r=1}^n \frac{1}{\sqrt{r}}}{\frac{n(n+1)}{2}} \quad \lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n \sqrt{\frac{r}{n}} \sum_{r=1}^n \frac{1}{\sqrt{\frac{r}{n}}}}{\frac{n^2}{2} \left(1 + \frac{1}{n}\right)}$

$$= 2 \times \int_0^1 \sqrt{x} \, dx \int_0^1 \frac{1}{\sqrt{x}} \, dx$$

$$= 2 \times \left[\frac{x^{3/2}}{3/2} \right]_0^1 \left[\frac{x^{1/2}}{1/2} \right]_0^1$$

$$= 2 \times \frac{2}{3} \times 2 = \frac{8}{3}$$

$\therefore k = 8$

3. $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^{1/a} \left\{ n^{a-1/a} + k^{a-1/a} \right\}}{n^{a+1}}$ is equal to **(Limit of a sum in Definite)**

Ans: 1

Sol: $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^{1/a} \left\{ n^{a-1/a} + k^{a-1/a} \right\}}{n^{a+1}}$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n} \cdot \left\{ \left(\frac{k}{n} \right)^{1/a} + \left(\frac{k}{n} \right)^a \right\}$$

$$= \int_0^1 (x^{1/a} + x^a) \, dx$$

$$= \left[\frac{x^{(1/a)+1}}{\frac{1}{a}+1} + \frac{x^{a+1}}{a+1} \right]_0^1$$

$$= \frac{a}{a+1} + \frac{1}{a+1} = 1$$

4. If $\int_0^{\pi/2} x^n \sin x \, dx = (3/4)(\pi^2 - 8)$, then the value of n is **(By parts)**

ANS: 3

SOL. Let $I_n = \int_0^{\pi/2} x^n \sin x \, dx$

Integrating by parts choosing $\sin x$ as the second function, we get

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$$I_n = [x^n (-\cos x)]_0^{\pi/2} - \int_0^{\pi/2} nx^{n-1} (-\cos x) dx$$

$$= 0 - n \int_0^{\pi/2} x^{n-1} \cos x dx$$

Again integrating by parts,

$$I_n = n[x^{n-1} \sin x]_0^{\pi/2} - n(n-1) \int_0^{\pi/2} x^{n-2} \sin x dx$$

$$\Rightarrow I_n = n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1) I_{n-2}$$

R.H.S. contains π^2 . So putting $n=3$, we get

$$\begin{aligned} I_3 &= 3 \left(\frac{\pi}{2} \right)^2 - 3 \times 2 I_1 \\ &= \frac{3\pi^2}{4} - 6 \int_0^{\pi/2} x \sin x dx \\ &= \frac{3\pi^2}{4} - 6 [x(-\cos x) + \sin x]_0^{\pi/2} \\ &= \frac{3\pi^2}{4} - 6(1) = \frac{3}{2}(\pi^2 - 8), \text{ which is true.} \end{aligned}$$

Hence, $n=3$.

Topic: Introduction to Definite integration

Sub Topic 2: Second Fundamental theorem of integral calculus - definition

5. The value of $\frac{e^{-\pi/4} + \int_0^{\pi/4} e^{-x} \tan^{50} x dx}{\int_0^{\pi/4} e^{-x} (\tan^{49} x + \tan^{51} x) dx}$ is **(BY PARTS)**

1) 51

2) 25

3) 50

4) 49

KEY:

3

SOL:

$$\begin{aligned} \int_0^{\pi/4} e^{-x} (\tan^{49} x + \tan^{51} x) dx &= \left(e^{-x} \cdot \frac{\tan^{50} x}{50} \right)_0^{\pi/4} + \int_0^{\pi/4} e^{-x} \frac{\tan^{50} x}{50} dx \\ &= \frac{e^{-\pi/4}}{50} + \frac{1}{50} \int_0^{\pi/4} e^{-x} \tan^{50} x dx \end{aligned}$$

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$$\frac{e^{-\frac{\pi}{4}} + \int_0^{\frac{\pi}{4}} e^{-x} \tan^{50} x \, dx}{\int_0^{\frac{\pi}{4}} e^{-x} (\tan^{49} x + \tan^{51} x) \, dx} = 50$$

6. If $\int_0^{\pi/2} x^n \sin x \, dx = \frac{3}{4}(\pi^2 - 8)$, then the value of n is equal to **(By parts)**

Key:3

Sol: let $I_n = \int_0^{\pi/2} x^n \sin x \, dx$

Integrating by parts choosing $\sin x$ as the second function, we get

$$\Rightarrow I_n = n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1) I_{n-2}$$

R.H.S contains π^2 , so putting $n = 3$, we get

$$I_3 = 3 \left(\frac{\pi}{2} \right)^2 - 3 \times 2 I_1$$

Hence, $n=3$

7. If the value of the definite integral $\int_0^1 {}^{207}C_7 x^{200} \cdot (1-x)^7 \, dx$ **(By parts)**

Key:8

Sol: Let $I = \int_0^x {}^{207}C_7 x^{200} \cdot (1-x)^7 \, dx$

$$= {}^{207}C_7 \cdot \frac{7}{201} \int_0^1 (1-x)^6 \cdot x^{201} \, dx$$

Integrating by parts again 6 more times, we get

$$\frac{(207)! \cdot 7!}{(207)! 7! \cdot 208} = \frac{1}{208} = \frac{1}{k} \text{ or } k = 208$$

8. Consider a real valued continuous function f such that $f(x) = \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + t f(t)) \, dt$. If M and m are maximum and minimum values of the function f, then the value of M/m is **(By parts)**

Key:3

Sol: we have $f(x) = \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + t f(t)) \, dt = \sin x + \pi \sin x + \int_{-\pi/2}^{\pi/2} t f(t) \, dt$

$$\therefore f(x) = (\pi + 1) \sin x + A$$

$$\therefore \frac{M}{m} = 3$$

Topic: Evaluation of definite integrals

Sub Topic 1: Evaluation of definite integrals by substitution

9. If $I_{m,n} = \int_0^1 x^{m-1} (1-x)^{n-1} dx$, for $m, n \geq 1$, and $\int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx = \alpha I_{m,n}, \alpha \in \mathbb{R}$, then α equals _____

Key:1

(Substitution Method)2021

Solution:

$$I_{m,n} = \int_0^1 x^{m-1} (1-x)^{n-1} dx \quad \text{Let } x = \frac{1}{t+1}$$

$$I_{m,n} = \int_0^1 \frac{-t^{n-1}}{(t+1)^{m+n}} dt = \int_0^\infty \frac{t^{n-1}}{(t+1)^{m+n}} dt \rightarrow (1)$$

$$\text{Similarly } I_{m,n} = \int_0^1 (1-x)^{m-1} x^{n-1} dx$$

$$\text{But } x = \frac{1}{y+1} \quad I_{m,n} = \int_0^\infty \frac{y^{m-1}}{(y+1)^{m+n}} dy \rightarrow (2)$$

$$(1) + (2) \Rightarrow 2I_{m,n} = \int_0^\infty \frac{t^{n-1}}{(t+1)^{m+n}} dt + \int_0^\infty \frac{y^{m-1}}{(y+1)^{m+n}} dy$$

$$= \int_0^\infty \frac{x^{n-1} + x^{m-1}}{(x+1)^{m+n}} dx = \int_0^1 \frac{x^{n-1} + x^{m-1}}{(x+1)^{m+n}} dx + \int_1^\infty \frac{x^{n-1} + x^{m-1}}{(x+1)^{m+n}} dx$$

$$\text{Put } x = \frac{1}{z} \Rightarrow \int_1^\infty \frac{x^{n-1} + x^{m-1}}{(x+1)^{m+n}} dx = \int_0^1 \frac{x^{n-1} + x^{m-1}}{(x+1)^{m+n}} dx$$

$$2I_{m,n} = 2 \int_0^1 \frac{x^{n-1} + x^{m-1}}{(x+1)^{m+n}} dx$$

$$\Rightarrow I_{m,n} = \int_0^1 \frac{x^{n-1} + x^{m-1}}{(x+1)^{m+n}} dx$$

10. If $\int_0^\pi (\sin^3 x) e^{-\sin^2 x} dx = \alpha - \frac{\beta}{e} \int_0^1 \sqrt{t} e^t dt$, then $\alpha + \beta$ is equal to _____

Key:5

(Substitution Method)2021

$$\text{Sol: } I = \int_0^\pi (\sin^3 x) e^{-\sin^2 x} dx = 2 \int_0^{\pi/2} (\sin^3 x) e^{-\sin^2 x} dx$$

$$2 \int_0^{\pi/2} (1 - \cos^2 x) \sin x \cdot e^{\cos^2 x - 1} dx$$

$$\cos^2 x = t \Rightarrow 2 \sin x \cos x dx = -dt$$

$$\text{And } \sin x dx = -\frac{dt}{\sqrt{t}}$$

$$\Rightarrow 2 \int_1^0 (1-t) e^{t-1} \cdot -\frac{dt}{\sqrt{t}}$$

apply integration by parts

$$\text{we get } I = 2 - \frac{3}{e} \int_0^1 e^x \sqrt{x} dx$$

$$11. \text{ Let } f_n = \int_0^{\frac{\pi}{2}} \left(\sum_{k=1}^n \sin^{k-1} x \right) \left(\sum_{k=1}^n (2k-1) \sin^{k-1} x \right) \cos x dx, n \in N.$$

Then $f_{21} - f_{20}$ is equal to

(SUBSTITUTION)

KEY:

41

SOL:

$$\text{Put } \sin x = t \Rightarrow \cos x dx = dt$$

$$f_n = \int_0^1 \left(\sum_{k=1}^n t^{k-1} \right) \left(\sum_{k=1}^n (2k-1) t^{k-1} \right) dt$$

$$f_2 - f_1 = 3 = 2(2) - 1$$

$$f_3 - f_2 = 5 = 2(3) - 1$$

$$f_{21} - f_{20} = 2(21) - 1 = 41$$

$$12. \text{ If } \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{\cot \theta d\theta}{\sqrt{\lambda \operatorname{cosec} \theta}} = 1 - \frac{1}{\sqrt{2}} \text{ then the value of } \lambda^2 = \underline{\hspace{2cm}}$$

Key: 4

(Substitution property)

$$\text{Sol: } \operatorname{cosec} \theta = t \Rightarrow \cot \theta d\theta = \frac{-dt}{t}$$

$$I = \int_2^1 \frac{-dt}{\sqrt{\lambda} \sqrt{t}} = \int_1^2 \frac{1}{\sqrt{\lambda}} t^{-3/2} dt = \frac{1}{\lambda} \left[\frac{t^{-3/2+1}}{-3/2+1} \right]_1^2$$

$$= \frac{-2}{\sqrt{\lambda} \sqrt{t}} \Big|_1^2 = \frac{1}{\sqrt{\lambda}} [2 - \sqrt{2}] \Rightarrow \frac{1}{\lambda} [-\sqrt{2} + 2] = \frac{\sqrt{2} - 1}{\sqrt{2}}$$

$$\Rightarrow \frac{\sqrt{2}}{\sqrt{\lambda}} (\sqrt{2} - 1) = \frac{\sqrt{2} - 1}{\sqrt{2}} \Rightarrow \lambda = 2$$

13 The value of $\int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin^4 x + \cos^4 x} dx$ is _____

Key: 2

(Substitution property)

Sol:

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin^4 x + \cos^4 x} dx &= \int_0^{\frac{\pi}{2}} \frac{2 \tan x \sec^2 x}{\tan^4 x + 1} dx = \tan^{-1} \left(\tan^2 x \right) \Big|_0^{\frac{\pi}{2}} \\ &= \tan^{-1}(\infty) - \tan^{-1}(0) = \frac{\pi}{2} = \frac{3.14}{2} = 1.57 \end{aligned}$$

14. If $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sin^4 x dx}{\sin^4 x + \cos^4 x} = \frac{\pi}{c} - \frac{1}{a\sqrt{b}} \ln 5$ then $\frac{10b+a}{c} = \underline{\hspace{2cm}}$

Key: 2

(Partial Fractions and Substitution property)

$$\text{Sol: } I = \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \left(1 - \frac{\cos^4 x - \sin^4 x}{\sin^4 x + \cos^4 x} \right) dx = \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \left(1 - \frac{\cos 2x}{1 - \frac{1}{2} \sin^2 2x} \right) dx$$

$$= \frac{1}{2} \left[x - \frac{1}{2\sqrt{2}} \log 5 \left| \frac{\sqrt{2} + \sin 2x}{\sqrt{2} - \sin 2x} \right| \right]_{\frac{\pi}{6}}^{\frac{\pi}{4}} = \frac{\pi}{24} - \frac{1}{4\sqrt{2}} \log 5$$

$$a = 4, b = 2, c = 24 \Rightarrow \frac{10b+a}{c} = \frac{24}{24} = 1$$

15. If $2^{2010} \frac{\int_0^1 x^{1004} (1-x)^{1004} dx}{\int_0^1 x^{1004} (1-x^{2010})^{1004} dx} = abcd$, where abcd is a four digit number

$$a + b + d =$$

Key: 4

(Substitution Method)

Sol: Let $I_2 = \int_0^1 x^{1004} (1-x^{2010})^{1004} dx$, Put $x^{1005} = t \Rightarrow 1005 x^{1004} dx = dt$

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$$\Rightarrow I_2 = \frac{1}{1005} \int_0^1 (1-t^2)^{1004} dt = \frac{1}{1005} \int_0^1 t(2-t)^{1004} dt$$

$$\text{Let } t = 2z \Rightarrow dt = 2dz$$

$$I_2 = \frac{1}{1005} \int_0^{\frac{1}{2}} 2^{1004} z^{1004} (2-2z)^{1004} \cdot 2dz$$

$$= \frac{2^{2009}}{1005} \int_0^{\frac{1}{2}} z^{1004} (1-z)^{1004} dz$$

$$I_1 = \int_0^1 x^{1004} (1-x)^{1004} dx = 2 \int_0^{\frac{1}{2}} x^{1004} (1-x)^{1004} \cdot x$$

$$\Rightarrow \frac{I_1}{I_2} = \frac{1005}{2^{2008}}$$

16. If $\int_0^1 \frac{4x^3(1+x^{8040})}{(1+x^4)^{2012}} dx = \frac{1}{n}$ then the sum of digits of n is

(Substitution)

Key: 4

Sol: put $u = 1 + x^4$ and $w = 1 - \frac{1}{u}$ then $du = 4x^3 dx$ and $dw = \frac{1}{u^2} du$

$$I = \int_1^2 \frac{1+(u-1)}{u^{2012}} du = \int_1^2 u^{-2012} du + \int_1^2 \left(1 - \frac{1}{u}\right) \cdot \frac{1}{u^2} du$$

$$= \frac{1}{2011}$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 1: Change of variables, interchanging limits, splitting limits Property,

Location of roots in an interval using definite integral,

Improper integral Property, Reflection property, Shifting property

17. Let $\{x\}$ and $[x]$ denote the fractional part of x and the greatest integer $\leq x$ respectively of a real number x . If $\int_0^n \{x\} dx$, $\int_0^n [x] dx$ and $10(n^2 - n)$, ($n \in \mathbb{N}, n > 1$) are three consecutive terms of a G.P., then n is equal to _____

Key: 21

(Step Function and Splitting rule) 2020

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Sol:

$$\left(\int_0^x [x] dx \right)^2 = \left(\int_0^x \{x\} dx \right)^2 10(x^2 - x) = n \left(\int_0^1 x dx \right)^2 10x(x-1)$$

$$\left(\int_0^x [x] dx - \int_0^x \{x\} dx \right)^2 = n \frac{1}{2} 10x(x-1) \quad \left(\frac{n^2}{2} - \frac{n}{2} \right)^2 = 5n^2(n-1)$$

$$\frac{n^2(n-1)^2}{4} = 5n^2(n-1) \quad n-1=20 \quad n=21$$

18. If $\int_{-a}^a (|x| + |x-2|) dx = 22, (a > 2)$ and $[x]$ denotes the greatest integer $\leq x$, then

$$\int_a^{-a} (x + [x]) dx \text{ is}$$

Key: 3

(Mod Function and Splitting rule)2021

Sol: $\int_{-a}^0 (-2x+2) dx + \int_0^2 (x+2+2) dx + \int_2^a (2x-2) dx = 22$

$$x^2 - 2x \Big|_0^{-a} + 2x \Big|_0^2 + x^2 - 2x \Big|_2^a = 22$$

$$a^2 + 2a + 4 + a^2 - 2a - (4 - 4) = 22$$

$$2a^2 = 18 \Rightarrow a = 3$$

$$\int_{-3}^3 (x + [x]) dx = -3 - 2 - 1 + 1 + 2 = -3$$

19. The value of the integral $\int_0^\pi |\sin 2x| dx$ is _____

Key : 2

(Mod Function and Splitting rule)2021

Solution:

$$\int_0^\pi |\sin 2x| dx = \frac{1}{2} \int_0^{2\pi} |\sin t| dt$$

$$= \frac{1}{2} \times 2 \int_0^\pi \sin t dt = 2$$

20. If $[.]$ represents the greatest integer function, the value of $\left| \int_0^{\sqrt{\frac{\pi}{2}}} [x^2] - \cos x dx \right|$ is _____

Key: 1

(Step Function and splitting rule)2021

Sol: $I = \int_0^{\sqrt{\pi/2}} [x^2 - \cos x] dx$

$$= \int_0^1 \left[[x^2] - \cos x \right] dx + \int_1^{\sqrt{\pi/2}} \left[[x^2] - \cos x \right] dx = \int_0^1 (-1) dx + 0 = -1 \quad \therefore |I| = 1$$

21. The value of $12 \int_0^3 |x^2 - 3x + 2| dx$ is _____

Key:22

(Spilting Of Limits)2023

Sol: $\int_0^1 (x^2 - 3x + 2) dx + \int_1^2 (-2 + 3x - x^2) dx + \int_2^3 (x^2 - 3x + 2) dx$

$$= \left(\frac{x^3}{3} - \frac{3x^2}{2} + 2x \right)_0^1 + \left(-2x + \frac{3x^2}{2} - \frac{x^3}{3} \right)_1^2 + \left(\frac{x^3}{3} - \frac{3x^2}{2} + 2x \right)_2^3$$

$$= \left(\frac{1}{3} - \frac{3}{2} + 2 \right) + \left(-4 + \frac{12}{2} - \frac{8}{3} \right) - \left(-2 + \frac{3}{2} - \frac{1}{3} \right) + \left(\frac{3^3}{3} - \frac{3(3)^2}{2} + 2(3) \right) - \left(\frac{2^3}{3} - \frac{3(2)^2}{2} + 2(2) \right)$$

$$Rv = (4 - 18 + 24) + (-48 + 72 - 32) - (-24 + 18 - 4) + (108 - 162 + 72) - (32 - 72 + 48)$$

22. Let $[t]$ denote the greatest integer $\leq t$. Then $\frac{2}{\pi} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8[\csc x] - 5[\cot x]) dx$ is equal to

KEY:14

(Greatest Integer Function)2023

SOL: $I = \frac{2}{\pi} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8[\csc x] - 5[\cot x]) dx$

$$I = \frac{2}{\pi} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8[\csc x] - 5[-\cot x]) dx$$

$$2I = \frac{2}{\pi} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (16[\csc x] + 5) dx \Rightarrow I = 14$$

23. Let $[t]$ denotes the greatest integer function. If

$$\int_0^{2.4} [x^2] dx = \alpha + \beta\sqrt{2} + \gamma\sqrt{3} + \delta\sqrt{5}, \text{ then } \alpha + \beta + \gamma + \delta \text{ is equal to}$$

KEY:6

(Greatest Integer Function)2023

SOL: $\int_0^{2.4} [x^2] dx = \int_0^1 0 + \int_1^{\sqrt{2}} 1 dx + \int_{\sqrt{2}}^{\sqrt{3}} 2 dx + \int_{\sqrt{3}}^2 3 dx + \int_2^{\sqrt{5}} 4 dx + \int_{\sqrt{5}}^{2.4} 5 dx$

$$= (\sqrt{2} - 1) + 2(\sqrt{3} - \sqrt{2}) + 3(2 - \sqrt{3}) + 4(\sqrt{5} - 2) + 5(2.4 - \sqrt{5})$$

$$= -1 - \sqrt{2} - \sqrt{3} - \sqrt{5} + 6 - 8 + 1.2$$

$$= 9 - \sqrt{2} - \sqrt{3} - \sqrt{5}$$

$$\alpha + \beta + \gamma + \delta = 9 - 1 - 1 - 1 = 6$$

24. If $m = \int_{-2}^0 \frac{|\sin x|}{\left[\frac{\lambda}{\pi}\right] + \frac{1}{2}} dx$ and $n = \int_0^2 \frac{|\sin x|}{\left[\frac{\lambda}{\pi}\right] + \frac{1}{2}} dx$ where $[.]$ is G.I.R then $m + n =$ _____

Key: 0

(Step Function and splitting rule)

Sol:

$$m = \int_{-2}^0 \frac{|\sin x|}{\left[\frac{\lambda}{\pi}\right] + \frac{1}{2}} dx \quad x = -\theta$$

$$= \int_0^2 \frac{|\sin(-\theta)|}{\left[\frac{\lambda}{\pi}\right] + \frac{1}{2}} d\theta = \int_0^2 \frac{|\sin \theta|}{-1 - \left[\frac{\theta}{\pi}\right] + \frac{1}{2}} d\theta$$

$$= - \int_0^2 \frac{|\sin \theta|}{\left[\frac{\theta}{\pi}\right] + \frac{1}{2}} d\theta = -n = m + n = 0$$

25. The value of $\int_{-2}^1 \left[x \left[1 + \cos \left(\frac{\pi x}{2} \right) \right] + 1 \right] dx$ is _____, where $[.]$ is G.I.F

Key: 2

(Step Function and splitting rule)

Sol:

$$-2 < x < -1, \left[1 + \cos\left(\frac{\pi x}{2}\right) \right] = 0$$

$$\left[x \left[1 + \cos\left(\frac{\pi x}{2}\right) \right] + 1 \right] = 1$$

$$-1 < x < 0, \left[1 + \cos\left(\frac{\pi x}{2}\right) \right] = 1 \text{ and } 0 < x+1 < 1 = \left[x \left[1 + \cos\left(\frac{\pi x}{2}\right) \right] + 1 \right] = 0$$

$$0 < x < 1 \left[1 + \cos\frac{\pi x}{2} \right] = 1 \text{ and } 1 < (x+1)2 = \left[x \left[1 + \cos\frac{\pi x}{2} \right] + 1 \right] = 1$$

$$I = \int_{-2}^{-1} 1 dx + 0 + \int_0^1 1 dx = 1 + 1 = 2$$

26. The value of $\int_{-1}^1 \left(\sin^{-1} \left[x^2 + \frac{1}{2} \right] + \cos^{-1} \left[x^2 - \frac{1}{2} \right] \right) dx$, where $[.] = GIF$

Key: 2

(Step Function and splitting rule)

Sol: $I = 2 \int_0^1 \sin^{-1} \left[x^2 + \frac{1}{2} \right] + \cos^{-1} \left[x^2 - \frac{1}{2} \right] dx$

$$\left[x^2 + \frac{1}{2} \right] = \begin{cases} 0, & 0 \leq x \leq \frac{1}{\sqrt{2}} \\ 1, & \frac{1}{\sqrt{2}} \leq x \leq 1 \end{cases} \Rightarrow \left[x^2 - \frac{1}{2} \right] = \begin{cases} -1, & 0 \leq x \leq \frac{1}{\sqrt{2}} \\ 0, & \frac{1}{\sqrt{2}} \leq x \leq 1 \end{cases}$$

$$\begin{aligned} I &= 2 \left(\int_0^{\frac{1}{\sqrt{2}}} \left(\sin^{-1}(0) + \cos^{-1}(-1) \right) dx + \int_{\frac{1}{\sqrt{2}}}^1 \left(\sin^{-1}(1) + \cos^{-1}(0) \right) dx \right) \\ &= 2 \left[(0 + \pi) x \right]_0^{\frac{1}{\sqrt{2}}} + 2 \left[\left(\frac{\pi}{2} + \frac{\pi}{2} \right) x \right]_{\frac{1}{\sqrt{2}}}^1 = 2 \left[\frac{\pi}{\sqrt{2}} + \frac{\pi}{2} \left(1 - \frac{1}{\sqrt{2}} \right) \right] = 2\pi = 6.28 \end{aligned}$$

27. $\int_{\pi/4}^{\pi/2} \left[\sin x + \left[\frac{2x}{\pi} \right] \right] dx$ where $[.]$ denotes the greatest integer function

Key: 0

(Step Function and Splitting rule)

Sol:

$$\text{Here, } I = \int_{\pi/4}^{\pi/2} \left[\sin x + \left\lfloor \frac{2x}{\pi} \right\rfloor \right] dx$$

$$\text{Also } \frac{\pi}{4} < x < \frac{\pi}{2} \Rightarrow \frac{1}{2} < \frac{2x}{\pi} < 1 \Rightarrow \left\lfloor \frac{2x}{\pi} \right\rfloor = 0$$

$$\text{So that } I = \int_{\pi/4}^{\pi/2} [\sin x] dx = 0$$

28. If $I = \int_0^2 x[2x]dx$, where $[.]$ denotes the greatest integer function, then the value of

$$\frac{4}{17}I =$$

Ans: 1

$$\text{Sol. } I = \int_0^2 x[2x]dx$$

$$= \int_0^{1/2} 0 \cdot dx + \int_{1/2}^1 x dx + \int_1^{3/2} 2x dx + \int_{3/2}^2 3x dx$$

$$= 0 + \left[\frac{x^2}{2} \right]_{1/2}^1 + [x^2]_{1/2}^{3/2} + \left[\frac{3x^2}{2} \right]_{3/2}^2$$

$$= \frac{1}{2} \left(1 - \frac{1}{4} \right) + \left(\frac{9}{4} - 1 \right) + \frac{3}{2} \left(4 - \frac{9}{4} \right)$$

$$= \frac{3}{8} + \frac{5}{4} + \frac{21}{8}$$

$$= \frac{34}{8} = \frac{17}{4}$$

$$\therefore \frac{4}{17}I = 1$$

29. If $I = \int_{-\pi/2}^{2\pi} \sin^{-1}(\sin x) dx = \frac{-\pi^2}{k}$ then $k =$

Ans: 8

(Splitting rule)

$$\text{Sol. } I = \int_{-\pi/2}^0 \sin^{-1}(\sin x) dx + \int_0^{\pi} \sin^{-1}(\sin x) dx + \int_{\pi}^{2\pi} \sin^{-1}(\sin x) dx$$

$$= -\frac{\pi^2}{8}$$

Since, $k = 8$

30. If the value of definite integral $\int_1^a x \cdot a^{-[\log_a x]} dx$ where $a > 1$, and $[\cdot]$ denotes the greatest integer, is $\frac{e-1}{2}$ then the value of $5[a]$ is ____

Key: 5

(Step Function and splitting rule)

Sol:

$$\text{Let } \log_a x = t \Rightarrow a^t = x \Rightarrow dx = a^t \log_e a$$

$$\therefore I = \ln a \int_0^1 a^t \cdot a^{-t} \cdot a^t dt = \ln a \int_0^1 a^{2t} dt = \frac{a^2 - 1}{2} = \frac{e-1}{2} \Rightarrow a = \sqrt{e}$$

31. If $\int_0^{20} [x]\{x\} dx = K$ then $\frac{K}{19}$ equal (Where $[\cdot]$ and $\{\cdot\}$ respectively denotes G.I.F and F.P.F)

ANS:5

(Greatest Integer function)

SOL

$$K = \int_0^{20} [x]\{x\} dx$$

$$K = \sum_{r=0}^{19} \int_r^{r+1} r\{x\} dx$$

$$\Rightarrow \frac{K}{19} = 5$$

32. The value of $\cot^{-1} \left(\int_0^{2\pi} [\cot^{-1} x] dx \right)$ where, $[\cdot]$ denotes the greatest integer function is

Key:1

(Greatest integer function)

$$\text{Sol: let } P = \int_0^{2\pi} [\cot^{-1} x] dx$$

$$= \int_0^{\cot^{-1} 1} 1 \cdot dx + \int_{\cot^{-1} 1}^{2\pi} 0 \cdot dx = \cot^{-1} 1 \quad \therefore \cot^{-1} P = 1$$

33. The value of $\int_0^{\frac{3\pi}{2}} \frac{|\tan^{-1} \tan x| - |\sin^{-1} \sin x|}{|\tan^{-1} \tan x| + |\sin^{-1} \sin x|} dx$ is equal to ____ (Modulus functions)

Key:0

$$\text{Sol: } \because \text{ integrand is discontinuous at } \frac{\pi}{2}, \int_0^{\pi/2} 0 \cdot dx + \int_{\pi/2}^{3\pi/2} 0 \cdot dx = 0$$

$$\because 0 < x < \frac{\pi}{2}, |\tan^{-1} \tan x| = |\sin^{-1} \sin x| \text{ and } \frac{\pi}{2} < x < \frac{3\pi}{2}$$

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$$|\tan^{-1} \tan x| \geq |\sin^{-1} \sin x|$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 2: Replacing x by $a-x$ – King's property, Replacing x by $a+b-x$

34. The value of the integral $\frac{48}{\pi^4} \int_0^\pi \left(\frac{3\pi x^2}{2} - x^3 \right) \frac{\sin x}{1 + \cos^2 x} dx$ is equal to ____.

Key: 6

(By parts and King's property)

$$\text{Sol. } I = \frac{48}{\pi^4} \int_0^\pi \left(\frac{3\pi}{2} x^2 - x^3 \right) \frac{\sin x}{1 + \cos^2 x} dx$$

$$I = \frac{48}{\pi^4} - \left[\left(\frac{3\pi}{2} x^2 - x^3 \right) \cdot \tan^{-1}(\cos x) \right]_0^\pi + \int_0^\pi (3\pi x - 3x^2) \cdot \tan^{-1}(\cos x) dx$$

$$I = \frac{48}{\pi^4} \left[\left(\frac{3\pi^3}{2} - \pi^3 \right) \frac{\pi}{4} - 0 \right] + 3 \int_0^\pi x(\pi - x) \tan^{-1}(\cos x) dx = 6 + 0 = 6 \quad (\because f(2a-x) \text{ property})$$

35. If $\int \frac{(x^2+1)e^x}{(x+1)^2} dx = f(x)e^x + C$, where C is a constant, then $\frac{d^3 f}{dx^3}$ at $x=1$ is equal to:

(A) $-\frac{3}{4}$

(B) $\frac{3}{4}$

(C) $-\frac{3}{2}$

(D) $\frac{3}{2}$

Ans: B

(By parts)

$$\text{Sol: } \int \left(\frac{x^2+1}{(x+1)^2} \right) e^x dx$$

$$= \int \left(\frac{x^2-1+2}{(x+1)^2} \right) e^x dx$$

$$= \int \left(\frac{x-1}{x+1} + \frac{2}{(x+1)^2} \right) e^x dx$$

$$= \int (f(x) + f'(x)) e^x dx$$

$$= f(x)e^x + c$$

$$\text{Where } f(x) = \frac{x-1}{x+1}$$

$$f'(x) = \frac{2}{(x+1)^2}$$

$$f''(x) = \frac{-4}{(x+1)^3}$$

$$= \frac{12}{(x+1)^4}$$

$$f''(1) = \frac{12}{16}$$

$$= \frac{3}{4}$$

36. Let $f(x)$ and $g(x)$ be two functions satisfying $f(x^2) + g(4-x) = 4x^3$ and

$g(4-x) + g(x) = 0$ then the value of $\int_{-4}^4 f(x^2) dx$ is _____

Key:512

(King's property)2021

Sol:

$$I = \int_{-4}^4 f(x^2) dx = 2 \int_0^4 f(x^2) dx = 2 \int_0^4 f((4-x)^2) dx = I$$

$$f(x^2) + g(4-x) = 4x^3$$

$$x = 4-x \quad f(4-x)^2 + g(x) = 4(4-x)^3$$

$$f(x^2) + f((4-x)^2) + g(x) + g(4-x) = 4(x^3 + (4-x)^3)$$

$$2I = 2 \int_0^4 (f(x^2) + f((4-x)^2)) dx$$

$$= 2 \int_0^4 4(x^3 + (4-x)^3) dx$$

$$2I = 8[64 - (0 - 64)] = 8 \cdot \left[\left(\frac{x^4}{4} - \frac{(4-x)^4}{4} \right) \right]_0^4$$

$$= 8(128) \quad I = 4 \times 128 = 512$$

37. The value of the integral $\frac{48}{\pi^4} \int_0^{\pi} \left(\frac{3\pi x^2}{2} - x^3 \right) \frac{\sin x}{1 + \cos^2 x} dx$ is equal to _____.

Key:6(Splitting of limits 0 to 2a as 0 to a in f(x) and 0 to 2a-x in f(2a-x))2022

$$\text{Sol. } I = \frac{48}{\pi^4} \int_0^{\pi} \left(\frac{3\pi x^2}{2} - x^3 \right) \frac{\sin x}{1 + \cos^2 x} dx$$

$$I = \frac{48}{\pi^4} - \left[\left(\frac{3\pi x^2}{2} - x^3 \right) \cdot \tan^{-1}(\cos x) \right]_0^{\pi} + \int_0^{\pi} (3\pi x - 3x^2) \cdot \tan^{-1}(\cos x) dx$$

$$I = \frac{48}{\pi^4} \left[\left(\frac{3\pi^3}{2} - \pi^3 \right) \frac{\pi}{4} - 0 \right] + 3 \int_0^{\pi} x(\pi - x) \tan^{-1}(\cos x) dx = 6 + 0 = 6 \quad (\because f(2a-x) \text{ property})$$

38. The value of $\frac{8}{\pi} \int_0^{\frac{\pi}{2}} \frac{(\cos x)^{2023}}{(\sin x)^{2023} + (\cos x)^{2023}} dx$ is _____

Key: **2** **(Replacing x by a-x - King's property)2023**

Sol: $I = \int_0^{\frac{\pi}{2}} \frac{(\cos x)^{2023}}{(\sin x)^{2023} + (\cos x)^{2023}} dx$

Using king's property $I = \int_0^{\frac{\pi}{2}} \frac{(\sin x)^{2023}}{(\cos x)^{2023} + (\sin x)^{2023}} dx$, $2I = \int_0^{\frac{\pi}{2}} 1 dx \Rightarrow I = \frac{\pi}{4}$

39. If $\int_0^{\pi} \frac{5^{\cos x} (1 + \cos x \cos 3x + \cos^2 x + \cos^3 x \cos 3x) dx}{1 + 5^{\cos x}} = \frac{k\pi}{16}$, then k is equal to _____

Key:13 **(Replacing x by a-x - King's property)2023**

Sol:

Key change : 13

Given key : 26

$$I = \int_0^{\pi} \frac{5^{\cos x} (1 + \cos x \cos 3x + \cos^2 x + \cos^3 x \cos 3x)}{1 + 5^{\cos x}} dx \dots (1)$$

$$= \int_0^{\pi} \frac{5^{-\cos x} (1 + \cos x \cos 3x + \cos^2 x + \cos^3 x \cos 3x)}{1 + 5^{-\cos x}} dx \dots (2)$$

(1) + (2)

$$\Rightarrow 2I = \int_0^{\pi} \frac{(5^{\cos x} + 1)}{5^{\cos x} + 1} (1 + \cos x \cos 3x + \cos^2 x + \cos^3 x \cos 3x) dx$$

$$\Rightarrow 2I = \int_0^{\pi} \left[1 + \frac{1}{2}(\cos 4x + \cos 2x) + \frac{1}{2}(1 + \cos 2x) \right] dx + \int_0^{\pi} \cos^3 x (4\cos^3 x - 3\cos x) dx$$

$$= \int_0^{\pi} \left[1 + \frac{1}{2}\cos 4x + \frac{1}{2}\cos 2x + \frac{1}{2} + \frac{\cos 2x}{2} \right] dx + 4 \int_0^{\pi} \cos^6 x dx - 3 \int_0^{\pi} \cos^4 x dx$$

$$= \left(x + \frac{\sin 4x}{8} + \frac{\sin 2x}{4} + \frac{x}{2} + \frac{\sin 2x}{4} \right)_0^{\pi} + 8 \left(\frac{5}{6} \times \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} \right) - 6 \left(\frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} \right)$$

$$= \pi + 0 + 0 + \frac{\pi}{2} + 0 + \frac{5\pi}{4} - \frac{9\pi}{8}$$

$$= \frac{3\pi}{2} + \frac{\pi}{8} = \frac{12\pi + \pi}{8} = \frac{13\pi}{8}$$

$$2I = \frac{13\pi}{8} \Rightarrow I = \frac{13\pi}{16}$$

$$\therefore k = 13$$

40. Let $f(x)$ be a continuous function such that $f(x) + f(2023 - x) = 0$ for all

$$x \in [0, 2023] \text{ then } \int_0^{2023} \frac{fx}{1 + e^{f(x)}} dx \text{ —————}$$

Key: 1012

(King's property)

Sol:

$$I = \int_0^{2023} \frac{fx}{1 + e^{f(2023-x)}} = \int_0^{2023} \frac{fx}{1 + e^{f(x)}}$$

$$I + I = \int_0^{2023} \left(\frac{1}{1 + e^{f(x)}} + \frac{e^{f(x)}}{e^{f(x)} + 1} \right) dx = \int_0^{2023} 1 dx$$

$$I = 1011.5$$

41. For $n > 0$ then the value of $\int_0^{2\pi} \frac{x \sin^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx$ is $a\pi^b$ then the $a^b = \text{—}$

Sol: 1

(King's property)

$$I = \int_0^{2\pi} \frac{x \sin^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx \because f(a-x) \text{ property}$$

$$\Rightarrow I = \pi \int_0^{2\pi} \frac{\sin^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx \because f(2a-x) \text{ property}$$

$$= 2\pi \int_0^{\pi} \frac{\sin^{2n} x dx}{\sin^{2n} x + \cos^{2n} x} = 4\pi \int_0^{\frac{\pi}{2}} \frac{\sin^{2n} x dx}{\sin^{2n} x + \cos^{2n} x} = 4\pi \cdot \frac{\pi}{4} = \pi^2$$

$$a^b = 1$$

42. Let $f(x) = \frac{4^x}{4^x + 2}$, $I_1 = \int_{f(1-a)}^{f(a)} xf(x(1-x))dx$ and $I_2 = \int_{f(1-a)}^{f(a)} f(x(1-x))dx$ where $f(a) > f(1-a)$
 then the value of $\frac{I_2}{I_1}$ is **(f(a + b - x) Property)**

Ans:2

Sol: $f(x) + f(1-x) = 1$

$$I_1 = \int_{f(1-a)}^{f(a)} x + (x(-x))dx$$

$$x + f(1-x) + f(x) = 1$$

$$f(1-x)$$

43. If $I = \int_{-1}^1 (1+x)^{1/2} (1-x)^{3/2} dx$, then the value of $\sec^3(I/2)$ is _____

ANS:8

(f(a + b - x) Property)

SOL. Using property $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$

$$I = \int_{-1}^1 (1-x)^{1/2} (1+x)^{3/2} dx$$

$$\Rightarrow 2I = \int_{-1}^1 (1+x)^{1/2} (1-x)^{1/2} [(1-x) + (1+x)] dx$$

$$\Rightarrow 2I = 2 \int_{-1}^1 \sqrt{1-x^2} dx$$

$$\Rightarrow I = 2 \int_0^1 \sqrt{1-x^2} dx (x = \sin \theta) \Rightarrow dx = \cos \theta d\theta$$

$$\Rightarrow I = 2 \int_0^{\pi/2} \cos^2 \theta d\theta = \frac{\pi}{2}$$

44. If $I = \int_0^{\pi} x (\sin^2(\sin x) + \cos^2(\cos x)) dx$, then $[I] = \underline{\hspace{1cm}}$, where $[.]$ denotes the greatest integer function

Key: 4

(King's property)

Sol: $I = \int_0^{\pi} (\pi - x) ((\sin^2(\sin x)) + \cos^2(\cos x)) dx \Rightarrow 2I = 2\pi \int_0^{\frac{\pi}{2}} (\sin^2(\sin x) + \cos^2(\cos x)) dx$

$$\Rightarrow I = \pi \int_0^{\frac{\pi}{2}} (\sin^2(\sin x) + \cos^2(\cos x)) dx = \pi \int_0^{\frac{\pi}{2}} (\sin^2(\cos x) + \cos^2(\sin x)) dx$$

$$\Rightarrow 2I = \pi \int_0^{\frac{\pi}{2}} 2 dx \Rightarrow I = \frac{\pi^2}{2}$$

45. Let $I_1 = \int_{-2}^2 \frac{x^6 + 3x^5 + 7x^4}{x^4 + 2} dx$ and $I_2 = \int_{-3}^1 \frac{2(x+1)^2 + 11(x+1) + 14}{x^4 + 2} dx$ then the value of $\frac{3}{50}(I_1 + I_2)$ is equal to **(f(x)=f(a+b)x)**

Key: 2

Sol: In I_2 , put $x + 1 = t$, then

$$I_2 = \int_{-2}^2 \frac{2t^2 + 11t + 14}{t^4 + 2} dt = \int_{-2}^2 \frac{2x^2 + 11x + 14}{x^4 + 2} dx$$

$$\begin{aligned} \therefore I_1 + I_2 &= \int_{-2}^2 \frac{x^6 + 3x^5 + 7x^4 + 2x^2 + 11x + 14}{x^4 + 2} dx \\ &= 2 \int_0^2 (x^2 + 7) dx = \frac{100}{3} \end{aligned}$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 3: Even - odd property

46. If $\int_{-0.15}^{0.15} |100x^2 - 1| dx = \frac{k}{3000}$, then k is equal to _____.

KEY: 575 (MODULUS & EVEN & ODD FUNCTION) 2023

SOL:

$$\text{Given } \int_{-0.15}^{0.15} |100x^2 - 1| dx = \frac{k}{3000},$$

$$\begin{aligned} I &= 2 \left(\int_0^{0.1} (1 - 100x^2) dx + \int_{0.1}^{0.15} (100x^2 - 1) dx \right) \\ &= 2 \left[0.1 - \frac{100}{3} (0.1)^3 + \frac{100}{3} \left((0.15)^3 - (0.1)^3 \right) \right] - 0.05 \\ &= 2 \left[\frac{0.2875}{3} \right] = 0.19166 = \frac{575}{3000} \end{aligned}$$

$$K = 575.$$

47. Given that

$$\int_1^y \sec^{-1} x \, dx = \lambda \text{ then } \int_{-y}^{-1} \sec^{-1} x - \tan^{-1}(\sqrt{x^2 - 1}) \, dx + \int_1^y \sec^{-1} x - \tan^{-1}(\sqrt{x^2 - 1}) \, dx = \text{---} (|y| \geq 1)$$

equals to $\pi(y-a) - b\lambda$ then $a+b = \text{---}$ **(By parts and inequality)**

Ans: 3

$$\begin{aligned} \int_{-y}^y \sec^{-1} x - \tan^{-1}(\sqrt{x^2 - 1}) \, dx &= \int_{-y}^{-1} \sec^{-1} x (\pi - \sec^{-1} x) \, dx \\ &\quad + \int_1^y \sec^{-1} x - \sec^{-1} x \, dx \end{aligned}$$

$$\begin{aligned} \text{Sol: } &= 2 \int_{-y}^{-1} \sec^{-1} x \, dx - \int_{-y}^{-1} \sec^{-1} x \\ &= 2 \int_1^y (\pi - \sec^{-1} x) \, dx - \pi(y-1) = \pi(y-1) - 2\lambda \\ \therefore a+b &= 3 \end{aligned}$$

48. For any real number x , let $[x]$ denotes the largest integer less than or equal to x . let

f be a real valued function defined on $[-10, 10]$ by $f(x) = \begin{cases} x - [x] & \text{if } [x] \text{ is odd} \\ 1 + [x] - x & \text{if } [x] \text{ is even} \end{cases}$ Then

the value of $\frac{\pi^2}{10} \int_{-10}^{10} f(x) \cos \pi x \, dx$ is

ANS:4

(Even and Odd Property)

$$\text{SOL: Given } f(x) = \begin{cases} x - [x] & \text{if } [x] \text{ is odd} \\ 1 + [x] - x & \text{if } [x] \text{ is even} \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} x - (2n-1), & \text{for } 2n-1 \leq x < 2n \\ 2n+1-x, & \text{for } 2n \leq x < 2n+1 \end{cases}$$

$f(x)$ and $\cos \pi x$ both are periodic with period 2 and both are even.

$$= \int_{-10}^{10} f(x) \cos \pi x \, dx = 2 \int_0^{10} f(x) \cos \pi x \, dx$$

$$= 10 \int_{-10}^{10} f(x) \cos \pi x \, dx$$

$$= 10 \left\{ (1-x) \left(\frac{\sin \pi x}{\pi} \right) - (-1) \left(\frac{-\cos \pi x}{\pi^2} \right) \right\}_0^1 + \left\{ (x-1) \left(\frac{\sin \pi x}{\pi} \right) - \left(\frac{\cos \pi x}{\pi^2} \right) \right\}_1^2$$

$$= 10 \left[\frac{1}{\pi^2} + \frac{1}{\pi^2} + \frac{1}{\pi^2} + \frac{1}{\pi^2} \right] = \frac{40}{\pi^2}$$

$$\Rightarrow \frac{\pi^2}{10} \int_{-10}^{10} f(x) \cos \pi x \, dx = 4$$

49. If $0 < \alpha < 1$ and $I = \int_{-1}^1 \frac{dx}{\sqrt{1-2\alpha x + \alpha^2}}$ then I equals to

(-a to a property - odd and even functions - half the integral limit property)

ANS:2

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$$\text{SOL} = \frac{1}{\alpha}[(1-\alpha)-(1+\alpha)] = 2$$

50. If $\int_{-1}^1 \frac{|x \cos \pi x|}{1+(1+|x|)^{\sin^{-1} x}} dx = k$ then $\left[\frac{1}{k}\right]$ is ([] is G.I.F)

(-a to a property - odd and even functions - half the integral limit property)

ANS3.

SOL Here $f(x) = |x \cos \pi x|$ & $h(x) = 1+|x|$ are both even and $g(x) = \sin^{-1} x$ is odd

$$\therefore I = \int_0^1 |x \cos \pi x| dx = \int_0^{\frac{1}{2}} x \cos \pi x dx + \int_{\frac{1}{2}}^1 x \cos \pi x dx = \frac{1}{\pi}$$

51. The value of the integral $\int_{-\frac{3\pi}{4}}^{\frac{5\pi}{4}} \frac{\cos x + \sin x}{1+e^{x-\frac{\pi}{4}}} dx$ is

(-a to a property - odd and even functions - half the integral limit property)

ANS0

Sol $\int_a^b f(x) = \int_a^b f(a+b-x)$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 4: Half the integral limit property (Splitting of limits 0 to 2a as 0 to a and a to 2a in f(x)), f(2a-x) property

52. $\int_0^{2\pi} x \ln \left(\frac{3+\cos x}{3-\cos x} \right) dx = \underline{\hspace{2cm}}$

(A) $\frac{\pi}{2} \ln 3$

(B) $\frac{\pi}{6} \ln 3$

(C) $\frac{\pi}{12} \ln 3$

(D) 0

Key: D

(Removal of x property)

Sol:

$$I = \int_0^{2\pi} (2\pi - x) \ln \left(\frac{3+\cos x}{3-\cos x} \right) dx \Rightarrow 2I = 2\pi \int_0^{2\pi} \ln \left(\frac{3+\cos x}{3-\cos x} \right) dx$$

$$\Rightarrow I = 2\pi \int_0^{\pi} \ln \left(\frac{3+\cos x}{3-\cos x} \right) dx = 2\pi \int_0^{\pi} \ln \left(\frac{3+\cos x}{3-\cos x} \right) dx = -I \Rightarrow I = 0$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 5: Periodic function property

53. Let $f: R \rightarrow R$ be a continuous function such that $f(x) + f(x+1) = 2$, for all $x \in R$. If

$$I_1 = \int_0^8 f(x) dx \text{ and } I_2 = \int_{-1}^3 f(x) dx, \text{ then the value of } I_1 + 2I_2 \text{ is equal to } \underline{\hspace{2cm}}$$

Key: 16 (Periodic Property)2021

Sol: $f(x) + f(x+1) = 2 \quad f(x+1) + f(x+2) = 2 \quad \Rightarrow f(x+2) = f(x)$

$$I_1 = 4 \int_0^2 f(x) dx \quad I_2 = 4 \int_{-1}^3 f(x) dx \quad x = t+1 \Rightarrow dx = dt$$

$$I_2 = \int_{-2}^2 f(t+1) dt = 2 \int_0^2 f(x+1) dt \quad I_1 + 2I_2 = 4 \int_0^2 f(x) + f(x+1) dx = 16$$

54. $\int_1^5 (|x-3| + |1-x|) dx$ is equal to

Key: 12 (Mod Function and Periodic Property)

Sol: We can represent the integrand as

$$f(x) = \begin{cases} 4-2x, & x \leq 1, \\ 2, & 1 < x < 3, \text{ we get} \\ 2x-4, & x \geq 3 \end{cases}$$

$$\int_1^3 (|x-3| + |1-x|) dx + \int_3^5 (|x-3| + |1-x|) dx$$

$$= \int_1^3 2 dx + \int_1^3 (2x-4) dx$$

$$= 2x \Big|_1^3 + (x^2 - 4x) \Big|_1^3$$

$$= 4 + 8 = 12$$

55. Let $f(x)$ be a non-negative continuous function defined on R such that

$$f(x) + f\left(x + \frac{1}{2}\right) = 3, \frac{1}{1008} \int_0^{2016} f(x) dx \text{ equals}$$

(Independence property of periodic function - Application 2 of periodic functions)

SOL Given that $f(x) + f\left(x + \frac{1}{2}\right) = 3 \dots \dots (1)$

$$f\left(x + \frac{1}{2}\right) + f(x+1) = 3 \left(\text{replace } x \text{ by } x + \frac{1}{2} \right) \dots \dots (2)$$

$$(1) - (2) \Rightarrow f(x) - f(x+1) = 0 \Rightarrow f(x+1) = f(x)$$

$\therefore f$ is periodic function with the period 1

$$= 2016 \left[\int_0^{\frac{1}{2}} f(x) dx + f\left(x + \frac{1}{2}\right) dx \right]$$

$$= 2016 \times \frac{3}{2} = 1008 \times 3$$

56. $\int_0^{\sqrt{3}} \frac{1}{1+x^2} \cdot \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx$

(Trigonometric Substitution)

A) $\frac{7}{72} \pi^2$

B) $\frac{3}{42} \pi^2$

C) $\frac{17}{72} \pi^2$

D) $7/72$

Key: A

Sol: Let $I = \int_0^{\sqrt{3}} \frac{1}{1+x^2} \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx$

$$\therefore I = \int_0^1 \frac{1}{1+x^2} \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx$$

$$\therefore I = \int_0^1 \frac{1}{1+x^2} \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx + \int_1^{\sqrt{3}} \frac{1}{1+x^2} \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx$$

$$\int_0^1 \frac{2 \tan^{-1} x}{1+x^2} dx + \int_1^{\sqrt{3}} \frac{\pi - 2 \tan^{-1} x}{1+x^2} dx$$

$$= \frac{\pi^2}{16} + \pi \left\{ \frac{\pi}{3} - \frac{\pi}{4} \right\} - \left\{ \frac{\pi^2}{9} - \frac{\pi^2}{16} \right\} = \frac{7}{72} \pi^2$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 6: Applications of inequality, Sandwich property in comparison of functions in definite, Boundedness and continuity property of definite integral,
Max-min inequality property of definite integral,
Root mean square value - RMSV of a definite integral

57. Let $f(x) = \max(|x+1|, |x+2|, |x+3|, \dots, |x+5|)$. Then $\int_{-6}^0 f(x) dx$ is equal to _____.

Key: 21

(MAXIMA AND MINIMA)2022

$$\begin{aligned} \text{Sol. } & \int_{-6}^{-3} |x+1| dx + \int_{-3}^0 |x+5| dx \\ &= -\int_{-6}^{-3} (x+1) dx + \int_{-3}^0 (x+5) dx \\ &= -\left(\frac{x^2}{2} + x\right)_{-6}^{-3} + \left(\frac{x^2}{2} + 5x\right)_{-3}^0 \\ &= -\left(\frac{9}{2} - 3 - 18 + 6\right) - \left(\frac{9}{2} - 15\right) \\ &= -\left(\frac{9-30}{2}\right) - \left(\frac{9-30}{2}\right) \\ &= 21 \end{aligned}$$

60. If $\int_0^1 \sqrt{1+x+x^3+x^4} dx \leq K$. then sum of the digits of $4K^2$ is _____

Key: 15 (RMSV of a definite integral)

$$\begin{aligned} \text{Sol: } & \left| \int_a^b f(x)g(x) dx \right|^2 \leq \left(\int_a^b f^2(x) dx \right) \left(\int_a^b g^2(x) dx \right) \\ & \int_0^1 \sqrt{(1+x)(1+x^3)} dx \leq \sqrt{\int_0^1 (1+x) dx} \times \sqrt{\int_0^1 (1+x^3) dx} \\ &= \sqrt{\left(x + \frac{x^2}{2}\right) \Big|_0^1} \sqrt{\left(x + \frac{x^4}{4}\right) \Big|_0^1} = \sqrt{\left(1 + \frac{1}{2}\right) \times \left(1 + \frac{1}{4}\right)} = \sqrt{\frac{3}{2} \times \frac{5}{2}} \\ & K = \sqrt{\frac{15}{4}} \Rightarrow 4K^2 = 15 \end{aligned}$$

61. If $I_n = \int_0^{\frac{\pi}{4}} \tan^n x dx$ and $a < I_9 < b$ then the value of $\frac{1}{a} + \frac{1}{b} + \frac{1}{ab} =$ _____

Sol: 398

(Max-Min Inequality Property using successive reduction)

$$0 < x < \frac{\pi}{4} \Rightarrow 0 < \tan x < 1$$

$$\Rightarrow \tan^n x > \tan^{n+2} x \Rightarrow I_{n+2} < I_n < I_{n-2} \Rightarrow I_{n+2} + I_n < 2I_n < I_n + I_{n-2}$$

$$\text{also } I_{n-2} + I_n = \frac{1}{n-1} \Rightarrow \frac{1}{2(n+1)} < I_n < \frac{1}{2(n-1)}$$

$$a = \frac{1}{20} \quad b = \frac{1}{18}$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{ab} = 20 + 18 + 360 = 398$$

62. $f: [0, 5] \rightarrow \mathbb{R}$, $y = f(x)$ such that $f''(x) = f''(5-x) \forall x \in [0, 5]$ $f'(0) = 1$ and $f'(5) = 7$, then evaluate $\int_1^4 f'(x) dx - 4$.

Ans: 8

(By parts)

Sol. $\int_1^4 f'(x) dx = [xf'(x)]_1^4 - \int_1^4 xf''(x) dx$

$$I = \int_1^4 xf''(x) dx = \int_1^4 (5-x)f''(5-x) dx$$

$$= 5 \int_1^4 f''(x) dx - I$$

$$I = \frac{5}{2} [f'(4) - f'(1)]$$

$$\text{So, } \int_1^4 f'(x) dx = \frac{3}{2} [f'(4) + f'(1)]$$

$$\text{Now, } f''(x) = f''(5-x) \Rightarrow f'(x) = -f'(5-x) + c$$

$$f'(0) + f'(5) = c \Rightarrow c = 8$$

$$\text{so } f'(x) + f'(5-x) = 8 \Rightarrow f'(4) + f'(1) = 8$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 7: Leibnitz's rule for differentiation under the integral sign

63. Let f be a differentiable function defined on $\left[0, \frac{\pi}{2}\right]$ such that $f(x) > 0$ and

$$f(x) + \int_0^x f(t) \sqrt{1 - (\log_e f(t))^2} dt = e, \quad \forall x \in \left[0, \frac{\pi}{2}\right]. \text{ Then } \left(6 \log_e f\left(\frac{\pi}{6}\right)\right)^2 \text{ is equal to } _$$

Key:-27

(LEIBNITZ'S RULE)2023

Sol:- $f(x) + \int_0^x f(t) \sqrt{1 - (\log_e f(t))^2} dt = e$

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By Leibnitz condition

$$f'(x) + f(x)\sqrt{1 - (\log_e f(x))^2} (1) - 0 = 0$$

$$f'(x) + f(x)\sqrt{1 - (\log_e f(x))^2} = 0$$

$$f'(x) = -f(x)\sqrt{1 - (\log_e f(x))^2}$$

$$\frac{1}{\sqrt{1 - (\log_e f(x))^2}} \frac{f'(x)}{f(x)} = -1$$

Integrate on both sides

$$\int \frac{1}{\sqrt{1 - (\log_e f(x))^2}} \frac{f'(x)}{f(x)} dx = -\int dx$$

$$\log_e f(x) = t$$

Differentiate on both sides w.r.t to x

$$\frac{f'(x)}{f(x)} dx = dt$$

$$\int \frac{1}{\sqrt{1 - t^2}} dt = -\int dx$$

$$\sin^{-1}(t) = -x + c$$

$$\sin^{-1}(\log_e f(x)) = -x + c \dots \dots (1)$$

$$\text{Put } x = 0 \Rightarrow \sin^{-1}(\log_e f(0)) = 0 + c$$

$$\therefore f(0) = e$$

$$\sin^{-1}(1) = c \Rightarrow c = \frac{\pi}{2}$$

From (1)

$$\sin^{-1}(\log_e f(x)) = -x + \frac{\pi}{2} \quad \log_e f(x) = \sin\left(\frac{\pi}{2} - x\right)$$

$$\log_e f(x) = \cos x$$

$$\text{Put } x = \frac{\pi}{6} \quad \log_e f\left(\frac{\pi}{6}\right) = \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$$

$$6 \log_e f\left(\frac{\pi}{6}\right) = 6 \left(\frac{\sqrt{3}}{2}\right) = 3\sqrt{3}$$

$$\text{S.O.B} \quad \left(6 \log_e f\left(\frac{\pi}{6}\right)\right)^2 = (3\sqrt{3})^2 = 27$$

64. If $\int_{\sin x}^1 t^2 f(t) dt = 1 - \sin x, x \in \left(0, \frac{\pi}{2}\right)$, then the value of $f\left(\frac{1}{\sqrt{3}}\right)$ is—

Key: 3

(Leibnitz's rule)

Sol:

$$\int_{\sin x}^1 t^2 f(t) dt = 1 - \sin x$$

Simillary w.r to x

$$0 - \sin^2 x \int (\sin x) \cos x = -\cos x$$

$$\int (\sin x) = \frac{1}{\sin^2 x}$$

$$f(x) = \frac{1}{x^2} \Rightarrow f\left(\frac{1}{\sqrt{3}}\right) = 3$$

65. $\lim_{x \rightarrow 0} \frac{\int_0^x \cos(t^2) dt}{x} =$

Key: 1

(Leibnitz's rule)

Sol: The limit is in indeterminate form $0/0$. The integral with a variable upper limit
$$\int_0^x \cos(t^2) dt \text{ has derivative}$$

$$\left(\int_0^x \cos(t^2) dt \right)' = \cos(x^2)$$

Therefore applying L hospital rule we obtain $\lim_{x \rightarrow 0} \frac{\int_0^x \cos(t^2) dt}{x} = \lim_{x \rightarrow 0} \frac{\cos(x^2)}{1} = 1$

66. $\lim_{x \rightarrow \infty} \frac{\int_0^x \left(\ln(t + \sqrt{1+t^2}) - \ln(1+t) \right) dt}{x+1}$ is equal to

Key: 2

(Leibnitz's rule)

Sol: The limit is in (∞, ∞) form. Therefore applying L' Hospital's Rule and FTCI we get

$$\begin{aligned} & \lim_{x \rightarrow \infty} \left[\ln(x + \sqrt{1+x^2}) - \ln(1+x) \right] \\ &= \lim_{x \rightarrow \infty} \ln \left(\frac{x + \sqrt{1+x^2}}{1+x} \right) = \lim_{x \rightarrow \infty} \ln \left(\frac{1 + \sqrt{\frac{1}{x^2} + 1}}{\frac{1}{x} + 1} \right) \ln = 2 \end{aligned}$$

67. If $f(x) = a \cos(\pi x) + b$, $f'\left(\frac{1}{2}\right) = \pi$ and $\int_{1/2}^{3/2} f(x) dx = \frac{2}{\pi} + 1$, then find the value of,

$$-\frac{12}{\pi} \left(\sin^{-1} a + \cos^{-1} b \right)$$

(Leibnitz's rule)

Ans: 6

sol: $f'(x) = -a\pi \sin(\pi x)$

$$\Rightarrow f'\left(\frac{1}{2}\right) = -a\pi \sin \frac{\pi}{2} = -a\pi = \pi \Rightarrow a = -1$$

$$\int (a \cos \pi x + b) dx = \left(\frac{a \sin \pi x}{\pi} + bx \right)_{1/2}^{3/2} = \left(\frac{-a}{\pi} + \frac{3b}{2} \right) - \left(\frac{a}{\pi} + \frac{b}{2} \right)$$

$$\Rightarrow \frac{2a}{\pi} + b = \frac{2}{\pi} + 1 \Rightarrow b = 1$$

$$\text{So, } -\frac{12}{\pi} \left(\sin^{-1}(-1) + \cos^{-1} 1 \right) = -\frac{12}{\pi} \left(-\frac{\pi}{2} + 0 \right) = 6$$

68. The function $f(x) = \int_1^x \{2(t-1)(t-2)^3 + 3(t-1)^2(t-2)^2\} dt$ attains its maximum at x is equal to

Ans: 1

(Leibnitz's rule)

Sol: $f'(x) = 2(x-1)(x-2)^3 + 3(x-1)^2(x-2)^2$

$$= (x-1)(x-2)^2 \{2(x-2) + 3(x-1)\}$$

$$= (x-1)(x-2)^2 (5x-7)$$

sign change of $f'(x)$ from +ve to -ve at $x = 1$

$\therefore$ maximum at $x = 1$.

69.

If $\int_0^{x^2(1+x)} f(t) dt = x$, then the value of $10f(2)$ must be

Ans: 2

(Leibnitz's rule)

Sol. Differentiating both sides w.r.t. x , then

$$f(x^2(1+x)) \times (2x + 3x^2) = 1$$

$$\text{At } x=1 \Rightarrow f(2) = \frac{1}{5} \therefore 10f(2) = 2$$

70.

If $f(x) = \int_a^x \frac{1}{f(x)} dx$ and $\int_a^1 \frac{1}{f(x)} dx = \sqrt{2}$, then the value of $f(2) =$

Ans: 2

(Leibnitz's rule)

Sol. Since, $f(x) = \int_a^x \frac{1}{f(x)} dx$

Differentiating both sides w.r.t. x , then

$$f'(x) = \frac{1}{f(x)} \Rightarrow 2f(x)f'(x) = 2$$

Integrating both sides, then

$$(f(x))^2 = 2x + c$$

$$\therefore f(x) = \sqrt{(2x + c)}$$

$$\text{But } \int_0^1 \frac{1}{f(x)} dx = \sqrt{2}$$

$$\text{And } f(1) = \int_a^1 \frac{1}{f(x)} dx = \sqrt{2}$$

$$\Rightarrow \sqrt{(2+c)} = \sqrt{2}$$

$$\therefore c = 0$$

$$\text{Then, } f(x) = \sqrt{2x}$$

71. If $\int_{\cos x}^1 t^2 f(t) dt = 1 - \cos x \quad \forall x \in \left(0, \frac{\pi}{2}\right)$, then $\left[f\left(\frac{\sqrt{3}}{4}\right) \right] = ([.] \text{ denotes the G.I.F})$

Key: 5

(Leibnitz's rule)

Sol: Conceptual

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 8: Miscellaneous properties

71. Let for $x \in \mathbb{R}$, $S_0(x) = x$, $S_k(x) = C_k x + k \int_0^x S_{k-1}(t) dt$, where

$C_0 = 1, C_k = 1 - \int_0^1 S_{k-1}(x) dx, k = 1, 2, 3, \dots$. Then $S_2(3) + 6C_3$ is equal to _____

KEY: 18

(IMPORTANT

EXPANSIONS)

$$\text{SOL: } s_1(x) = c_1 \cdot x + \int_0^x t dt = c_1 \cdot x + \left[\frac{t^2}{2} \right]_0^x = c_1 \cdot x + \frac{x^2}{2} = \frac{x}{2} + \frac{x^2}{2} = \frac{x^2 + x}{2}$$

$$c_1 = 1 - \int_0^1 x dx = 1 - \frac{1}{2} = \frac{1}{2} \quad s_2(x) = c_2 \cdot x + 2 \int_0^x \left(\frac{t^2 + t}{2} \right) dt = c_2 x + \frac{x^3}{3} + \frac{x^2}{2} \quad c_2 = 1 - \int_0^1 \frac{x^2 + x}{2} dx$$

$$= 1 - \frac{1}{2} \left(\frac{1}{3} + \frac{1}{2} \right) = 1 - \frac{5}{12} = \frac{7}{12} \quad s_2(x) = \frac{7}{12} x + \frac{x^3}{3} + \frac{x^2}{2} \Rightarrow s_2(3) = \frac{7}{4} + 9 + \frac{9}{2} = \frac{61}{4}$$

$$c_3 = 1 - \int_0^1 \left(\frac{7}{12} x + \frac{x^3}{3} + \frac{x^2}{2} \right) dx = 1 - \left(\frac{7}{12} \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{3} \right) \quad c_3 = 1 - \left(\frac{7}{24} + \frac{1}{12} + \frac{1}{6} \right)$$

$$= 1 - \left(\frac{7+2+4}{24} \right) = 1 - \frac{13}{24} = \frac{11}{24} \quad s_3(3) + 6c_3 = \frac{61}{4} + 6 \cdot \frac{11}{24} = \frac{72}{4} = 18$$

72. Let $p(x)$ be a real polynomial of degree 3 which vanishes at $x = -3$. Let $p(x)$ have

local minima at $x = 1$ local maxima at $x = -1$ and $\int_{-1}^1 P(x) dx = 18$, then the sum of all

the coefficients of the polynomial $p(x)$ is equal to

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Key:8

(Max-min inequality property of definite integral)2021

Sol: $P'(x) = k(x^2 - 1)$ $P(x) = k\left(\frac{x^3}{3} - x\right) + c$

$$P(-3) = 0 \Rightarrow -6k + c = 0 \Rightarrow c = 6k$$

$$P(x) = k\left(\frac{x^3}{3} - x + 6\right)$$

$$\int_{-1}^1 P(x) dx = k \left[(6x) \right]_{-1}^1 \Rightarrow 12k = 18$$

$$P(x) = \frac{x^3}{2} - \frac{3x}{2} + 9$$

$$\therefore \text{Sum of coefficients} = \frac{1}{2} - \frac{3}{2} + 9 = 8$$

73. If p, q, r are real and

$$\int_0^1 (px^2 + qx + r) dx = \frac{1}{6} (f(a) + f(b) + \lambda f(c)) \text{ then } a + b + \lambda c = \underline{\hspace{2cm}}$$

Key: 3

(Miscellaneous)

Sol: $\int_0^1 (px^2 + qx + r) dx = \frac{p}{3} + \frac{q}{2} + r$

$$f(0) = r, f(1) = p + q + r, f\left(\frac{1}{2}\right) = \frac{p}{4} + \frac{q}{2} + r$$

$$a = 0, b = 1, \lambda = 4, c = \frac{1}{2}$$

$$a + b + \lambda c = 0 + 1 + 2 = 3$$

74. Let $f(x) = x^3 - \frac{3x^1}{2} + x + \frac{1}{4}$

Then the values of $\left(\int_{1/4}^{3/4} f(f(x)) dx \right)^{-11}$ is

(Simplification)

Key:3

Sol: Given $f(x) = x^3 - \frac{3x^2}{2} + x + \frac{1}{4} = \frac{1}{4}(4x^3 - 6x^2 + 4x + 1)$

$$= \frac{1}{4} [x^4 - (1-x)^4] + \frac{2}{4}$$

$$\therefore f(x) + f(1-x) = \frac{2}{4} + \frac{2}{4} = 1$$

$$\therefore I^{-1} = 4$$

Topic: Evaluation of definite integrals

Sub Topic 3: Reduction formulae in definite integrals (Wallis' Formula)

75. Let $I_n = \int_1^e x^{19} (\log|x|)^n dx$, where $n \in \mathbb{N}$. If $(20)I_{10} = \alpha I_9 + \beta I_8$, for natural numbers α and β , then $\alpha - \beta$ is equal to _____

Key: 1

(By parts and Successive Reduction)2021

Sol: $I_x = \int_1^e x^{19} (\log|x|)^n dx \Rightarrow I_{10} = \int_1^e x^{19} (\log)^{10} dx$

$$I_{10} = (\log x)^{10} \frac{x^{20}}{20} \Big|_1^e - \frac{10}{20} \int_1^e x^{19} (\log x)^9 dx. \text{ Integration by parts}$$

$$I_{10} = \frac{e^{20}}{20} - \frac{10}{20} I_9 \Rightarrow 20I_{10} = 9I_8 + 10I_9$$

$$I_9 = \frac{e^{20}}{20} - \frac{9}{20} I_8$$

$$\alpha = 10, \beta = 9, \alpha - \beta = 1$$

76. For $m, n > 0$, let $\alpha(m, n) = \int_0^2 t^m (1+3t)^n dt$. If $11\alpha(10, 6) + 18\alpha(11, 5) = p(14)^6$, then p is equal to _____

KEY:

32

(REDUCTION)2023

SOL: $\alpha(m, n) = \int_0^2 \underset{\downarrow II}{t^m} \underset{\downarrow I}{(1+3t)^n} dt$

$$\frac{t^{m+1}}{m+1} (1+3t)^n \Big|_0^2 - \int_0^2 \frac{t^{m+1}}{m+1} n(1+3t)^{n-1} (3) dt$$

$$\alpha(m, n) = \frac{2^{m+1} (7)^n}{m+1} - \frac{n}{m+1} \int_0^2 t^m (1+3t)^{n-1} (3t+1-1) dt$$

$$= \frac{2^{m+1} 7^n}{m+1} - \frac{n}{m+1} \alpha(m, n) + \frac{n}{m+1} \alpha(m, n-1) - \frac{(m+1+n)}{m+1} \alpha(m, n) - \frac{n}{m+1} (\alpha(m, n-1))$$

$$= \frac{2^{m+1} 7^n}{m+1}$$

$$(m+n+1) \alpha(m, n) - n\alpha(m, n-1) = 2^{m+1} 7^n$$

$$\alpha(m, n) = \int_0^2 \underset{\downarrow I}{t^m} \underset{\downarrow II}{(1+3t)^n} dt$$

$$\frac{(1+3t)^{n+1}}{3(n+1)} t^m - \int_0^2 \frac{(1+3t)^{n+1}}{3(n+1)} m t^{m-1} dt$$

$$\left. \frac{(1+3t)^{n+1} t^m}{3(n+1)} \right|_0^2 - \frac{m}{3(n+1)} \int_0^2 (1+3t)^n t^{m-1} (1+3t) dt$$

$$\alpha(m, n) = \frac{7^{n+1} 2^m}{3(n+1)} - \frac{m}{3(n+1)} \alpha(m-1, n) - \frac{m}{(n+1)} \alpha(m, n)$$

$$\frac{(m+n+1) \alpha(m, n)}{n+1} + \frac{m}{3(n+1)} \alpha(m-1, n) = \frac{7^{n+1} 2^m}{3(n+1)}$$

$$\Rightarrow 3(m+n+1) \alpha(m, n) + m \alpha(m-1, n) = 7^{n+1} 2^m \quad \dots(1)$$

$$\Rightarrow 3(m+n+1) \alpha(m, n) - 3n \alpha(m, n-1) = 3 \cdot 2^{m+1} 7^n \quad \dots(2)$$

$$(2) - (1)$$

$$\Rightarrow m \alpha(m-1, n) + 3n \alpha(m, n-1) = 7^n \cdot 2^m (7-6)$$

$$11\alpha(10, 6) + 18\alpha(11, 5) = 11\alpha(10, 6) + 18\alpha(11, 5) = 7^6 \cdot 2^4$$

$$\begin{aligned} \text{By taking } m=11, n=6 & \quad = (14)^6 \cdot 2^5 \\ & \quad = 32 \cdot (14)^6 \end{aligned}$$

By comparing given expression we get $p=32$

77. If $I_n = \int_0^\pi x^n \sin x dx$ and $I_5 + 20I_3 = \pi^k$, then the value of k is **(Reduction property)**

ANS:5

SOL. $I_n = \int_0^\pi x^n \sin x dx$

$$= [-x^n \cos x]_0^\pi + n \int_0^\pi x^{n-1} \cos x dx$$

$$= \pi^n + n[x^{n-1} \sin x]_0^\pi - n(n-1) \int_0^\pi x^{n-2} \sin x dx$$

$$\Rightarrow I_n = \pi^n + n \times 0 - n(n-1) I_{n-2}$$

Putting $n=5$, we get

$$I_5 = \pi^5 - 20I_3$$

$$\Rightarrow I_5 + 20I_3 = \pi^5$$

ADVANCE MODEL QUESTIONS:**EXERCISE - III
(MULTI CORRECT)****Topic:** Introduction to Definite integration**Sub Topic 1:** Definite integral as a limit of an infinite Reimann Sum

1. Let $f(x) = \lim_{n \rightarrow \infty} \left(\frac{n^n (x+n) \left(x + \frac{n}{2}\right) \dots \left(x + \frac{n}{n}\right)}{n! \left(x^2 + n^2\right) \left(x^2 + \frac{n^2}{4}\right) \dots \left(x^2 + \frac{n^2}{n^2}\right)} \right)^{\frac{x}{n}}$, for all $x > 0$. Then

[2016-P2 (Q43)]

A. $f\left(\frac{1}{2}\right) \geq f(1)$

B. $f\left(\frac{1}{3}\right) \leq f\left(\frac{2}{3}\right)$

C. $f'(2) \leq 0$

D. $\frac{f'(3)}{f(3)} \geq \frac{f'(2)}{f(2)}$

Ans: B C

(Limits using definite integration)

Sol: $f(x) = \lim_{n \rightarrow \infty} \left(\frac{\prod_{r=1}^n \left(1 + \frac{rx}{n}\right)}{\prod_{r=1}^n \left(1 + \left(\frac{rx}{n}\right)^2\right)} \right)^{\frac{x}{n}}$

$$= e^{\int_0^x \ln(1+xy) dy - \ln(1+(xy)^2) dy} = e^{\int_0^x \ln(1+t) dt - \ln(1+t^2) dt}$$

$$f'(x) = f(x) \ln\left(\frac{1+x}{1+x^2}\right)$$

For $x \in (0,1)$ it is increasing function

$$f'(2) = f(2) \ln\left(\frac{3}{5}\right) < 0$$

$$\frac{f'(3)}{f(3)} = \ln\left(\frac{2}{5}\right), \frac{f'(2)}{f(2)} = \ln\left(\frac{3}{5}\right)$$

2. For $a \in \mathbb{R}, |a| > 1$, let $\lim_{n \rightarrow \infty} \left(\frac{1 + \sqrt[3]{2} + \dots + \sqrt[3]{n}}{n^{7/3} \left(\frac{1}{(an+1)^2} + \frac{1}{(an+2)^2} + \dots + \frac{1}{(an+n)^2} \right)} \right) = 54$. Then the possible value(s) of a is/are [2019-P2 (Q6)]
- A. -9 B. -6 C. 7 D. 8

Ans: A D

(Limit sum, & limits)

Sol: $\lim_{x \rightarrow \infty} \frac{\sqrt[3]{1} + \sqrt[3]{2} + \dots + \sqrt[3]{n}}{n^{7/3} \left(\frac{1}{(xa+1)^2} + \frac{1}{(xa+2)^2} + \dots + \frac{1}{(xa+n)^2} \right)} = 54$

$$= \lim_{x \rightarrow \infty} \frac{\sum_{r=1}^n r^{1/3}}{n^{7/3} \left(\sum_{r=1}^n \frac{1}{(na+r)^2} \right)} = \lim_{x \rightarrow \infty} \frac{\sum_{r=1}^n \left(\frac{r}{n} \right)^{1/3}}{\sum_{r=1}^n \left(\frac{1}{\left(a + \frac{r}{n} \right)^2} \right)}$$

$$= \frac{\int_0^1 x^{1/3} dx}{\int_0^1 \frac{dx}{\left(a + x \right)^2}} = \frac{\frac{3}{4} x^{4/3} \Big|_0^1}{-\left(\frac{1}{a+x} \right) \Big|_0^1} = \frac{\frac{3}{4}}{-\left(\frac{1}{a+1} - \frac{1}{a} \right)} = 54$$

$$\Rightarrow \frac{3}{4 \left(\frac{1}{a(a+1)} \right)} = 54 \Rightarrow a = 8 \text{ or } a = -9$$

3. The value of $\lim_{n \rightarrow \infty} \left(\frac{\sqrt{n}}{(3+4\sqrt{n})^2} + \frac{\sqrt{n}}{\sqrt{2}(3\sqrt{2}+4\sqrt{n})^2} + \dots + \frac{1}{49n} \right)$ is less than or equal to
- A. $\frac{1}{14}$ B. $\frac{2}{7}$ C. $\frac{3}{7}$ D. $\frac{1}{20}$

Ans: A B C

Sol. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{\sqrt{r}}{\sqrt{r}(3\sqrt{r}+4\sqrt{n})^2}$

Put $\frac{r}{n} = x \Rightarrow \frac{1}{n} = dx$

$$= \int_0^1 \frac{dx}{\sqrt{x}(3\sqrt{x}+4)^2} = \frac{1}{14}$$

4. If $\lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n \sqrt{r} \sum_{r=1}^n \frac{1}{\sqrt{r}}}{\sum_{r=1}^n r} = \frac{k}{3}$ then

A. $k < 0$

B. $k > 0$

C. The number of factors of k is perfect square.

D. $k \geq 4$

Ans: B C D

Hint: $\lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n \sqrt{r} \sum_{r=1}^n \frac{1}{\sqrt{r}}}{\frac{n(n+1)}{2}} \lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n \sqrt{\frac{r}{n}} \sum_{r=1}^n \frac{1}{\sqrt{\frac{r}{n}}}}{\frac{n^2}{2} \left(1 + \frac{1}{n}\right)}$

$$= 2 \times \int_0^1 \sqrt{x} dx \int_0^1 \frac{1}{\sqrt{x}} dx = 2 \times \left[\frac{x^{3/2}}{\frac{3}{2}} \right]_0^1 \left[\frac{x^{1/2}}{\frac{1}{2}} \right]_0^1 = 2 \times \frac{2}{3} \times 2 = \frac{8}{3} \therefore k = 8$$

Topic: Introduction to Definite integration**Sub Topic 2:** Second Fundamental theorem of integral calculus - definition

5. If $\int_1^3 x^2 F'(x) dx = -12$ and $\int_1^3 F''(x) dx = 40$, then the correct expression(s) is(are)

[2015-P2 (Q60)]

A. $9f'(3) + f'(1) - 32 = 0$

B. $\int_1^3 f(x) dx = 12$

C. $9f'(3) - f'(1) + 32 = 0$

D. $\int_1^3 f(x) dx = -12$

Ans: C D (Second Fundamental theorem of integral calculus - definition of definite integral)

Sol: $\int_1^3 f(x) dx = \int_1^3 x F'(x) dx$

$$= \left[\frac{x^2}{2} F(x) \right]_1^3 - \frac{1}{2} \int_1^3 x^2 F''(x) dx$$

$$= \frac{9}{2} F(3) - \frac{1}{2} F(1) + 6 = -12$$

$$40 = \left[x^3 f'(x) \right]_1^3 - 3 \int_1^3 x^2 F'(x) dx$$

$$40 = 27 F'(3) - F'(1) + 36 \quad \dots\dots(i)$$

$$f'(x) = F(x) + x F'(x)$$

$$f'(3) = F(3) + 3 F'(3)$$

$$f'(1) = F(1) + F'(1)$$

$$9 f'(3) - f'(1) + 32 = 0$$

Topic: Evaluation of definite integrals

Sub Topic 1: Evaluation of definite integrals by substitution

6. Let $f(x) = 7 \tan^8 x + 7 \tan^6 x - 3 \tan^4 x - 3 \tan^2 x$ for all $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then the correct expression(s) is(are) **[2015-P2 (Q55)]**

A. $\int_0^{\pi/4} x f(x) dx = \frac{1}{12}$

B. $\int_0^{\pi/4} f(x) dx = 0$

(Substitution method)

C. $\int_0^{\pi/4} x f(x) dx = \frac{1}{6}$

D. $\int_0^{\pi/4} f(x) dx = 1$

Ans: A B

Sol: $f(x) = (7 \tan^6 x - 3 \tan^2 x)(\tan^2 x + 1)$

$$\int_0^{\pi/4} f(x) dx = \int_0^{\pi/4} (7 \tan^6 x - 3 \tan^2 x) \sec^2 x dx$$

$$\Rightarrow \int_0^{\pi/4} f(x) dx = 0$$

$$\int_0^{\pi/4} x f(x) dx = \left[x \int f(x) dx \right]_0^{\pi/4} - \int_0^{\pi/4} \left[\int f(x) dx \right] dx$$

$$\int_0^{\pi/4} x f(x) dx = \frac{1}{12}$$

7. Consider the equation

$$\int_1^e \frac{(\log_e x)^{\frac{1}{2}}}{x \left(a - (\log_e x)^{\frac{3}{2}} \right)^2} dx = 1, \quad a \in (-\infty, 0) \cup (1, \infty).$$

Which of the following statement(s) is/are TRUE?

[2022-P1 (Q9)]

- A. No a satisfies the above equation
- B. An integer a satisfies the above equation
- C. An irrational number a satisfies the above equation
- D. More than one a satisfy the above equation

Ans: C D

(Properties of evaluation definite integrals)

Sol: $\int_1^e \frac{(\ln x)^{\frac{1}{2}}}{x \left(a - (\ln x)^{\frac{3}{2}} \right)^2} dx = 1$

$$\ln x = t \Rightarrow x = e^t \begin{cases} x=1 \Rightarrow t=0 \\ x=e \Rightarrow t=1 \end{cases}$$

$$\Rightarrow \int_0^1 \frac{t^{\frac{1}{2}} \cdot e^t}{e^t \left(a - t^{\frac{3}{2}} \right)^2} dt = 1$$

$$\text{Now } a - t^{\frac{3}{2}} = z \begin{cases} t=0 \Rightarrow z=a \\ t=1 \Rightarrow z=a-1 \end{cases}$$

$$\frac{-3}{2} t^{\frac{1}{2}} dt = dz$$

$$\Rightarrow \int_a^{a-1} \frac{-2}{z^2} dz = 1$$

$$\Rightarrow \frac{-2}{3} \int_a^{a-1} \frac{1}{z^2} dz = 1$$

$$\Rightarrow \left[\frac{1}{z} \right]_a^{a-1} = \frac{3}{2}$$

$$\frac{1}{a-1} - \frac{1}{a} = \frac{3}{2}$$

$$\frac{a-a+1}{a(a-1)} = \frac{3}{2}$$

$$a^2 - a = \frac{3}{2}$$

$$3a^2 - 3a - 2 = 0$$

$$a = \frac{3 + \sqrt{9+24}}{6} \Big| a = \frac{3 - \sqrt{9+24}}{6}$$

$$a = \frac{3 + \sqrt{33}}{6} \Big| a = \frac{3 - \sqrt{33}}{6}$$

$$a = 1.45 \Big| a = -0.45$$

8. For $0 < a < 1, -1 < x < 1$. Let $f(x) = \frac{a-x}{1-ax}$ and If $\int_0^a \frac{1-(f(x))^6}{1-x^2} dx = f(a)$ then

A. $f(a)$ is increasing function

B. $f(a)$ is an odd function

C. $f(1) = \frac{23}{15}$

D. $f(x) = 0$ has exactly one real root.

Ans: B C

Sol: Let $t = \frac{a-x}{1-ax} \Rightarrow x = \frac{t-a}{at-1} \Rightarrow dx = \frac{a^2-1}{(at-1)^2} dt \quad \therefore 1-x^2 = 1 - \left(\frac{t-a}{at-1}\right)^2 = \frac{(a^2-1)(t^2-1)}{(at-1)^2}$

$$\therefore \int_0^a \frac{1-t^6}{t^2-1} dt = \int_0^a \frac{t^6-1}{t^2-1} dt = \frac{a^5}{5} + \frac{a^3}{3} + a = f(a) \quad f(1) = \frac{1}{5} + \frac{1}{3} + 1 = \frac{3+5+\sqrt{5}}{15} = \frac{23}{15}$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 1: Change of variables, interchanging limits, splitting limits Property,

Location of roots in an interval using definite integral, Improper integral Property, Reflection property, Shifting property

9. Let α and $f(\alpha)$ be the eccentricity of the ellipse $\frac{x^2}{3b^2-2a^2} + \frac{y^2}{2b^2-a^2} = 1$ ($3b^2 > 2a^2$) and

$$\frac{x^2}{2b^2-a^2} + \frac{y^2}{b^2} = 1 \quad (2b^2 > a^2) \text{ then}$$

A. $f(\alpha) = \frac{\alpha}{\sqrt{1-\alpha^2}}$

B. $\int_0^{\frac{1}{2}} f(f(f(f(\alpha)))) d\alpha = \frac{1}{4}$

$$C. \int e^{\alpha} (f(\alpha) - f''(\alpha)) d\alpha = e^{\alpha} \left(\frac{1}{(1-\alpha^2)^{\frac{3}{2}}} + \frac{\alpha}{\sqrt{1-\alpha^2}} \right) + c$$

$$D. f(\alpha) = \frac{\alpha}{2\sqrt{1-\alpha^2}}, b \in \mathbb{R} - \{0\}$$

Ans: A B C

$$\text{SOL: } [f(\alpha) = \frac{\alpha}{\sqrt{1-\alpha^2}}$$

$$\int e^{\alpha} (f(\alpha) - f''(\alpha)) dx = e^{\alpha} (f(\alpha) - f'(\alpha)) + c$$

$$= e^{\alpha} \left(\frac{\alpha}{\sqrt{1-\alpha^2}} + \frac{1}{\sqrt{(1-\alpha^2)^3}} \right) + c$$

$$f(f(f(f(\alpha)))) = \frac{\alpha}{\sqrt{1-4\alpha^2}} \Rightarrow \int_0^{\frac{1}{2}} \frac{\alpha}{\sqrt{1-4\alpha^2}} d\alpha$$

$$= \frac{1}{4}$$

$$10. \text{ Let } f(x) = \int_0^1 |t-x| dt \text{ then}$$

- A. $f(x)$ is a constant function
- B. $f(x)$ is continuous for all $x \in \mathbb{R}$
- C. $f(x)$ is differentiable for all $x \in \mathbb{R}$
- D. $f(x)$ is not differentiable at $x=0, 1$

Ans: B C

$$\text{Sol: } f(x) = \begin{cases} \int_0^1 (t-x) dt & x \leq 0 \\ \int_0^x (x-t) dt + \int_x^1 (t-x) dt & 0 < x < 1 \\ \int_0^1 (x-t) dt & x \geq 1 \end{cases} \quad f(x) = \begin{cases} \frac{1}{2} - x & x \leq 0 \\ \frac{1}{2} - x + x^2 & 0 < x < 1 \\ x - \frac{1}{2} & x \geq 1 \end{cases}$$

Which is continuous and differentiable for all x .

$$= \frac{4}{\pi} \cdot \frac{\pi}{4} = 1.$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 4: Half the integral limit property (Splitting of limits 0 to 2a as 0 to a and a to 2a in f(x)), f(2a-x) property

11. The option(s) with the values of a and L that satisfy the following equation is(are)

$$\frac{\int_0^{4\pi} e^t (\sin^6 at + \cos^4 at) dt}{\int_0^{\pi} e^t (\sin^6 at + \cos^4 at) dt} = L ? \quad [2015-P2 (Q53)]$$

A. $a = 2, L = \frac{e^{4\pi} - 1}{e^{\pi} - 1}$

B. $a = 2, L = \frac{e^{4\pi} + 1}{e^{\pi} + 1}$

C. $a = 4, L = \frac{e^{4\pi} - 1}{e^{\pi} - 1}$

D. $a = 4, L = \frac{e^{4\pi} + 1}{e^{\pi} + 1}$

Ans: A C (Splitting of limits 0 to 2a as 0 to a in f(x) and 0 to 2a-x in f(2a-x))

Sol: Let $\int_0^x e^t (\sin^6 at + \cos^4 at) dt = A$

$$I = \int_{\pi}^{2\pi} e^t (\sin^6 at + \cos^4 at) dt$$

$$\text{Put } t = \pi + x$$

$$dt = dx$$

for $a = 2$ as well as $a = 4$

$$I = e^{\pi} \int_0^{\pi} e^x (\sin^6 ax + \cos^4 ax) dx$$

$$I = e^{\pi} A$$

$$\text{Similarly } \int_{2\pi}^{3\pi} e^t (\sin^6 at + \cos^4 at) dt = e^{2\pi} A$$

$$\text{So, } L = \frac{A + e^{\pi} A + e^{2\pi} A + e^{3\pi} A}{A} = \frac{e^{4\pi} - 1}{e^{\pi} - 1}$$

For both $a=2, 4$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 6: Applications of inequality, Sandwich property in comparison of functions in

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definite, Boundedness and continuity property of definite integral, Max-min inequality property of definite integral, Root mean square value - RMSV of a definite integral

12. Which of the following inequalities is/are TRUE?

[2020-P1 (Q12)]

A. $\int_0^1 x \cos x dx \geq \frac{3}{8}$

B. $\int_0^1 x \sin x dx \geq \frac{3}{10}$

C. $\int_0^1 x^2 \cos x dx \geq \frac{1}{2}$

D. $\int_0^1 x^2 \sin x dx \geq \frac{2}{9}$

Ans: A B D

(property of comparison of functions in definite integral-Application of inequality)

Sol: A) $\int_0^1 x \cos x dx \geq \int_0^1 x \left(1 - x \frac{x^2}{2!}\right) dx$
 $\geq \frac{3}{8}$

B) $\int_0^1 x \sin x dx \geq \int_0^1 x \left(x - \frac{x^3}{3!}\right) dx$
 $\geq \frac{3}{10}$

C) $\int_0^1 x^2 \cos x dx \leq \int_0^1 x^2 dx$
 $\leq \frac{1}{3}$

D) $\int_0^1 x^2 \sin x dx \geq \int_0^1 x \left(x - \frac{x^3}{3!}\right) dx$
 $\geq \frac{2}{9}$

13. Let $f'(x) = \frac{192x^3}{2 + \sin^4 \pi x}$ for all $x \in \mathbb{R}$ with $f\left(\frac{1}{2}\right) = 0$. If $m \leq \int_{1/2}^1 f(x) dx \leq M$, then the possible values of m and M are

[2015-P2 (Q56)]

A. $m = 13, M = 24$

B. $m = \frac{1}{4}, M = \frac{1}{2}$

C. $m = -11, M = 0$

D. $m = 1, M = 12$

Ans: D (Max-min inequality property of definite integral - Applications of inequality)

$$\text{Sol: } \frac{192}{3} \int_{1/2}^x t^3 dt \leq f(x) \leq \frac{192}{2} \int_{1/2}^x t^3 dt$$

$$16x^4 - 1 \leq f(x) \leq 24x^4 - \frac{3}{2}$$

$$\int_{1/2}^1 (16x^4 - 1) dx \leq \int_{1/2}^1 f(x) dx \leq \int_{1/2}^1 \left(24x^4 - \frac{3}{2}\right) dx$$

$$1 < \frac{26}{10} \leq \int_{1/2}^1 f(x) dx \leq \frac{39}{10} < 12$$

14. Let $f: \mathbb{R} \rightarrow (0, 1)$ be a continuous function. Then, which of the following function(s) has(have) the value zero at some point in the interval $(0, 1)$? [2017-P1 (Q40)]

A. $x^9 - f(x)$

B. $x - \int_0^{\frac{\pi}{2}-x} f(t) \cos t dt$

C. $e^x - \int_0^x f(t) \sin t dt$

D. $f(x) + \int_0^{\frac{\pi}{2}} f(t) \sin t dt$

Ans: A B (Max-min inequality property of definite integral - Applications of inequality)

Sol: [A] no solution for $e^x = \int_0^x f(t) \sin t dt, x \in (0, 1)$

A LHS is > 1 and RHS is less than 1

[B] let $h(x) = x^9 - f(x)$

$$h(0) = -f(0) < 0$$

$$h(1) = 1 - f(1) > 0$$

By IVT (As $h(x)$ is cont. in $[0, 1]$)

$h(x) = 0$ will be for some $x \in (0, 1)$

[C] both are +ve function so no solution

[D] $h(x) = x - \int_0^{\frac{\pi}{2}-x} f(t) \cos t dt$

$$h(0) = -ve$$

$$h(1) = +ve$$

so as $h(x)$ cont. in $[0, 1]$

so $h(x)=0$ for some $x \in (0,1)$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 7: Leibnitz's rule for differentiation under the integral sign

15. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = (x-1)(x-2)(x-5)$. Define $F(x) = \int_0^x f(t) dt, x > 0$.

Then which of the following options is/are correct?

[2019-P2 (Q7)]

- A. F has a local minimum at $x = 1$ (Newton Leibniz)
- B. F has a local maximum at $x = 2$
- C. F has two local maxima and one local minimum in $(0, \infty)$
- D. $F(x) \neq 0$ for all $x \in (0, 5)$

Ans: A B D

Sol: $F(x) = \int_0^x f(t) dt$

$$F'(x) = f(x) = (x-1)(x-2)(x-5)$$

$\Rightarrow x = 1, 5$ is point of local minima for $x > 0$

$x = 2$ is point of local maxima for $x > 0$

$$F(2) = \int_0^2 f(t) dt < 0 \Rightarrow F(x) < 0 \forall x \in (0, 5)$$

16. Let $f: (0, \infty) \rightarrow \mathbb{R}$ be given by $f(x) = \int_{\frac{1}{x}}^x e^{-\left(t+\frac{1}{t}\right)} \frac{dt}{t}$. Then [2014-P1 (Q43)]

- A. $f(x)$ is monotonically increasing on $[1, \infty)$ (Newton & Max & Min)
- B. $f(x)$ is monotonically decreasing on $(0, 1)$
- C. $f(x) + f\left(\frac{1}{x}\right) = 0$, for all $x \in (0, \infty)$
- D. $f(2^*)$ is an odd function of x on $\mathbb{R}$

Ans: A C D

Sol: $f'(x) = \frac{2e^{-\left(x+\frac{1}{x}\right)}}{x}$

Which of the increasing in $(1, \infty]$

Also, $f(x) + f\left(\frac{1}{x}\right) = 0$

$g(x) = f(2^x) = \int_{2^{-x}}^{2^x} \frac{e^{-\left(t+\frac{1}{t}\right)}}{t} dt$

$g(-x) = \int_{2^x}^{2^{-x}} \frac{e^{-\left(t+\frac{1}{t}\right)}}{t} dt = -g(x)$

Hence, an odd function

17. Let $f: [a, b] \rightarrow [1, \infty)$ be a continuous function and let $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$g(x) = \begin{cases} 0 & \text{if } x < a, \\ \int_a^x f(t) dt & \text{if } a \leq x \leq b, \\ \int_a^b f(t) dt & \text{if } x > b. \end{cases}$$

[2014-P1 (Q45)]

- A. $g(x)$ is continuous but not differentiable at a
 B. $g(x)$ is differentiable on $\mathbb{R}$
 C. $g(x)$ is continuous but not differentiable at b
 D. $g(x)$ is continuous and differentiable at either a or b but not both

Ans: A C

(Newton Leibniz Continuity and differentiability)

Sol: Since $f(x) \geq 1 \forall x \in [a, b]$

For $g(x)$

LHD at $x=a$ is zero

and RHD at $(x=a) = \lim_{x \rightarrow a^+} \frac{\int_a^x f(t) dt - 0}{x - a} = \lim_{x \rightarrow a^+} f(x) \geq 1$

Hence $g(x)$ is not differentiable at $x=a$

Similarly LHD at $x=b$ is greater than 1

$g(x)$ is not differentiable at $x=b$

18. If $g(x) = \int_{\sin x}^{\sin(2x)} \sin^{-1}(t) dt$, then

[2017-P2 (Q44)]

- A. $g'\left(\frac{\pi}{2}\right) = -2\pi$ B. $g'\left(-\frac{\pi}{2}\right) = 2\pi$
 C. $g'\left(\frac{\pi}{2}\right) = 2\pi$ D. $g'\left(-\frac{\pi}{2}\right) = -2\pi$

Ans: A (Leibnitz's rule for differentiation under the integral sign)

Sol: $g(x) = \int_{\sin x}^{\sin(2x)} \sin^{-1}(t) dt$

$$g'(x) = 2(\cos 2x) \sin^{-1}(\sin 2x) - (\cos x) \sin^{-1}(\sin x)$$

$$g'\left(\frac{\pi}{2}\right) = 2(-1)(0) - \cos\left(\frac{\pi}{2}\right)(1) = 0$$

$$g'\left(-\frac{\pi}{2}\right) = 0 \text{ no option is matching.}$$

19. Let $f(x) = \int_{\pi^2/4}^{x^2} \frac{\sin x}{1 + \cos^2 \sqrt{t}} dt$ then

- A. $f'\left(\frac{\pi}{2}\right) = \pi$ B. $f'\left(-\frac{\pi}{2}\right) = \pi$ C. $f'\left(\frac{3\pi}{2}\right) = -3\pi$ D. $f'(\pi) = \int_{\pi^2}^{\pi^2/4} \frac{dx}{1 + \cos^2 \sqrt{x}}$

Ans: A B C D

Sol: $f'(x) = \sin x \times \frac{2x}{1 + \cos^2 x} + \cos x \int_{\pi^2/4}^{x^2} \frac{1}{1 + \cos^2 \sqrt{t}} dt$

20. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = (x+1)(x+2)(x+5)$. Define $F(x) = \int_0^x f(t) dt, x < 0$.

Then which of the following options is/are correct?

- A. F has a local minimum at $x = -1$
 B. F has a local maximum at $x = -2$
 C. F has two local maxima and one local minimum in $(-\infty, 0)$
 D. $F(x) \neq 0$ for all $x \in (-5, 0)$

Ans: A B D

Sol: $F(x) = \int_0^x f(t) dt$

$$F'(x) = f(x) = (x+1)(x+2)(x+5)$$

$\Rightarrow x = -1, -5$ is point of local minima for $x < 0$

$x = -2$ is point of local maxima for $x < 0$

$$F(2) = \int_0^2 f(t) dt < 0 \Rightarrow F(x) < 0 \forall x \in (0, 5)$$

21. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = (x-1)(x-4)(x-25)$. Define

$$F(x) = \int_0^{x^2} f(t) dt, x \in (-5, 5). \text{ Then which of the following options is/are correct?}$$

A. F has a local minimum at $x = 1$

B. F has a local maximum at $x = 2$

C. F has two local maxima and one local minimum in $(-5, 5)$

D. $F(x) = 0$ for all $x \in (0, 5)$

Ans: A B

Sol: $F(x) = \int_0^{x^2} f(t) dt$

$$F'(x) = (x^2 - 1)(x^2 - 4)(x^2 - 25) \cdot 2x$$

$$F(2) = \int_0^4 (x-1)(x-4)(x-25) < 0$$

22. The points of extremum of $\int_0^{x^2} \left(\frac{t^2 - 5t + 4}{2 + e^t} \right) dt$ are

A. $x = -2$

B. $x = 1$

C. $x = 0$

D. $x = -1$

Ans: A B C D

Sol: Let $f(x) = \int_0^{x^2} \left(\frac{t^2 - 5t + 4}{2 + e^t} \right) dt$

$$f'(x) = \left(\frac{x^4 - 5x^2 + 4}{2 + e^{x^2}} \right) \times 2x$$

For extremum $f'(x) = 0$

23. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be given by $f(x) = \int_{\frac{4}{x^2}}^{x^2} e^{-\left(\frac{t+2}{2t}\right)} dt$. Then

- A. $f(x)$ is monotonically increasing on $[1, \infty)$
- B. $f(x)$ is monotonically decreasing on $(0, 1)$
- C. $f(x) < 0$ for all $x \in (0, \infty)$
- D. $f(x)$ is an even non negative function of x on $\mathbb{R}$

Ans: B D

Sol: $f'(x) = 2e^{-\left(\frac{x^2+2}{2x^2}\right)} \left(2x - \frac{8}{x^3}\right) \Rightarrow R^+$ there is only one minima, $x = \sqrt{2}$

24. Let $f : [2, 6] \rightarrow [2, \infty)$ be a continuous function and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$g(x) = \begin{cases} 0 & \text{if } x < 2, \\ \int_2^x f(t) dt & \text{if } 2 \leq x \leq 6, \\ \int_2^6 f(t) dt & \text{if } x > 6. \end{cases} \text{ Then}$$

- A. $g(x)$ is continuous but not differentiable at 2
- B. $g(x) = 0$ has two real roots in $(2, 6)$
- C. $g(x)$ is continuous but not differentiable at 6
- D. $g(x)$ is continuous and differentiable at either 2 or 6 but not both

Ans: A C

Sol: Since $f(x) \geq 1 \forall x \in [2, 6]$

For $g(x)$

LHD at $x=2$ is zero

$$\text{and RHD at } (x=2) = \lim_{x \rightarrow 2^+} \frac{\int_2^x f(t) dt - 0}{x - 2} = \lim_{x \rightarrow 2^+} f(x) \geq 2$$

Hence $g(x)$ is not differentiable at $x=a$

Similarly LHD at $x=6$ is greater than 1

$g(x)$ is not differentiable at $x=6$

EXERCISE - IV
(SINGAL CORRECT)

Topic: Introduction to Definite integration

Sub Topic 1: Definite integral as a limit of an infinite Reimann Sum

1. For $a \in \mathbb{R}$ (the set of all real numbers), $a \neq -1$,

$$\lim_{n \rightarrow \infty} \frac{1^a + 2^a + \dots + n^a}{(n+1)^{a-1} [(na+1) + (na+2) + \dots + (na+n)]} = \frac{1}{60} \quad \text{Then } a = \quad [2013-P2 (Q45)]$$

- A. 5 B. 7 C. $\frac{-15}{2}$ D. $\frac{-17}{2}$

(INTRODUCTION TO DEFINITE INTEGRALS)

Ans: B

Sol: Required limit = $\frac{\int_0^1 x^a dx}{\int_0^1 (a+x) dx} = \frac{\frac{2}{(2a+1)(a+1)}}{\frac{2}{120}} = \frac{2}{120}$

$$\Rightarrow a = 7 \text{ or } -\frac{17}{2}$$

2. If $\lim_{n \rightarrow \infty} \frac{1^a + 2^a + \dots + n^a}{(n+1)^{a-1} [(na+1) + (na+2) + \dots + (na+n)]} = \frac{1}{60}$ where $a \in \mathbb{R}$ (the set of all real numbers), $a \neq -1$, then the values of 'a' are

- 1) 5 2) $\frac{-15}{7}$ 3) 7 4) $\frac{-17}{2}$

Key: 4

Sol:

$$\lim_{n \rightarrow \infty} \frac{1^a + 2^a + \dots + n^a}{(n+1)^{a-1} [(na+1) + (na+2) + \dots (na+n)]} = \frac{1}{60}$$

$$\lim_{n \rightarrow 60} \frac{\sum_{r=1}^n (r)^2}{(n+1)^{a-1} \left[\sum_{r=1}^n (na+r) \right]} = \frac{1}{60}$$

$$\lim_{n \rightarrow 60} \frac{n^a \sum_{r=1}^n \left(\frac{r}{a} \right)^n}{(n+1)^{a-1} \sum_{r=1}^n \left(a + \frac{r}{n} \right)} = \frac{1}{60}$$

$$\frac{\int_0^1 x^a dx}{\int_0^1 (a+x) dx} = \frac{1}{60}$$

By that we get quadratic equation

$$2a^2 + 3a - 119 = 0$$

values of 'a' are $7, -\frac{17}{2}$

3. $\lim_{n \rightarrow \infty} \left[\frac{1}{1+n^2} + \frac{2}{2+n^2} + \dots + \frac{n}{n^2+n^2} \right]$ is

1) $\log 2$

2) $2\log 2$

3) $\frac{\log 2}{2}$

4) $\frac{\log 2}{3}$

Key: 3

Sol:

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{r}{n^2 + r^2}$$

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n^2} \left(\frac{r}{1 + \left(\frac{r^2}{n^2} \right)} \right)$$

$$\int_0^1 \frac{x}{1+x^2} dx$$

multiply & divide by 2

$$= \frac{1}{2} \log 2$$

4. The value of $\lim_{n \rightarrow \infty} \frac{1}{n^3} \left(\sqrt{n^2 + 1} + 2\sqrt{n^2 + 2^2} + \dots + n\sqrt{n^2 + n^2} \right)$ is
- 1) $\frac{\sqrt{2}-1}{5}$ 2) $\frac{\sqrt{2}-1}{3}$ 3) $\frac{2\sqrt{2}-1}{5}$ 4) $\frac{2\sqrt{2}-1}{3}$

Key: 4

Sol:

$$\lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{r=1}^n r \sqrt{n^2 + r^2}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \sqrt{1 + \left(\frac{r}{n}\right)^2} \cdot \frac{r}{n}$$

$$\int_0^1 \sqrt{1+x^2} \cdot x dx$$

Let's assume $\sqrt{1+x^2} = t$

$x dx = t dt$

$$\int_1^{\sqrt{2}} t^2 dt$$

$$= \frac{2\sqrt{2}-1}{3}$$

Topic: Introduction to Definite integration

Sub Topic 2: Second Fundamental theorem of integral calculus - definition

5. The value(s) of $\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx$ is/are [2010-P1 (Q41)]

A. $\frac{22}{7} - \pi$ B. $\frac{2}{105}$ C. 0 D. $\frac{71}{15} - \frac{3\pi}{2}$

Ans A **(INTRODUCTION TO DEFINITE INTEGRATION)**

Sol: $\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx = \int_0^1 \left(x^6 - 4x^5 + 5x^4 - 4x^3 + 4x - \frac{4}{1+x^2} \right) dx$

$$= \left[\frac{x^7}{7} - \frac{2x^6}{3} + x^5 - \frac{4x^4}{3} + 4x^2 - 4 \tan^{-1} x \right]_0^1 - \pi$$

$$= \frac{1}{2} - \frac{2}{3} + 1 - \frac{4}{3} + 4 - \pi = \frac{22}{7} - \pi$$

Topic: Evaluation of definite integrals

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Sub Topic 1: Evaluation of definite integrals by substitution

6. The following integral $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2 \operatorname{cosec} x)^{17} dx$ is equal to [2014-P2 (Q42)]

A. $\int_0^{\log(1+\sqrt{2})} 2(e^u + e^{-u})^{16} du$ B. $\int_0^{\log(1+\sqrt{2})} (e^u + e^{-u})^{17} du$ (substitution)

C. $\int_0^{\log(1+\sqrt{2})} (e^u - e^{-u})^{17} du$ D. $\int_0^{\log(1+\sqrt{2})} 2(e^u - e^{-u})^{16} du$

Ans: A

Sol: $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2 \operatorname{cosec} x)^{17} dx$

Let $e^u + e^{-u} = 2 \operatorname{cosec} x, x = \frac{\pi}{4} \Rightarrow u = \ln(1 + \sqrt{2}), x = \frac{\pi}{2} \Rightarrow u = 0$

$\Rightarrow \operatorname{cosec} x + \cot x = e^u$ and $\operatorname{cosec} x - \cot x = e^{-u} \Rightarrow \cot x = \frac{e^u - e^{-u}}{2}$

$(e^u - e^{-u}) dx = -2 \operatorname{cosec} x \cot x dx$

$\Rightarrow -\int (e^u + e^{-u})^{17} \frac{(e^u - e^{-u})}{2 \operatorname{cosec} x \cot x} du$

$= -2 \int_{\ln(1+\sqrt{2})}^0 (e^u - e^{-u})^{16} du$

$= \int_0^{\ln(1+\sqrt{2})} 2(e^u + e^{-u})^{16} du$

7. $\int \frac{(x^{3x} + x^x)(\log x + 1)}{x^{4x} + 1} dx = ?$ ('c' is integration constant)

A) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x^{2x} - 1}{\sqrt{2} x^x} \right) + c$ B) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x^x + 1}{\sqrt{2} x^x} \right) + c$

C) $\sqrt{2} \tan^{-1} \left(\frac{x^{2x} + 1}{x^x} \right) + c$ D) $\sqrt{2} \tan^{-1} (x^x + 1) + c$

Key: A

Sol: $I = \int \frac{(x^{2x} + 1)x^x(1 + \ln x)}{(x^x)^4 + 1} dx$ put $x^x = t$

Then $I = \int \frac{t^2 + 1}{t^4 + 1} dt = \int \frac{1 + \frac{1}{t^2}}{\left(t - \frac{1}{t}\right)^2 + 2} dt = \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{t^2 - 1}{\sqrt{2}t} \right)$

8. The following integral $\int_0^{\frac{\pi}{4}} (2 \sec x)^{2023} dx$ is equal to

A. $\int_0^{\log(1+\sqrt{2})} 2(e^u + e^{-u})^{2022} du$

B. $\int_0^{\log(1+\sqrt{2})} (e^u + e^{-u})^{2023} du$

C. $\int_0^{\log(1+\sqrt{2})} (e^u - e^{-u})^{2023} du$

D. $\int_0^{\log(1+\sqrt{2})} 2(e^u - e^{-u})^{2022} du$

Ans: A

Sol: $\int_0^{\frac{\pi}{4}} (2 \sec x)^{2023} dx$

Let $e^u + e^{-u} = 2 \sec x, x = \frac{\pi}{4} \Rightarrow u = \ln(1 + \sqrt{2}), x = 0 \Rightarrow u = 0$

$(e^u - e^{-u}) dx = 2 \sec x \tan x dx$

$\therefore$ Given integral

$= \int_0^{\ln(1+\sqrt{2})} 2(e^u + e^{-u})^{2022} du$

9. If $\int_0^4 \frac{(y^2 - 4y + 5) \sin(y - 2)}{2y^2 - 8y + 11} dy = \lambda$ then $[\lambda + 2] =$ (where $[.]$ denotes the greatest integer function)

A. 3

B. 1

C. 0

D. 2

Key: C

Hint $y - 2 = Z$ the $I = \int_{-2}^2 \frac{Z^2 + 1}{2Z^2 - 7} \sin Z dz = 0$

10. $A = \int_1^{\sin \theta} \frac{t dt}{1+t^2}; B = \int_1^{\csc \theta} \frac{dt}{t(1+t^2)}$ then the value of $\begin{vmatrix} A & A^2 & B \\ e^A e^B & B^2 & -1 \\ 1 & A^2 + B^2 & -1 \end{vmatrix}$ is _____

A. $A+B$

B. $2A+B$

C. $\sin \theta + \operatorname{cosec} \theta$

D. 1

Key: A

Sol. put $t = \frac{1}{x}$ in A we get $A = -B \Rightarrow A+B=0$

$$\therefore \begin{vmatrix} A & A^2 & -A \\ 1 & A^2 & -1 \\ 1 & 2A^2 & -1 \end{vmatrix} = 0 = A+B$$

Topic: Evaluation of definite integrals**Sub Topic 2:** Properties of evaluation of definite integrals**Mini Topic 2:** Replacing x by $a-x$ – King's property, Replacing x by $a+b-x$

11. The value of $\int_{\sqrt{\ln 2}}^{\sqrt{\ln 3}} \frac{x \sin x^2}{\sin x^2 + \sin(\ln 6 - x^2)} dx$ is [2011-P1 (Q48)]

A. $\frac{1}{4} \ln \frac{3}{2}$

B. $\frac{1}{2} \ln \frac{3}{2}$

C. $\ln \frac{3}{2}$

D. $\frac{1}{6} \ln \frac{3}{2}$

Ans: A

(Evaluation of definite integrals)

Sol: Put $x^2 = t$

$$\Rightarrow x dx = dt/2$$

$$\therefore I = \int_{\ln 2}^{\ln 3} \frac{\sin t \cdot \frac{dt}{2}}{\sin t + \sin(\ln 6 - t)} \quad \dots\dots(i)$$

$$\text{Using, } \int_a^b f(x) dx = \int_a^b f(a+b-x) dx$$

$$= \frac{1}{2} \int_{\ln 2}^{\ln 3} \frac{\sin(\ln 2 + \ln 3 - t)}{\sin(\ln 2 + \ln 3 - t) + \sin(\ln 6 - (\ln 2 + \ln 3 - t))} dt$$

$$= \frac{1}{2} \int_{\ln 2}^{\ln 3} \frac{\sin(\ln 6 - t)}{\sin(\ln 6 - t) + \sin(t)} dt$$

$$\therefore I = \int_{\ln 2}^{\ln 3} \frac{\sin(\ln 6 - t)}{\sin(\ln 6 - t) + \sin t} dt \quad \dots\dots(ii)$$

On adding Eqs. (i) and (ii), we get

$$2I = \frac{1}{2} \int_{\ln 2}^{\ln 3} \frac{\sin t + \sin(\ln 6 - t)}{\sin(\ln 6 - t) + \sin t} dt$$

$$\Rightarrow 2I = \frac{1}{2} (t)_{\ln 2}^{\ln 3} = \frac{1}{2} (\ln 3 - \ln 2) \quad \therefore \quad I = \frac{1}{4} \ln \left(\frac{3}{2} \right)$$

12. Area of region bounded by the curve $y = e^x$ and lines $x = 0$ and $y = e$ is

1) $e - 1$ 2) $\int_1^e \ln(e + 1 - y) dy$ 3) $e - \int_0^1 e^x dx$ 4) $e - \int_0^1 \ln y dy$

Key: 2

Sol: The shaded region represents the area which is equal to

$$\int_1^e \ln y dy = \int_1^e \ln(e + 1 - y) dy$$

$$U \sin g \int_a^b f(x) \int_a^b f(a + b - x) dx$$

Of it can be when considered w.r.t x as

$$e - \int_0^1 e^x dx = e - [e^x]_0^1 = e - [e - 1] = 1 \neq e - 1$$

13. If $f(x) = \frac{2 - x \cos x}{2 + x \cos x}$ and $g(x) = \log_e x$ then the value of the integral $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} g(f(x)) dx =$

1) $\log_e 3$ 2) $\log_e 1$ 3) $\log_e e$ 4) $\log_e 2$

Key: 2

Sol:

$$f(x) = \frac{2 - x \cos x}{2 + x \cos x}; g(x) = \log e^x$$

$$I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} g(f(x)) dx$$

$$I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \log_e \left(\frac{2 - x \cos x}{2 + x \cos x} \right) dx$$

$$\text{Apply } \int_b^a f(x) dx = \int_b^a f(a+b-x) dx$$

$$I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \log_e \left(\frac{2 - x \cos x}{2 + x \cos x} \right) dx$$

$$I = \log e^1$$

14. $\int_0^{\frac{\pi}{2}} \sin x \sin 2x \sin 3x \sin 4x dx =$

A) $\frac{\pi}{4}$

B) $\frac{\pi}{8}$

C) $\frac{\pi}{16}$

D) $\frac{\pi}{32}$

Key: C

Sol: $I = \int_0^{\frac{\pi}{2}} \sin x \sin 2x \sin 3x \sin 4x dx$

$$= \int_0^{\frac{\pi}{2}} \cos x \sin 2x \sin 3x \sin 4x dx$$

$$2I = \int_0^{\frac{\pi}{2}} \cos 2x \sin 2x \sin 4x dx = \frac{\pi}{8}$$

15. Let $I_1 = \int_{\sec^2 z}^{2-\tan^2 z} x f(x(3-x)) dx$ and let $I_2 = \int_{\sec^2 z}^{2-\tan^2 z} f(x(3-x)) dx$ where 'f' is a continuous function and 'z' is any real number, then $\frac{I_1}{I_2} =$

A. $\frac{3}{2}$

B. $\frac{1}{2}$

C. 1

D. $\frac{2}{3}$

Ans: A

Sol: $I_1 = \int_{\sec^2 z}^{2-\tan^2 z} x f(x(3-x)) dx$

$$I_2 = \int_{\sec^2 z}^{2-\tan^2 z} x f(x(3-x)) dx$$

$$2I_1 = \int_{\sec^2 z}^{2-\tan^2 z} (3 - x f((3-x)x)) dx$$

$$2I_1 = \int_{\sec^2 z}^{2-\tan^2 z} 3f(x(3-x)) dx$$

$$= 3 \int_{\sec^2 z}^{2-\tan^2 z} ((x)(3-x)) dx$$

$$2I_1 = 3I_2$$

$$\Rightarrow \frac{I_1}{I_2} = \frac{3}{2}$$

16. $\int_{-2\pi}^{2\pi} \frac{\sin^6 x}{(\sin^6 x + \cos^6 x)(1 + e^{-x})} dx =$

A. 2π

B. π

C. $\pi/2$

D. 4π

Ans: B

Sol: $\int_{-a}^a f(x) dx = \int_0^a (f(x) + f(-x)) dx$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 3: Even - odd property

17. $\int_{-\pi/2}^{\pi/2} \left(x^2 + \ln \frac{\pi+x}{\pi-x} \right) \cos x dx$ is

[2012-P2 (Q47)]

A. 0

B. $\frac{\pi^2}{2} - 4$

C. $\frac{\pi^2}{2} + 4$

D. $\frac{\pi^2}{2}$

Ans: B

SUB TOPIC: Properties of evaluation definite integrals)

Sol:
$$\int_{-\pi/2}^{\pi/2} \left(x^2 + \ln \left(\frac{\pi+1}{\pi-1} \right) \right) \cos x \, dx = 2 \int_0^{\pi/2} x^2 \cos x \, dx + 0 \left(\because \ln \left(\frac{\pi+1}{\pi-1} \right) \text{ is an odd function} \right)$$

$$= 2 \left[\left(x^2 \sin x \right)_0^{\pi/2} - \int_0^{\pi/2} 2x \sin x \, dx \right] = 2 \left(\frac{\pi^2}{4} - 0 \right) - 4 \int_0^{\pi/2} x \sin x \, dx$$

$$= \frac{\pi^2}{2} - 4 \left[\left(-x \cos x \right)_0^{\pi/2} + \int_0^{\pi/2} \cos x \, dx \right]$$

$$= \frac{\pi^2}{2} - 4$$

18. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2 \cos x}{1+e^x} \, dx$ is equal to [2016-P2 (Q40)]

A. $\frac{\pi^2}{4} - 2$ B. $\frac{\pi^2}{4} + 2$ C. $\pi^2 - e^{\frac{\pi}{2}}$ D. $\pi^2 + e^{\frac{\pi}{2}}$

Ans: A (If f(x) is odd and definite integral of f(t) from 0 to x is even)

Sol:
$$= \int_0^{\pi/2} \left(\frac{x^2 \cos x}{1+e^x} + \frac{x^2 \cos x}{1+e^{-x}} \right) dx$$

$$= \int_0^{\pi/2} \frac{x^2 \cos x + x^2 e^x \cos x}{1+e^x} dx$$

$$= \int_0^{\pi/2} x^2 \cos x \, dx$$

$$= \left(x^2 \sin x \right)_0^{\pi/2} - \int_0^{\pi/2} 2x \sin x \, dx$$

$$= \frac{\pi^2}{4} - 2 \left[\left(x(-\cos x) \right)_0^{\pi/2} - \cos x \, dx \right]$$

$$= \frac{\pi^2}{4} - 2 \left[-(0-0) + (\sin x)_0^{\pi/2} \right]$$

$$= \frac{\pi^2}{4} - 2$$

19.

$$\text{Let } I_1 = \int_{-2}^2 \frac{x^6 + 3x^5 + 7x^4}{x^4 + 2} dx \text{ and}$$

$$I_2 = \int_{-3}^1 \frac{2(x-1)^2 + 11(x-1) + 14}{(x-1)^4 + 2} dx, \text{ then the value of}$$

$$I_1 + I_2 \text{ is}$$

A) 8

B) 200/3

C) 100/3

D) None of these

KEY: C

SOL:

In I_2 , put $x+1=t$, then

$$I_2 = \int_{-2}^2 \frac{2t^2 + 11t + 14}{t^4 + 2} dt = \int_{-2}^2 \frac{2x^2 + 11x + 14}{x^4 + 2} dx$$

$$\therefore I_1 + I_2 = \int_{-2}^2 \frac{x^6 + 3x^5 + 7x^4 + 2x^2 + 11x + 14}{x^4 + 2} dx$$

$$= \int_{-2}^2 \frac{(x^2 + 3x + 7)(x^4 + 2) + 5x}{x^4 + 2} dx$$

$$= \int_{-2}^2 (x^2 + 3x + 7) dx + 5 \int_{-2}^2 \frac{x}{x^4 + 2} dx$$

$$= 2 \int_0^2 (x^2 + 7) dx = \frac{100}{3}$$

(The other integrals are zero, being integrals of odd functions.)

$$\int_{-4}^4 \frac{\sin^{-1}(\sin x) + \cos^{-1}(\cos x)}{(1+x^2) \left(1 + \left\lceil \frac{x^2}{17} \right\rceil\right)} dx = \log \left(\frac{(1+x^2)}{a} \right)$$

20..

$+b\pi \tan^{-1} \left(\frac{c-\pi}{1+c\pi} \right)$ (where, $\lceil \cdot \rceil$ denotes greatest integer function), then number of ways in

which $a - (2b + c)$ Distinct object can distributed among $\frac{a-5}{c}$ persons equally, is

A) $\frac{9!}{(3!)^3}$

B) $\frac{12!}{(4!)^3}$

C) $\frac{15!}{(5!)^3}$

D) $\frac{10!}{(6!)} \times 3!$

KEY: A

SOL: Here, $-4 \leq x \leq 4 \Rightarrow 0 \leq \frac{x^2}{17} < 1$

$$\therefore \left\lceil \frac{x^2}{17} \right\rceil = 0 \text{ and } \sin^{-1}(\sin x) \text{ is an odd function.}$$

$$\text{Let, } I = \int_{-4}^4 \frac{\sin^{-1}(\sin x)}{(1+x^2) \times 1} dx + \int_{-4}^4 \frac{\cos^{-1}(\cos x)}{(1+x^2) \times 1} dx$$

$$= 0 + 2 \int_0^\pi \frac{\cos^{-1}(\cos x)}{(1+x^2)} dx$$

$$\begin{aligned}
 I &= 2 \left[\int_0^\pi \frac{x}{1+x^2} dx + \int_\pi^4 \frac{2\pi}{(1+x^2)} dx \right] \\
 &= \log(1+\pi^2) + 2\pi (\tan^{-1} x)_\pi^4 - [\log(1+16) - \log(1+\pi^2)] \\
 &= \log(1+\pi^2) + 2(\tan^{-1} 4 - \tan^{-1} \pi) - \log \frac{17}{1+\pi^2} \\
 &= \log \frac{(1+\pi^2) \cdot (1+\pi^2)}{17} + 2\pi \tan^{-1} \left[\tan^{-1} \left(\frac{4-\pi}{1+4\pi} \right) \right] \\
 &= \log \left(\frac{(1+\pi^2)^2}{17} \right) + 2\pi \tan^{-1} \left(\frac{4-\pi}{1+4\pi} \right)
 \end{aligned}$$

On comparing with $\log \left(\frac{(1+\pi^2)^2}{a} \right) + b\pi \tan^{-1} \left(\frac{c-\pi}{1+c\pi} \right)$, we get

$$a = 17, b = 2 \text{ and } c = 4$$

$$\therefore a - (2b + c) = 17 - 8 = 9 \text{ and } \frac{a-5}{c} = 3$$

Thus, the number of ways to distribute 9 distribute – distinct bijective into 3 persons equally is $\frac{9!}{(3!)^3}$.

21. If $f(x) = \begin{cases} e^{\cos x} \sin x; & \text{for } |x| \leq 2 \\ 2; & \text{otherwise} \end{cases}$, then $\int_{-2}^3 f(x) dx$ is
- 1) 0 2) 1 3) 2 4) 3

Key: 3

Sol:

$$I = \int_{-2}^3 f(x) dx$$

$$I = \int_{-2}^2 e^{\cos x} \sin x dx + \int_2^3 2 dx$$

$$\text{Let } f(x) = e^{\cos x} \sin x$$

$$f(-x) = -f(x)$$

$$I = \int_2^3 2 dx$$

$$I = 2$$

22. Value of $\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{\pi + 4x^3}{2 - \cos\left(|x| + \frac{\pi}{3}\right)} dx$

$$1) \frac{4\pi}{\sqrt{3}} \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) \quad 2) \frac{4\pi}{\sqrt{2}} \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) \quad 3) \frac{2\pi}{\sqrt{3}} \tan^{-1}\left(\frac{1}{2}\right) \quad 4) \frac{4\pi}{\sqrt{3}} \tan^{-1}\left(\frac{1}{2}\right)$$

Key: 4

Sol:

$$\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{\pi + 4x^3}{2 - \cos\left(|x| + \frac{\pi}{3}\right)} dx$$

$$= \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{\pi + 4x^3}{2 - \cos\left(|x| + \frac{\pi}{3}\right)} dx + 4 \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{x^3}{2 - \cos\left(|x| + \frac{\pi}{3}\right)} dx$$

$$u \sin g \int_{-\alpha}^{\alpha} f(x) dx = \int_0^a f(x) dx \quad f(-x) = -f(x) \quad f(-x) = f(x)$$

$$= 2\pi \int_0^{\frac{\pi}{3}} \frac{dx}{2 - \cos\left(x + \frac{\pi}{3}\right)}$$

$$x + \frac{\pi}{3} = t, dx = dt$$

$$= 2\pi \int_{\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{dx}{2 - \cos t}$$

$$\text{after solving we get } \frac{4\pi}{\sqrt{3}} \tan^{-1}\left(\frac{1}{2}\right)$$

Topic: Evaluation of definite integrals**Sub Topic 2:** Properties of evaluation of definite integrals**Mini Topic 4:** Half the integral limit property (Splitting of limits 0 to 2a as 0 to a and a to 2a in f(x)), f(2a-x) property

23. The value of $\int_{-2}^1 \left[x \left[1 + \cos\left(\frac{\pi x}{2}\right) \right] + 1 \right] dx$, where $[.]$ denotes the greatest integer function, is

A) 1

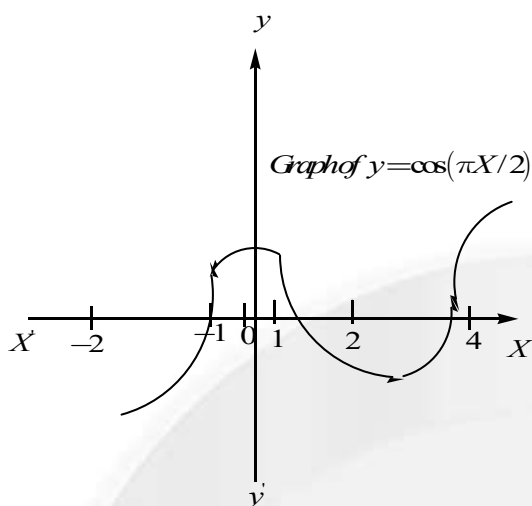
B) $\frac{1}{2}$

C) 2

D) None of these

KEY: C

SOL:



$$\begin{aligned}
 &\text{From graph, } \int_{-2}^1 x \left[1 + \cos \frac{\pi x}{2} + 1 \right] dx \\
 &= \int_{-2}^{-1} [x(1 + (-1)) + 1] dx + \int_{-1}^1 [x(1 + 0) + 1] dx \\
 &= (x)^{-1}_{-2} + \int_{-1}^1 [x + 1] dx = (-1 - (-2)) + \int_{-1}^0 0 dx + \int_0^1 1 dx = 2
 \end{aligned}$$

24. If $\int_0^x [x] dx = \int_0^x x dx$, $x \notin \text{integer}$ (where $[.]$ & $\{.\}$ denotes the greatest integer and fractional parts respectively then the value of $4\{x\}$ is
- 1) 4 2) 8 3) 2 4) 1

Key: 3

Sol: Given $\int_0^x [x] dx = \int_0^x x dx$

$$\begin{aligned}
 \int_0^x [x] dx &= \int_0^1 0 dx + \int_1^2 1 dx + \dots + \int_{[x]-1}^{[x]} (([x]-1)) dx + \int_{[x]}^{[x]} [x] dx \\
 &= 1 + 2 + \dots + ([x]-1) + [x](x - [x]) \\
 &= \frac{([x]-1)([x]-1+1)}{2} + [x]\{x\}
 \end{aligned}$$

$$\int_0^{[x]} x dx = \frac{x^2}{2} \Big|_0^{[x]} = \frac{[x]^2}{2}$$

Form we have

$$\frac{([x]-1)([x])}{2} + [x]\{x\} = \frac{[x]^2}{2}$$

$$\text{by that we get } \{x\} = \frac{1}{2}$$

$$4\{x\} = 2$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 6: Applications of inequality, Sandwich property in comparison of functions in definite, Boundedness and continuity property of definite integral, Max-min inequality property of definite integral, Root mean square value - RMSV of a definite integral

25. Let $f: \left[\frac{1}{2}, 1\right] \rightarrow \mathbb{R}$ (the set of all real numbers) be a positive, non-constant and differentiable function such that $f'(x) < 2f(x)$ and $f\left(\frac{1}{2}\right) = 1$. Then the value of

$$\int_{1/2}^1 f(x) dx \text{ lies in the interval}$$

[2013-P1 (Q43)]

(property of comparison of functions in definite integral-Application of inequality)

- A. $(2e-1, 2e)$ B. $(e-1, 2e-1)$
 C. $\left(\frac{e-1}{2}, e-1\right)$ D. $\left(0, \frac{e-1}{2}\right)$

Ans: D

Sol: Given $f'(x) - 2f(x) < 0$

$$\Rightarrow f(x) < ce^{2x}$$

$$\text{Put } x = \frac{1}{2} \Rightarrow c > \frac{1}{e}$$

$$\text{Hence } f(x) < e^{2x-1}$$

$$\Rightarrow 0 < \int_{1/2}^1 f(x) dx < \int_{1/2}^1 e^{2x-1} dx$$

$$0 < \int_{1/2}^1 f(x) dx < \frac{e-1}{2}$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 7: Leibnitz's rule for differentiation under the integral sign

26. Let $f : [0, 2] \rightarrow \mathbb{R}$ be a function which is continuous on $[0, 2]$ and is differentiable on $(0, 2)$ with $f(0) = 1$. let $F(x) = \int_0^{x^2} f(\sqrt{t}) dt$ for $x \in [0, 2]$. If $F'(x) = f'(x)$ for all $x \in (0, 2)$, then $F(2)$ equals [2014-P2 (Q44)]

- A. $e^2 - 1$ B. $e^4 - 1$ (Newton-Leibniz function binding)
C. $e - 1$ D. e^4

Ans: B

Sol: $F(0) = 0$

$$F'(x) = 2x f(x) = f'(x)$$

$$f(x) = e^{x^2+c}$$

$$f(x) = e^{x^2} (\because f(0) = 1)$$

$$F(x) = \int_0^{x^2} e^x dx$$

$$F(x) = e^{x^2-1} (\because F(0) = 0)$$

$$\Rightarrow F(2) = e^4 - 1$$

27. If $f(X) = \int_x^{x^2} \frac{dt}{(\log t)^2}$, $X \neq 0, x \neq 1$, then $f(x)$ is

- A) monotonically increasing in $(2, \infty)$ B) monotonically increasing in $(1, 2)$
C) monotonically increasing $(2, \infty)$ D) monotonically decreasing in $(0, 1)$

KEY: A

SOL:

$$f'(X) = \frac{2X}{(\log X^2)^2} - \frac{1}{(\log X)^2} = \frac{1}{(\log X)^2} \left[\frac{X}{2} - 1 \right]$$

$$\Rightarrow f'(X) > 0 \Rightarrow X \in (2, \infty) \Rightarrow f'(X) < 0 \Rightarrow X \in (0, 2)$$

28. Number of real roots for which $\int_0^x \frac{t \cdot dt}{1+t^2} = 2x^2 - 120$ are

- 1) 2 2) 3 3) 4 4) 1

Key: 4

Sol:

$$\int_0^x \frac{t \cdot dt}{1+t^2} = 2x^2 - 120$$

by lebnitz rule

$$\frac{x}{1+x^2} = 4x$$

$$x(4x^2 + 3) = 0$$

 $x = 0$ is only real root

29. The value of $\int_0^{\sin^2 x} \sin^{-1}(\sqrt{t}) dt + \int_0^{\cos^2 x} \cos^{-1}(\sqrt{t}) dt$ is

1) π

2) $\frac{\pi}{2}$

3) $\frac{\pi}{4}$

4) None

Key: 3

Sol:

$$\text{Let } \int_0^{\sin^2 x} \sin^{-1}(\sqrt{t}) dt + \int_0^{\cos^2 x} \cos^{-1}(\sqrt{t}) dt$$

$$\phi'(x) = 0$$

$$\phi'(x) = k(\text{constant})$$

$$\phi(x) = \frac{\pi}{4}$$

30. Let $f: (0, \infty) \rightarrow R$ and $F(x) = \int_0^x f(t) dt$. If $F(x^2) = x^2(1+x)$, then

A. $f(4) = 1$

B. $f(1) = 4$

C. $f(4) = 7$

D. $f(1) = \frac{5}{2}$

Ans: D

Sol: $F'(x) = f(x)$

$$F'(x^2) 2x = 2x(1+x) + x^2$$

$$f(x) = \frac{2x(1+x) + x^2}{2x} = \frac{2(1+x) + x}{2}$$

31. If $G(x, t) = \begin{cases} x(t-1), & \text{where } x \leq t \\ t(x-1), & \text{where } t < x \end{cases}$ and if t is continuous function of x in $[0, 1]$.

Let $g(x) = \int_0^1 f(t)G(x, t)dt$, then

- A. $g(0) = 1$ B. $g(0) = 0$ C. $g(1) = 1$ D. $g''(x) = f'(x)$

Ans: B

Sol: For $x = 0$

$$G(0, t) = 0, t \geq 0$$

$$g(1) = \int_0^1 f(t)G(0, t)dt = \int_0^1 0 \cdot dt = 0$$

And for $x = 1$

$$G(1, t) = 0, t < 1$$

$$g(1) = \int_0^1 f(t)G(1, t)dt = 0$$

$$\text{Also, } g(x) = \int_0^x f(t)G(x, t)dt + \int_x^1 f(t)G(x, t)dt$$

$$x = (x-1) \int_0^x t f(t)dt = x \int_x^1 (t-1)f(t)dt$$

$$\therefore g''(x) = xf(x) + \{0 - (x-1)f(x)\} \\ = f(x)$$

Topic: Evaluation of definite integrals

Sub Topic 3: Reduction formulae in definite integrals (Wallis' Formula)

31. Let $I_n = \int_0^\infty e^{-x} (\sin x)^n dx$, $n \in N, n > 1$ then $\frac{I_{2008}}{I_{2006}}$ equals

- A. $\frac{2007 \times 2006}{2008^2 + 1}$ B. $\frac{2008 \times 2007}{2008^2 + 1}$ C. $\frac{2006 \times 2004}{2008^2 - 1}$ D. $\frac{2008 \times 2007}{2008^2 - 1}$

Ans: B

Sol: $I_n = \int_0^\infty e^{-x} (\sin x)^n dx$

$$= \left[\sin^n x (-e^{-x}) \right]_0^\infty + \int_0^\infty n \sin^{n-1} x \cos x e^{-x} dx$$

$$\begin{aligned}
&= 0 + n \int_0^{\infty} (\sin^{n-1} x \cos x) e^{-x} dx \\
&= n \left[(\sin^{n-1} x \cos x) (-e^{-x}) \right]_0^{\infty} - n \int_0^{\infty} \{ -\sin^n x + (n-1) \sin^{n-2} x \cos^2 x \} (-e^{-x}) dx \\
&= 0 + n \int_0^{\infty} e^{-x} \{ -\sin^n x + (n-1) \sin^{n-2} x (1 - \sin^2 x) \} dx \\
&= n \int_0^{\infty} e^{-x} \{ (n-1) \sin^{n-2} x - n \sin^n x \} dx \\
&= n(n-1) I_{n-2} - n^2 I_n
\end{aligned}$$

We have $(1+n^2)I_n = n(n-1)I_{n-2}$

$$\text{Then } \frac{I_n}{I_{n-2}} = \frac{n(n-1)}{n^2+1}$$

Topic: Evaluation of definite integrals

Sub Topic 4: BY PARTS

32. Let $f: [-1, 2] \rightarrow [0, \infty)$ be a continuous function such that $f(x) = f(1-x)$ for all $x \in [-1, 2]$. Let $R_1 = \int_1^2 x f(x) dx$, and R_2 be the area of region bounded by $y = f(x)$, $x = -1$, $x = 2$ and x-axis. Then
- 1) $R_1 = 2R_2$ 2) $R_1 = 3R_2$ 3) $2R_1 = R_2$ 4) $3R_1 = R_2$

Key: 3

Sol:

$$\begin{aligned}
R_1 &= \int_1^2 x f(x) dx = \int_1^2 (1-x) f(1-x) dx \\
&= \int_1^2 f(1-x) dx - \int_1^2 x f(1-x) dx \\
&= R_2 - R_1 \\
2R_1 &= R_2
\end{aligned}$$

33. If $y = f(x)$ makes +ve intercept of 2 and 0 unit with x and y and enclose an area of $\frac{3}{4}$

square unit with the axes then $\int_0^2 x f'(x) dx$ is

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1) $\frac{3}{2}$

2) 1

3) $\frac{5}{4}$

4) $-\frac{3}{4}$

Key: 4

Sol:

$$\int_0^2 x f'(x) dx = \frac{3}{4}$$

$$\int_0^2 x f'(x) dx = x \int_0^2 f'(x) dx - \int_0^2 f(x) dx$$

$$= [x.f(x)]_0^2 - \frac{3}{4} = 2f(2) - \frac{3}{4} = \frac{-3}{4}$$

EXERCISE - V

(NUMERICAL ANSWER TYPE)

Topic: Introduction to Definite integration

Sub Topic 1: Definite integral as a limit of an infinite Reimann Sum

1. For each positive integer n , define a function $f(x)$ on $(0, 1)$ as

$$f_n(x) = \begin{cases} 0 & \text{if } x = 0 \\ \sin \frac{r\pi}{2n} & \text{if } \frac{r-1}{n} < x \leq \frac{r}{n}, r = 1, 2, 3, \dots, n \end{cases} \quad \text{then the value of } \pi \left(\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx \right)$$

Ans: 2

Sol: $\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \int_{\frac{r-1}{n}}^{\frac{r}{n}} \sin \frac{r\pi}{2n} dx$$

$$= \int_0^1 \sin \frac{\pi x}{2} dx = \frac{2}{\pi} \therefore \pi \left(\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx \right) = 2$$

2. $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^{1/a} \left\{ n^{a-\frac{1}{a}} + k^{a-\frac{1}{a}} \right\}}{n^{a+1}}$ is equal to

Ans: 1

Sol: $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^{1/a} \left\{ n^{a-\frac{1}{a}} + k^{a-\frac{1}{a}} \right\}}{n^{a+1}}$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n} \cdot \left\{ \left(\frac{k}{n} \right)^{1/a} + \left(\frac{k}{n} \right)^a \right\} = \int_0^1 (x^{1/a} + x^a) dx$$

$$= \left\{ \frac{x^{(1/a)+1}}{\frac{1}{a}+1} + \frac{x^{a+1}}{a+1} \right\}_0^1 = \frac{a}{a+1} + \frac{1}{a+1} = 1$$

Topic: Introduction to Definite integration

Sub Topic 2: Second Fundamental theorem of integral calculus - definition

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3. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f(0) = 0$, $f\left(\frac{\pi}{2}\right) = 3$ and $f'(0) = 1$.
If

$$g(x) = \int_x^{\frac{\pi}{2}} [f'(t) \operatorname{cosec} t - \cot t \operatorname{cosec} t f(t)] dt$$

For $x \in \left(0, \frac{\pi}{2}\right)$, then $\lim_{x \rightarrow 0} g(x) =$ [2017-P1 (Q45)]

Ans: 2 (Second Fundamental theorem of integral calculus - definition of definite integral)

Sol: $g(x) = \int_x^{\pi/2} [f'(t) \operatorname{cosec} t - \cot(t) \operatorname{cosec}(t) f(t)] dt$

$$\Rightarrow g(x) = f(t) \operatorname{cosec}(t) \Big|_x^{\pi/2} = 3 - \frac{f(x)}{\sin x}$$

$$\Rightarrow \lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} \left(3 - \frac{f(x)}{\sin x} \right)$$

$$\Rightarrow \lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} \left(3 - \frac{f(x)}{\sin x} \right)$$

$$= 3 - f'(0) = 2$$

4. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that its derivative f' is continuous and $f(\pi) = -6$. If $F: [0, \pi] \rightarrow \mathbb{R}$ is defined by $F(x) = \int_0^x f(t) dt$, and if

$$\int_0^{\pi} (f'(x) + F(x)) \cos x dx = 2,$$

Then the value of $f(0)$ is [2020-P2 (Q17)]

Ans: 4 (Second fundamental theorem of integral calculus-definition of definite integrals, constant multiple rule of definite integrals)

Sol: $\int_0^{\pi} (f'(x) + F(x)) \cos x dx$

$$\int_0^{\pi} [F''(x) \cos x - F'(x) \sin x + F'(x) \sin x + F(x) \cos x] dx$$

$$\begin{aligned}
 &= \int_0^{\pi} (F'(x) \cos x + F(x) \sin x) \\
 &= -F'(\pi) - F'(0) = -f'(\pi) - f(0) = 2 \\
 &f(0) = 4
 \end{aligned}$$

Topic: Evaluation of definite integrals

Sub Topic 1: Evaluation of definite integrals by substitution

5. If $\alpha = \int_0^1 (e^{9x+3\tan^{-1}x}) \left(\frac{12+9x^2}{1+x^2} \right) dx$ [2015-P2 (Q46)]

Where $\tan^{-1}x$ takes only principal values, then the value of $\left(\log_e |1+\alpha| - \frac{3\pi}{4} \right)$ is

Ans: 9 (Substitution method)

Sol: $\alpha = \int_0^1 e^{(9x+3\tan^{-1}x)} \left(\frac{12+9x^2}{1+x^2} \right) dx$

Put $9x+3\tan^{-1}x = t$

$$\Rightarrow \left(9 + \frac{3}{1+x^2} \right) dx = dt$$

$$\Rightarrow \alpha = \int_0^{9+\frac{3\pi}{4}} e^t dt = e^{9+\frac{3\pi}{4}} - 1$$

$$\Rightarrow \left(\log_e |1+\alpha| - \frac{3\pi}{4} \right) = 9$$

6. The value of the integral

$$\int_0^{\frac{1}{2}} \frac{1+\sqrt{3}}{\left((x+1)^2 (1-x)^6 \right)^{\frac{1}{4}}} dx \text{ is } \underline{\hspace{2cm}} \quad [2018-P2 (Q7)]$$

Ans: 2 (Evaluation of definite integrals)

Sol: $I_2 = \int_0^{1/2} \frac{1+\sqrt{3}}{\sqrt{1-x^2} (1-x)} dx$

Let $x = \sin \theta$

$$I_2 = \int_0^{\pi/6} \frac{1+\sqrt{3}}{\left(\sin \frac{\theta}{2} - \cos \frac{\theta}{2}\right)^2} d\theta = \int_0^{\pi/2} \frac{(1+\sqrt{3}) \sec^2 \frac{\theta}{2}}{\left(\tan \frac{\theta}{2} - 1\right)^2} d\theta$$

Let $\tan \frac{\theta}{2} = y$

$$\Rightarrow I = \int_0^{2-\sqrt{3}} \frac{2(1+\sqrt{3}) dy}{(y-1)^2} = 2(1+\sqrt{3}) \left| \frac{-1}{y-1} \right|_0^{2-\sqrt{3}} = 2(1+\sqrt{3}) \left(\frac{\sqrt{3}-1}{2} \right) = 2$$

7. Let $A_n = \int_{\frac{1}{n+1}}^{\frac{1}{n}} \frac{\tan^{-1} nx}{\sin^{-1} nx} dx$ and $\lim_{n \rightarrow \infty} n^2 A_n = L$ then $8L =$

Key: 4

Sol: $A_n = \int_{\frac{1}{n+1}}^{\frac{1}{n}} \frac{\tan^{-1} nx}{\sin^{-1} nx} dx$ put $nx = t$

$$A_n = \frac{1}{n} \int_{\frac{n}{n+1}}^1 \frac{\tan^{-1} t}{\sin^{-1} t} dt$$

Now $L = \lim_{n \rightarrow \infty} n^2 A_n = \lim_{n \rightarrow \infty} \frac{\int_{n/n+1}^1 \frac{\tan^{-1} t}{\sin^{-1} t} dt}{\frac{1}{n}}$

Now use L Hospital rule Then $L = \frac{1}{2}$

8. If $\int_{\pi/4}^{\pi/3} \frac{(\sin^3 \theta - \cos^3 \theta - \cos^2 \theta)(\sin \theta + \cos \theta + \cos^2 \theta)^{2010}}{(\sin \theta)^{2012} (\cos \theta)^{2012}} d\theta = \frac{(a + \sqrt{b})^n - (1 + \sqrt{c})^n}{d}$

Where a, b, c and d are all positive integers. Find the value $(a + b + c + d)$.

Ans: 2021

Sol: $I = \int_{\pi/4}^{\pi/3} \frac{(\sin^3 \theta - \cos^3 \theta - \cos^2 \theta)(\sin \theta + \cos \theta + \cos^2 \theta)^{2010}}{\sin^2 \theta \cdot \cos^2 \theta (\sin \theta)^{2010} (\cos \theta)^{2010}} d\theta$

$$= \int_{\pi/4}^{\pi/3} (\tan \theta \sec \theta - \cot \theta \operatorname{cosec} \theta - \operatorname{cosec}^2 \theta) (\sec \theta + \operatorname{cosec} \theta + \cot \theta)^{2010} d\theta$$

Put $\sec \theta + \operatorname{cosec} \theta + \cot \theta = t;$

$$(\sec \theta \cot \theta - \cot \theta \operatorname{cosec} \theta - \operatorname{cosec}^2 \theta) d\theta = dt$$

$$I = \int_{\pi/4}^{\pi/3} t^{2010} dt = \frac{1}{2008} t^{2008}$$

$$= \frac{1}{2008} (\sec \theta + \operatorname{cosec} \theta + \cot \theta)^{2008} \Big|_{\pi/4}^{\pi/3}$$

$$= \frac{1}{2008} \left[\left(2 + \frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} \right)^{2008} - (\sqrt{2} + \sqrt{2} + 1)^{2008} \right]$$

$$= \frac{1}{2008} \left[(2 + \sqrt{3})^{2008} - (1 + \sqrt{8})^{2008} \right]$$

Hence, $a = 2, b = 3, c = 8$ and $d = 2008$.

$\Rightarrow$ Sum = 2021 (correct answer is 2024)

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 2: Replacing x by $a-x$ – King's property, Replacing x by $a + b - x$

9. If $I = \frac{2}{\pi} \int_{-\pi/4}^{\pi/4} \frac{dx}{(1 + e^{\sin x})(2 - \cos 2x)}$ then $27I^2$ equals _____ [2019-P1 (Q17)]

Ans: 4

(Properties of D.I, substitution)

Sol: $I = \frac{2}{\pi} \int_{-\pi/4}^{\pi/4} \frac{dx}{(1 + e^{\sin x})(2 - \cos 2x)}$ (i)

$$x = -t$$

$$I = \frac{2}{\pi} \int_{-\pi/4}^{\pi/4} \frac{e^{\sin t}}{(1 + e^{\sin t})(2 - \cos 2t)} dt$$
(ii)

Add (i) and (ii)

$$I = \frac{1}{\pi} \int_{-\pi/4}^{\pi/4} \frac{1}{2 - x \cos 2t} dt = \frac{1}{\pi} \int_{-\pi/4}^{\pi/4} \frac{\sec^2 t}{1 + 3 \tan^2 t} dt = \frac{1}{\pi} \cdot \frac{2\pi}{3\sqrt{3}}$$

$$3\sqrt{3}I = 2 \Rightarrow 27I^2 = 4$$

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10. The value of the integral $\int_0^{\pi/2} \frac{3\sqrt{\cos \theta}}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^5} d\theta$ equals _____ [2019-P2 (Q13)]

Ans: 0.5

(substitution)

Sol: $I = \int_0^{\pi/2} \frac{3\sqrt{\cos \theta} d\theta}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^5}$

$$\Rightarrow 2I = 3 \int_0^{\pi/2} \frac{d\theta}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^4} = \int_0^{\pi/2} \frac{3d\theta}{\cos^2 \theta (1 + \sqrt{\tan \theta})^4}$$

$$\Rightarrow I = \frac{3}{2} \int_0^{\pi/2} \frac{\sec^2 \theta d\theta}{(1 + \sqrt{\tan \theta})^4}$$

Let $1 + \sqrt{\tan \theta} = t \Rightarrow \frac{1}{2\sqrt{\tan \theta}} \sec^2 \theta d\theta = dt$

$$I = \frac{3}{2} \int_1^{\infty} \frac{2(t-1)}{t^4} dt = \frac{3}{2} \left[\frac{2}{-2t^2} + \frac{2}{3t^3} \right]_1^{\infty}$$

$$= \frac{3}{2} \left(\frac{2}{2} - \frac{2}{3} \right) = \frac{1}{2} = 0.5$$

11. $I = \int_0^1 \sqrt[3]{2x^3 - 3x^2 + 3x - 1} dx = \underline{\hspace{2cm}}$

Key:0

Sol: $= \int_0^1 \sqrt[3]{x^3 - (1-x)^3} dx$

$$= - \int_0^1 \sqrt[3]{(1-x)^3 - x^3} dx$$

$$= -I$$

$$\therefore I = 0$$

12. The value of $\int_0^1 \frac{1}{(5+2x-2x^2)(1+e^{2-4x})} dx$ is $\frac{1}{2\sqrt{a}} \ln \frac{\sqrt{a}+b}{\sqrt{a}-b}$ then $a-2b =$

Key:9

Sol: $I = \int_0^1 \frac{dx}{(5+2x-2x^2)(1+e^{2-4x})} \dots (1)$

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$$= \int_0^1 \frac{dx}{[5 + 2(1-x) - 2(1-x)^2][1 + e^{2-4(1-x)}]} \quad \text{by } \int_a^b f(x) dx = \int_a^b f(a+b-x) dx$$

$$= \int_0^1 \frac{dx}{(5 + 2x - 2x^2)(1 + e^{-2+4x})}$$

$$= \int_0^1 \frac{e^{2-4x} dx}{(5 + 2x - 2x^2)(1 + e^{2-4x})} \dots (2)$$

$$(1) + (2)$$

$$\Rightarrow 2I = \int_0^1 \frac{e^{2-4x} + 1}{(5 + 2x - 2x^2)(1 + e^{2-4x})} = \int_0^1 \frac{dx}{(5 + 2x - 2x^2)}$$

$$= -\frac{1}{2} \int_0^1 \frac{dx}{\left(x^2 - x - \frac{5}{2}\right)} = -\frac{1}{2} \int_0^1 \frac{dx}{\left(x - \frac{1}{2}\right)^2 - \left(\frac{\sqrt{11}}{2}\right)^2} = \frac{1}{2} \cdot \frac{1}{\sqrt{11}} \left[\ln \left(\frac{\frac{\sqrt{11}}{2} + x - \frac{1}{2}}{\frac{\sqrt{11}}{2} - x + \frac{1}{2}} \right) \right]_0^1 = \frac{1}{2\sqrt{11}} \ln \left(\frac{\sqrt{11} + 1}{\sqrt{11} - 1} \right)$$

13. If $I = \frac{6}{\pi} \int_{-\pi/4}^{\pi/4} \frac{1}{(1 + e^{\sin x})(2 - 2\sin^2 x)} dx$ then $3I^2$ equals _____

Ans: 4

Sol: $I = \frac{6}{\pi} \int_{-\pi/4}^{\pi/4} \frac{dx}{(1 + e^{\sin x})(2 - \cos 2x)} \dots (i)$

$$x = -t$$

$$I = \frac{6}{\pi} \int_{-\pi/4}^{\pi/4} \frac{e^{\sin t}}{(1 + e^{\sin t})(2 - \cos 2t)} dt \dots (ii)$$

Add (i) and (ii)

$$I = \left(\frac{1}{\pi} \int_{-\pi/4}^{\pi/4} \frac{1}{2 - x \cos 2t} \right)^3 dt = \left(\frac{1}{\pi} \int_{-\pi/4}^{\pi/4} \frac{\sec^2 t}{1 + 3 \tan^2 t} \right)^3 dt = \frac{1}{\pi} \cdot \frac{2\pi \cdot 3}{3\sqrt{3}}$$

$$\Rightarrow 3I^2 = 4$$

14. If $I_1 = \int_0^1 \frac{dx}{e^x(1+x)}$ and $I_2 = \int_0^{\pi/4} \frac{e^{\tan^2 \theta}}{(2 - \tan^2 \theta)} \frac{\sin \theta d\theta}{\cos^3 \theta}$, then $\frac{eI_1}{I_2}$ is equal to :....2

Ans: 2

Sol: $I_1 = \int_0^1 \frac{dx}{e^x(1+x)}$ and $I_2 = \int_0^{\pi/4} \frac{e^{\tan^2 \theta}}{2 - \tan^2 \theta} \cdot \frac{\sin \theta}{\cos^3 \theta} d\theta$

$$I_2 = \frac{1}{2} \int_0^1 \frac{e^x}{2-x} dx = \frac{1}{2} \int_0^1 \frac{e^{1-x}}{2-(1-x)} dx = \frac{e}{2} \int_0^1 \frac{dx}{e^x(1+x)}$$

$$\therefore \frac{I_1}{I_2} = \frac{2}{e} \Rightarrow \frac{eI_1}{I_2} = 2$$

15. The value of the integral $\int_0^{\pi/2} \frac{3\sqrt{\cos \theta}}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^4} d\theta$ equals _____

Ans: 1.5

Sol: $I = \int_0^{\pi/2} \frac{3\sqrt{\cos \theta} d\theta}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^4}$

$$\Rightarrow 2I = 3 \int_0^{\pi/2} \frac{d\theta}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^3} = \int_0^{\pi/2} \frac{3d\theta}{\cos^2 \theta (1 + \sqrt{\tan \theta})^3}$$

$$\Rightarrow I = \frac{3}{2} \int_0^{\pi/2} \frac{\sec^2 \theta d\theta}{(1 + \sqrt{\tan \theta})^3}$$

Let $1 + \sqrt{\tan \theta} = t \Rightarrow \frac{1}{2\sqrt{\tan \theta}} \sec^2 \theta d\theta = dt$

$$I = \frac{3}{2} \int_1^{\infty} \frac{2}{t^2} - \frac{2}{t^3}$$

$$= \frac{3}{2} \left(\frac{-2}{t} - \frac{1}{t^3} \right)_1^{\infty} = \frac{3}{2} (2 - 1) = \frac{3}{2} = 1.5$$

16. The value of the integral $\int_0^{\pi/2} \frac{k\sqrt{\cos \theta}}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^5} d\theta$ then k equals _____

Ans: 3

Sol: $I = \int_0^{\pi/2} \frac{k\sqrt{\cos \theta} d\theta}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^5}$

$$\Rightarrow 2I = k \int_0^{\pi/2} \frac{d\theta}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^4} = \int_0^{\pi/2} \frac{kd\theta}{\cos^2 \theta (1 + \sqrt{\tan \theta})^4}$$

$$\Rightarrow I = \frac{k}{2} \int_0^{\pi/2} \frac{\sec^2 \theta d\theta}{(1 + \sqrt{\tan \theta})^4}$$

$$\text{Let } 1 + \sqrt{\tan \theta} = t \Rightarrow \frac{1}{2\sqrt{\tan \theta}} \sec^2 \theta d\theta = dt$$

$$I = \frac{k}{2} \int_1^\infty \frac{2(t-1)}{t^4} dt = \frac{k}{2} \left[\frac{2}{-2t^2} + \frac{2}{3t^3} \right]_1^\infty = \frac{k}{6} = \frac{1}{2}$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 4: Half the integral limit property (Splitting of limits 0 to 2a as 0 to a and a to 2a in f(x)), f(2a-x) property

17. For any real number x, let $[x]$ denote the largest integer less than or equal to x. If

$$I = \int_0^{10} \left[\sqrt{\frac{10x}{x+1}} \right] dx,$$

Then the value of 9I is _____ [2021-P2 (Q19)]

Ans: 182 (Properties of evaluation definite integrals)

Sol: $I = \int_0^{10} \left[\sqrt{\frac{10x}{x+1}} \right] dx$

$$\left[\sqrt{\frac{10x}{x+1}} \right] = n \Rightarrow \frac{n^2}{10-n^2} \leq x < \frac{(n+1)^2}{10-(n+1)^2} \text{ where } n \in I$$

$$\text{For } n=1, 0 \leq x < 1/9$$

$$n=1; 1/9 \leq x < 2/3$$

$$n=2; 2/3 \leq x < 9, n=3, x \geq 9$$

$$\begin{aligned} \Rightarrow I &= \int_0^{1/9} 0 \cdot dx + \int_{1/9}^{2/3} 1 \cdot dx + \int_{2/3}^9 2 \cdot dx + \int_9^{10} 3 \cdot dx \\ &= \left(\frac{2}{3} - \frac{1}{9} \right) + 2 \left(9 - \frac{2}{3} \right) + 3(10-9) = \frac{182}{9} = 9I = 182 \end{aligned}$$

18. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \begin{cases} [x], & x \leq 2 \\ 0, & x > 2 \end{cases}$,

Where $[x]$ is the greatest integer less than or equal to x . If $I = \int_{-1}^2 \frac{xf(x^2)}{2+f(x+1)} dx$, then the value of $(4I - I)$ is **[2015-P1 (Q46)]**

Ans: 0 (Splitting limits - Application 1 of piecewise function property)

$$\text{Sol: } I = \int_{-1}^0 \frac{x \cdot 0}{2+0} dx + \int_0^1 \frac{x \cdot 0}{2+1} dx + \int_1^{\sqrt{2}} \frac{x \cdot 1}{2+0} dx + 0 = \frac{1}{4}$$

$$\Rightarrow 4I - I = 0$$

1. If $\int_{\pi}^{2\pi} \cos^{-1}(\cos x) dx = \lambda\pi^2 + \mu\pi$ then $|\lambda - \mu| = \underline{\hspace{2cm}}$.

KEY: 7

$$\text{SOL: } \int_{\pi}^{2\pi} (2\pi - 5) dx = (2\pi - 5)\pi = 2\pi^2 - 5\pi$$

$$\therefore \lambda = 2, \mu = -5.$$

19. The value of $I = \int_{\pi/2}^{\pi/2} \frac{\cos x dx}{1 + 2[\sin^{-1}(\sin x)]}$ (where, $[\cdot]$ denotes greatest integer function) is

KEY: 0

SOL:

$$\begin{aligned} \text{Here, } I &= \int_{-\pi/2}^{\pi/2} \frac{\cos x dx}{1 + 2[\sin^{-1}(\sin x)]} \\ &= \int_{-\pi/2}^{-1} \frac{\cos x}{-3} dx + \int_{-1}^0 \frac{\cos x}{-1} dx + \int_0^1 \cos x dx + \int_1^{\pi/2} \frac{\cos x}{3} dx \\ &= \frac{1}{3} \int_{-1}^{-\pi/2} \cos x dx + \int_0^{-1} \cos x dx + \int_0^1 \cos x dx + \frac{1}{3} \int_1^{\pi/2} \cos x dx \\ &= \frac{1}{3} \int_0^{\pi/2} \cos t (-dt) + \int_0^1 \cos(t) \cdot (-dt) \\ &\quad + \int_0^1 \cos x dx + \frac{1}{3} \int_1^{\pi/2} \cos x dx \\ &= 0 \end{aligned}$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 5: Periodic function property

20. For any real number x , let $[x]$ denote the largest integer less than or equal to x . Let f be a real valued function defined on the interval $[-10, 10]$ by

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$$f(x) = \begin{cases} x - [x] & \text{if } [x] \text{ is odd,} \\ 1 + [x] - x & \text{if } [x] \text{ is even} \end{cases}$$

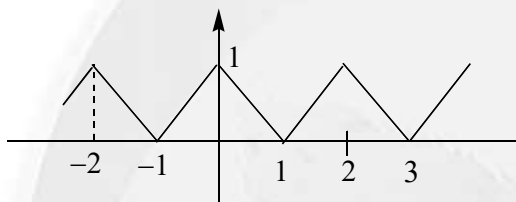
Then the value of $\frac{\pi^2}{10} \int_{-10}^{10} f(x) \cos \pi x dx$ is

[2010-P1 (Q52)]

Ans: 4

(PROPERTIES OF EVALUATION OF DEFINITE INTEGRALS)

Sol: $f(x) = \begin{cases} x-1 & 1 \leq x < 2 \\ 1-x & 0 \leq x < 1 \end{cases}$



$f(x)$ is periodic with period 2

$$\therefore I = \int_0^{10} f(x) \cos \pi x dx$$

$$= 2 \int_0^{10} f(x) \cos \pi x dx = 2 \times 5 \int_0^2 f(x) \cos \pi x dx$$

$$= 10 \left[\int_0^1 (1-x) \cos \pi x dx + \int_1^2 (x-1) \cos \pi x dx \right] = 10(I_1 + I_2)$$

$$I_2 = \int_1^2 (x-1) \cos \pi x dx \quad \text{put } x-1 = t$$

$$I_2 = - \int_0^1 t \cos \pi t dt$$

$$I_1 = \int_0^1 (1-x) \cos \pi x dx = - \int_0^1 x \cos(\pi x) dx$$

$$\therefore I = 10 \left[-2 \int_0^1 x \cos \pi x dx \right]$$

$$= -20 \left[x \frac{\sin \pi x}{\pi} + \frac{\cos \pi x}{\pi^2} \right]_0^1$$

$$= -20 \left[-\frac{1}{\pi^2} - \frac{1}{\pi^2} \right] = \frac{40}{\pi^2} \therefore \frac{\pi^2}{10} I = 4$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 6: Applications of inequality, Sandwich property in comparison of functions in definite, Boundedness and continuity property of definite integral, Max-min inequality property of definite integral, Root mean square value - RMSV of a definite integral

21. The total number of distinct $x \in [0, 1]$ for which $\int_0^x \frac{t^2}{1+t^4} dt = 2x - 1$ is **[2016-P1 (Q52)]**

Ans: 1 (Max-min inequality property of definite integral - Applications of inequality)

Sol: $\int_0^x \frac{t^2}{1+t^4} dt = 2x - 1$

Let $f(x) = \int_0^x \frac{t^2}{1+t^4} dt$

$\Rightarrow f'(x) = \frac{x^2}{1+x^4} - 2 < 0 \forall x \in [0, 1]$

Now, $f(0) = 1$ and $f(1) = \int_0^1 \frac{t^2}{1+t^4} dt - 1$

As $0 \leq \frac{t^2}{1+t^2} - 2 < 0 \forall x \in [0, 1]$

$\Rightarrow \int_0^1 \frac{t^2}{1+t^4} dt < \frac{1}{2} \Rightarrow f(1) < 0$

$\Rightarrow f(x) = 0$ has exactly one root in $[0, 1]$

- 22. Let $f: [0, 1] \rightarrow \mathbb{R}$ be an integrable function such that $\int_0^1 xf(x) dx = 1$ then the minimum of $3 \int_0^1 f^2(x) dx$ is

Ans: 5

Sol: $(f(x) - x)^2 \geq 0 \Rightarrow f^2(x) - 2xf(x) + x^2 \geq 0$

$\int_0^1 f^2(x) dx \geq 2 \int_0^1 xf(x) dx - \int_0^1 x^2 dx = 2 - \frac{1}{3} = \frac{5}{3}$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 7: Leibnitz's rule for differentiation under the integral sign

23. For $X \in \mathbb{R}$, let $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then the minimum value of the function

$$f: \mathbb{R} \rightarrow \mathbb{R} \text{ defined by } f(x) = \int_0^{x \tan^{-1} x} \frac{e^{(t-\cos t)}}{1+t^{2023}} dt \text{ is} \quad [2023-P2 (Q8)]$$

Ans: 0

(PROPERTIES OF EVALUATION OF DEFINITE INTEGRALS)

Sol: $f'(x) = \frac{e^{[x \tan^{-1} x - \cos(x \tan^{-1} x)]}}{1+(x \tan^{-1} x)^{2023}} \times \left(\frac{x}{1+x^2} + \tan^{-1} x \right)$

$$f'(x) = g(x) \cdot h(x)$$

$$\text{where } g(x) = \frac{e^{[x \tan^{-1} x - \cos(x \tan^{-1} x)]}}{1+(x \tan^{-1} x)^{2023}} > 0 \forall x$$

$$\text{and } h(x) = \frac{x}{1+x^2} + \tan^{-1} x < 0 \text{ for } x < 0$$

$$= 0 \quad x = 0$$

$$> 0 \quad x > 0$$

$\therefore f(x)$ has minimum at $x=0$

$$\text{And } f(x)_{\min} = f(0) = 0$$

24. Let $F(x) = \int_x^{x^2 + \frac{\pi}{6}} 2 \cos^2 t dt$ for all $x \in \mathbb{R}$ and $f: \left[0, \frac{1}{2}\right] \rightarrow [0, \infty)$ be a continuous

function. For $a \in \left[0, \frac{1}{2}\right]$, if $F'(a) + 2$ is the area of the region bounded by

$x=0, y=0, y=f(x)$ and $x=a$, then $f(0)$ is

[2015-P1 (Q48)]

Ans: 3

(Leibnitz's rule for differentiation under the integral sign)

Sol: $F'(a) + 2 = \int_0^a f(x) dx$

Differentiating w.r.t. a

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$$F''(a) = f(a)$$

$$F'(x) = 2 \cos^2 \left(x^2 + \frac{\pi}{6} \right) 2x - 2 \cos^2 x$$

$$F''(x) = 4 \cos^2 \left(x^2 + \frac{\pi}{6} \right) - 16x^2 \cos \left(x^2 + \frac{\pi}{6} \right) \sin \left(x^2 + \frac{\pi}{6} \right) + 4 \cos x \sin x$$

$$F''(0) = f(0) = 4 \cos^2 \frac{\pi}{6} = 3$$

Topic: Evaluation of definite integrals

Sub Topic 3: Reduction formulae in definite integrals (Wallis' Formula)

25. If $\int_0^{\pi} x \sin^5 x \cos^6 x dx = \frac{k\pi}{693}$ then $k = \dots\dots\dots$

Key. 8

Sol. Let $I = \int_0^{\pi} x \sin^5 x \cos^6 x dx = \int_0^{\pi} (\pi - x) \sin^5 (\pi - x) \cos^6 (\pi - x) dx = \int_0^{\pi} (\pi - x) \sin^5 x \cos^6 x dx$

$$= \int_0^{\pi} \pi \sin^5 x \cos^6 x dx - \int_0^{\pi} x \sin^5 x \cos^6 x dx$$

$$\Rightarrow 2I = \pi \cdot 2 \int_0^{\pi/2} \sin^5 x \cos^6 x dx \Rightarrow I = \pi \left[\frac{4}{11} \cdot \frac{2}{9} \cdot \frac{1}{7} \right] = \frac{8\pi}{693}.$$

Topic: Evaluation of definite integrals

Sub Topic 4 : BY PARTS

26. The value of $\int_0^1 4x^3 \left\{ \frac{d^2}{dx^2} (1-x^2)^5 \right\} dx$ is [2014-P1 (Q55)]

Ans: 2

(Byparts)

Sol: $\int_0^1 4x^3 \frac{d^2}{dx^2} (1-x^2)^5 dx$

$$= \left[4x^3 \frac{d}{dx} (1-x^2)^5 \right]_0^1 - \int_0^1 12x^2 \frac{d}{dx} (1-x^2)^5 dx$$

$$= \left[4x^2 \times 5(1-x^2)^4 (-2x) \right]_0^1 - 12 \left[\left[x^2 (1-x^2)^5 \right]_0^1 - \int_0^1 2x (1-x^2)^5 dx \right]$$

$$= 0 - 0 - 12[0 - 0] + 12 \int_0^1 2x (1-x^2)^5 dx$$

$$= 12 \times \left[-\frac{(1-x^2)^6}{6} \right]_0^1$$

$$= 12 \left[0 + \frac{1}{6} \right] = 2$$

27. $\frac{\int_0^{\pi/2} \left(\frac{\theta}{\sin \theta} \right)^2 d\theta}{\int_0^{\pi/2} \log \sin \theta d\theta}$ equals

Ans: 2

- Sol: $I = \int_0^{\pi/2} \left(\frac{\theta}{\sin \theta} \right)^2 d\theta = \int_0^{\pi/2} \theta^2 \operatorname{cosec}^2 \theta d\theta$
 - $= \left[\theta^2 (-\cot \theta) \right]_0^{\pi/2} - \int_0^{\pi/2} 2\theta (-\cot \theta) d\theta$
 - (integrating by parts)
 - $\left[-\lim_{\theta \rightarrow \pi/2} \theta^2 \cot \theta \right] + 2 \int_0^{\pi/2} \theta \cot \theta d\theta$
 - $= 0 + 2 \left[0 \log \sin \theta \right]_0^{\pi/2} - \int_0^{\pi/2} \log \sin \theta$
 - (Integrating by parts)
 - $= 2 \left[-\lim_{\theta \rightarrow 0} \ln \sin \theta - k \right]$ Where $k = \int_0^{\pi/2} \ln \sin \theta d\theta = -2k$.

28. Let $f: \mathbb{R} \rightarrow [-3, 3]$ be a differentiable function such that $f\left(\frac{\pi}{2}\right) = -2$

$$g(x) = \int_x^{\frac{\pi}{2}} [f'(t) \sin t + f(t) \cos t] dt \text{ for } x \in \left(0, \frac{\pi}{2}\right], \text{ then } \lim_{x \rightarrow 0} g(x) =$$

Key:2

Sol: $g(x) = -f(x) \sin x + C$

$$g\left(\frac{\pi}{2}\right) = 0 \Rightarrow +2 + C = 0$$

$$\Rightarrow C = -2$$

$$\Rightarrow g(x) = -f(x) \sin x - 2$$

$$\Rightarrow \lim_{x \rightarrow 0} g(x) = -2$$

30. $f: [0, 5] \rightarrow \mathbb{R}$, $y = f(x)$ such that $f''(x) = f''(5-x) \forall x \in [0, 5]$ $f'(0) = 1$ and $f'(5) = 7$, then evaluate $\int_1^4 f'(x) dx - 4 \dots \dots \dots$

Ans: 8

Sol. $\int_1^4 f'(x) dx = [xf'(x)]_1^4 - \int_1^4 xf''(x) dx$

$$I = \int_1^4 xf''(x) dx = \int_1^4 (5-x)f''(5-x) dx$$

$$= 5 \int_1^4 f''(x) dx - I$$

$$I = \frac{5}{2} [f'(4) - f'(1)]$$

$$\text{So, } \int_1^4 f'(x) dx = \frac{3}{2} [f'(4) + f'(1)]$$

$$\text{Now, } f''(x) = f''(5-x) \Rightarrow f'(x) = -f'(5-x) + c$$

$$f'(0) + f'(5) = c \Rightarrow c = 8$$

$$\text{so } f'(x) + f'(5-x) = 8 \Rightarrow f'(4) + f'(1) = 8$$

31. The value of $\int_0^1 kx^3 \left\{ \frac{d^2}{dx^2} (1-x^2)^5 \right\} dx = 2$ then k is equals to _____

Ans: 4

Sol: $\int_0^1 4x^3 \frac{d^2}{dx^2} (1-x^2)^5 dx$

$$= \left[4x^3 \frac{d}{dx} (1-x^2)^5 \right]_0^1 - \int_0^1 12x^2 \frac{d}{dx} (1-x^2)^5 dx$$

$$= \left[4x^2 \times 5(1-x^2)^4 (-2x) \right]_0^1 - 12 \left[\left[x^2 (1-x^2)^5 \right]_0^1 - \int_0^1 2x (1-x^2)^5 dx \right]$$

$$= 0 - 0 - 12[0 - 0] + 12 \int_0^1 2x (1-x^2)^5 dx$$

$$= 12 \times \left[-\frac{(1-x^2)^6}{6} \right]_0^1$$

$$= 12 \left[0 + \frac{1}{6} \right] = 2$$

32. Given that $U_n = \{x(1-x)\}^n$ and $n \geq 2$ and $\frac{d^2 U_n}{dx^2} = n(n-1)U_{n-2} - 2n(2n-1)U_{n-1}$, further

if $V_n = \int_0^1 e^x \cdot U_n dx$, then for $n \geq 2$, we have $V_n + K_1 n(2n-1) \cdot V_{n-1} + K_2 n(n-1) \cdot V_{n-2} = 0$.

Find $(K_1 + K_2)$.

Ans: 1

Sol: $U_n = \{x(1-x)\}^n$

$$\frac{dU_n}{dx} = x \underbrace{[x(1-x)]^{n-1}}_I \cdot \underbrace{(1-2x)}_{II}$$

$$= \frac{d^2U_n}{dx^2} = x[x(1-x)]^{n-1}(-2) + (1-2x)^2 n(n-1)[x(1-x)]^{n-2}$$

$$= n[x(1-x)]^{n-1}(-2) + n(n-1)[x(1-x)]^{n-2} + (4x^2 - 4x)n(n-1)[x(1-x)]^{n-2}$$

$$= n(n-1)U_{n-2} + 4(x^2 - 4x)n(n-1)(x-x^2)^{n-2} - 2n(x-x^2)^{n-1}$$

$$= n(n-1)U_{n-2} - 2n(x-x^2)^{n-1}[2n-2+1]$$

$$= n(n-1)U_{n-2} - 2n(2n-1)(x-x^2)^{n-1}$$

$$V_n = \int_0^1 e^x \cdot U_n dx = U_n \cdot e^x \Big|_0^1 - \int_0^1 \underbrace{\frac{dU_n}{dx}}_I \cdot \underbrace{e^x}_{II} dx$$

$$= -n(1-2x)[x(1-x)]^{n-1} \Big|_0^1 - \int_0^1 \frac{d^2U_n}{dx^2} \cdot e^x dx + \int_0^1 e^x [n(n-1)U_{n-2} - 2n(2n-1)U_{n-1}] dx$$

$$V_n = n(n-1)V_{n-1} - 2n(2n-1)V_{n-1}$$

i.e., $V_n + 2n(2n-1)V_{n-1} - n(n-1)V_{n-1} = 0$

•
•

Topic: Evaluation of definite integrals

33. The absolute value of $\frac{\pi}{\log(2-\sqrt{3})} \int_{\frac{\pi}{6}}^{\frac{1}{2}} \operatorname{cosec} \pi x dx$ is _____

KEY: 1

SOL: $I = \frac{\pi}{\log(2-\sqrt{3})} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \operatorname{cosec} \pi \cdot \frac{dt}{\pi} \quad (\because \pi x = t)$

$$= \frac{\pi}{\log(2-\sqrt{3})} \cdot \frac{1}{\pi} \cdot \log |\operatorname{cosec} t - \cot t| \Big|_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$= \frac{1}{\log(2-\sqrt{3})} [\log|1-0| - \log|2-\sqrt{3}|]$$

$$= -1$$

$$\therefore I=1$$

34. Let

$$f(x) = x^3 + 3x^2 - 6x + 11, g(x) = \ln|x|. \text{ If } A = f(g'(-1)), B = g\left(f'(-1)\right), C = \int_{-2}^1 f'(x)g'(f(x))dx,$$

$$D = \frac{d}{dt} \int_1^{t^2} g(x)dx \text{ evaluated at } t=3 \text{ and } A(B+C) - D = k \ln(3) \text{ then the value of } K \text{ is } \underline{\hspace{2cm}}$$

KEY: 7

SOL:

$$A = f(-1) = 19$$

$$B = g(-9) = 2 \ln 3$$

$$C = \int_{-2}^1 \frac{3(x^2 + 2x - 2)}{x^3 + 3x^2 - 6x + 11} dx = \ln 9 - \ln 27 = -\ln 3$$

$$D = \frac{d}{dt} \int_1^{t^2} g(x)dx = 6g(9) = 12 \ln 3$$

$$A(B+C) - D = 7 \ln 3 \Rightarrow k = 7$$

Topic: Evaluation of definite integrals

Sub Topic 5 : INVERSE OF A FUNCTION

35. The greatest integer less than or equal to

$$\int_1^2 \log_2(x^3 + 1) dx + \int_1^{\log_2 9} (2^x - 1)^{\frac{1}{3}} dx \text{ is } \underline{\hspace{2cm}} \quad [2022-P2 (Q3)]$$

Ans: 5

(Properties of evaluation definite integrals)

Sol:

$$\int_1^2 \log_2(x^3 + 1) dx + \int_1^{\log_2 9} (2^x - 1)^{\frac{1}{3}} dx$$

$$\underbrace{\hspace{10em}}_{f(x)} \quad \underbrace{\hspace{10em}}_{f^{-1}(x)}$$

$$bd - ac = 2 \log_2 9 - 1$$

$$= 4 \log_2 3 - 1$$

$$= 4 \left\{ \frac{0.477}{0.301} \right\} - 1$$

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$$= 5.33$$

$$[5.33] = 5$$

36. The least integer greater than

$$\int_{\pi/6}^{\pi/3} \sin x \, dx + \int_{1/2}^{\sqrt{3}/2} \sin^{-1} x \, dx \text{ is } \underline{\hspace{2cm}}$$

ANS: 1

$$\int_a^b f(x) dx + \int_{f(a)}^{f(b)} f^{-1}(x) dx = b f(b) - a f(a)$$

$$= \frac{\pi}{3} \cdot \frac{\sqrt{3}}{2} - \frac{\pi}{6} \cdot \frac{1}{2} = \frac{\pi}{12} (2\sqrt{3} - 1)$$

$$= 0.6(app)$$

EXERCISE - VI (MATRIX MATCH)

Topic: Evaluation of definite integrals

Sub Topic 1: Evaluation of definite integrals by substitution

1. If $2^{2010} \frac{\int_0^1 x^{1004} (1-x)^{1004} dx}{\int_0^1 x^{1004} (1-x^{2010})^{1004} dx} = abcd$, where abcd is a four digit number and

a, b, c, d are the digits lying from 0 to 9. Then

Match the following from Column – I to Column -II

	Column – I		Column – II
(A)	$a + b + d =$	(p)	2
(B)	$a^2 - b^2 + d^2 =$	(q)	4
(C)	$b + c - d =$	(r)	8
(D)	$b^2 + c^2 + d^2 =$	(s)	16

Key: $A \rightarrow Q; B \rightarrow S; C \rightarrow P; D \rightarrow Q$

Sol: 1. Let $I_2 = \int_0^1 x^{1004} (1-x^{2010})^{1004} dx$, Put $x^{1005} = t \Rightarrow 1005 x^{1004} dx = dt$

$$\Rightarrow I_2 = \frac{1}{1005} \int_0^1 (1-t^2)^{1004} dt = \frac{1}{1005} \int_0^1 t(2-t)^{1004} dt$$

Let $t = 2z \Rightarrow dt = 2dz$

$$I_2 = \frac{1}{1005} \int_0^{\frac{1}{2}} 2^{1004} z^{1004} (2-2z)^{1004} \cdot 2dz = \frac{2^{2009}}{1005} \int_0^{\frac{1}{2}} z^{1004} (1-z)^{1004} dz$$

$$I_1 = \int_0^1 x^{1004} (1-x)^{1004} dx = 2 \int_0^{\frac{1}{2}} x^{1004} (1-x)^{1004} x \Rightarrow \frac{I_1}{I_2} = \frac{1005}{2^{2008}}$$

2. **Column – I**

Column – II

A) If $f(\pi) = 2$ and $\int_0^{\pi} (f(x) + f''(x)) \sin x dx = 5$, then p) 2

$f(0)$ is equal to (It is given that $f(x)$ is continuous in $[0, \pi]$)

B) The value of $\int_{1/e}^{\tan x} \frac{t}{1+t^2} dt + \int_{1/e}^{\cot x} \frac{1}{t(1+t^2)} dt$, where $x \in \left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ q) 1/2

is equal to

C) $\lim_{n \rightarrow \infty} \left(\sin \frac{\pi}{2n} \cdot \sin \frac{2\pi}{2n} \cdot \sin \frac{3\pi}{2n} \dots \sin \frac{(n-1)\pi}{n} \right)^{1/n}$ is equal to r) 3

D) If $f(x) = \begin{cases} 0 & , \text{ where } x = \frac{n}{n+1}, n=1,2,3,\dots \\ 1 & , \text{ else where} \end{cases}$ s) 1

then the value of $\int_0^2 f(x) dx$

A. A – r ; B – s ; C – q ; D – p

B. A – r ; B – s ; C – r ; D – q

C. A – r ; B – s ; C – p ; D – q

D. A – s ; B – r ; C – q ; D – p

key. A

sol. 5. A)

$$B) \cap = \int_{1/e}^{\tan x} \frac{t}{1+t^2} dt + \int_{1/e}^{\cot x} \frac{1}{t(1+t^2)} dt$$

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Put $t = \frac{1}{x}$

$$\cap = \int_{1/e}^{\tan x} \frac{t}{1+t^2} dt + \int_e^{\tan x} \frac{t}{\frac{1}{x}\left(1+\frac{1}{x^2}\right)} dx \cdot \left(-\frac{1}{x^2}\right) dx$$

$$\cap = \int_{1/e}^{\tan x} \frac{t}{1+t^2} + \int_{\tan x}^e \frac{x}{1+x^2} dx = \int_{1/e}^e \frac{t}{1+t^2} dt = \left(\frac{1}{2} \ln(1+t^2)\right)_{1/e}^e = 1$$

C)

$$D) \int_0^2 f(x) dx = \int_0^{1/2} 1 \cdot dx + \int_{1/2}^{2/3} 1 \cdot dx + \int_{2/3}^{3/4} 1 \cdot dx + \dots + \int_{\frac{n-1}{n}}^{\frac{n}{n+1}} 1 \cdot dx + \dots + \int_1^2 1 \cdot dx$$

$$= \left(\frac{1}{2}\right) + \left(\frac{2}{3} - \frac{1}{2}\right) + \left(\frac{3}{4} - \frac{2}{3}\right) + \dots + \left(\frac{n}{n+1} - \frac{n-1}{n}\right) + \dots + 1 = \frac{n}{n+1} + \dots + 1 \quad \text{as } n \rightarrow \infty$$

Taking limit $n \rightarrow \infty$ We get $\int_0^2 f(x) dx = 1 + 1 = 2$

3. A function F is defined by $F(x) = \int_1^x \frac{e^t}{t} dt \quad \forall x > 0$.

COLUMN-I

COLUMN-II

A) $\int_1^x \frac{e^t}{t+2} dt$

p) $F(x) - \frac{e^x}{x} + e$

B) $\int_1^x \frac{e^{3t}}{t} dt$

q) $xe^{\frac{1}{x}} - e - F\left(\frac{1}{x}\right)$

C) $\int_1^x \frac{e^t}{t^2} dt$

r) $e^{-2} [F(x+2) - F(3)]$

D) $\int_1^x \frac{1}{t} e^t dt$

s) $F(3x) - F(3)$

t) $F(3x) - e^x + 1$

Key. A – r

B – s

C – p

D – q

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sol. a) put $t + 2 = z$

b) put $3t = z$

c) Integrate by parts

d) Integrate by parts

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 1: Change of variables, interchanging limits, splitting limits Property,

Location of roots in an interval using definite integral, Improper integral Property, Reflection property, Shifting property

4. **Column – I**

Column – II

A) Let $f(x) = \min(|x|, 1 - |x|, 1/4), \forall x \in R$,

p) 59/6

then value of $\int_{-1}^1 f(x) dx$ is equal to

B) Let $f(x) = \int_0^x (t^2 - t + 1) dt, \forall x \in [3, 4]$, then the difference q) 1/14

between the greatest and the least values of the function is

C) The integral $\int_{\pi/4}^{5\pi/4} (|\cos t| \sin t + |\sin t| \cos t) dt$ r) 0

has the value equal to

D) $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{\sqrt{n}}{\sqrt{r}(3\sqrt{r} + 4\sqrt{n})^2}$ is equal to s) 3/8

A. A – s ; B – q ; C – r ; D – p

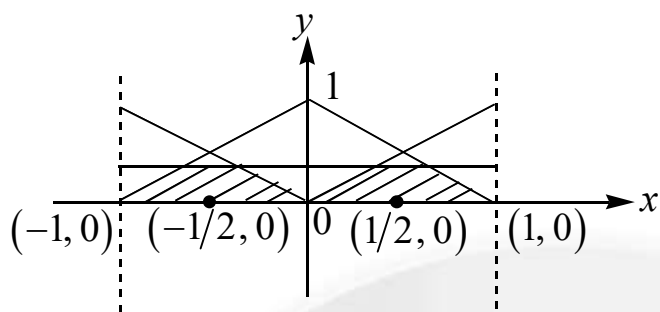
B. A – s ; B – r ; C – p ; D – q

C. A – s ; B – p ; C – r ; D – q

D. A – p ; B – r ; C – s ; D – q

Key. C

sol. A) $\int_{-1}^1 f(x) dx = 2 \int_0^1 f(x) dx = 2 \left(\int_0^{1/4} x dx + \int_{1/4}^{3/4} \frac{1}{4} dx + \int_{3/4}^1 (1-x) dx \right)$



$$= 2 \left(\frac{1}{2} \times \frac{1}{4} \times \frac{1}{4} + \frac{1}{4} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{4} \times \frac{1}{4} \right) = \frac{3}{8}$$

B) $f'(x) = x^2 - x + 1 > 0 \forall x \in \mathbb{R}$

$\therefore f(x)$ is increasing in $x \in (3, 4)$ so minimum at $x = 3$ and maximum at $x = 4$

$$\therefore \min f(x) = \int_0^3 (t^2 - t + 1) dt = \left[\frac{t^3}{3} - \frac{t^2}{2} + t \right]_0^3 = \frac{15}{2}$$

$$\max f(x) = \int_0^4 (t^2 - t + 1) dt = \left[\frac{t^3}{3} - \frac{t^2}{2} + t \right]_0^4 = \frac{52}{3}$$

$$\therefore \max f(x) - \min f(x) = \frac{59}{6}$$

c) Conceptual

$$\text{d) } I = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{\sqrt{n}}{\sqrt{r} \cdot n \left(3\sqrt{\frac{r}{n}} + 4 \right)^2} = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{\sqrt{\frac{r}{n}} \left(3\sqrt{\frac{r}{n}} + 4 \right)^2} \cdot \frac{1}{n} = \int_0^1 \frac{1}{\sqrt{x} (3\sqrt{x} + 4)^2} dx$$

$$\text{Put } z = 3\sqrt{x} + 4 \Rightarrow I = \frac{2}{3} \int_4^7 \frac{dz}{z^2} = \frac{1}{14}$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 2: Replacing x by $a-x$ – King's property, Replacing x by $a+b-x$

5. Match the following

[2010-P2 (Q38)] **(D)**

List-I	List-II
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P	A line from the origin meets the lines $\frac{x-2}{1} = \frac{y-1}{-2} = \frac{z+1}{1}$ and $\frac{x-\frac{8}{3}}{2} = \frac{y+3}{-1} = \frac{z-1}{1}$ at P and Q respectively. If length $PQ = d$, then d^2 is	1	-4
Q	The values of x satisfying $\tan^{-1}(x+3) - \tan^{-1}(x-3) = \sin^{-1}\left(\frac{3}{5}\right)$ are	2	0
R	Non-zero vectors $\vec{a}, \vec{b}$ and $\vec{c}$ satisfy $\vec{a} \cdot \vec{b} = 0$, $(\vec{b} - \vec{a}) \cdot (\vec{b} + \vec{c}) = 0$ and $2 \vec{b} + \vec{c} = \vec{b} - \vec{a} $. If $\vec{a} = \mu\vec{b} + 4\vec{c}$, then the possible values of μ are	3	4
S	Let f be the function on $[-\pi, \pi]$ given by $f(0) = 9$ and $f(x) = \sin\left(\frac{9x}{2}\right) / \sin\left(\frac{x}{2}\right)$ for $x \neq 0$. The value of $\frac{2}{\pi} \int_{-\pi}^{\pi} f(x) dx$ is	4	5
		5	6

A. $P \rightarrow 5; Q \rightarrow 1, 3; R \rightarrow 2, 4; S \rightarrow 3$

B. $P \rightarrow 3; Q \rightarrow 1, 2; R \rightarrow 3, 4; S \rightarrow 2$

C. $P \rightarrow 5; Q \rightarrow 1, 3; R \rightarrow 3, 4; S \rightarrow 2$

D. $P \rightarrow 5; Q \rightarrow 1, 3; R \rightarrow 5, 4; S \rightarrow 3$

Ans: A (PROPERTIES OF EVALUATION OF DEFINITE INTEGRALS)

Sol: Let the line through origin is $\frac{x}{\lambda} = \frac{y}{\mu} = \frac{z}{1}$

$$\Rightarrow x = \lambda z, y = \mu z \dots\dots(1)$$

To find point of intersection of line (1) and line $\frac{x-2}{1} = \frac{y-1}{-2} = \frac{z+1}{1} \dots\dots(2)$

We have $\frac{\lambda z - 2}{1} = \frac{\mu z - 1}{-2} = z + 1$

$$\Rightarrow z = \frac{3}{\lambda - 1} = \frac{-1}{\mu + 2}$$

$$\Rightarrow \lambda + 3\mu + 5 = 0 \dots\dots\dots(3)$$

To find point of intersection of line (1) and line $\frac{x-\frac{8}{3}}{2} = \frac{y+3}{-1} = \frac{z-1}{1} \dots\dots\dots(4)$

We have $\frac{\lambda z - \frac{8}{3}}{2} = \frac{\mu z + 3}{-1} = \frac{z-1}{1}$

$$\Rightarrow z = \frac{2}{3(\lambda-2)} = \frac{-2}{\mu+1}$$

$$\Rightarrow 3\lambda + \mu = 5 \dots\dots(5)$$

Solving (3) and (5), $\lambda = \frac{5}{2}$ and $\mu = -\frac{5}{2}$

$\therefore z = 2, x = 5, y = -5$ for point P

And $z = \frac{4}{3}, x = \frac{10}{3}, y = -\frac{10}{3}$ for point Q

$$\therefore PQ^2 = \frac{4}{9} + \frac{25}{9} + \frac{25}{9} = 6$$

(B)

$$\tan^{-1}(x+3) - \tan^{-1}(x-3) = \sin^{-1}(3/5)$$

$$\Rightarrow \tan^{-1}\left(\frac{x+3-x-3}{1+x^2-9}\right) = \tan^{-1}(3/4)$$

$$\Rightarrow \frac{6}{x^2-8} = \frac{3}{4} \Rightarrow x^2 = 16$$

$$\therefore x = \pm 4$$

(C)

Since $\vec{a} \cdot \vec{b} = 0$

$\therefore$ Let $\vec{b} = \lambda_1 \hat{i}, \vec{a} = \lambda_2 \hat{j}$

Now $2|\vec{b} + \vec{c}| = |\vec{b} - \vec{a}|$ & $\vec{a} = \mu\vec{b} + 4\vec{c}$

$$\Rightarrow 2\left|\lambda_1 \hat{i} + \frac{\lambda_2 \hat{j} - \lambda_1 \mu \hat{j}}{4}\right| = \lambda_1 \hat{i} - \lambda_2 \hat{j}$$

$$|\lambda_1(4-\mu)\hat{i} + \lambda_2 \hat{j}| = 2|\lambda_1 \hat{i} + \lambda_2 \hat{j}|$$

Squaring

$$\lambda_1^2(4-\mu)^2 + \lambda_2^2 = 4\lambda_1^2 + 4\lambda_2^2$$

$$\Rightarrow 3\lambda_2^2 = (12 + \mu^2 - 8\mu)\lambda_1^2 \dots\dots(1)$$

Also $(\vec{b} - \vec{a}) \cdot (\vec{b} + \vec{c}) = 0$

$$\Rightarrow (\lambda_1 \hat{i} - \lambda_2 \hat{j}) \cdot \left(\lambda_1 \hat{i} + \frac{\lambda_2 \hat{j} - \lambda_1 \mu \hat{i}}{4} \right) = 0 \Rightarrow \frac{\lambda_1^2 (4 - \mu) - \lambda_2^2}{4} = 0$$

$$\Rightarrow \lambda_2^2 = \lambda_1^2 (4 - \mu) \dots\dots(2) \quad \text{From (1) \& (2)}$$

$$12 + \mu^2 - 8\mu = 12 - 3\mu$$

$$\Rightarrow \mu^2 - 5\mu = 0 \Rightarrow \mu = 0, 5$$

$$(D) \quad I = \frac{2}{\pi} \int_{\pi}^{\frac{\pi}{2}} \frac{\sin \frac{9x}{2}}{\sin \frac{x}{2}} dx = \frac{4}{\pi} \int_0^{\frac{\pi}{2}} \frac{\sin \frac{9x}{2} \cos \frac{x}{2}}{\sin \frac{x}{2} \cos \frac{x}{2}} dx = \frac{8}{\pi} \int_0^{\frac{\pi}{2}} \frac{\sin \frac{9x}{2} \cos \frac{x}{2}}{\sin x} dx$$

$$I = \frac{4}{\pi} \int_0^{\frac{\pi}{2}} \frac{\sin 5x + \sin 4x}{\sin x} dx \dots\dots(i)$$

$$(\text{using } \int_0^b f(x) dx = \int_0^b f(a+b-x) dx)$$

$$= \frac{4}{\pi} \int_0^{\frac{\pi}{2}} \frac{\sin 5x - \sin 4x}{\sin x} dx \dots\dots(ii)$$

Add (i) and (ii)

$$I = \frac{4}{\pi} \int_0^{\frac{\pi}{2}} \frac{\sin 5x}{\sin x} dx$$

Consider

$$I_k - I_{k-2} = \frac{4}{\pi} \int_0^{\frac{\pi}{2}} \frac{\sin kx - \sin(k-2)}{\sin x} dx = \frac{8}{\pi} \int_0^{\frac{\pi}{2}} \frac{\cos(k-1)x \sin x}{\sin x} dx$$

$$I_k = I_{k-2}$$

$$\text{so } I_5 - I_3 \Rightarrow I_5 = I_1 = \frac{4}{\pi} \int_0^{\frac{\pi}{2}} dx = 4$$

Aliter

$$\text{Let } I = \frac{2}{\pi} \int_{-\pi}^{\frac{\pi}{2}} \frac{\sin(9x/2)}{\sin(x/2)} dx$$

$$I = \frac{4}{\pi} \int_0^{\frac{\pi}{2}} \frac{\sin(9x/2)}{\sin(x/2)} dx \dots\dots(1) \quad (\because f(x) \text{ is even function})$$

$$I = \frac{4}{\pi} \int_0^{\frac{\pi}{2}} \frac{\cos(9x/2)}{\cos(x/2)} dx \dots\dots(2)$$

$$(\text{using } \int_0^b f(x) dx = \int_0^b f(a+b-x) dx)$$

Add (1) \& (2)

$$\begin{aligned}
 I &= \frac{4}{\pi} \int_0^{\pi} \frac{\sin 5x}{2 \sin(x/2) \cos(x/2)} dx = \frac{4}{\pi} \int_0^{\pi} \frac{\sin 5x}{\sin x} dx \\
 \Rightarrow I &= \frac{8}{\pi} \int_0^{\pi/2} \frac{\sin 5x}{\sin x} dx \\
 \Rightarrow I &= \frac{8}{\pi} \int_0^{\pi/2} \left(\frac{16 \sin^5 x - 20 \sin^3 x + 5 \sin x}{\sin x} \right) dx \\
 \Rightarrow I &= \frac{8}{\pi} \int_0^{\pi/2} \left(16x \frac{3 \times 1 \times \pi}{4 \times 2 \times 2} - 20 \times \frac{1 \times \pi}{2 \times 2} + \frac{5\pi}{2} \right) \\
 \Rightarrow I &= \frac{8}{\pi} \left[3\pi - 5\pi - \frac{5\pi}{2} \right] \\
 \Rightarrow I &= 4
 \end{aligned}$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 3: Even - odd property

6. Match the following

[2014-P2 (Q58)]

List-I		List-II	
P	The number of polynomials $f(x)$ with non-negative integer Coefficients of degree ≤ 2 , satisfying $f(0)=0$ and $\int_0^1 f(x)dx=1$, is	1	8
Q	The number of points in the interval $[-\sqrt{13}, \sqrt{13}]$ at which $f(x) = \sin(x^2) + \cos(x^2)$ attains its maximum value, is	2	2
R	$\int_{-2}^2 \frac{3x^2}{(1+e^x)} dx$ equals	3	4
S	$\left(\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos 2x \log \left(\frac{1+x}{1-x} \right) dx \right)$ equals $\left(\int_0^{\frac{1}{2}} \cos 2x \log \left(\frac{1+x}{1-x} \right) dx \right)$	4	0

- | | | | | | | | | | |
|----|---|---|---|---|----|---|---|---|---|
| | P | Q | R | S | | P | Q | R | S |
| A. | 3 | 2 | 4 | 1 | B. | 2 | 3 | 4 | 1 |
| C. | 3 | 2 | 1 | 4 | D. | 2 | 3 | 1 | 4 |

Ans: D (a+b-x) properties find variable through equations D.I properties

Sol: (P) $f(x) = ax^2 + bx, \int_0^1 f(x) dx = 1 \Rightarrow 2a + 3b = 6$

$\Rightarrow (a, b) \equiv (0, 2) \text{ and } (3, 0) \quad (Q) f(x) = \sqrt{2} \cos\left(x^2 - \frac{\pi}{4}\right)$

for maximum value, $x^2 - \frac{\pi}{4} = 2n\pi \Rightarrow x^2 = 2n\pi + \frac{\pi}{4}$

$\Rightarrow x = \pm \sqrt{\frac{\pi}{4}}, \sqrt{\frac{9\pi}{4}} \text{ as } x \in [-\sqrt{3}, \sqrt{13}]$

(R) $\int_0^2 \left(\frac{3x^2}{1+e^x} + \frac{3x^2}{1+e^{-x}} \right) dx = \int_0^2 3x^2 dx = 8$

(S) $\int_{-1/2}^{1/2} \cos 2x \ln\left(\frac{1+x}{1-x}\right) dx = 0$ as it is an odd function

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 4: Half the integral limit property (Splitting of limits 0 to 2a as 0 to a and a to 2a in f(x)), f(2a-x) property, Greatest integer function

7. In the following [] represents G.I.F

Column-I	Column-II
A) $\int_0^{\frac{\pi}{2}} \left[\frac{1+2\sin^2 x}{1+\sin^2 x} \right] dx$	p) $\frac{\pi}{4}$
B) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} [\cos x - \cos^2 x] dx$	q) 0
C) $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \left[\frac{\sec x + \cos ecx - \sec x \cos ecx}{2} \right] dx$	r) π
D) $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \left[\frac{\cot x - \tan x}{2} \right] dx = \text{---}$	s) $\frac{\pi}{2}$
	t) $\frac{\pi}{3}$

KEY:

SOL:

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A) When $0 \leq x \leq \frac{\pi}{2}$ $\frac{1+2\sin^2 x}{1+\sin^2 x} \in \left[1, \frac{3}{2}\right]$

B) When $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ $\cos x - \cos^2 x \in \left[0, \frac{1}{4}\right]$

$$\therefore f(x) \in \left[\sqrt{2}-1, 1-\frac{1}{\sqrt{3}}\right]$$

C) When $\frac{\pi}{6} \leq x \leq \frac{\pi}{4}$, Then $\left[\frac{1}{1+\sin x + \cos x}\right] = 0$

D) In $\frac{\pi}{6} \leq x \leq \frac{\pi}{4}$, $\frac{\cot x - \tan x}{2} = \cot 2x$

$$0 \leq \cot 2x \leq \frac{1}{\sqrt{3}} \quad [\cot 2x] = 0$$

Topic: Evaluation of definite integrals

8. Match the following

[2011-P2 (Q59)] (C)

	List-I	List-II	
P	If $\vec{a} = \hat{j} + \sqrt{3}\hat{k}$, $\vec{b} = -\hat{j} + \sqrt{3}\hat{k}$ and $\vec{c} = 2\sqrt{3}\hat{k}$ form a triangle, then the internal angle of the triangle between $\vec{a}$ and $\vec{b}$ is	1	$\frac{\pi}{6}$
Q	If $\int_a^b (f(x) - 3x) dx = a^2 - b^2$, then the value of $f\left(\frac{\pi}{6}\right)$ is	2	$\frac{2\pi}{3}$
R	The value of $\frac{\pi^2}{\ln 3} \int_{\frac{1}{6}}^{\frac{5}{6}} \sec(\pi x) dx$ is	3	$\frac{\pi}{3}$
S	The maximum value of $\left \operatorname{Arg}\left(\frac{1}{1-z}\right) \right $ for $ z =1$, $z \neq 1$ is given by	4	π
		5	$\frac{\pi}{2}$

A. $P \rightarrow 2; Q \rightarrow 1, 2, 3, 4, 5; R \rightarrow 4; S \rightarrow$

B. $P \rightarrow 3; Q \rightarrow 1, 2; R \rightarrow 1, 2, 3, 4; S \rightarrow 2$

C. $P \rightarrow 5; Q \rightarrow 1, 2, 3; R \rightarrow 3, 4; S \rightarrow 2$

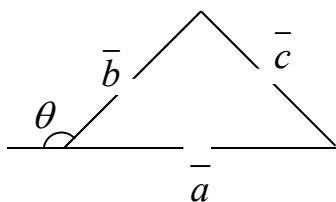
D. $P \rightarrow 2, 3, 4, 5; Q \rightarrow 1, 3; R \rightarrow 5, 4; S \rightarrow 3$

Ans: A

(Properties of evaluation definite integrals)

Sol: (P) $\vec{a} + \vec{b} = \vec{c}$

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$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{-1+3}{2 \times 2} = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}$$

$$\text{Required angle} = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

$$(Q) \int_a^b f(x) dx = \frac{b^2 - a^2}{2} \Rightarrow f\left(\frac{\pi}{6}\right) = \text{can take many values}$$

$$(R) \frac{\pi^2}{\ln 3} \int_{7/6}^{5/6} (\sec \pi x) dx = \frac{\pi^2}{\ln 3} \frac{[\ln |\sec \pi x + \tan \pi x|]_{7/6}^{5/6}}{\pi}$$

$$= \frac{\pi}{\ln 3} \left(\ln \left| \frac{3}{\sqrt{3}} \right| - \ln \left| \frac{1}{\sqrt{3}} \right| \right) = \pi$$

$$(S) \left| \text{Arg} \left(\frac{1}{1-z} \right) \right| < \frac{\pi}{2}$$

Hence maxima does not exist.

EXERCISE - VII
(PASSAGE TYPE)

Topic: Evaluation of definite integrals

Sub Topic 1: Evaluation of definite integrals by substitution

PARAGRAPH FOR QUESTIONS 1 AND 2

Given that for each $a \in (0, 1)$, $\lim_{h \rightarrow 0^+} \int_h^{1-h} t^{-a} (1-t)^{a-1} dt$ exists. Let this limit be $g(a)$. is differentiable on $(0, 1)$.

1. The value of $g\left(\frac{1}{2}\right)$ is [2014-P2 (Q53)]

A. π B. 2π C. $\frac{\pi}{2}$ D. $\frac{\pi}{4}$ (D.I)

Ans: A

$$\begin{aligned} \text{Sol: } g\left(\frac{1}{2}\right) &= \lim_{h \rightarrow 0^+} \int_h^{1-h} t^{-1/2} (1-t)^{-1/2} dt \\ &= \int_0^1 \frac{dt}{\sqrt{t-t^2}} = \int_0^1 \frac{dt}{\sqrt{\frac{1}{4} - \left(t - \frac{1}{2}\right)^2}} = \sin^{-1} \left(\frac{t - \frac{1}{2}}{\frac{1}{2}} \right) \Bigg|_0^1 \\ &= \sin^{-1} 1 - \sin^{-1}(-1) = \pi \end{aligned}$$

2. The value of $g'\left(\frac{1}{2}\right)$ is [2014-P2 (Q54)]

A. $\frac{\pi}{2}$ B. π C. $-\frac{\pi}{2}$ D. 0 (D.I derivatives)

Ans: D

Sol: We have $g(a) = g(1-a)$ and g is differentiable

$$\text{Hence } g'\left(\frac{1}{2}\right) = 0$$

Paragraph For Questions 1 and 2:

Let f be a differentiable function f satisfies the relation

$$f(xy) = f(x) + f(y) + xy - x - y \quad \forall \quad x, y > 0, f'(1) = 4 \quad \text{Then answer the following}$$

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1. $\int \frac{f(x)}{x} dx =$ ('c' is integration constant)

- A) $3(\ln x)^2 + x + c$ B) $3\ln x + x + c$ C) $\frac{3}{2}\ln x + x + c$ D) $\frac{3}{2}(\ln x)^2 + x + c$

2. $\int_2^3 (e^{f(x)-x} + x) dx$

- A) 0 B) $\frac{75}{4}$ C) $\frac{1}{4}$ D) $\frac{75}{2}$

Key: 51) D, 52) B

Sol: $f(x) = 3\ln x + x$

PARAGRAPH FOR QUESTIONS 1 AND 2

Given that for each $a \in (0, 2)$, $\lim_{h \rightarrow 0^+} \int_h^{2-h} t^{-a} (2-t)^{-a} dt$ exists. Let this limit be $g(a)$. is differentiable on $(0, 2)$.

1. The value of $g\left(\frac{1}{2}\right)$ is

- A. π B. 2π C. $\frac{\pi}{2}$ D. $\frac{\pi}{4}$

Ans: A

Sol: $g\left(\frac{1}{2}\right) = \lim_{h \rightarrow 0^+} \int_h^2 t^{-1/2} (2-t)^{-1/2} dt = \frac{\pi}{2}$

$= \sin^{-1} 1 - \sin^{-1}(-1) = \pi$

2. The value of $g'(1)$ is

- A. $\frac{\pi}{2}$ B. π C. $-\frac{\pi}{2}$ D. 0

Ans: D

Sol: We have $g(a) = g(2-a)$ and g is differentiable

Hence $g'(1) = 0$

2. The value of the expression $\frac{\int_0^a x^4 \sqrt{a^2 - x^2} dx}{\int_0^a x^2 \sqrt{a^2 - x^2} dx}$ is equal to ($a \neq 0$)

A) $\frac{a^2}{6}$

B) $\frac{3a^2}{2}$

C) $\frac{3a^2}{4}$

D) $\frac{a^2}{2}$

Ans. D

$$\text{Sol: } \int_0^a x^4 \sqrt{a^2 - x^2} dx = \left[\frac{-x^3 (a^2 - x^2)^{3/2}}{3} \right]_0^a + a^2 \cdot \frac{3}{6} \int_0^a x^2 \sqrt{a^2 - x^2} dx = \frac{a^2}{2} \int_0^a x^2 \sqrt{a^2 - x^2} dx$$

Topic: Evaluation of definite integrals

Sub Topic 2: Properties of evaluation of definite integrals

Mini Topic 2: Replacing x by $a-x$ – King's property, Replacing x by $a+b-x$

PARAGRAPH FOR QUESTIONS 1 AND 2

Let $g_i : \left[\frac{\pi}{8}, \frac{3\pi}{8} \right] \rightarrow \mathbb{R}, i=1,2$ and $f : \left[\frac{\pi}{8}, \frac{3\pi}{8} \right] \rightarrow \mathbb{R}$ be functions such that

$g_1(x) = 1, g_2(x) = |4x - \pi|$ and $f(x) = \sin^2 x$, for all $x \in \left[\frac{\pi}{8}, \frac{3\pi}{8} \right]$ Define

$$S_i = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} f(x) \cdot g_i(x) dx, i=1,2$$

1. The value of $\frac{16S_1}{\pi}$ is _____ [2021-P2 (Q11)]

Ans: 2 (Properties of evaluation definite integrals)

$$\text{Sol: } g_i : \left[\frac{\pi}{8}, \frac{3\pi}{8} \right] \rightarrow \mathbb{R}, i=1,2, f : \left[\frac{\pi}{8}, \frac{3\pi}{8} \right] \rightarrow \mathbb{R}$$

$$g_1 = 1, g_2 = |4x - \pi|, f(x) = \sin^2 x$$

$$S_i = \int_{\pi/8}^{3\pi/8} f(x) \cdot g_i(x) dx$$

$$S_1 = \int_{\pi/8}^{3\pi/8} \sin^2 x dx = \int_{\pi/8}^{3\pi/8} \sin^2 \left(\frac{\pi}{2} - x \right) dx \Rightarrow 2S_1 = \int_{\pi/8}^{3\pi/8} 1 dx$$

$$\Rightarrow S_1 = \frac{1}{2} \left(\frac{3\pi}{8} - \frac{\pi}{8} \right) = \frac{\pi}{8} \Rightarrow \frac{16S_1}{\pi} = 2$$

2. The value of $\frac{48S_2}{\pi^2}$ is _____

[2021-P2 (Q12)]

Ans: 1.5 (Properties of evaluation definite integrals)

$$\text{Sol: } S_2 = \int_{\pi/8}^{3\pi/8} \sin^2 x |4x - \pi| dx = \int_{\pi/8}^{3\pi/8} \sin^2 \left(\frac{\pi}{2} - x \right) \left| 4 \left(\frac{\pi}{2} - x \right) - \pi \right| dx$$

$$= \int_{\pi/8}^{3\pi/8} \cos^2 x |4\pi - 1| dx$$

$$\Rightarrow 2S_2 = \int_{\pi/8}^{3\pi/8} |4x - \pi| dx = 2 \int_{\pi/8}^{\pi/4} (\pi - 4x) dx$$

$$S_2 = \left(\pi x - 2x^2 \right) \Big|_{\pi/8}^{\pi/4} = \pi \left(\frac{\pi}{4} - \frac{\pi}{8} \right) - 2 \left(\frac{\pi^2}{16} - \frac{\pi^2}{64} \right)$$

$$S_2 = \frac{4\pi^2}{8} - \frac{3\pi^2}{32} = \frac{\pi^2}{32} = \frac{48S_2}{\pi^2} = \frac{48}{\pi^2} \times \frac{\pi^2}{32} = \frac{3}{2}$$

$$\frac{48S_2}{\pi^2} = 1.5$$

Topic: Evaluation of definite integrals

SUBTOPIC : BY PARTS

PASSAGE :

Information for questions a, b

If a_1, a_2 and a_3 are the three values of a which satisfy the equation

$$\int_0^{\pi/2} (\sin x + a \cos x)^3 dx - \frac{4a}{\pi - 2} \int_0^{\pi/2} x \cos x dx = 2 \text{ then}$$

1) $2(a_1 + a_2 + a_3) =$

A) 2

B) -3

C) 3

D) 1

Ans: B

2) $4 \sum a_i^2 =$

A) 21

B) -23

C) 23

D) -21

Ans: A

Sol for a and B:

$$\text{Let } I_1 = \int_0^{\pi/2} (\sin x + a \cos x)^3 dx \text{ and } I_2 = \int_0^{\pi/2} x \cos x dx$$

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$$I_1 = \int_0^{\pi/2} (\sin^3 x + a^3 \cos^3 x + 3a \sin^2 x \cos x + 3a^2 \sin x \cos^2 x) dx$$

$$= \frac{2}{3} (1 + a^3) + 3a \left(1 - \frac{2}{3}\right) + 3a^2 \left(1 - \frac{2}{3}\right)$$

$$\therefore I_1 = \frac{2a^3}{3} + a^2 + a + \frac{2}{3}$$

$$I_2 = x \sin x + \cos x \Big|_0^{\pi/2}$$

$$a_1 + a_2 + a_3 = -\frac{3}{2} \Rightarrow \sum a_1 a_2 = -\frac{3}{2}$$

$$\therefore \sum a_1^2 = \frac{9}{4} + \frac{6}{2} = \frac{21}{4}$$

$$\therefore 1000 \sum a_1^2 = 1000 \times \frac{21}{4} = 250 \times 21 = 5250$$

Topic: Evaluation of definite integrals

Passage-1

If $f: \mathbb{R} \setminus \{-1\} \rightarrow \mathbb{R}$ and f is differentiable function that satisfies the equation

$f(x + f(y) + xf(y)) = y + f(x) + y f(x)$, $\forall x, y \in \mathbb{R} - \{-1\}$ and $f(x) \neq x$, then

1. $\int_0^1 \left\{ f(x) + f\left(\frac{1}{x}\right) \right\} dx$ equals

- (A) 2 (B) -2 (C) -1 (D) 1

Key: 2)C

Sol: Differentiating both side with respect to x and then w.r.t. to y and then dividing the result obtained in both cases.

$$\text{We get } f'(x) = \pm \frac{(1+f(x))}{1+x} \Rightarrow \frac{1+f(x)}{c} = (1+x)^{\pm 1}$$

Now, putting $x = 0$, $y = 0$, we get

$$f(c-1) = (c-1)$$

$$\Rightarrow c = 0, 1$$

$$\therefore f(x) = -\frac{x}{1+x}$$